

Subject: Defense Approval

To: Department of Manufacturing Engineering and Management  
De La Salle University, Manila

From:   
Dr. Ira C. Valenzuela  
Thesis Adviser

Date: April 08, 2025

---

The undergraduate thesis entitled **Vision-Based Mini-Production Line for Grading and Sorting of Iceberg Lettuce** prepared by **Alberto, Alexi; Forones, Romell; Galang, Gabriel; Magallanes, Fernando Jr.; Mandigma, Elmo**, in partial fulfilment of the requirements for the Degree of Bachelor of Science in Manufacturing Engineering and Management is hereby accepted with a grade of \_\_\_\_\_.

Endorsed by:



Dr. Ira C. Valenzuela  
Thesis Adviser

Panel of Advisers

Approved by the Committee on Oral Defense dated Month Day, Year  
2<sup>nd</sup> Term AY 2024-2025



Dr. Renann G. Baldovino  
Panel Chairperson



Dr. Robert Kerwin C.  
Billones  
Panel Member



Dr. Ronnie S. Concepcion II  
Panel Member

Accepted and approved in partial fulfilment of the requirements for the Degree of Bachelor of Science in Manufacturing Engineering and Management.



Engr. Richard Josiah C. Tan Ai  
Thesis Coordinator



Dr. Ryan Rhay P. Vicerra  
Department Chairman



**ANIMO REPOSITORY SUBMISSION CONSENT FORM A**  
**UG/GS ETD Not Covered by [DLSU Intellectual Property Policy Section 6.2 \(2021\)](#)**

The consent form below provides authorization for the permanent addition of ETDs to the Animo Repository for discoverability and wider dissemination of knowledge. A scanned copy of this form should be included as part of the preliminary pages of the ETD final PDF version.

I/we am/are executing this form in accordance with [Section 6.2 of the DLSU Intellectual Property Policy \(2021\)](#).

I/We hereby grant De La Salle University Libraries ("the Libraries") the royalty-free, non-exclusive license to manage and archive the work(s) identified below ("the submission") through the [Animo Repository](#) platform, subject to the provision of applicable laws.

Title of work : Vision-Based Mini-Production Line for Grading and Sorting of Iceberg Lettuce

---

I/We also grant the Libraries the right to make the full version of this submission accessible in electronic format for educational and research purposes only, based on the release permission below.

**Release Permission** (Please check only ONE box. Please consult your thesis adviser before ticking any box.)

- ☐ **Available to library patrons immediately (no embargo)**
- ☒ **Available to library patrons after the embargoed period** (Choose this option if you and your thesis adviser intend to publish this work and/or protect your intellectual property. The metadata of your work will still be searchable in the Animo Repository during the embargo period.)
- ☒ 1 year    ☐ 2 years    ☐ 3 years
- ☐ **Not available to library patrons** (For storage only. However, the metadata of your work will still be searchable in the Animo Repository.)

I/We agree that the Libraries may, without changing the content, convert and migrate this submission to any medium or format, and keep more than one copy for purposes of security, back-up, and preservation.

I/we am/are giving my/our consent for the Libraries to collect, store, and share my/our personal information, including my/our personal name, course, department, college, and institution, through the Animo Repository. I/we further agree that the personal data collected will only be used for the academic community and the general public to search, retrieve, and locate my/our work in the Animo Repository for educational and research purposes.

I/we understand that the security of the collected personal information shall be the responsibility of the Libraries and should comply with the requirements of RA 10173. I/we further consent that the personal data collected and included in the metadata in the Animo Repository will be permanently stored.

I/We further agree that this consent will remain in place unless I/we terminate it via signed communication with the Libraries.

I/We confirm that:

- The version I/we submitted is the final version approved by my/our thesis committee / panel.
- The submission is my/our original work and I/we have the right to grant the license above mentioned to the Libraries.



- The submission, does not, to the best of my/our knowledge, infringe upon any third-party's copyright.
- If the submission contains material for which I/we do not hold copyright, I/we will present a certification that I/we have obtained the unrestricted permission of the copyright holder to grant this license to the Libraries, subject to the limitations of copyright and fair use under the IP Code. I/we further certify that such third-party material is clearly identified and acknowledged within the text or content of this submission.
- In the event of a subsequent dispute over the copyrights to material contained in this submission, I/we agree to indemnify and hold harmless the Libraries and its employees or agents for any uses of the material authorized by this license.
- This submission properly cites other sources of information and contains permitted works from original sources (e.g., text excerpts, images, sound or video content, interview transcripts, and observational studies).
- If the submission is a commissioned work, I/we have fulfilled all obligations required by De La Salle University and/or other funding agencies.
- The submission does not contain any confidential, personal, or sensitive information belonging to others.
- I/We have read and understood the contents of the [DLSU Intellectual Property Policies](#) and have taken the necessary steps to ensure adherence to the rules set forth in the above-mentioned policies.
- The information I/we have provided about the submission is accurate to the best of my/our knowledge and belief.

By signing below, I/we confirm my/our acceptance of this consent.

|                                                                                                    |                                                            |
|----------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| <div>Fernando N. Magallanes Jr.</div> <div>Student's _____</div> <div>Printed Name</div>           | <div>April 10, 2025</div> <div>_____</div> <div>Date</div> |
| <div>Alexi S. Alberto</div> <div>Student's Signature _____</div> <div>_____</div>                  | <div>April 10, 2025</div> <div>_____</div> <div>Date</div> |
| <div>Romell M. Forones</div> <div>Student's _____</div> <div>Printed Name</div>                    | <div>April 10, 2025</div> <div>_____</div> <div>Date</div> |
| <div>Juan Gabriel L. Galang</div> <div>Student's Signature _____</div> <div>_____</div>            | <div>April 10, 2025</div> <div>_____</div> <div>Date</div> |
| <div>Elmo F. Manuila</div> <div>Student's Signature over Printed Name _____</div> <div>_____</div> | <div>April 10, 2025</div> <div>_____</div> <div>Date</div> |
| <div>Student's Signature over Printed Name _____</div> <div>_____</div>                            | <div>_____</div> <div>_____</div> <div>Date</div>          |
| <div>Student's Signature over Printed Name _____</div> <div>_____</div>                            | <div>_____</div> <div>_____</div> <div>Date</div>          |



CLEARED FOR ANIMO REPOSITORY SUBMISSION BY:

Check the appropriate column for your answer.

*"As the thesis adviser, I have discussed with my advisee/s the contents of and reason for this form."*

|                                                                                                                   |                                           |
|-------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| <div>Dr. Ira C. Valenzuela</div> <div>Thesis Adviser</div> <div>(Signature over Printed Name)</div>               | <div>April 10, 2025</div> <div>Date</div> |
| <div>Dr. Kenneth G. Baldovino</div> <div>Thesis Panel Member</div> <div>(Signature over Printed Name)</div>       | <div>April 10, 2025</div> <div>Date</div> |
| <div>Dr. Ronnie S. Concepcion II</div> <div>Thesis Panel Member</div> <div>(Signature over Printed Name)</div>    | <div>April 10, 2025</div> <div>Date</div> |
| <div>Dr. Robert Kerwin C. Biliñes</div> <div>Thesis Panel Member</div> <div>(Signature over Printed Name)</div>   | <div>April 10, 2025</div> <div>Date</div> |
| <div>Engr. Richard Josiah C. Tan Ai</div> <div>Thesis Panel Member</div> <div>(Signature over Printed Name)</div> | <div>April 10, 2025</div> <div>Date</div> |
| <div>Dr. Ryan Rhay P. Vicerra</div> <div>Research Coordinator</div> <div>(Signature over Printed Name)</div>      | <div>April 10, 2025</div> <div>Date</div> |
| <div></div> <div>Department Chair</div> <div>(Signature over Printed Name)</div>                                  | <div></div> <div>Date</div>               |

I agree to have my name included and searchable in the Animo Repository.

I want to stay anonymous.

☒☐☒☐☒☐☒☐☒☐☒☐



# De La Salle University



A Thesis Presented to the  
Department of Manufacturing Engineering and Management  
Gokongwei College of Engineering  
De La Salle University

In partial fulfillment of the requirements for the course  
**MEM Thesis 2 (THSMFE2)**

**Vision-Based Mini-Production Line for Grading and Sorting of Iceberg Lettuce**

By:

**Alberto, Alexi S.**

**Forones, Romell M.**

**Galang, Juan Gabriel L.**

**Magallanes, Fernando Jr. N.**

**Mandigma, Elmo F.**

**Dr. Ira C. Valenzuela**

Thesis Adviser

**April 2025**



## ABSTRACT

In the Philippines, iceberg lettuce (*Lactuca sativa* var. *capitata*) is a priority crop that lacks an efficient postharvest system, with quality grading and sorting still performed manually by farmers. To optimize post-processing time and maintain quality, an automated grading and sorting system for iceberg lettuce was developed.

This study aimed to design a vision-based quality grading and sorting system within a mini-production line, integrating computer vision and automation to enhance postharvest handling efficiency. Multi-gene symbolic regression programming (MSRGP) was employed to develop predictive models for chlorophyll-*a* content, moisture content, and fresh head weight. The resulting models exhibited varying levels of accuracy, with  $R^2$  values of 0.72 for chlorophyll-*a*, 0.431 for moisture content, and 0.565 for fresh head weight. Prediction accuracies for these parameters were 89.97%, 85.38%, and 92.80%, respectively, with the highest accuracy observed in moisture content prediction. The three quality parameters were utilized as inputs for fuzzification in a Mamdani-Type 1 fuzzy logic system, which classified iceberg lettuce into three grades: Grade 1 (high quality), Grade 2 (medium quality), and Grade 3 (low quality). The fuzzy logic prediction model demonstrated an average classification accuracy of 90%, correctly grading 27 out of 30 test samples. The automated sorting conveyor system, which accommodates five lettuce heads per batch, exhibited a repeatability rate of 100% at 70% speed and 92% at full operational speed upon five trials. Lastly, the throughput time achieved 8 lettuces per minute based on the system time study.

The findings highlight the potential of computer vision and fuzzy logic-based automation in optimizing postharvest systems, enhancing the farm-to-market supply chain for iceberg lettuce in the Philippines.



## ACKNOWLEDGEMENTS

The authors of this research, the Chlorophyters, would like to express their deepest gratitude to Dr. Ira Valenzuela-Estropia for her never-ending support, guidance, and trust throughout this research. They are especially grateful for the late nights she dedicated to guiding them, her patience and understanding throughout the whole process, and her constant encouragement, which motivated them to strive to do better. Special thanks to our panel chair, Dr. Renann Baldovino, and panel members, Dr. Ronnie Concepcion II and Dr. Robert Billones, for their insightful, constructive feedback and guidance.

We also like to acknowledge Anthony Velasquez for guiding us throughout the laboratory experimentation and genetic programming. Special thanks to the DLSU College of Science Microbiology Laboratory faculty, most especially, Dr. Llewelyn Espiritu, for providing us with the necessary tools for our iceberg lettuce extraction. Special thanks to the DLSU Department of Chemical Engineering for allowing us to use their research laboratory for our iceberg lettuce extraction. Thank you for DOST-SEI for partially funding this project to help a scholar who is a proponent of the study.

Special thanks to InfinityTools Mfg. Corp.'s, Engr. Rogelio Galang and Mrs. Hazel Galang, for accommodating us with the assembly of our prototype and providing the necessary tools. Special thanks to Engr. Dominic Taguibao for helping us with the prototype's assembly and providing invaluable insight on how to tackle various approaches. Special thanks to Engr. Venus Lague for helping us with the calculations regarding the conveyor design considerations and Engr. Clarisse Sta. Maria for assisting us in executing our methodology. Last but not the least, special thanks to our family, friends, and the whole DMEM faculty.



## TABLE OF CONTENTS

|                                                                                     |      |
|-------------------------------------------------------------------------------------|------|
| ABSTRACT .....                                                                      | i    |
| ACKNOWLEDGEMENTS .....                                                              | ii   |
| TABLE OF CONTENTS .....                                                             | iii  |
| LIST OF FIGURES .....                                                               | viii |
| LIST OF TABLES .....                                                                | xi   |
| Chapter 1 Introduction .....                                                        | 1    |
| 1.1 Background of the Study .....                                                   | 1    |
| 1.2 Statement of the Problem .....                                                  | 3    |
| 1.3 Significance of the Study .....                                                 | 3    |
| 1.4 Objectives .....                                                                | 5    |
| 1.4.1 General Objective .....                                                       | 5    |
| 1.4.2 Specific Objectives .....                                                     | 5    |
| 1.5 Scope and Delimitations .....                                                   | 5    |
| 1.5.1 Scope .....                                                                   | 5    |
| 1.5.2 Delimitations .....                                                           | 6    |
| 1.6 Engineering Standards .....                                                     | 7    |
| Chapter 2 Review of Related Literature .....                                        | 8    |
| 2.1 Literature Survey Strategy .....                                                | 8    |
| 2.1 Quality Parameters and Handling Practices for Postharvest Iceberg Lettuce ..... | 21   |
| 2.1.1 Postharvest Practices for Iceberg Lettuce .....                               | 26   |
| 2.1.2 Quality Grading Standards Related to Iceberg Lettuce .....                    | 29   |
| 2.1.3 Industry Standards for Postharvest Handling Systems .....                     | 32   |
| 2.2 Emerging Technologies on Postharvest Grading and Sorting Systems .....          | 35   |
| 2.3 Computer Vision in Postharvest Grading and Sorting Systems .....                | 37   |





|                                                    |    |
|----------------------------------------------------|----|
| 2.4 Relevant Patents .....                         | 41 |
| 2.5 Synthesis.....                                 | 42 |
| Chapter 3 Theoretical & Conceptual Framework ..... | 44 |
| 3.1 Theoretical Framework .....                    | 44 |
| 3.1.1 Iceberg Lettuce Postharvest Quality .....    | 45 |
| 3.1.2 Extraction of Quality Parameters .....       | 46 |
| 3.1.2.1 UV-VIS Spectrometry.....                   | 46 |
| 3.1.2.2 Oven Drying Method .....                   | 46 |
| 3.1.2.3 Weight Measurement .....                   | 47 |
| 3.1.3 Computer Vision .....                        | 47 |
| 3.1.3.1 RGB Color Space .....                      | 47 |
| 3.1.3.2 Hue Color Space.....                       | 48 |
| 3.1.3.3 Image Processing Techniques .....          | 48 |
| 3.1.3.4 Segmentation and Thresholding.....         | 48 |
| 3.1.3.5 Data Augmentation .....                    | 49 |
| 3.1.4 Computational Intelligence .....             | 49 |
| 3.1.4.1 Genetic Programming .....                  | 50 |
| 3.1.4.2 Fuzzy Logic.....                           | 51 |
| 3.1.5 Performance Metrics .....                    | 52 |
| 3.1.5.1 Regression Metrics.....                    | 52 |
| 3.1.5.2 Throughput Time .....                      | 53 |
| 3.1.5.3 Repeatability.....                         | 54 |
| 3.2 Conceptual Framework .....                     | 55 |
| Chapter 4 Methodology.....                         | 57 |



|                                                                            |    |
|----------------------------------------------------------------------------|----|
| 4.1 Engineering Standards and Design Constraints .....                     | 57 |
| 4.1 System Overview Flowchart .....                                        | 58 |
| 4.2 MSRGP Implementation Procedures .....                                  | 61 |
| 4.2.1 Extraction Processes.....                                            | 61 |
| 4.2.1.1 Image Capturing Process.....                                       | 61 |
| 4.2.1.2 Chlorophyll- <i>a</i> Extraction.....                              | 62 |
| 4.2.1.3 Moisture Content Extraction .....                                  | 63 |
| 4.2.2 MSRGP Modelling and Accuracy Testing.....                            | 65 |
| 4.2.3 MATLAB GPTIPS Toolbox .....                                          | 65 |
| 4.2.4 MSRGP Model Considerations and Results .....                         | 66 |
| 4.2.4.1 MSRGP models using RGB-HSV color space .....                       | 67 |
| 4.2.4.1.1 MSRGP output for chlorophyll- <i>a</i> using RGB-HSV input ..... | 67 |
| 4.2.4.1.2 MSRGP output for moisture content using RGB-HSV input .....      | 69 |
| 4.2.4.1.3 MSRGP models for fresh head weight using RGB-HSV input .....     | 71 |
| 4.2.4.2 MSRGP models using HSV only color space.....                       | 73 |
| 4.2.4.2.1 MSRGP output for chlorophyll- <i>a</i> using HSV input only..... | 73 |
| 4.2.4.2.2 MSRGP output for moisture content using HSV input only.....      | 75 |
| 4.2.4.2.3 MSRGP output for fresh head weight using HSV input only .....    | 77 |
| 4.3 Fuzzy Logic Implementation Procedures.....                             | 79 |
| 4.3.1 Fuzzy Logic Modelling and Accuracy Testing.....                      | 79 |
| 4.3.2 MATLAB Fuzzy Logic Designer .....                                    | 80 |
| 4.3.3 Fuzzy Logic Model Considerations and Results.....                    | 82 |
| 4.4 Mini-Production Line Implementation Procedures.....                    | 85 |
| 4.4.1 Design Considerations.....                                           | 85 |
| 4.4.1.1 Mechanical Design.....                                             | 85 |



|                                                       |     |
|-------------------------------------------------------|-----|
| 4.4.1.1.1 Imaging Chamber .....                       | 85  |
| 4.4.1.1.2 Sorting Conveyor .....                      | 87  |
| 4.4.1.1.3 Conveyor Belt Selection.....                | 90  |
| 4.4.1.1.4 Dimensional Calculation .....               | 91  |
| 4.4.1.1.5 Motor Selection for Sorting Mechanism ..... | 93  |
| 4.4.1.2 Electronic Design .....                       | 95  |
| 4.4.1.2.1 Ultrasonic Sensor .....                     | 95  |
| 4.4.1.2.2 Arduino Mega 2560 .....                     | 95  |
| 4.4.1.2.3 Circuit Breaker .....                       | 95  |
| 4.4.1.2.4 AC-to-DC Power Supply .....                 | 96  |
| 4.4.1.2.5 Control Box .....                           | 96  |
| 4.4.1.2.6 Cameras .....                               | 96  |
| 4.4.2 MATLAB Controls Flowchart .....                 | 96  |
| 4.4.3 Developed Prototype .....                       | 98  |
| Chapter 5 RESULTS AND DISCUSSION.....                 | 102 |
| 5.1 Resulting Datasets .....                          | 102 |
| 5.1.1 Training Dataset .....                          | 102 |
| 5.1.2 Testing Dataset .....                           | 107 |
| 5.2 MSRGP Output Models and Accuracy .....            | 111 |
| 5.3 Fuzzy Logic Model and Accuracy .....              | 119 |
| 5.4 Throughput Time Analysis.....                     | 120 |
| 5.5 Repeatability Testing.....                        | 122 |
| 5.6 Results Synthesis.....                            | 123 |
| Chapter 6 CONCLUSIONS AND RECOMMENDATIONS .....       | 126 |



|                                                 |     |
|-------------------------------------------------|-----|
| 6.1 Conclusion.....                             | 126 |
| 6.2 Recommendations .....                       | 127 |
| REFERENCES.....                                 | 129 |
| APPENDIX A. Bill of Materials.....              | 140 |
| APPENDIX B. Raw Datasets.....                   | 142 |
| APPENDIX C. Source Code.....                    | 149 |
| APPENDIX D. Operations Manual .....             | 172 |
| APPENDIX E. Maintenance Manual.....             | 174 |
| APPENDIX F. Research Ethics Clearance From..... | 175 |



## LIST OF FIGURES

|                                                                                                                                                                                                                        |    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| Figure 2.1 Bibliometric Visualization for SCOPUS-indexed keyword "postharvest" produce AND "computer vision" .....                                                                                                     | 9  |
| Figure 2.2 Morphological features of Iceberg lettuce. a.) Young leaf - venation 1 pinnate; 2 flabellate. b.) Head -overlapping of leaves 0 none; 3 partly; 5 half; 7 complete. [49].....                               | 21 |
| Figure 2.3 Overall Appearance of whole iceberg lettuce during storage. [59] .....                                                                                                                                      | 24 |
| Figure 2.4 Postharvest Practices as enumerated by food and agricultural organization (FAO) [71].....                                                                                                                   | 27 |
| Figure 2.5 Iceberg lettuce varieties in the Philippines. [74].....                                                                                                                                                     | 28 |
| Figure 2.6 Visual representation of quality factors of iceberg lettuce based on PNS/BAFPS 19:2005 a.) Known sample of without defects and with defects; b.) continuation of known defects for iceberg lettuce [9]..... | 30 |
| Figure 2.7 Horizontal Belt Conveyor Setup Convention .....                                                                                                                                                             | 34 |
| Figure 3.1. The Theoretical Framework.....                                                                                                                                                                             | 44 |
| Figure 3.2 Example of a MSRGP model [119] .....                                                                                                                                                                        | 51 |
| Figure 3.3 The Conceptual Framework.....                                                                                                                                                                               | 56 |
| Figure 4.1. Overall system process flowchart .....                                                                                                                                                                     | 58 |
| Figure 4.2. MSRGP block diagram for the current study .....                                                                                                                                                            | 59 |
| Figure 4.3 Fuzzy logic block diagram for the current study .....                                                                                                                                                       | 60 |
| Figure 4.4 Mini-production line block diagram .....                                                                                                                                                                    | 60 |
| Figure 4.5. Image Capturing Process for Iceberg Lettuce Samples .....                                                                                                                                                  | 61 |
| Figure 4.6 Chlorophyll extraction process. Adapted from [114] .....                                                                                                                                                    | 62 |
| Figure 4.7. Moisture Content Extraction Process .....                                                                                                                                                                  | 64 |
| Figure 4.8 MSRGP Modelling Flowchart .....                                                                                                                                                                             | 65 |
| Figure 4.9 Predicted vs. actual scatterplot for train dataset on Model ID 241 .....                                                                                                                                    | 69 |
| Figure 4.10 Predicted vs. actual scatterplot for train dataset on Model ID 2248.....                                                                                                                                   | 71 |
| Figure 4.11 Predicted vs. actual scatterplot for train dataset on Model ID 967.....                                                                                                                                    | 73 |



|                                                                                                                                                                                                                                                                                                                    |    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| Figure 4.12 Predicted vs. actual scatterplot for train dataset on Model ID 1147 .....                                                                                                                                                                                                                              | 75 |
| Figure 4.13 Predicted vs. actual scatterplot for train dataset on Model ID 2852 .....                                                                                                                                                                                                                              | 77 |
| Figure 4.14 Predicted vs. actual scatterplot for train dataset on Model ID 3460 .....                                                                                                                                                                                                                              | 79 |
| Figure 4.15 Fuzzy Logic Modelling Flowchart .....                                                                                                                                                                                                                                                                  | 80 |
| Figure 4.16 Membership functions of current study .....                                                                                                                                                                                                                                                            | 81 |
| Figure 4.17 Fuzzy surface graphs for the current study .....                                                                                                                                                                                                                                                       | 84 |
| Figure 4.18. Imaging Chamber CAD model .....                                                                                                                                                                                                                                                                       | 85 |
| Figure 4.19. Camera Placement on Imaging Chamber CAD Model .....                                                                                                                                                                                                                                                   | 86 |
| Figure 4.20. Image of the imaging chamber with a batch of lettuce .....                                                                                                                                                                                                                                            | 87 |
| Figure 4.21. Light intensity chart for imaging chamber of the current study .....                                                                                                                                                                                                                                  | 87 |
| Figure 4.22. Isometric view of CAD drawing the sorting conveyor with known<br>placement of sorting arms. ....                                                                                                                                                                                                      | 88 |
| Figure 4.23. Isometric view of servo mount with servo motor in place .....                                                                                                                                                                                                                                         | 88 |
| Figure 4.24. Isometric view of 3D-printed sorter arm to secure in place the MG966R<br>servo motor on the conveyor. ....                                                                                                                                                                                            | 89 |
| Figure 4.25. Isometric view of a 3D-printed sorter arm with the MG966R servo motor<br>placed on conveyor. ....                                                                                                                                                                                                     | 89 |
| Figure 4.26. Isometric view of the 83.40 mm guide rail .....                                                                                                                                                                                                                                                       | 90 |
| Figure 4.27. Dimensional Requirements for the conveyor at the image-capturing station<br>.....                                                                                                                                                                                                                     | 92 |
| Figure 4.28. Midpoint-to-point distance used for the conveyor at the image-capturing<br>station .....                                                                                                                                                                                                              | 93 |
| Figure 4.29. Overall system control logic flowchart.....                                                                                                                                                                                                                                                           | 98 |
| Figure 4.30. Developed prototype of the current study.....                                                                                                                                                                                                                                                         | 99 |
| Figure 4.31. Known driven and driver of the developed prototype. Shown is the a.)<br>driven side of the prototype displaying the imaging chamber and start position for<br>passing iceberg lettuce; b.) drive side detailing the placement of the crates that will<br>catch the iceberg lettuce upon sorting. .... | 99 |



|                                                                                                                                                                                                                                                   |     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| Figure 4.32. Imaging chamber with lettuce sample .....                                                                                                                                                                                            | 100 |
| Figure 4.33. Sorter system of the current study. Shown in a.) is the transition of the iceberg lettuce from going to the grading conveyor to sorting conveyor; b.) is the top view of sorting conveyor with sorting arm actuated into crate. .... | 101 |
| Figure 5.1 Boxplots of train dataset values from the current study based on quality grades for: a) chlorophyll- <i>a</i> , b) moisture content, and c) fresh head weight .....                                                                    | 107 |
| Figure 5.2 Boxplots of test dataset values from the current study based on quality grades for: a) chlorophyll- <i>a</i> , b) moisture content, and c) fresh head weight.....                                                                      | 110 |
| Figure 5.3 Test data predictions for MSRGP models for a.) chlorophyll- <i>a</i> b.) moisture content and c.) fresh head weight with RGB-HSV values as inputs.....                                                                                 | 113 |
| Figure 5.4 Test data predictions for MSRGP models for a.) chlorophyll- <i>a</i> b.) moisture content and c.) fresh head weight with HSV values as input only.....                                                                                 | 116 |



## LIST OF TABLES

|                                                                                                                                                                                               |    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| Table 2.1 Summary of Search Strategy and Results .....                                                                                                                                        | 8  |
| Table 2.2 Most cited research papers related to "postharvest" produce AND "computer vision" .....                                                                                             | 10 |
| Table 2.3 Most cited research papers related to ("postharvest" produce AND "computer vision") AND ("quality" AND classification AND "grading") .....                                          | 17 |
| Table 2.4 Most cited research papers related to ("postharvest" produce AND "computer vision") AND ("lettuce" AND "quality" AND classification AND grading) AND (sorting AND automation) ..... | 19 |
| Table 2.5 Some related works that used fresh head weight of iceberg lettuce as a postharvest quality parameter.....                                                                           | 22 |
| Table 2.6 Some related works that used chlorophyll concentrations of iceberg lettuce as a postharvest quality parameter.....                                                                  | 24 |
| Table 2.7 Philippine national standard for iceberg lettuce composition and quality factors .....                                                                                              | 29 |
| Table 2.8 Philippine national standard for iceberg lettuce size classification .....                                                                                                          | 31 |
| Table 2.9 Summary of reviewed quality parameter, its perceived grading and comparison of it to PNS for iceberg lettuce .....                                                                  | 31 |
| Table 2.10 Specifications of Flat Belt Drive for Conveyor Belt Systems [15].....                                                                                                              | 33 |
| Table 2.11 Some related works of postharvest grading and sorting systems on their sensing, smart, sustainable domains.....                                                                    | 36 |
| Table 2.12 Some related works of postharvest grading systems.....                                                                                                                             | 37 |
| Table 2.13 Some related works of computer vision for postharvest lettuce quality classification.....                                                                                          | 39 |
| Table 2.14 Summary of Relevant Patents Regarding Postharvest Handling Systems....                                                                                                             | 41 |
| Table 2.15 Synthesis table of review of related literature of current study .....                                                                                                             | 43 |
| Table 3.1 Relevant lettuce quality parameters .....                                                                                                                                           | 45 |
| Table 3.2 Grading criteria of iceberg lettuce for current study .....                                                                                                                         | 46 |





|                                                                                                                 |    |
|-----------------------------------------------------------------------------------------------------------------|----|
| Table 3.3 Summary of Genetic Programming (GP) components and evolutionary phases .....                          | 50 |
| Table 3.4 Summary of fuzzy logic terms and description.....                                                     | 52 |
| Table 4.1 Engineering standards relevant to the current study .....                                             | 58 |
| Table 4.2 List of items for chlorophyll- <i>a</i> extraction process .....                                      | 63 |
| Table 4.3 List of items for moisture content extraction process .....                                           | 64 |
| Table 4.4 MSRGP run parameters for the current study .....                                                      | 66 |
| Table 4.5 Top 3 considered models for MSRGP chlorophyll- <i>a</i> from pareto report using RGB-HSV input .....  | 67 |
| Table 4.6 Model ID 241 GP tree structural properties .....                                                      | 68 |
| Table 4.7 Top 3 considered models for MSRGP moisture content from pareto report using RGB-HSV input .....       | 69 |
| Table 4.8 Model ID 2248 GP tree structural properties .....                                                     | 70 |
| Table 4.9 Top 3 considered models for MSRGP fresh head weight from pareto report using RGB-HSV input .....      | 71 |
| Table 4.10 Model ID 967 GP tree structural properties .....                                                     | 72 |
| Table 4.11 Top 3 considered models for MSRGP chlorophyll- <i>a</i> from pareto report using HSV input only..... | 73 |
| Table 4.12 Model ID 1147 GP tree structural properties .....                                                    | 74 |
| Table 4.13 Top 3 considered models for MSRGP moisture content from pareto report using HSV input only .....     | 75 |
| Table 4.14 Model ID 2852 GP tree structural properties .....                                                    | 76 |
| Table 4.15 Top 3 considered models for MSRGP fresh head weight from pareto report using HSV input only .....    | 77 |
| Table 4.16 Model ID 3460 GP tree structural properties .....                                                    | 78 |
| Table 4.17 Associative matrix for the current study .....                                                       | 82 |
| Table 4.18. Conveyor Specifications .....                                                                       | 90 |
| Table 4.19. Selection of Servo Motors.....                                                                      | 93 |



|                                                                                              |     |
|----------------------------------------------------------------------------------------------|-----|
| Table 5.1 Study's training dataset from extraction processes .....                           | 103 |
| Table 5.2 Study's testing dataset from extraction processes.....                             | 108 |
| Table 5.3 Chosen MSRGP models of RGB-HSV $R^2$ Values and Complexities.....                  | 112 |
| Table 5.4 Known error metrics for MSRGP models of RGB-HSV input for the current study .....  | 114 |
| Table 5.5. Chosen MSRGP models of HSV inputs only $R^2$ Values and Complexities              | 114 |
| Table 5.6 Known error metrics for MSRGP models of HSV input only for the current study ..... | 116 |
| Table 5.7 Summary comparison table of the output MSRGP models considered for the study ..... | 117 |
| Table 5.8 MSRGP Testing Accuracy .....                                                       | 118 |
| Table 5.9 Fuzzy logic testing accuracy for the current study.....                            | 119 |
| Table 5.10. Overall throughput time per station for the current study .....                  | 120 |
| Table 5.11. System repeatability trials for the current study .....                          | 121 |
| Table 5.12 Repeatability test at 70% conveyor speed.....                                     | 122 |
| Table 5.13 Repeatability test at 100% conveyor speed.....                                    | 123 |
| Table 5.14 Summary table of results and deliverables for the current study .....             | 124 |
| Table 5.15 Comparison table of existing studies to current study's metrics.....              | 125 |



## Chapter 1

### Introduction

#### 1.1 Background of the Study

Farmers have extensive practical knowledge about assessing the quality of their produce, however this expertise is qualitative. The quality of fruits and vegetables is determined by five key attributes: visual appearance (size, shape, color, defects), texture, flavor (taste and aroma), nutritional values, and safety. Fruits and vegetables are initially evaluated for quality based on appearance [1]. Quality assessment of fruits and vegetables often uses destructive methods to measure physical and chemical attributes for sensory evaluation [1]. Utilizing manual on-site sorting approaches has higher labor costs and lower efficiency [2]. It is highly dependent on trained human resources to inspect the outer appearance of produce to determine the grade quality [3]. The deployment of AI systems helps overcome visual variability, inaccuracy, and fatigue in quality grading [2].

Computer vision has transformed the agricultural postharvest system by providing enhanced ways for assessing quality, sorting, and grading agricultural produce. Computer vision systems use high-resolution cameras and machine learning algorithms to correctly identify the quality of fruits and vegetables' rot or discoloration [4]. This technique allows farmers and distributors to reduce post-harvest losses by segregating and sorting produce based on quality, ensuring that only the best items reach the market. Furthermore, computer vision systems can aid in optimizing storage conditions, monitoring temperature and humidity levels, and estimating the shelf life of harvested commodities, hence extending product freshness and lifespan.

In addition, computer vision is helping automate post-harvest activities. Computer vision-driven automated sorting and grading systems can replace labor-intensive manual sorting, resulting in significant cost savings and greater efficiency [5]. These systems boost postharvest productivity by precisely categorizing agricultural produce based on size, shape, color, and other quality factors. Farmers and stakeholders can make educated



decisions to prevent waste and maximize the economic value of their crops by integrating real-time data analytics and remote monitoring capabilities, contributing to a more sustainable and efficient agricultural supply chain [6].

Deep learning-based models have shown significant promise in automatic fruit grading in recent years. FruitVision, a deep learning-based model for automatically grading various fruits, achieved high accuracy across different fruit types [7]. Computer vision has been widely used for the inspection and grading of both fruits and vegetables, offering the potential to automate manual grading practices and standardize techniques [8]. This technology has been applied across various agricultural products, including grains, and has been used to classify different wheat varieties and detect quality attributes [8].

Integrating Computer Vision Systems (CVS) offers a viable solution for the quality grading of fruits and vegetables utilizing contactless and non-destructive methods. RGB color cameras are used for these types of systems as they reproduce the vision of the human eye by using three monochromatic filters centered in red, green, and blue at wavelengths of 700 nm, 546 nm, and 435 nm. This setup enables the detection of external qualities such as color, shape, size, texture, and defects [1].

This study will specifically focus on iceberg lettuce (*Lactuca sativa* var. *capitata*) as the primary produce for testing. Quantifying qualitative standards for assessing the quality of iceberg lettuce will enhance the efficiency and consistency of quality grading. Subjective attributes such as appearance, texture, and freshness can be transformed into measurable values using computer vision to measure three parameters: chlorophyll-*a* concentration, moisture content, and weight estimation. This approach addresses the limitations of traditional methods, which are heavily reliant on human inspection and can lead to inconsistencies. Establishing a quantitative grading system contributes to improving the overall quality control process in the agricultural sector.



## 1.2 Statement of the Problem

The Philippine National Standard for Head Lettuce (PNS/BAFPS 19:2005) outlines qualitative grading criteria for attributes such as size, freshness, color, firmness, and defect tolerance such as decay, discoloration and insect damage [9]. However, these standards rely heavily on subjective human judgment, lacking quantitative thresholds for critical parameters like chlorophyll-*a* concentration, and moisture content. Manual grading methods that depend on visual inspection are prone to inconsistencies due to human variability, fatigue, and training disparities, leading to inefficiencies in post-harvest sorting and significant labor costs [2]. For instance, Grade 1 lettuce must be "fairly firm" and "green," however these terms lack measurable definitions which can result in ambiguous classification during manual sorting [9].

This subjectivity creates challenges in ensuring uniform compliance with export and domestic market requirements, contributing to post-harvest losses and reduced economic value. While PNS/BAFPS 19:2005 provides a framework, the absence of quantitative metrics hinders scalability and automation. Recent advancements in computer vision and AI offer solutions to translate qualitative attributes into quantifiable data—such as using RGB cameras to assess color uniformity (700–435 nm wavelength filters) or machine learning models to detect defects. However, the Philippine agricultural sector has yet to integrate these technologies into its grading systems, perpetuating reliance on manual practices, especially for small-scale farms [2]. Addressing this gap requires bridging qualitative standards with quantitative measurement tools to enhance grading accuracy, reduce waste, and improve market competitiveness.

## 1.3 Significance of the Study

In 2018, the Philippines confronted formidable challenges with heightened food insecurity and significant post-harvest losses within the agricultural sector. This research undertakes the development of a specialized lettuce grading system tailored to the



Philippine context. The primary objective is to streamline lettuce farming practices, aligning production dynamics with consumer demand while optimizing post-harvest processes for heightened efficiency and economic viability, which is also in line with what the Food and Agriculture Organization (FAO) of the United Nations has mentioned in their article [10]. The research seeks to enhance economic sustainability within the lettuce farming sector by equipping farmers with a comprehensive tool designed to improve the quality of lettuce.

At the societal level, the study's lettuce grading system and post-harvest processing line aim to alleviate food insecurity significantly. The strategic implementation of these systems is expected to result in an increased supply of high-quality produce in consumer markets, particularly benefiting underserved populations. Improved post-harvest handling practices help maintain fresh produce's nutritional and sensory qualities, ensuring higher-quality food for consumers [11]. Furthermore, the focus on locally sourced produce promotes sustainable agricultural practices, contributing to a more resilient and health-oriented food ecosystem.

Foreseeing broader implications, the research outcomes transcend the immediate lettuce farming context. The envisaged grading system, once developed, could serve as a paradigmatic model applicable to analogous agricultural contexts and diverse crop portfolios. A United Nations Centre for Sustainable Agricultural Mechanization study found that adopting improved post-harvest technologies and grading systems can minimize these losses and enhance the overall food supply chain [12]. By systematically addressing the intricate challenges embedded in the local iceberg lettuce supply chain, this study aligns with the global imperative of mitigating food waste and fortifying agricultural resilience in response to evolving challenges.



## 1.4 Objectives

### 1.4.1 General Objective

The study aims to develop a vision-based quality grading and sorting system for iceberg lettuce (*Lactuca sativa var. capitata*) within a mini-production line, integrating computer vision and automation to enhance postharvest handling efficiency.

### 1.4.2 Specific Objectives

- 1) To develop and validate a non-invasive method for estimating chlorophyll-*a* concentrations, moisture content, and fresh head weight of freshly harvested iceberg lettuces using multi-gene symbolic regression programming (MSRGP).
- 2) To create an equivalent quality classification model that maps out three quality grading criteria (high, medium, and low) for iceberg lettuce.
- 3) To develop and evaluate a conveyor belt sorting system achieving five lettuces per batch sampling.

## 1.5 Scope and Delimitations

### 1.5.1 Scope

This study will cover the conceptualization and development of an intelligent postharvest grading and sorting system, notably featuring a vision-based quality grading system integrated into a sorting system for iceberg lettuce based on a grading criterion. The scope is further detailed in the outline below.

- Lettuce samples collected and considered as freshly harvested by the current study for extraction processes is within the period of two weeks after it is pruned from the farm.
- Focus exclusively on the postharvest grading and sorting of iceberg lettuce based on three key quality parameters: chlorophyll-*a*, moisture content and fresh head weight.
- Develop a vision-based quality grading system that converts the measured quality parameters into discrete grade levels aligned with PNS/BAFPS 19:2005.



- Design and validate a system that categorizes and sorts iceberg lettuce into distinct quality bins based solely on the extracted grading criteria.

## 1.5.2 Delimitations

This study excludes other potential quality indicators of iceberg lettuce, such as defects, diseases, and other quality parameters. Consequently, the postharvest system of iceberg lettuce does not consider the different stages of the agricultural life cycle of iceberg lettuce and will focus only on grading and sorting. The study will not review other equipment or material compatibility for our system detailed in methods and will only consider the efficiency and calibration of these items. The delimitations are further detailed in the outline below.

- The study is limited to the grading and sorting processes within the postharvest handling of iceberg lettuce and does not address other phases, such as packaging, transportation, or storage. Other plausible food sorting systems, such as for tomatoes, will also not be considered as a part of the study's system and modelling.
- Only the three quality parameters (chlorophyll-a, moisture content, and fresh head weight) are considered, excluding other potential quality indicators like morphological defects, disease presence, or additional bio- or phytochemical properties.
- The study will not consider the complexities of full-scale mechanical sorting automation beyond relating the grading criterion to sorting output and consideration of compatibility with other equipment or materials aside from those outlined in the current study's methods.
- The study will not incorporate an embedded system for grading and sorting; a computer running MATLAB will be utilized to perform these functions.
- The study will not consider varying camera angles; the cameras will exclusively focus on the front view of the iceberg lettuce.





- The age of the iceberg lettuce samples used in the dataset will strictly be limited to 1–2 weeks post-harvest to account for the transit period before reaching the study for analysis; samples outside this timeframe will not be included in the dataset.

## **1.6 Engineering Standards**

ANSI-CEMA 402-2003 [14] and PNS-PAES 302-2000 [15] tackles the conveyor belt configuration for postharvest handling. NEMA Standards detail the actuators needed for the current study includes both servo and stepper motors for industrial use [16]. 3-A Sanitary Standards cover the food-grade safety of the conveyor belt to handle iceberg lettuce [17]. Lastly, Philippine National Electrical Code covers the needed design and safety requirements for controls and electrical design that the current study will consider [18]. An overview of these standards can be seen in Table 4.1.



## Chapter 2

### Review of Related Literature

The study mainly focuses on five main categories namely (1) postharvest iceberg lettuce quality, (2) computer vision, (3) quality classification, (4) quality grading, and (5) sorting automation. These topics will be discussed in succeeding sections of this chapter.

#### 2.1 Literature Survey Strategy

The literature survey strategy employed is a systematic review to evince the most prominent related works to the current research field being studied. It justifies areas where research is lacking and gaps that the authors have considered for their study. Shown in Table 2.1 reflects the filtered peer-reviewed literature extracted from SCOPUS journal database along the input parameters done for survey. The indexing was done for all field textual content within the year 2018-2024 extracting all document types except books and short survey. The initial keyword of “postharvest produce” AND “computer vision” provided documents with a collective h-index of 46. This indicates high impact and performance of the papers indexed on these keywords. Narrowing this to another keyword related to the study, “quality” AND classification AND grading’, yielded 33 documents with a 14 h-index. Lastly, another keyword further targets the study topic, “sorting automation”, yielded documents with a 4 h-index.

Table 2.1 Summary of Search Strategy and Results

| Parameters         | Keyword 1                                         | Keyword 2                                     | Keyword 3             |
|--------------------|---------------------------------------------------|-----------------------------------------------|-----------------------|
| Search Field       | all textual content fields                        |                                               |                       |
| Keyword Vector     | "postharvest" produce<br>AND "computer<br>vision" | "quality" AND classification<br>AND "grading" | sorting<br>automation |
| Year Inclusions    | 2018-2024                                         |                                               |                       |
| Language Inclusion | English                                           |                                               |                       |
| Document type      | Article, Review, Conference Paper, Book Chapters  |                                               |                       |
| Results            | 588                                               | 33                                            | 9                     |



Shown in Figure 2.1 is a bibliometric visualization on the initial keyword passed through the SCOPUS database. It provides a basis on the co-occurrence of all keywords indexed and its relationship with other keywords within the document results. This was performed through VOS viewer with the following settings: Counting method done was full counting to provide that the weights of the keywords are the same in each related document; The set minimum number of occurrences of a keyword set to a minimum of 15 with the top 30 most occurred keywords extracted from this setting will visualize the strengths of each occurring keywords and its relations.

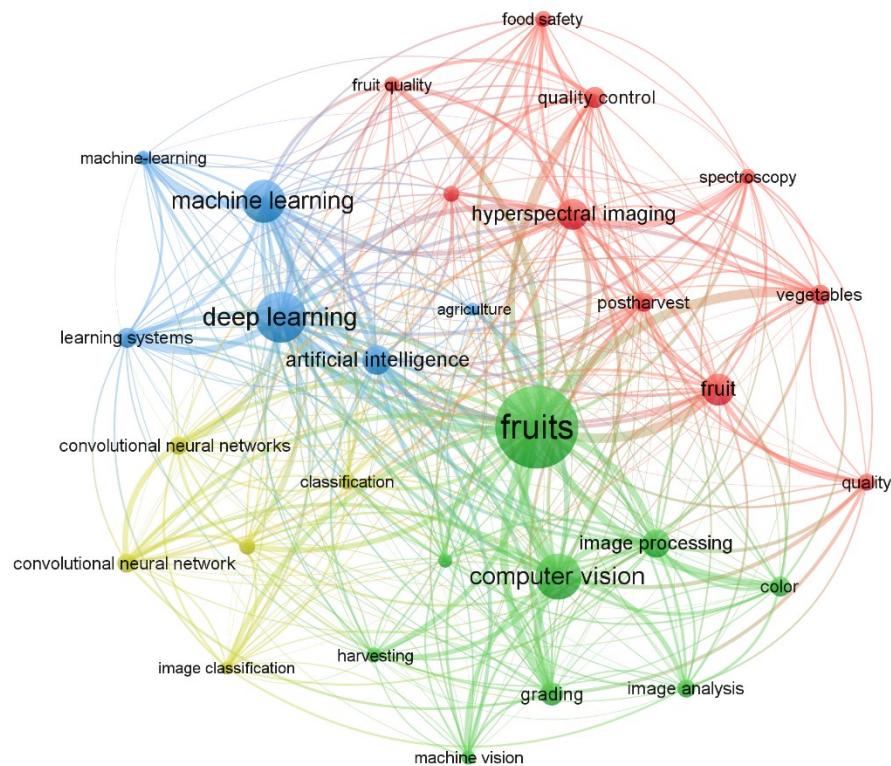


Figure 2.1 Bibliometric Visualization for SCOPUS-indexed keyword "postharvest" produce AND "computer vision"

With the SCOPUS keyword results, below are the most cited research papers from the keyword "postharvest" produce AND "computer vision" which enumerates the



paper's findings and contributions made related to the topic and how these correlates to current study on remarks as shown in Table 2.2.

From Table 2.2, it was revealed that the most papers cited are reviews mainly for higher forms of spectral imaging for computer intelligence wherein the trend covers use of neural networks to define intended features from postharvest produce and conduct classification afterwards. Literature revolves also on retrieving internal qualities of the produce with mentions of internal bruising and the nutritional content information being correlated with the extracted images from RGB color spaces and HSV values. Most research uses non-conventional cameras and setups to see the internal qualities of the postharvest produce with minimal mentions of image filters or feature extraction process done which may not be suitable for cost-constraint applications.

Table 2.2 Most cited research papers related to "postharvest" produce AND "computer vision"

| Author, Publication Date      | Number of Citations | Main Contribution                                                                                                                                                                                                                                                                                                                                                                                                            | Remarks related to the study                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bhargava and Bansal, 2018 [4] | 354                 | This paper provided a comprehensive overview of the This review covered pre-processing segmentation, feature extraction, classification computer vision techniques that addressed fruits and vegetable quality based on color, texture, size, shape and defects. The mention of quality inspection involves five steps namely: image acquisition, pre-processing, image segmentation, feature extraction and classification. | Color features for postharvest processing and handling stages. Most widely used visual features in image retrieval and indexing, Use of RGB, HIS, and CIELab color spaces are commonly used for postharvest quality inspections. Color features like correlogram, coherence vector, moments, and histograms. Color moments are cited to be effective with values on mean, standard deviation and skewness. |
| Rehman et al., 2019 [19]      | 312                 | This review provided insights on what type of machine learning (ML) algorithms should be used specifically for certain agricultural applications summarized in terms of crop, ML task, data/sensor type, application area, and accuracy. The following ML algorithms were discussed: Naïve-Bayes, Discriminant analysis, K-nearest neighbors, SVM. These were used primarily as under supervised                             | The use of a logistic regression model-based pixel classification was used for postharvest fruit detection and size estimation from citrus. Support vector machine (SVM) were also applied for crop quality inspection. The use of multispectral                                                                                                                                                           |



|                             |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             |     | learning with feature extraction and annotation done.                                                                                                                                                                                                                                                                                                                                                                                                                          | features centered at 450-800 nm with respective bandwidths ranging from 40- 80 nm were done for bi-colored apples.                                                                                                                                                                                                                                                                                                                     |
| Williams et al., 2019 [20]  | 263 | This paper is concerned with the development and performance of a robotic kiwifruit harvesting system with deep learning (DL) network based on its computer vision. The mention of post-processing techniques to achieve detection of the kiwifruit involved the use of Basler AC1920 cameras suited for outdoor lighting condition with a working distance of 900 mm in consideration of the robot arms picking the kiwifruit.                                                | The systems adapted VGG-net16, a fully convolutional network (FCN) with additional illumination under robust conditions. The network was trained on 63 200-by-200-pixel hand-labelled images annotations. It highlighted a 89.6% detection rate for given fruit during training with testing capable of reaching 70% accuracy.                                                                                                         |
| Signoroni et al., 2019 [21] | 253 | The paper reviewed DL architectures may be combined with hyperspectral imaging and acquisition techniques like spectroscopy for higher wavelength and color bands capturing other than the common red, green, blue (RGB) imaging. Traditional learning utilizes this imaging by feature extraction for linear classifiers while DL solutions capitalizes on labeled data for supervised learning without the need to identify further features within the hyperspectral image. | The use of hyperspectral imaging techniques has been known to aid of postharvesting controls in crop products. These include the use of nutrient inspection, early pathogenic diagnosis, fruit ripening indicators, segregation of damaged fruits and presence of plant diseases. Common range used for agriculture and food revolves within Near Infrared Spectroscopy range of 700-2500nm. This is linked to chemometric principles. |
| Lu et al., 2020 [22]        | 183 | The paper reviewed the use of hyperspectral imaging for evaluating quality and safety of crop produce specific on imaging based on reflectance, transmittance, fluorescence, absorption and the Raman effect.                                                                                                                                                                                                                                                                  | The well-known RGB computer vision was developed rapidly specifically for quality grading and defect detection within the food industry for crop products like apples and tomatoes. The use of combined filters in the range of near infrared provided efficiency for further applications on said crops with imaging at multiple bands of 3-10.                                                                                       |



|                                  |     |                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Subeesh and Mehta, 2021 [23]     | 171 | The paper investigated recent research applications of artificial intelligence (AI) and Internet of Things (IoT) for the digitization of agriculture. The paper made significant claims that use of industry 4.0 technologies enhances input use for frames with automation being leveraged in real-time.                | The mention of convolutional neural networks (CNN) has been the forefront in agriculture automation with limiting factors based on the quality and quantity of data being inputted for its decision-making. Mentions of usage of computer vision on farm machinery include task revolving navigation and performance evaluation under for development of unmanned aerial vehicles (UAV). |
| Hameed et al., 2018 [24]         | 163 | The paper reviews classification techniques often applied for fruit and vegetables noting on the following points: key challenges, state-of-the-art algorithms, data acquisition methods, data pre-processing, feature extraction and performance metrics.                                                               | The use of classifiers models using spatial pyramid pooling, nearest neighbor, kernel support vector machine and linear support vector machines were utilized mainly on quality assessment applications.                                                                                                                                                                                 |
| Wang et al., 2018 [25]           | 126 | The paper is concerned with the use of DL techniques to detect internal mechanical damage of blueberries on the basis of hyperspectral transmittance data. The results provided utilized ResNet CNN architecture that achieved the highest average accuracy of 88.44%. Inference takes 5.2-6.2 milliseconds for sorting. | A comparative study was done to further sift through ML algorithms suited for real-time inferencing with 5 traditional machine learning algorithms as classifiers to complement the ResNet model. The total number of blueberries subjected for hyperspectral transmittance were 737 with it being cut to see internal damage as ground truth.                                           |
| Tripathi and Maktedar, 2020 [26] | 119 | This paper provided an extensive survey of research regarding computer vision and image processing techniques focused properly on mathematical framework, feature descriptor, defect detection on multiple datasets of fruits and vegetables elaborately.                                                                | A detailed review of postharvest quality evaluation systems based on computer vision and composition of organically produced fruits revealed the key quality parameters are physiochemical properties, storage performance, microbiological, sensory and nutritional quality. Handling of such produce                                                                                   |





|                           |     |                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           |     |                                                                                                                                                                                                                                                                                                                                                                         | mentioned as meeting traceability requirements.                                                                                                                                                                                                                                                                                                                                 |
| Bauer et al., 2019 [27]   | 117 | The paper developed the use of aerial phenotyping for the quality classification based on 3 size categories for iceberg lettuce. The use of normalized difference vegetation index imagery was used to determine crop counting and crop quality assessment. The proposed lettuce counting was validated through manual counting with a R-squared correlation of 0.9997. | The image processing done for the computer vision application involved increasing contrast between foreground and background through contrast limited adaptive histogram equalization (CLAHE). This is then removed of noise and partitioned, utilized a sliding window to traverse the sectioned image. Then, a CNN classifier quantifies the lettuce number and lettuce size. |
| Bai et al., 2023 [28]     | 111 | The paper reviewed in detail the current research trend of vision-based technology for agricultural robots. The use of global navigation satellite systems, light detection and ranging, inertial measurement unit, ultra-wide band and vision sensor have been the focus of instruments aiding autonomous navigation for these robots.                                 | Challenges that revolved on the development of vision-based navigation still need to address improvement of variability on illumination conditions based on given environment. The use of appropriate image preprocessing methods can be a means to prevent adverse environmental conditions.                                                                                   |
| Nyalala et al., 2019 [29] | 107 | The paper developed a computer vision system to estimate mass and volume of cherry tomatoes through extraction of both 2D and 3D features from depth images. The proposed techniques were able to achieve an average accuracy of 94% and 95% using radial basis function-support vector machine model for volume estimation and mass estimation respectively.           | The use of image pre-processing in terms of background removal, distance thresholding, smoothing and binarization were employed. 2D features extracted from the image were the following: Area, perimeter, major axis length, minor axis length, eccentricity, and radial distance for both mass estimation while 3D features of surface area and volume were extracted.        |
| Li et al., 2019 [30]      | 107 | The paper provided a novel approach on early detecting decay on apples through combination of hyperspectral imaging and an improved watershed segmentation method. Based on the results, it was able to distinguish healthy apples to decayed ones                                                                                                                      | The use of gradient image was performed through sobel, prewitt, canny differential filters for its noise sensitivity in the images. The watershed segmentation for the                                                                                                                                                                                                          |



|                                  |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  |     | with 89% accuracy under four key wavelengths of 781, 813, 962 and 1000 nm.                                                                                                                                                                                                                                                                                                                                                                                                                                     | decayed regions on apples utilized the mentioned filters for morphological gradient for reconstructing gradient image before applying watershed transform.                                                                                                                                                                                          |
| Schifferstein et al., 2019 [31]  | 107 | The paper investigated consumer expectations of the expected coloring of fresh produce with specific focus on carrots. The results of the paper showcase how different colors of carrots elicit expected taste and texture of the carrot where attractiveness where associated to healthy produce while uncommon color were anticipated to have less nutrition and bad taste.                                                                                                                                  | Although the paper did not provide specific computer vision techniques, it focused on color information and its manipulation. The use of simple luminance histogram provides easy and accurate estimation of freshness perception. The use of luminance to describe the amount of light reflected was heavily emphasized.                           |
| Zhang et al., 2018 [32]          | 106 | The paper reviewed the use of produce grading and quality systems over the past decades more attuned on its commercialization of the technology. The paper evinced that whole surface detection, discrimination between defects and stems, robustness of features and algorithms, all hinder the system's inspection accuracy and technology transfer.                                                                                                                                                         | Solutions for these involve rotating grading systems, use of mirrors or multiple cameras for whole surface view and use of 3D features to distinguish stems/calyxes from defects; weed out unobvious defects during postharvest processing stage using higher or multi-spectral imaging; more tailored feature selection with evaluation criterion. |
| Verboven et al., 2020 [33]       | 105 | The paper investigated the use of industrial digitalization for modelling for food processing. The use of digital twin in food applications were highlighted in terms of the following: sensor networks that measure essential variables and properties of the product in real time; use of cloud-based data storage and high-performance computing within the platform to connect sensors; use of the platform as to perform computations for testing, design, optimization and control for decision support. | Use of digital twinning for postharvest has been cited as a complementary data that quantifies quality loss within fresh horticultural produce. Technology also provides benefits on actionable data on said perishables.                                                                                                                           |
| Nturambirwe and Opara, 2020 [34] | 99  | The paper reviewed the use of machine learning (ML) methodologies for effective evaluation of food commodities' quality specific around defect detection. The use of hue, saturation and intensity (HIS) values have been preferred for defect detection. A challenge presented is the standardization of                                                                                                                                                                                                      | Postharvest handling was attributed to bruising and defects known on fruits and vegetables. A suggestion was to measure and index the potential bruise damage these handling systems may                                                                                                                                                            |





|                          |    |                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          |    | such ML techniques beyond research investigations.                                                                                                                                                                                                                                                                                                                                                                     | produce under mechanical loading with a suggestion made to grade damaged fruit based on damage severity, both internal and external.                                                                                                                                                                                                                                                                                   |
| Abasi et al., 2018 [35]  | 96 | The paper examined sensing devices that are low-cost, low-weight, portable, user-friendly and rapid inspection of food quality through non-destructive means. It was able to show three distinct device types dedicated for this application namely: devices based on image processing technology; devices based on acoustic ultrasound and impact force test methods; devices using available portable spectrometers. | For postharvest horticultural produce, it features devices with higher-level wavelength imaging technology like that of reflectance spectroscopy for assessment of chlorophyll content, soluble solids content (SSC) and firmness for apples. Other devices considered laser diode modules for SSC and firmness while others combined ML techniques of ANN and SVM to supplement image processing on object detection. |
| Onwude et al., 2020 [36] | 91 | The paper is concerned with the improvements made to the food loss prevention technology utilized within supply chains. It highlights the use of imaging technology, spectroscopy, X-ray imaging, multi-sensors, electronic nose, radio frequency identification, printed sensors, acoustic impulse response and mathematical models.                                                                                  | For the use of computer vision in postharvest supply chains, it was mentioned that applications were limited to surface detection. Features related to surface properties based on interaction of light were primarily discussed.                                                                                                                                                                                      |
| Du et al., 2020 [37]     | 87 | This paper reviewed imaging techniques for bruise detection for postharvest loss prevention. The imaging techniques revolved on the following: hyper-/multi-spectral, nuclear magnetic, thermal, structured-illumination reflectance.                                                                                                                                                                                  | A method proposed for imaging systems in general was to divide the image in several segments on the visible region and non-visible regions and applying principal component analysis for dimensionality reduction and classify images at a certain wavelength range via ML methods.                                                                                                                                    |
| Gill et al., 2022 [38]   | 86 | The paper reviewed the use of high-throughput phenotyping (HTP) techniques and ML for plant stress phenotyping. HTP varies from the use of ground-based imaging to remote sensing for defining biotic and abiotic factors related to crop growth and                                                                                                                                                                   | The mention of the following ML algorithms provides rapid adoption for HTP platforms in agriculture for commonly used vegetation indices:                                                                                                                                                                                                                                                                              |



|                         |    |                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         |    | quality. The techniques utilize feature extraction, identification, classification and prediction criteria to identify pertinent data for use plant quality determination                                                                                                                                                                                       | linear discriminant analysis (LDA), SVM, Logistic Regression, Random Forest (RF), Linear Regression; Partial Least Square Regression (PLSR). A mention of an automated video tracking platform was used for trait phenotyping for biotic and abiotic stress and other morphological traits in lettuce for resistance to aphids and other sucking insects.                                          |
| Meenu et al., 2021 [39] | 86 | The paper reviewed the use of digital signal processing (DIP) for quality assessment of various food products. Diverse range of ML and DL approaches for determining quality parameters of food materials. The use of HSV, RGB and CIELab color spaces were frequently mentioned as basis for color and morphological features on reviewed horticultural crops. | Salient to DIP is the assurance of a good quality image acquisition system that was emphasized in the review. It mentions proper visibility of relevant features with proper light and reflecting conditions with proper calibration methods. It was highlighted that light sources of known wavelength should be used for determination of light intensity and scattering area from crop samples. |

Table 2.3 details the research papers that mention specifically lettuce quality and its usage of computer vision as a means of grading the crop produce within their study. From the table, many research efforts revolve on weight estimation and correlation of the nutritional and physiochemical characteristics of the postharvest produce into the given quality classification. The grading falls in under 3 categories for weight estimation. For quality grading, it depends on how many classes are needed to define distinguishable difference from the postharvest crop for market selling which ranges from 3-16 categories. Mention of applications for Industry 4.0 in combination with CV grading to alert stakeholders of state of system have also been in the trend lately for traceability requirements.



Table 2.3 Most cited research papers related to ("postharvest" produce AND "computer vision") AND ("quality" AND classification AND "grading")

| Author, Publication Date  | Number of Citations | Main Contribution                                                                                                                                                                                                                                                                                                                                                       | Remarks related to the study                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Nyalala et al., 2019 [29] | 107                 | The paper developed a computer vision system to estimate mass and volume of cherry tomatoes through extraction of both 2D and 3D features from depth images. The proposed techniques were able to achieve an average accuracy of 94% and 95% using radial basis function-support vector machine model for volume estimation and mass estimation respectively.           | Grading of cherry tomatoes are based heavily on the mass and volume estimation that the study proposed with the assumption that the density of the crop is constant. The study outlines that 95% accuracy is the set acceptable limit for such estimations. A needed t-test is then employed to indicate that errors from measured and estimated volume and mass are not significantly different, thus within the predefined 5% error. |
| Lazaro et al., 2019 [40]  | 45                  | The paper investigated the use of an (near-field communication) NFC tag with a colorimeter aperture to determine the color of typical tropical fruits as an indication of their maturity. The developed system utilized HSV and CIELab values to analyze the ripeness state of the fruit with linear discriminant analysis (LDA) and KNN provided mean accuracy of 80%. | The experimental results show that the ripeness grade is based on time and environmental condition function particularly on its storage temperature under the hue and saturation values. Two classes were considered for the fruit test where less than six days after purchase is considered in good state and more than six days is considered in a bad state grade.                                                                 |
| Melesse et al., 2022 [41] | 43                  | The paper is concerned with the application of digital twin for monitoring banana quality changes throughout its postharvest storage. A thermal camera was used in order to retrieve the surface and physiological changes of the fruit. The study applied a CNN architecture to achieve 99% prediction accuracy among four grade classes.                              | The use of a Wi-Fi gateway and MQTT message broker as the communication protocol was utilized for data ingestion towards SAP edge services for real-time classification under four grades, fresh, good, bad and rotten. CNN was trained under 500 images per category with the features                                                                                                                                                |



|                               |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | being extracted focused on segmented thermal readings of the fruit.                                                                                                                                                                                                                                                                                                                |
| Chakraborty et al., 2023 [42] | 41 | The paper investigated the use of modern sensors and devices for handling deep learning image analytics for their vision-based sorting and weight-based grading of citrus fruits machine. The results of their custom classification model achieved a 97% accuracy and the weight-based grading to be 91% accuracy.                                                                                                                                                       | The machine was able to sort within surface classification of the citrus fruit based on the grading done by manual sorters and grading down within three weight categories where large was based to be greater than 140 grams; medium to be between 140 to 100 grams; and small to be less than 100 grams.                                                                         |
| Utai et al., 2019 [43]        | 40 | The paper is concerned with mass estimation for irregularly shaped mango cultivar using fruit dimensions of length, width, thickness and area that is fed within an artificial neural network (ANN). The paper showcased that utilizing length, width and thickness as parameters provided a 99% accuracy for the task. The paper evidenced that 2-D vision systems provided efficient means to lessen equipment requirements, computational time and memory consumption. | The size classes for the mango as determinant of its grade are as follows: small, less than 300 grams; medium, between 300-399 grams; large, greater than 399 grams. The paper provided more insight on the grading done will depend on the feature extraction under image processing analysis that should be considered for both axi-symmetric and irregular shaped crop produce. |
| Sabzi et al., 2020 [44]       | 21 | The paper investigated the use of computer vision to conduct a pH value estimation from three varieties of oranges. Color and texture-based features on the gray level co-occurrence matrix were extracted for the prediction values where in regression coefficient of 95.7% was achieved on one orange variety.                                                                                                                                                         | The time it takes to conduct estimation is as follows for the following steps: 0.15s for image segmentation, 0.13s for feature extraction and selection; 0.14s for multi-layer perception (MLP) neural network regression. This totals for 0.42s of inference speed proper for industrial applications.                                                                            |

Lastly, Table 2.4 covers the research papers that specify how iceberg lettuce quality was retrieved through the fields of computer vision and automation. The studies indicate that current efforts are in line with merging applications of sorting on custom



built postharvest handling systems with vision-based quality grading. Other accessories aiding with the computational intelligence to infer proper grading of crop produce like that of modern sensors and industry 4.0 technologies to aid in the decision-making process of the sorting has also been noted for a real-time application of the quality assessment.

Table 2.4 Most cited research papers related to ("postharvest" produce AND "computer vision") AND ("lettuce" AND "quality" AND classification AND grading) AND (sorting AND automation)

| Author, Publication Date      | Number of Citations | Main Contribution                                                                                                                                                                                                                                                                                                                                                                   | Remarks related to the study                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Melesse et al., 2022 [41]     | 43                  | The paper is concerned with the application of digital twin for monitoring banana quality changes throughout its postharvest storage. A thermal camera was used to retrieve the surface and physiological changes of the fruit. The study applied a CNN architecture to achieve 99% prediction accuracy among four grade classes. No mention of sorting or automation was provided. | The study did not provide any sorting mechanism present in line with the application of the digital twinning of the outputs from the CV system. The automation mentioned covers the data ingestion to be near real-time for monitoring and intended execution of physical actuation done through the cloud.                                                                                                |
| Chakraborty et al., 2023 [42] | 41                  | The paper investigated the use of modern sensors and devices for handling deep learning image analytics for their vision-based sorting and weight-based grading of citrus fruits machine. The results of their custom classification model achieved a 97% accuracy and the weight-based grading to be 91% accuracy.                                                                 | The sorting system presented consists of two channels after the washing process of the citrus with designated solenoid relays to actuate plunger linkages for action and gate mechanism that will place the citrus to designated sorting chutes. The plunger is considered a four-bar mechanism where kinematic synthesis was conducted to verify compatibility with respect to intended precision points. |
| Utai et al., 2019 [43]        | 40                  | The paper is concerned with mass estimation for irregularly shaped mango cultivar using fruit dimension of length, width, thickness and area that is fed within an ANN. The paper showcased that utilizing length, width and thickness as parameters provided a 99%                                                                                                                 | The paper did not provide a specific sorting or mechanical mechanism for sorting mango fruits. The mention of the advantage of top-view                                                                                                                                                                                                                                                                    |



|                                |    |                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                |    | accuracy for the task. The paper evidenced that 2-D vision systems provided efficient means to lessen equipment requirements, computational time and memory consumption.                                                                                                                                                            | picture for only one camera is recommended to reduce computing time and operational cost.                                                                                                                                                                                                                                                                                    |
| Sola-Guirado et al., 2020 [46] | 14 | The paper is concerned with the development of a low-cost CV analysis system for olive greens in-field to determine its maturity, grade and bruise index. The results show that the system is reliable for weight estimation with a 0.9 r-squared coefficient and the prediction of one of its maturity indices to be 91% accuracy. | The automation given was through the CV that enabled calculation of maturity through a defined maturity index group as well as its bruising. The paper referenced another system for sorting that utilized pneumatic valves for the green olives under falling angle, belt speed, valve position and camera angle to produce a 98% success rate in separation through color. |
| Xu and Lu, 2024 [47]           | 4  | The paper examined the use of YOLOv8 for segmenting and tracking of sweet potatoes along a roller conveyor to perform full-surface quality assessment and volume estimation upon the crop produce travel within the handling system. It was able to garner 91.7% accuracy for grading among four classes.                           | The system was built on a custom-designed roller conveyer belt specific to adding a LiDAR-RGBD camera on top of a chain-gear-rack transmission system simplified to rotate the sweet potatoes that are irregular shaped to provide real-time measurements on all parts of the crop. The system, however, did not provide any sorting mechanical mechanism.                   |

Given the literature search strategy provided above, there is ongoing research endeavors among experts that tries to advance industry 4.0 technologies and practices under postharvest handling systems particular for grading and sorting of priority crops. Most papers are heavy on utilizing deep learning algorithms and less feature extraction were provided. The need for research to develop a system that is autonomous through computer vision technology focusing on a different artificial intelligence approach should be a point of interest for the current study for iceberg lettuce as there are minimal studies that are tackling these technologies specific for this lettuce variety.



## 2.1 Quality Parameters and Handling Practices for Postharvest Iceberg Lettuce

A detailed review for the morphological, biochemical and physicochemical quality attributes of the lettuce variety iceberg lettuce, or crisphead lettuce, (*Lactuca sativa* var. *capitata*) along with the vegetables' conventional handling practices is examined. Iceberg lettuce physical appearance is often described as that of a tight head of crisp light green leaves with excellent texture often used in salad dishes and in fast food chains [48]. A genetic resource study made by [49] described iceberg lettuce to have thick crisp leaves and flabellate leaf venation and is edible to be eaten raw. The head may be completely developed with overlapping leaves. It is a cool-season vegetable which grows best in temperatures between 7 and 24 °C. These are the morphological parameters defined by the general appearance that can be quantified and are a significant determinant for growth and postharvest quality [50]. Figure 2.2 showcase the different morphology parameters of the lettuce variety mentioned in literature.

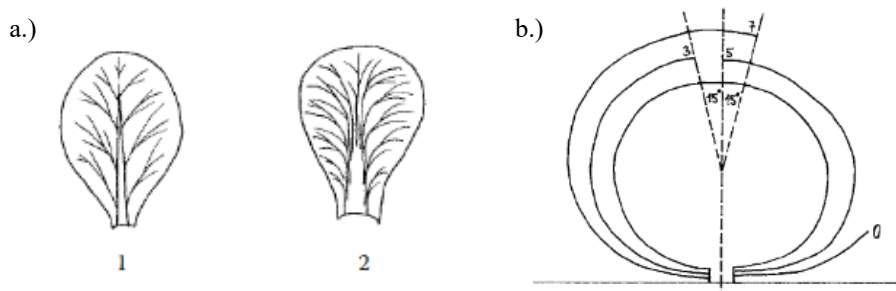


Figure 2.2 Morphological features of Iceberg lettuce. a.) Young leaf - venation 1 pinnate; 2 flabellate. b.) Head -overlapping of leaves 0 none; 3 partly; 5 half; 7 complete. [49]

Many studies cover the different length, size and shape of the iceberg lettuce leaves as a determinant of its quality both in terms of its growth and postharvest properties. Studies made by [51] [52] made insights that growth phase of the lettuce cultivar particular to its head formation has a correlation significant to its environment where in it must be grown in high altitude and under low temperature conditions. Seasonality can also affect its weight where farmers tend to farm it during colder seasons of September to May.





The roundness of the lettuce can also provide insights to its growth where studies have evaluated iceberg lettuce based on its longitudinal diameter and equatorial diameter for its quality. In a study to determine the influence of cultivar, fertilizer and irrigation on iceberg lettuce morphology, the following traits have been measured: head height, head diameter, head weight, number of leaves, fresh head weight, stem height, stem diameter, and fresh stem height [53]. Another study focused its findings on using head weight to determine irrigation scheduling with emphasis to have maximum marketable yield [54]. Lastly, another study covered head mass of iceberg cultivars alongside its mineral composition for postharvest quality. Table 2.5 covers these related works more thoroughly, where head weight was a significant determinant of iceberg lettuce cultivars overall. This is then a quality parameter that should be explored more in detail relevant to the current study.

Table 2.5 Some related works that used fresh head weight of iceberg lettuce as a postharvest quality parameter

| Author, Publication Date | Main Statements of head weight as a quality parameter                                                                                                                                                                                                                                                                                                                                                                                                                                 | Remarks in relation to current study                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Simko et al., 2015 [51]  | This book chapter revolves around lettuce yield gain within the year of 2014 and past. It provides a comprehensive overview of current demands within the north american market with notable increase of yields by 102% from 1970 to 2010. This is attributed to the popularity of packaged salads. Head formation and higher head weight are promoted by high intensity and lower temperatures. Provided cultivars have been recommended to be planted within March-July date range. | Emphasis has been made that iceberg lettuce cores and darker green outer leaves are not used for salad processing. Other signs of leaf blanching that is not also part of processing are described to be yellow to yellow-green in color. One mitigation for these unattractive leaf traits are to introduce genes addressing chlorophyll deficiency during production that increases leaf length and number of leaves per plant. |
| Tudela et. al, 2017 [52] | Quality related parameters of fresh-cut iceberg lettuce are outlined as follows: head weight, dry matter, respiration rate, off-odors, ethanol content, acetaldehyde content, cut edge browning, hue angle. Through pearson correlation coefficient, respiration rate of fresh-cut iceberg lettuce significantly increase in June compared to data obtained in November to May although no difference in head weight throughout seasons.                                              | This study investigated varying season climates as influence for the quality of fresh-cut lettuce retrieved from different cultivars and harvest based on lettuce producing regions in Murcia, Spain. The climatic factors related to quality loss were growing season mean temperature, oxygen and carbon dioxide content within field atmosphere                                                                                |





|                              |                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Johnson et al.,<br>2016 [54] | Lettuce quality on yield trials are based on head weight, total biomass yield and marketable yield. As basis of the treatment is efficiency of the irrigation scheduling, primary consideration was based on the field yield given same assumed daily parameters.                                                                                                                | This study tackled decision-support models for scheduling of cool-season vegetable crops specific for iceberg and broccoli. The models utilized daily reference evapotranspiration data and estimates of crop fractional cover to guide irrigation. The results showed that the model-based treatments lessen water usage by 26-51% than standard practices within California, USA.  |
| Aliste et al.,<br>2020 [55]  | The quality attributes assessment of the iceberg lettuce were evaluated in total yield, weight, longitudinal diameter, and equatorial diameter for assessment of the productivity of lettuce growth while external color, internal color, water content, mineral composition were considered for visual appeal and known nutritional quality of iceberg lettuce for consumption. | This study tackled the use of agro-wastewater with known pesticides reused for iceberg lettuce grown within a greenhouse in Murcia, Spain. The paper evinces that after treatment of the wastewater of metallic catalyst provided no significant difference in quality parameters of lettuce grown using treated agro-wastewater compared to those irrigated with non-treated water. |

Another quality parameter cited in many literature for iceberg lettuce is its chlorophyll concentration. It is a biochemical attribute where higher chlorophyll concentrations have been cited to have better lettuce growth which is favorable to market postharvest. A study mentions how pigment measurements of chlorophyll levels indicate slow degradation of the lettuce quality overtime as it losses its nutrients due to biochemical and enzymatic activities resulting also to browning effects [57]. Color serves still as one of the crucial quality indicators for freshness of vegetables especially for postharvest marketing [58]. Shown in Figure 2.3 is the overall appearance of whole iceberg lettuce given set time periods where the study made by [59] provided insight how chlorophyll values ( $\text{mg } 100\text{g}^{-1}$ ) drop significantly from 0.38a to 0.20b between day 0 and day 71. Quality of the iceberg lettuce has been cited in the study that visible chlorophyll degradation of 30% is seen after 17 days from fresh harvest. This can be then correlated with the appearance of the lettuce shown.

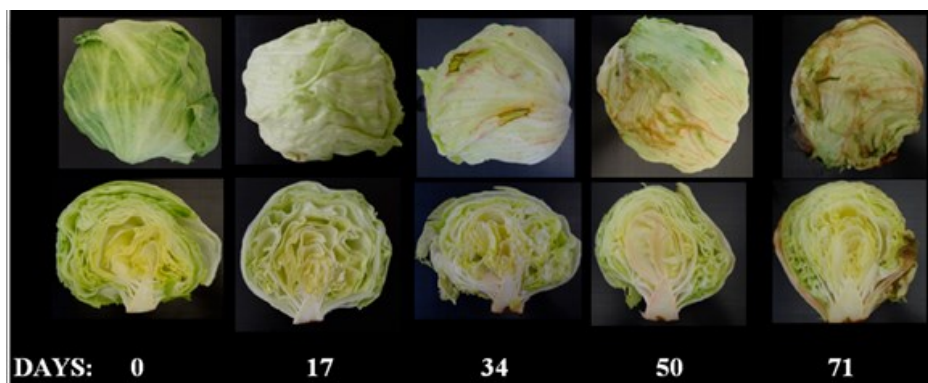


Figure 2.3 Overall Appearance of whole iceberg lettuce during storage. [59]

Some studies about this quality parameter are focused on knowing mainly the effects of light-emitting diodes (LEDs) treatment to postharvest quality to minimize senescence or stress during storage. It has been cited in a study that chlorophyll fluorescence measurements can provided insight on the storage potential and quality of iceberg lettuce by aiding photosynthetic effects within plant [60]. This is also supported by another study that postharvest performance of whole head lettuce was reflected in improved chlorophyll fluorescence values [61]. This resulted in better shell life of iceberg lettuce that increased visual quality values and delayed browning on leaf edges.

Table 2.6 tackles further use cases of chlorophyll concentrations of iceberg lettuce as a quality parameter and can also be applied in the current study for postharvest quality use.

Table 2.6 Some related works that used chlorophyll concentrations of iceberg lettuce as a postharvest quality parameter

| Author, Publication date           | Main Statements of Chlorophyll concentrations as a quality parameter                                                                                                                                                                                                                                                                                                                            | Remarks in relation to current study                                                                                                                                                                                                                                                                                                                 |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Woltering and Witkowska, 2016 [61] | The following analyses were done to portray effects of lighting to lifecycle of iceberg lettuce: Overall visual quality ratings, computer image analysis, chlorophyll fluorescence measurements, net photosynthetic rate and dark respiration. Discussion specific to light stored provided more higher sugar and starch level values but lower variable chlorophyll indicated in fluorescence. | The study tackled the light quality and intensity affecting quality attributes of fresh-cut iceberg lettuce during growth and storage conditions. Particular, the use of high-intensity monochromatic light has shown increase of carbohydrate levels that correlates to shelf-life extension. Ethylene production were also discussed in regards to |



|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | wounding and stress within environment                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Falcioni et al., 2017 [62]      | In this paper, lettuce was classified into different pigmentations based on set parameters through ATR-FTIR spectroscopy and machine learning models. Some of these parameters include chemical contents of the plant such as chlorophyll- <i>a</i> and <i>b</i> , carotenoids, flavonoids, anthocyanins. Results of the study show that spectral signatures of lettuce varieties differ from each other. This allows the model to accurately predict and classify the specific metabolites in lettuces.                                                                                                                                                                       | Different kinds of lettuce had different specific spectral signatures that allows machine learning to classify them. This offers an alternative to classification of the lettuce which is faster, and does not require as much intensive data collection.                                                                                                                                                                                                                             |
| Concepcion II et al., 2021 [63] | This study makes use of non-destructive and onsite measurements in closed environment agricultural systems. The use of UV-Vis spectrophotometry, fourier transform infrared spectroscopy, and integrated computer vision were used. The effects of different photosynthetic light treatments on aquaponic lettuce canopy reveals concentration changes of various photopigments and nutrients. Hybrid Decision Tree Multi-Gene Regression Genetic Programming (DT-MSRGP) model, provide spectro-textural-morphological signatures, demonstrated high predictive accuracy for chlorophyll <i>a</i> and <i>b</i> , -carotene, anthocyanin, lutein, and vitamin C concentrations. | It was found that different phytopigments under different light treatments, $\beta$ -carotene and anthocyanin act as a natural sunscreen in protecting lettuce from light stress. Red-blue light was found to induce chlorophyll- <i>b</i> concentration and white light promoted other pigments and vitamin C. The study recognizes the use of DT-MSRGP model for accurately predicting phytopigment and nutrient concentrations during lettuce head development and harvest stages. |
| Cömert et al., 2020 [64]        | The use of the CIELab color space where $L^*$ value indicates lightness, $a^*$ indicates redness/greenness, $b^*$ indicates yellowness/blueness. Hue angle was used as a qualitative attribute to define color difference in reference to grey color for fresh produce pigmentation. This was utilized for chlorophyll concentration and determination for total antioxidant capacity.                                                                                                                                                                                                                                                                                         | This study focused on use of visual perception as an indicator of food quality and grade. The results indicate the color hues analyzed by image analysis can be related with total antioxidant capacity of fruits and vegetables, but with some limitations and can be used as a guide for food selection to increase daily antioxidant intake. Most fruits and vegetables having hue values above $180^\circ$ and below $20^\circ$ , have high antioxidant capacity.                 |

Significant literature points lastly to moisture content, a physicochemical attribute, upon harvest as a postharvest quality parameter that determines an iceberg lettuce's weight and diameter as well as higher aerobic bacteria count that may reduce



lettuce head quality [65]. Limited studies have covered this attribute of iceberg lettuce but it is known that due to the nature of fresh produce being highly perishable, moisture content may lead to discoloration like that of pinking after a 20 days of harvest [66]. Preharvest water stress, an abiotic factor, can have an impact on yield and postharvest quality as seen in a study made by [67] where irrigation deficit provided less pinking but also less maturity growth of the lettuce head in terms of its diameter with high moisture content ranging 93-95%.

## **2.1.1 Postharvest Practices for Iceberg Lettuce**

Postharvest processes tackle the following actions: handling, grading, sorting, trimming, cleaning, packaging, and transporting of agricultural crop produce as shown in Figure 2.4. In order to optimize crop quality produce into the market, it entails coordination and integration of tasks from respective stakeholders from these actions to increase profitability while being within industry standards and regulations [68].

Specific to postharvest handling, it covers the goods' quality, freshness, marketability, and profitability [69]. Grading, sorting, quality evaluation, monitoring, flaw identification, recognition, and maturity detection are all included in it. Historically, manual labor has been employed mostly for various agricultural post-production phases such as grading, sorting, and quality assessment. However, mistakes are frequently made during this process because of faulty and unreliable techniques. As a result, agricultural manual labor can be enhanced by technological developments [70]. Furthermore, Alfred et al., reports that 33% of agricultural waste resulted from inadequate logistics and storage [63].

The Food and Agriculture Organization (FAO) of the United Nations has made a guideline for stakeholders within the horticultural supply chain management that evinces postharvest handling operations of lettuce [71]. The article enumerated different good practices that should be carried out when handling crop produce to be packed and delivered to the market. This is important to prevent crop harvest loss which is a dominant issue faced within the post-production agricultural lifecycle.



FAO has stated in the article that the sorting process can be simplified into the separation of good and bad quality lettuce produced specifically on the following definition of what defines good quality lettuce: Good quality lettuce must have fresh, fairly tender, well-trimmed leaves and must be free from decay, discoloration, foreign material, insects, and damage caused by coarse stalks or other mechanical means [72]. Typical sorters have first-hand experience in how to sort the produce. Still, recent studies have mentioned that this is inefficient in making proper sorting based on quality grade and human error as this process is a tedious activity [73].

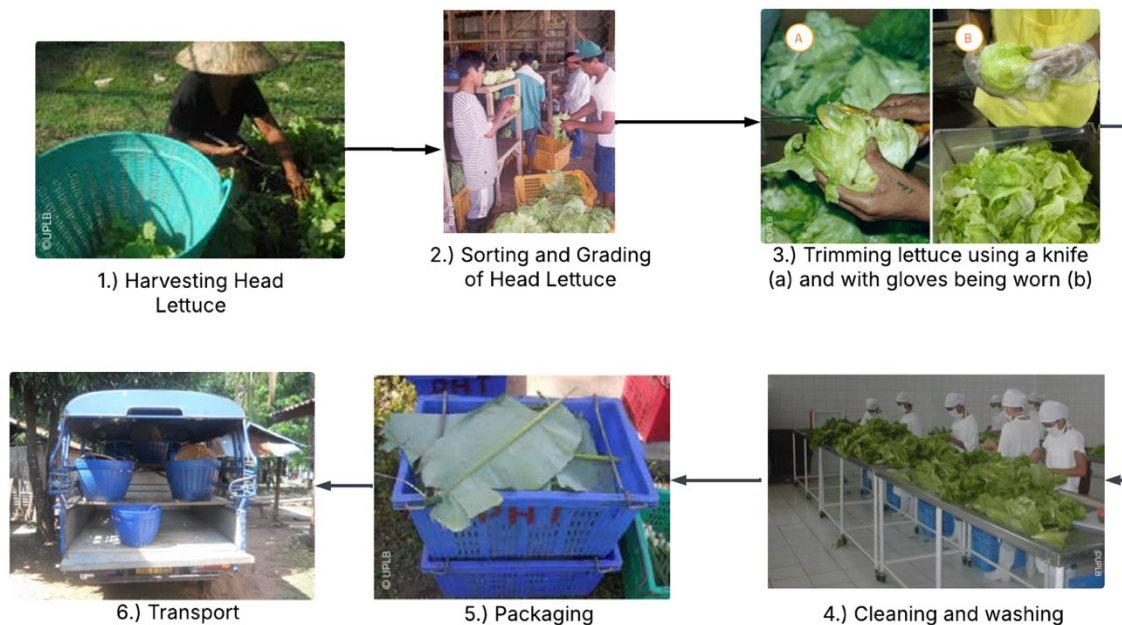


Figure 2.4 Postharvest Practices as enumerated by food and agricultural organization (FAO) [71]

In the Philippines, Benguet, Bukidnon and Cavite are the provinces with top lettuce production with over 3,634.12 megatons being harvest and sold yearly. Known crisphead or iceberg varieties within the Philippines are Great Lakes 54, Great Lakes 659 and West Lake Green. Provided in Figure 2.5 are photo samples of iceberg lettuce varieties as shown by the Bureau of Plant Industry (BPI) lettuce production guide where on the left side of the figure features the Great Lakes variety while on the right covers West Lake greens. [74].





Figure 2.5 Iceberg lettuce varieties in the Philippines. [74]

In the Philippines, a study made by [75] provides known cost within the postharvest supply chain covering hauling, transporting, trimming and packaging for local lettuce produce. Open-field technologies are prevalent in the country where lettuce varieties of higher caliber are unable to be sold on spot markets due to high investment cost to maintain cultivation and low marketing channels to differentiate high quality produce.

Manufacturing practices have been pointed out to be a primary safety concern for fresh-cut processing. Similar postharvest losses were given by the Philippine Department of Agriculture (DA) in their Philippine Vegetable Industry Roadmap. In their report, highland vegetables that includes lettuce produce, account for 20-30% losses due to improper trimming and sorting [77].

To address these concerns of postharvest losses, the National Economic and Development Authority (NEDA), along with the Department of Science and Technology (DOST), has enumerated the different agendas that should be tackled within 2022-2028 in their report of the Harmonized National Research and Development Agenda. (HNRDA) [78]. The report has tackled within the section for agriculture, aquatic, and natural resources the need for postharvest, primary processing, and product development that aims to alleviate issues related to postharvest of high-priority crops and further improve on ensuring quality produce is being sold into the Philippine markets.



## 2.1.2 Quality Grading Standards Related to Iceberg Lettuce

Manual labor has traditionally been utilized primarily for various agricultural post-production phases such as grading, sorting, and quality assessment. However, errors are common during this process due to inadequate and unreliable techniques. As a result, technological progress can be employed to improve manual agricultural activities [70].

In the local scene, for lettuce quality parameters that define the corresponding grade for iceberg lettuce, the Bureau of Agriculture and Fisheries Standard (BAFS) has defined the Philippine national standard (PNS) on different factors that should be considered for grading and classification of iceberg lettuce in their article PNS/BAFPS 19:2005 [9]. Summarized in Table 2.7 are the different descriptors for the essential composition and quality factors that are pertinent for grading. This grading follows similar standards indicated within the ASEAN region [79] and regulations from the European Union (EU) [80]. These factors mainly revolve around the general appearance of iceberg lettuce to minimize defects that dissuade consumers from buying iceberg lettuce. Other quality parameters like that of biochemical and nutritional contents are not considered.

Table 2.7 Philippine national standard for iceberg lettuce composition and quality factors

| Quality Factors     | Description                                                                                                                                 |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Clean               | head is free from dirt, filth and other foreign matter including live and dead insects                                                      |
| Closely trimmed     | the base is trimmed closely to the point of attachment of the outer leaves and does not have more than two (2) wrapper leaves and one cap   |
| Fairly-well trimmed | the base is trimmed off closely to the point of attachment of the outer leaves, does not have more than four (4) wrapper leaves             |
| Fresh               | the head as a whole has normal succulent leaves, and the wrapper leaves are not more than slightly wilted                                   |
| Green               | one-half or more of the exterior surface of the head, exclusive of the wrapper leaves, has at least a light green color                     |
| Reasonably trimmed  | the base is trimmed closely to the point of attachment of the outer leaves and does not have more than six (6) wrapper leaves               |
| Solidity            | head is fairly firm, not compact, nor soft or spongy but with good head formation and edible content; yields to slight or moderate pressure |



A visual representation of quality factors for iceberg lettuce can be seen in Figure 2.6. Emphasis should be made that the figure depicts an iceberg lettuce without any defect in the topmost left based on visual cues. This is the reference point for the different defects that lessen the grading of the lettuce sample. Other standard references from Association of Southeast Asian Nations (ASEAN) Standard 50 [81] and United Nations Economic Commission for Europe (UNECE) Standard FFV-22 [82] for iceberg lettuce standards provide similar standard and description for quality control and size grading. The only different is that these standards did not accompany visuals for their definition of defects within the lettuce.

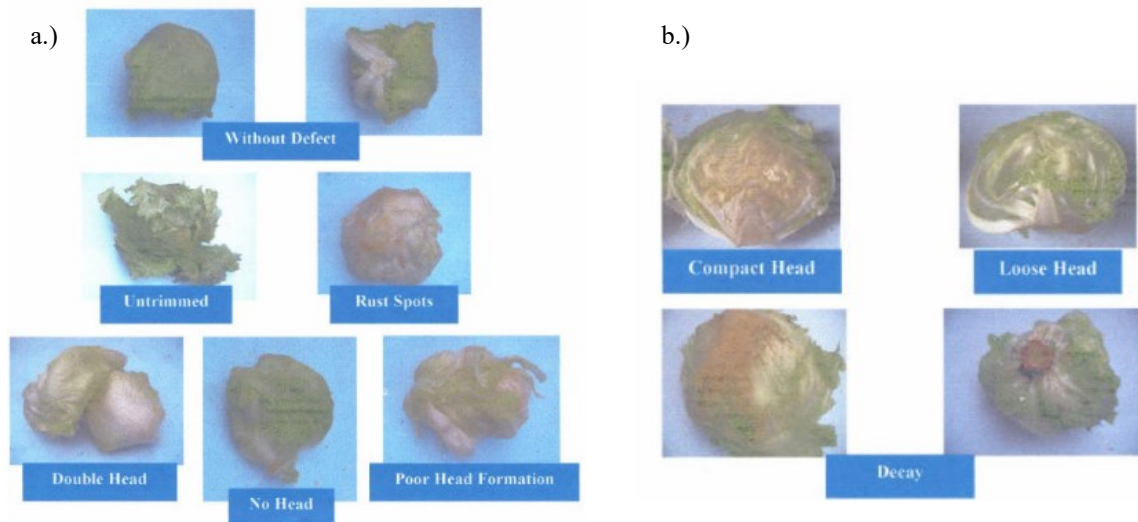


Figure 2.6 Visual representation of quality factors of iceberg lettuce based on PNS/BAFPS 19:2005 a.) Known sample of without defects and with defects; b.) continuation of known defects for iceberg lettuce [9]

Aside from the morphological factors of the lettuce appearance, another factor considered for PNS would be the lettuce size, as indicated in Table 2.8. The classification is defined in four categories with its corresponding size range descriptor. As mentioned, head weight is salient to the quality parameter of iceberg lettuce as this defines crop maturity and nutritional content.





Table 2.8 Philippine national standard for iceberg lettuce size classification

| Size Classification | Description   |
|---------------------|---------------|
| Small               | 200-300 grams |
| Medium              | 301-450 grams |
| Large               | 451-600 grams |
| Extra Large         | >600 grams    |

With the provided PNS standard and previous knowledge of quality parameters for fresh head weight, chlorophyll-*a* and moisture content, Table 2.9 details how a grading criteria can be made from these quality attributes and relate it to existing national standard. that evaluates iceberg lettuce quality through lettuce composition and factors mentioned in Table 2.7. It enumerates the basic requirements to be included for crop grading. This grading also has tolerance levels considered specifically the following mentioned: For grade 1, should ten percent of iceberg lettuce fail to meet the basic requirements, including two percent for defects causing serious damage and two percent for heads affected by decay on any portion of the head exclusive of the wrapper leaves.

–For grade 2, tolerance would be twelve percent of iceberg lettuce that fail to meet the requirements of this grad which include four percent for soft heads, two percent for decay, and six percent for head having permanent defects; For grade 3, twelve percent fail to meet the requirements of this grade which include four percent for heads which are affected by decay on any portion of the head exclusive of the wrapper leaves and eight percent for heads having permanent defects. These tolerances should be defined more in these quality parameters for grading iceberg lettuce for reviewed parameters. These will be considered for the current study.

Table 2.9 Summary of reviewed quality parameter, its perceived grading and comparison of it to PNS for iceberg lettuce

| Quality Parameter                       | Grade 1 (High Quality)                              | Grade 2 (Medium Quality)                                 | Grade 3 (Low Quality)                                        | Comparison with the Philippine National Standard (PNS)                                      |
|-----------------------------------------|-----------------------------------------------------|----------------------------------------------------------|--------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Moisture Content [65], [66], [83], [84] | 94–96% or higher. Based on PNS, can be described as | Between 90–94%. Slightly below preferred range. Slightly | 90% or lower. Described as pale/discolored leaves with faded | PNS guidelines for lettuce quality emphasize that optimal moisture contributes to freshness |



|                                              |                                                                                                            |                                                                                                                |                                                                                                                               |                                                                                                                                                                                                          |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                              | Crisp, compact firm, and turgid leaves.<br><br>Considered to be optimal range.                             | dry but still vibrant<br><br>Considered to be acceptable range.                                                | color. May show signs of decay.<br><br>Considered to be suboptimal range.                                                     | and shelf life. Although exact numerical thresholds may vary, the industry often expects high-quality lettuce to fall within a narrow moisture range like the Grade 1 criteria.                          |
| Chlorophyll- <i>a</i> [62], [63], [85], [86] | 0.3691 mg/g or higher. High Concentration Reflects strong photosynthetic capacity and vibrant green color. | Between 0.2884-0.3681 mg/g. Acceptable level, though may indicate slight senescence, less known green pigment. | 0.241 mg/g or lower. Pale or discolored leaves. Suggests nutrient deficiency or advanced senescence, resulting in pale color. | While PNS does not specify chlorophyll- <i>a</i> in mg/g, visual standards (e.g., RGB-HSV values) correlate with pigment concentration and is associated with the descriptors mentioned in the standard. |
| Fresh Weight [9], [53], [54], [82]           | 451-600 grams or higher. Indicates full maturity, the solidity and desirable marketability                 | Between 450-201 grams. Acceptable but potentially less consistent in head formation                            | 200 grams or lower. May indicate immaturity or yield loss, impacting economic value                                           | PNS/BAFS standard sets size classifications for iceberg lettuce correlating the measured weight to established market-grade categories.                                                                  |

Known quality parameters and the national standards provided align closely to one another. Industry expectation for postharvest for iceberg lettuce would want to have market-grade lettuce characterized by high moisture retention, strong indication of chlorophyll-*a* concentration, and full maturity [87] [88] . Defects act as a basic guide to accept/reject postharvest iceberg lettuce for consumer sell but overall, it does not cover internal qualities that relate it to other market values. The provided studies help undermine visual and structural attributes that are quantifiable in terms of its biochemical, physicochemical parameter towards postharvest grading and sorting.

### 2.1.3 Industry Standards for Postharvest Handling Systems

The main component for any agricultural sorting and grading machine will be a conveyor belt passing the fruits and vegetables in each file for processing. In the Philippines, Agricultural Machinery Testing and Evaluation Center (AMTEC) has established a national standard for the different components necessary for flat belts that is used to drive belt conveyor applications [15]. Provided in Table 2.10 is a summary of



the technical specifications needed for utilizing flat belts to drive a conveyor machine for agricultural purposes like that of transporting leafy vegetables like iceberg lettuce in a single line.

Table 2.10 Specifications of Flat Belt Drive for Conveyor Belt Systems [15]

| Specifications           | Details                                                                                  |
|--------------------------|------------------------------------------------------------------------------------------|
| Belt Width and Thickness | 10-200 millimeters nylon cord belts adjustable with 2.5 mm thickness for medium capacity |
| Minimum Belt Speed       | 152 millimeters/min                                                                      |
| Minimum Pulley Diameter  | 102 millimeters                                                                          |
| Number of Belt Ply       | 3                                                                                        |
| Power Rating             | 21 watts                                                                                 |
| Arc of Contact Factor    | 0.69 on 90-degree contact                                                                |
| Service Factor           | 1.4 for squirrel cage AC motors used typically for conveyors.                            |

Other calculations that must be done by the system designer would be to consider what type of material and conveyor is being used for the specific application. The safety requirements on the other hand entail that the drivers are enclosed to prevent foreign material contact with regular inspection made on belt tightness. Lastly, with the provided initial belt specifications, the belt length calculation then needed to be retrieved as final variable.

Light conveyor belts are often in the form of plastic modular belts which uses plastic interlocking belts for material handling. This is often mentioned by manufacturers also as the conveyor belt type typically used for agricultural produce handling and sorting [89]. Described in Figure 3 is the typical configuration of light conveyor belt based on the ANSI-CEMA 402-2003 standard [14]. This provides the setup of tensioners, pulley, drive, and return and snub rollers on a flat horizontal setup. The length of the belt may vary on applications but a general placement in terms of length ration is provided within this standard with know variables that was mentioned in the previous paragraph for light conveyor belts like power requirement, speed and pulley diameter.

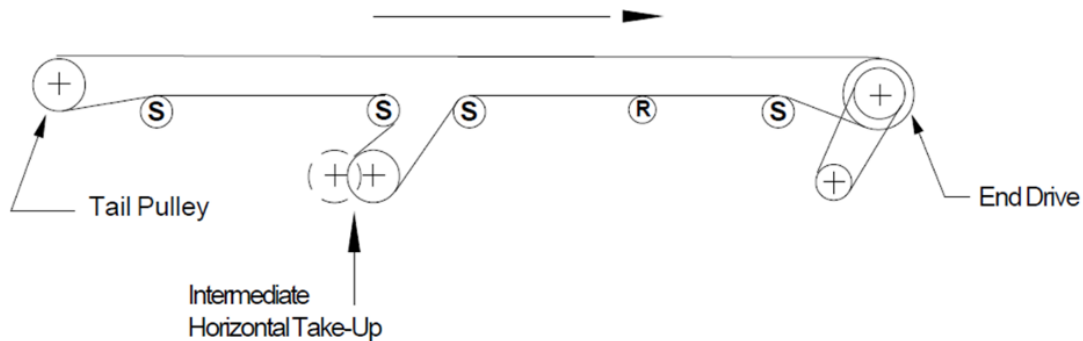


Figure 2.7 Horizontal Belt Conveyor Setup Convention

Based on the standards above, similar applications of sorting devices with conveyor systems applied to related works like that of Joaquin and Avila [90] has defined their sorting machine to be optical in a single-line type enumerating that they can process up to 720 to 800 pieces of Carabao mangoes in their sorting machine that has a power requirement of 33 watts and have an overall length dimension of 2720 millimeters.

Similar applications have also mentioned the use of ISO standard 21183-1 for their conveyor belt design consideration. A study made for automated pili fruit sorting device [91] has utilized this as basis for their conveyor apparatus which describes light conveyor belt systems characteristics and applications.

Lastly, other peripherals for the conveyor systems revolve around the mounting brackets to hold firm the system into ground as well as the controls to start the setup and program the speed of the motor and other actuators. The use of PLC has been used in many literatures for the controls being deployed in Flexible Manufacturing System (FMS) towards automation on programmable workstations [92] where the main objective of the system is to control motor motion at five stations including three intermediate working and two mandatory stations. Ladder logic is imposed to various degrees to instruct the controller how to move the corresponding motor [93]. All technologies stated above will be considered in current study.



## 2.2 Emerging Technologies on Postharvest Grading and Sorting Systems

Postharvest handling systems is defined as all operations needed to aid on the harvesting of the crop in the field to delivering the final product to the consumer [94]. The study aims to examine current research trends on sorting and grading within the agri-food sector with emphasis on its sensing, smart and sustainable (S<sup>3</sup>) domains. With the adoption of machineries and Industry 4.0 solutions, enterprises within agri-food sector must now consider the following domains of these technologies. Based on [95], the following domains are defined as follows: Sensing domain refers to the capability of a system to detect events, acquire data and measure changes that occur in a physical environment; Smart domain refers to the capability of a system to incorporate actuation and control functions in order to describe and analyze situations and make decisions based on available data; Lastly, the sustainable concept is applied to products to balance or co-optimize, their performance considering social, economic and environmental objectives.

Table 2.11 provide a summary of the related works that covered postharvest grading and sorting systems along with their known sensing, smart and sustainable domains. Most of the literature cover Level 3-4 sensing solutions where in the modules executes automatically then informs the human of its decision and others without any human oversight. This is predominant with the use of cameras as a image or video capturing feature for their target crops and enable a classification algorithm for grading. Most of the studies also cover a Level 3 smart solution where human is out of the operation but intervenes if necessary. This is evident with the sorting made immediately once algorithm determines the perceived grade of the product and sorts it through given actuators presented. Lastly, the papers have Level 3 sustainable solutions where in design principles and values seek to improve pre-existing products towards a more user-friendly and long-lasting product. This is evident with how each study aimed to improve performance and reduction of cost, expenses and even ergonomics compared to conventional grading and sorting methods within agri-food industries.



Table 2.11 Some related works of postharvest grading and sorting systems on their sensing, smart, sustainable domains

| Author, Publication Date                   | Target Crop, Objective of the Study                                                                                                                        | Components used for Sensing Domain                                                                                                                                                                                                      | Components used for Smart Domain                                                                                                                                   | Impact on Sustainable Domain                                                                                                                                                                                                |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Nandi et al., 2015 [96]                    | Indian mango varieties: To sort and predict maturity level through machine vision-based using Support vector machine (SVM) classifiers in a conveyor belt. | A charge coupled device (CCD) camera placed on an artificially illuminated image collection chamber. A light sensor for consistent illumination proper.                                                                                 | Lab VIEW toolbox; Programmable logic controller (PLC) to actuate solenoid valves and control light sources. Conveyor controlled at constant 2 meters/second (m/s). | The paper aimed to improve the performance and reduction of computation to meet real time target of manual sorting of the crop.                                                                                             |
| Novelero, J., and Dela Cruz, J., 2022 [91] | Pili Nut or fruits of similar properties; To sort produce by its maturity using Linear-SVM classifier on a conveyor belt.                                  | An A4Tech digital camera to take snippets of pili fruit. Morphological features from segmented images were extracted for image classification.                                                                                          | MATLAB toolbox; Arduino Uno to actuate a SG90 servo motor. It has a serviceability of sorting 720 pili fruits in an hour.                                          | Price for development was compared to existing systems and was noted to be more than 20 times lower than that of conventional grading sorting system.                                                                       |
| A. C. Joaquin and R. P. Avila 2024 [90]    | Philippine mango; To develop a sorting machine complaint to BAFS/PNS and other standards through a multiple linear regression (MLR) model.                 | 1080p/30 frames per second (fps) with autofocus camera; Two stage approach where initial item is rejected/accepted if it has any defects. Grading is then sorted using its size. Infrared proximity sensor to detect presence of fruit. | Laptop supplying controls. Pneumatic eject arms are actuated through an Arduino Mega within designated sorting bins                                                | A partial budget analysis covered that the sorting machine is more beneficial that traditional sorting methods known. A comparative analysis between three standards was also made to reveal a suited standard for sorting. |
| Chakraborty et al., 2023 [42]              | Citrus Fruits; To develop a machine for washing, vision-based sorting using a lightweight CNN “Sort Net” and weight-based grading of citrus fruits.        | A RAV17 camera used for the image acquisition system; Initial reject/accept through known defects of citrus fruit; Load cell used for the weight measurement                                                                            | Arduino Mega controlling solid-state relays (SSRs) to actuate solenoid valves; Conveyor and washing tank was also control through microcontroller.                 | An ergonomic assessment was made to compare the use of the machine to that of the laborers performing postharvest handling the crop.                                                                                        |



## 2.3 Computer Vision in Postharvest Grading and Sorting Systems

Based on [98] computer vision (CV) is a scientific discipline that includes methods for acquiring, processing, analyzing and understanding images of the real to produce numerical or symbolic information that a computer can process. In agri-food sector, it has been used primarily for nondestructive analysis of agricultural products to inspect and classify quality in terms of external quality, size, color. Presented in Table 2.12 are related works reviewed in relation to current study. Main interest lies on what image processing and machine learning algorithms or methods used for their study and provided accuracy of the algorithm used and how fast inference or throughput presented.

Based on the papers, it is given that segmentation is required to isolate the given crop produce using either Otsu's Thresholding or Watershed Algorithm. All grading systems considered either the weight of the crop or based it off through its ripeness values. Defects were only a determination if it is suitable to graded. Traditional machine learning techniques are still favored for repeatability while deep learning algorithms are steadily also being applied to grading applications due to less feature extraction needed to be applied to dictate grading. Target accuracy should be within above 90% accuracy at least. Repeatability should be within 80%. Throughput or inference time varies between crop produce and computer vision model.

Table 2.12 Some related works of postharvest grading systems

| Author, Publication Date | Main Contributions of the Study                                                                                                                               | Computer Vision Algorithm/Method Used                                                                                                                                                                   | Known Accuracy, Repeatability and Inference Time/Throughput                                                                                                                                              |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Wang et al., 2018 [99]   | The paper focused on the design of a sorting system for fresh white button mushrooms to detail a proper means to grade the crop based on its pileus diameter. | Watershed method to identify boundaries of crop; Canny operator for edge detection of features and region of interest; Iterate imaging processing using Graham algorithm to scan convex hull of produce | - Overall Accuracy: 97.42%.<br>- Repeatability was not explicitly stated by mention of damage rate and undetected rate to be 0.05%, 0.96% respectively.<br>- Throughput is to be 102.41 mushrooms/minute |
| Zhang et al., 2021 [100] | The study presents a novel apple grading and sorting system designed for infield                                                                              | Double Class- Linear Discriminant Classifier to segment fruit from background of                                                                                                                        | - Overall Accuracy: 99% accuracy for sorting for different varieties of apples.                                                                                                                          |





|                                      |                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      | use, which integrates fruit singulation, rotation, and transportation mechanisms with imaging-based grading and sorting technologies.                                                                                                                                                       | conveyor; Thresholding algorithms to classify pixels into different color categories for the apples; Size estimation was then done by calculating maximum equatorial diameter.                                                                                                                                                        | <ul style="list-style-type: none"> <li>- Repeatability of grading was 81% across different lanes of sorting on multiple runs.</li> <li>- Throughput is to be 7.5 apples per second covering the entire singulation.</li> </ul>                                                                                                                                                                                 |
| Ismail, N. and Malik, O., 2022 [101] | The study developed a Raspberry Pi-based system with a camera and touchscreen for real-time fruit grading, enabling non-destructive, cost-effective visual inspection for both apples and bananas to determine if it is unripe, ripe, overripe.                                             | <p>For Preprocessing: Gaussian Filter, CLAHE (Contrast-Limited Adaptive Histogram Equalization) to enhanced image contrast and noise reduction. Segmentation using watershed extract foreground in training.</p> <p>EfficientNet with transfer learning techniques was trained on three different batches with data augmentation.</p> | <ul style="list-style-type: none"> <li>-Test Accuracy: Apples, 96.7%; Bananas 93.68%</li> <li>-Repeatability on inter lane variability suggested to be between 93-97% on actual real-time testing.</li> <li>-Throughput was not explicitly reported aside from mention of it being real-time.</li> </ul>                                                                                                       |
| Chakraborty et al., 2023 [42]        | The study developed the use of a Raspberry Pi 4B for real-time citrus fruit grading, combining vision-based sorting based on surface defects and weight-based grading among three categories. Additionally, an ergonomic assessment was done knowing that 82.9% compared to manual grading. | Captured image was resized to 125x125 pixels and 10 frames were averages to reduce noise. A low-latency CNN architecture "Sortnet" of 4 convolutional layers, 2 max-pooling layers optimized for edge devices                                                                                                                         | <ul style="list-style-type: none"> <li>-Test accuracy was 97.6% accuracy. Real-time operation was 97%. (This was for defects and not in regards to its weight that used a load cell)</li> <li>- 97.7 intra-lane consistence under 4m/min belt speed and 160mm field of view.</li> <li>-Inference time was stated to be 0.32 seconds per citrus fruit. A throughput of 102 fruit/minute was covered.</li> </ul> |
| Azarmdel et al., 2020 [102]          | The study developed a machine vision system to classify mulberries into three ripeness                                                                                                                                                                                                      | Otsu's Thresholding was used to segment mulberry from background. A feature selection optimization was done through                                                                                                                                                                                                                   | -ANN with CONS feature resulted to highest test accuracy of 98.3%. Mean-square error (MSE) of $3.1 \times 10^{-3}$ .                                                                                                                                                                                                                                                                                           |



|  |                                                                                                                                                                |                                                                                                                                                                                                 |                                                                                                                                                                                                                                           |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | levels, unripe, ripe, overripe, using image processing and through traditional machine learning of ANN and SVM. This was to address postharvest marketability. | correlation-based Feature Selection (CFS) and Consistency subset (CONS) methods.<br><br>Classification models of artificial neural network (ANN) and support vector machine (SVM) was utilized. | -Repeatability was not explicitly stated by implied to be high due to MSE values to be extremely low. No intra-or inter- lane testing done as system had only one lane.<br><br>- Throughput or inference time was not mentioned in study. |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Other reviewed literature revolved around specific instances of computer vision use for current study's target crop, iceberg lettuce. Shown in Table 2.13 is a summary of the examined works highlighting the contributions or insights made by the study, its computer vision techniques or algorithms used and known accuracy and other metrics. Based on these findings, it can be summarized that trend for the use of CIELab color space is predominant as features for the postharvest lettuce quality classification. Known correlations of lettuce quality based of moisture content, chlorophyll contents and fresh weight has been relevant to existing literature and will be a consideration for current study.

Table 2.13 Some related works of computer vision for postharvest lettuce quality classification

| Author, Publication Date   | Main Contributions of the Study                                                                                                                                                                                          | Computer Vision Algorithm/Method Used                                                                                                                                                                | Known Accuracy and other Metrics                                                                                                 |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Cavallo et al., 2018 [103] | The study focused on determining the quality of shredded iceberg lettuce within its packaging. This is done by classifying samples of the <i>a</i> and <i>b</i> components color histograms from the CIELab color space. | 3-Nearest Neighbours classification approach was employed where feature were subjected to a Leave-1-out validation method. Image segmentation was done by a CNN separate package from fresh produce. | The performance showed a 97.9% segmentation accuracy for the CNN and a 83% quality classification among five quality categories. |
| Bauer et al., 2019 [27]    | The study investigated the use of deep learning methods for ultra-scale aerial phenotyping of iceberg lettuce. The use of normalized difference vegetation index (NDVI) imagery was used to                              | A custom CNN applied for both lettuce detection and classification. CLAHE applied to enhance contrast between crop and background. Non-Maximum suppression to eliminate duplicate detections and a   | Overall accuracy of 98% with an $R^2$ -value of 0.98 achieved on correlation of system to manual and automated counting and an   |



|                                  |                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                  |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  | determine for crop counting and crop quality assessment.                                                                                                                                                                                                        | sliding window used on NDVI to detect lettuce heads.                                                                                                                                                                                                           | R <sup>2</sup> -value of 0.988 for large regions of the field.                                                                                                                                                                   |
| Xu et al., 2022 [104]            | The study introduced an image-based 4-category grading system for lettuce freshness based on moisture content, visual cues based off chlorophyll degradation and weight of lettuce.                                                                             | Lettuce images acquired were pre-processed by resizing to 224x224 and RGB values were normalized. The deep learning residual network model was then compared to AlexNet, VGG16, GoogleNet, ResNet50.                                                           | Validation accuracy of 96.59% for proposed deep learning network and with a test accuracy of 95.60%. This outperformed all model comparisons.                                                                                    |
| Concepcion II et al., 2021 [105] | The study introduced a vision-based lettuce phenotype model (VIPHLET) for optimization of nutrient delivery in an aquaponic system. The adaptive fertigation was based on the fresh weight, chlorophyll a, chlorophyll-b and vitamin C contents of the lettuce. | Extraction of chlorophyll a and b was done using UV-Vis spectrophotometry at 645 and 663 nm. CIELab color space was used for vegetation segmentation. The study employed a multigene symbolic regression genetic programming (MSRGP) for predictive modelling. | Overall efficiency based on the VIPHLET model introduced to system was to be 99.678% with a known 7.21% improvement over manual fertigation.<br><br>This was also cited by [106] that model can be used for postharvest lettuce. |
| Wang et al., 2024 [107]          | The study develop a field-programmable gate array (FPGA)-based system for lettuce quality assessment from the detection of nitrogen and potassium nutrient levels of hyperspectral imaging.                                                                     | A lightweight and optimized version of FasterNet, a deep learning model was introduced. Principal Component Analysis (PCA) and Successive Projections Algorithm (SPA) was done to select wanted hyperspectral band for nutrient detection.                     | Overall classification accuracy of 91.87% was achieved. The inference time was 83 milliseconds per hyperspectral image. Nitrogen detection was that of 93.1% and potassium detection accuracy was that of 92.2%.                 |

In conclusion, integrating computer vision and applied machine learning techniques is pivotal for intelligent postharvest handling systems [108]. Non-invasive high throughput methods, such as vision-based quality grading and adaptive controls, are identified as essential. Future directives include addressing challenges in model



explainability, generalizability, and usability in an industrial context. The potential for image processing techniques, convolutional neural networks and genetic algorithm in postharvest applications is acknowledged, emphasizing the need for large, labeled datasets and features for effective implementation.

## 2.4 Relevant Patents

The examination of similar design constructs and ideas that are already patented as intellectual property related to the study has been summarized in Table 2.14. The focus for such patents is based on postharvest handling systems with mention of computer vision, industry 4.0 technologies and artificial intelligence for the aiding of the quality grading and sorting of horticultural fresh produce. Pertinent information about the patent and its patent number are presented.

Table 2.14 Summary of Relevant Patents Regarding Postharvest Handling Systems

| Patent                                                                                       | Innovation                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Publication Number |
|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| Food Imaging And Processing Systems And Methods [109]                                        | The patent defined a food processing system with an imaging chamber within the X-ray spectrum where food is passed along a conveyor line. The use of a control system control the I/O system iuth a corresponding robotic arm to feed food into a cutting mechanism for predefined cuts. The crop produce cited for processing was corn cobs with the intent to discard corn ears.                                                                                         | US2024208098A1     |
| Automatic vision guided intelligent fruits and vegetables processing system and method [110] | The patent defined a fully automated vision-based processing system for removal of unwanted parts of fruits and vegetables, with example given on strawberry decaying. The system utilizes water and air jets/nozzles for cutting and controlling position and orientation of the fruits. It provides a defect classification through deep learning models that examines the strawberry under the roller conveyor, top-view, for any of its calyx to be removed.           | US11531317B2       |
| Vision-based grading with automatic weight calibration [111]                                 | The patent claimed the use of an adjusting image-to-weight function to derive estimated weights from capture image. It covers that the calibration of such system will be done through weighers and the weight grading of the system can be processed individually or in batches for selected food items, with shrimp as an example item. It covers also sorting through downstream from imaging system uses ejection actuators to place it into dedicated lanes and bins. | US9886752B2        |
| Small fruit and vegetable pneumatic                                                          | The patent claimed a small automatic pneumatic fruit and vegetable sorting machine. It details the following elements to establish a proper machine vision pneumatic                                                                                                                                                                                                                                                                                                       | CN110681603A       |



|                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |
|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| automatic sorting machine [112]                                                              | sorting: conveyor belt, speed regulating motor, V-shaped guide frame, pneumatic actuators and a machine vision box with appropriate computer and controller as the main parts of the machine. The machine covers also infrared sensor to detect the crop that is timed synchronously for the pneumatic actuation.                                                                                                                                                                 |             |
| Multi-index comprehensive spherical fruit and vegetable quality grading-sorting device [113] | The patent claimed a quality grading and sorting device characterized by the following: a feeding mechanism, a sub-channel transmission belt, a multispectral imaging system, a screen mechanism, and size sorting conveyor belt. It stated that use of multi-index quality grading through a multiple spectral camera that is divided through a camera obscura and there are two channels that provide the screening, and the sorting is done through adjustable roller baffles. | CN111069094 |

## 2.5 Synthesis

Provided in Table 2.15 is a compilation of the different related literature that prove relevant to the current study. First a comprehensive overview of postharvest lettuce quality parameters reveal that studies employed methods have a grading criterion for postharvest lettuce quality in terms of moisture content, chlorophyll-*a* and fresh head weight. This will be the parameters that the study will consider when making their own artificial intelligence (AI) modelling for lettuce quality. Covering the S<sup>3</sup> domains, emerging technologies within postharvest grading sorting systems are evident with the use of alternative industry items to provide a cheaper but still a reliable alternative for industry systems. The domains also tackle cost-benefit analysis and ergonomic assessments as basis for improvement of automation over manual labor. Benefits also included connectivity within the user with studies tackling communications associated with Industry 4.0. Industry standards and relevant patents also point out to these remarks. Lastly, the industry and national standards for making a sorting systems and materials handling was also covered where in designs of their system will be considered. Further discussion of performance evaluation is needed.

The gap that this study fills in the current research literature delves on the novelty of the use of a grading criterion with consideration made by PNS for iceberg lettuce quality classification. Specifically, the use of moisture content, chlorophyll-*a* and fresh



head weight parameters for classification will be utilized for the computer vision system that will be integrated into a conveyor belt as a sorting system and post processing iceberg lettuce varieties within the Philippines.

Table 2.15 Synthesis table of review of related literature of current study

| Author Citations                                                                             | Key Claim and Contributions                                                                                                                                                                                                            | Relevance to current study                                                                                                                                                                                                |
|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [46], [47], [65], [66], [83], [62], [63], [71], [75], [84], [85], [86] [9], [53], [54], [82] | Studies employed non-destructive methods for predicting postharvest lettuce quality parameters and act as a grading criteria. Known values of moisture content, chlorophyll a and fresh weight act as determinants of lettuce quality. | The knowledge of the methodologies for the extraction and use of these values towards grading of postharvest lettuce will serve as the basis for the current study.                                                       |
| [42],[96], [91], [97],                                                                       | Studies covered the sensing, smart and sustainable domains of their use of emerging technologies towards postharvest handling systems.                                                                                                 | The current study will abide to the S <sup>3</sup> domains seeking to lower cost on an existing postharvest sorting system made in the Philippines.                                                                       |
| [42], [101], [102], [103], [104], [105],                                                     | Studies provided an in-depth application of using computer vision techniques for grading and sorting to evaluate the performance of their developed systems.                                                                           | The use of inferring lettuce quality through CV techniques and algorithm that are interpretable like that of MSRGP and integrating it into a sorting system that is cheap and reliable will be explored in current study. |
| [15], [90], [89], [91], [14], [93], [111], [112],                                            | Industry and national standards of grading postharvest lettuce quality and materials handling that should be observed with developing postharvest handling systems.                                                                    | The current study will follow and be guided with the standards reviewed for the development of postharvest lettuce grading and sorting system.                                                                            |



## Chapter 3

### Theoretical & Conceptual Framework

#### 3.1 Theoretical Framework

The lettuce postharvest quality parameters, computer vision, color space, image processing, quality parameter extraction methods and statistical analysis are explicitly discussed in the following subsections. These concepts will be salient to the development of our proposed system. The journal articles show which part of the respective papers were applied to the study, which includes the non-invasive methods for extracting chlorophyll-*a* [39], the use of an artificial neural network model and genetic algorithm [120] for evaluating chlorophyll-*a* levels and moisture content in iceberg lettuce, the application of computer vision techniques on sorting and grading of fruits and vegetables [96] [90], and parameters to be set to the created model for classifying and grading the lettuce [121]., which discusses related literatures that were found to be useful and applicable to the study. The journal articles show which part of the respective papers were applied to the study, which includes the non-invasive methods for extracting chlorophyll-*a* [39], the use of an artificial neural network model and genetic algorithm [120] for evaluating chlorophyll-*a* levels and moisture content in iceberg lettuce, the application of computer vision techniques on sorting and grading of fruits and vegetables [96] [90], and parameters to be set to the created model for classifying and grading the lettuce [121].

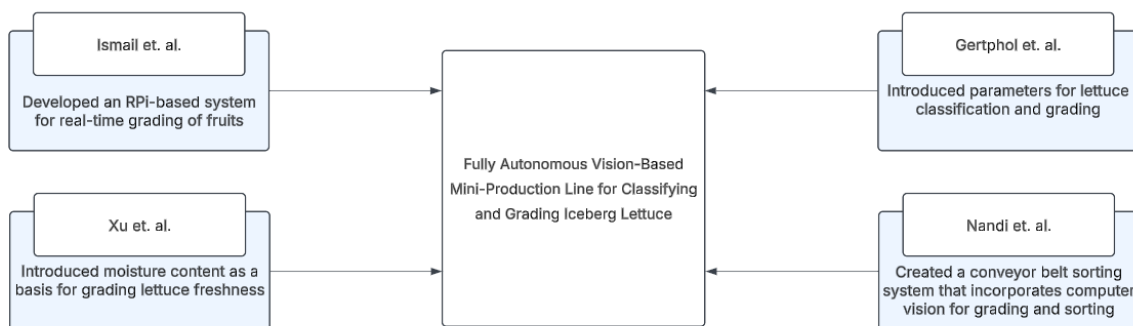


Figure 3.1. The Theoretical Framework





### 3.1.1 Iceberg Lettuce Postharvest Quality

Iceberg lettuce (*Lactuca sativa* var. *capitata*) is a crisphead lettuce variety known for its dense, tightly packed, and spherical head of pale green leaves. The head may be completely developed with overlapping leaves characterized as flabellate leaf venation [49]. The study aims to get the west lake green variety of the iceberg lettuce that is commonly bought in Philippine lettuce market and matches the description. Table 3.1 covers the quality parameters that the study investigates for the correlation of these parameters as a grading criterion for postharvest lettuce quality.

Moisture content is the amount of water present within its leaves, expressed in percentage (%). It has been discussed to have a significant correspondence to the freshness and crispness of the iceberg lettuce leaves [65]- [67]. Chlorophyll-*a* is the pigment responsible for the green coloration of lettuce leaves and can be quantified through milligrams per grams (mg/g) [61]- [64]. It has been covered that this is a visual indicator for food quality and grade. Lastly, fresh head weight is defined as the total weight of the lettuce head at point of extraction from harvest and is quantified in terms of weight expressed in grams (g). It has been evinced that it fresh weight attracts its marketable yield and is representative of the levels of its nutritional compositions [51]- [56]. Corresponding to these studies are the different range values that is also mentioned in Table 2.9 were it is consolidated in Table 3.2. This will be salient in determination of the ground truth of the different grading levels that is to be associated in fuzzy logic development.

Table 3.1 Relevant lettuce quality parameters

| Parameter                              | Unit | Description                                                                                                                             |
|----------------------------------------|------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Moisture Content (MC)                  | %    | The percentage of water present in iceberg lettuce. Essential for maintaining freshness, crispness                                      |
| Chlorophyll- <i>a</i> (Chl- <i>a</i> ) | mg/g | The primary photosynthetic pigment is responsible for the green coloration of lettuce leaves, indicating freshness and nutrient status. |
| Fresh Head Weight (W)                  | g    | The total weight of an iceberg lettuce head at harvest, reflecting its maturity, density, and market quality                            |



Table 3.2 Grading criteria of iceberg lettuce for current study

| Quality Parameter     | Grade 1 (High Quality) | Grade 2 (Medium Quality)   | Grade 3 (Low Quality) |
|-----------------------|------------------------|----------------------------|-----------------------|
| Moisture Content      | 94–96% or higher       | Between 90–94%             | 90% or lower          |
| Chlorophyll- <i>a</i> | 0.3691 mg/g or higher  | Between 0.2884-0.3681 mg/g | 0.241 mg/g or lower   |
| Fresh Weight          | 451-600 g or higher    | Between 450-201 g          | 200 g or lower        |

### 3.1.2 Extraction of Quality Parameters

The current study will employ the following techniques to retrieve quantities for their chosen lettuce quality parameters and act as features for the chosen artificial intelligence model.

#### 3.1.2.1 UV-VIS Spectrometry

Spectrometry is an analytical technique for quantifying light absorbance of a chemical substance over a certain wavelength. Biochemical components in plant tissues, such as chlorophyll-*a* in lettuce can be quantified using this method. UV-Vis spectrophotometry specifically measures the absorbance of light at specific wavelengths to determine the concentration of target compounds and will be considered for the study. Chlorophyll *a* has strong absorption peaks at 663 nm ( $A_{663}$  for red light) and 645 nm ( $A_{645}$  for blue light) which are used to calculate its concentration using the Arnon equation provided at equation 3.1 based on the protocol made by [114].

$$\text{Chlorophyll-}a \text{ (mg/g)} = (12.7 * A_{663}) - (2.59 * A_{645}) \quad 3.1$$

#### 3.1.2.2 Oven Drying Method

The oven drying method is a standard laboratory technique for determining moisture content. This method involves heating a sample at a controlled temperature of 80°C ideally at least for 8 hours [115]. Determination of that method is done when a constant weight is achieved, ensuring that all free water has evaporated. The moisture content (MC) percentage is calculated using the equation 3.2.  $W_i$  is the initial fresh weight of the sample lettuce leaf and  $W_f$  is the initial weight after oven drying.



$$\text{Moisture Content (\%)} = \frac{W_i - W_f}{W_i} \times 100$$

3.2

### 3.1.2.3 Weight Measurement

The fresh head weight of lettuce is typically measured using precision electronic weighing scales down to the second significant digit. This ensures accurate classification into market categories as stated by PNS of iceberg lettuce [9]. It is expressed in W for fresh head weight in grams.

### 3.1.3 Computer Vision

Computer vision (CV) is a scientific discipline that includes methods for acquiring, processing, analyzing and understanding images of the real to produce numerical or symbolic information that a computer can process as stated by [98]. It covers use of segmentation, classification image captioning and analysis. In the current study, it is salient to the grading criterion to correlate quality parameters from the pixels of the image to inspect and classify quality in terms of mentioned postharvest quality parameters.

#### 3.1.3.1 RGB Color Space

The RGB color space is one of the most commonly used color representation models in computer vision, particularly in food industry applications such as sorting, grading, and quality analysis. It consists of three primary color channels: Red (R), Green (G), and Blue (B), which combine to produce a wide range of colors [4], [39], [64]. RGB is widely utilized in automated food inspection systems due to its compatibility with digital imaging devices such as cameras and sensors. In food sorting, RGB values help distinguish ripeness levels, detect surface defects, and classify products based on color features. However, challenges arise in varying lighting conditions, as RGB is not perceptually uniform, meaning small changes in illumination can significantly alter the captured color. Therefore, standardized lighting setups are essential to maintain consistent image acquisition across different imaging chambers in food processing environments. The RGB color space is a widely used color representation model in computer vision and food quality analysis, designed to approximate human color perception.



### 3.1.3.2 Hue Color Space

The HSV (Hue, Saturation, and Value) color space is based on the tint, shade and tone of the image. HSV separates color into chromatic content (hue) from intensity (value), varying values from 0 to 1. making it particularly useful for image processing in varying lighting conditions. In this model, Hue (H) represents the dominant wavelength (color type, e.g., green, red), Saturation (S) defines the purity or intensity of the color, and Value (V) indicates brightness. These values can be retrieved from the RGB values of the image. In the current study where multiple imaging chambers will be considered, it is essential to keep values within a certain error limit in terms of hue differences. Provide in equations 3.2, 3.3, 3.4 are the conversion values for RGB into HSV color space based on [116]. This color space has been used to detect fruit ripeness and quality based on [39] [40].

$$Value (V) = \max (R, G, B) \quad 3.2$$

$$Saturation = \frac{V - \min(R, G, B)}{V} \quad 3.3$$

$$Hue = \begin{cases} 1 + \frac{G - B}{V - \min (R, G, B)} \text{ for } V = R \\ 3 + \frac{B - R}{V - \min (R, G, B)} \text{ for } V = G \\ 5 + \frac{R - G}{V - \min (R, G, B)} \text{ for } V = B \end{cases} \quad 3.4$$

### 3.1.3.3 Image Processing Techniques

Image processing involves identifying pixel patterns forming objects or regions of interest and extracting relevant information from them. Preprocessing is often required as a previous step to improve specific image characteristics, enhance image quality, remove noise, or correct errors due to the acquisition process. Below are image processing techniques to be considered for the study.

### 3.1.3.4 Segmentation and Thresholding

Segmentation is the process of dividing an image into different regions of interest to facilitate further analysis. There are several methods for segmentation, including



thresholding, pixel-based, and region-based techniques. Thresholding is one of the simplest methods and involves finding gray-level values between peaks in the histogram to separate the background and objects. Otsu's thresholding minimizes intra-class variance between the foreground and background. The threshold value is calculated based on Equation 3.5 where  $\sigma_B^2$  represents class variance and  $\omega$  represents the probability distributions of the foreground and background pixels. Given that the current study will consider a constant background, it is easy for Otsu's thresholding to conduct segmentation.

$$\text{Thresholding Value} = \arg \min [\sigma_B^2 (\omega)] \quad 3.5$$

### 3.1.3.5 Data Augmentation

Data augmentation is a technique used to artificially expand the training dataset in machine learning and deep learning models, improving model generalization and performance. It involves applying transformations to existing images, such as rotation, flipping, cropping, and color adjustments, to create new variations of the same data [98]. For the current study, this will be applied to expand the dataset of the lettuce images to add features for the AI model.

### 3.1.4 Computational Intelligence

Computational intelligence (CI) is a computer learning implementation from provided data in order for a computer system to perform tasks that are usually associated with human intelligence or expertise without being explicitly instructed. It is typically defined as an autonomous agency with the ability to perform cognitive functions in contrast to the natural intelligence displayed by humans, such as learning from reasoning more commonly associated with artificial intelligence (AI) [98]. Under this scope is the use of genetic programming (GP) and fuzzy logic. GP is used for the current study to define regression models of our quality parameters to make proper estimations based on empirical data between variables. Fuzzy logic then combines these estimations to have crisp discrete values that can be translated into the study's desired grading levels.



### 3.1.4.1 Genetic Programming

Genetic Programming (GP) is an evolutionary computation technique inspired by the principles of natural selection, designed to evolve computer programs or mathematical expressions automatically. It operates on a population-based approach, where candidate programs—often represented as syntax trees—undergo processes analogous to biological evolution, such as selection, crossover (reproduction), and mutation summarized in Table 3.3. The process begins by generating a diverse initial population of mathematical expressions composed of basic functions such as addition, subtraction and trigonometric functions combined with terminals of variables and constants. [117]. These expressions are structured as trees, and their fitness is evaluated based on how well they solve a given problem. High-performing trees are more likely to be selected for reproduction, where they exchange subtrees (crossover) or undergo mutation to produce new, potentially better offspring. [118].

Table 3.3 Summary of Genetic Programming (GP) components and evolutionary phases

| GP Programming Terms |                                                                                                                     | GP Evolutionary Phases                    |                                                                                                                                                                                        |
|----------------------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Terminal Set         | This involves the specification of the dependent and independent variables.                                         | 1.) Initialization and Fitness Evaluation | A population of randomly generated programs is created. Population is evaluated by function set.                                                                                       |
| Function Set         | This specifies the nature of the functions used to solve the problem.                                               | 2.) Selection                             | The best performing programs are selected for reproduction based on their fitness measure value                                                                                        |
| Fitness Measure      | The fitness measure defines whether the GP results are good or bad fit.                                             | 3.) Reproduction and Replacement          | Selected programs are used as parents to generate offsprings using operators. Offsprings replace least-fit programs in populations, creating a new generation.                         |
| Termination Criteria | parameters that control the GP runtime by setting the maximum generations, population size, and maximum tree depth. | 4.) Termination                           | GP process resumes until termination criteria are satisfied. Development of favorable optimized program is thus achieved by evolving population of programs over multiple generations. |

Under the GP scope is multi-gene symbolic regression genetic programming (MSRGP). It is a known evolutionary algorithm used for symbolic regression to discover



a mathematical expression that best fits a dataset; it is a more advanced kind of genetic programming (GP) since MSRGP allows multiple functions or genes to evolve at the same time. In MSRGP, populations are represented by a set of genes, each of which can be independent expressions; this algorithm evolves the genes with the help of genetic operations (mutation, crossover, and selection) to optimize how well the genes fit a dataset. Equation 3.6 represents the multi gene symbolic model where the following symbols are defined as:  $y$  is the predict value,  $n$  is the number of genes,  $w_i$  is the weight of gene  $i$ ;  $f_i(x)$  is the expression level of gene  $i$ ;  $x$  is the input data;  $b$  is the bias term.

$$y = \sum_{i=1}^b w_i f_i(x) + b \quad 3.6$$

In MSRGP, a multigene individual is made up of more than one gene; for each gene, a GP syntax tree is made from which the SR equation is developed as a linear combination of the tree sets; the tree sets are not necessarily linear hence, the equation formed can capture non-linear relationships and have similar linear model simplicity as linear regression [119]. Shown in Figure 2.1Figure 3.2 are syntax trees where regression equations are developed as a combination of the tree sets and thus can capture non-linear relationships.

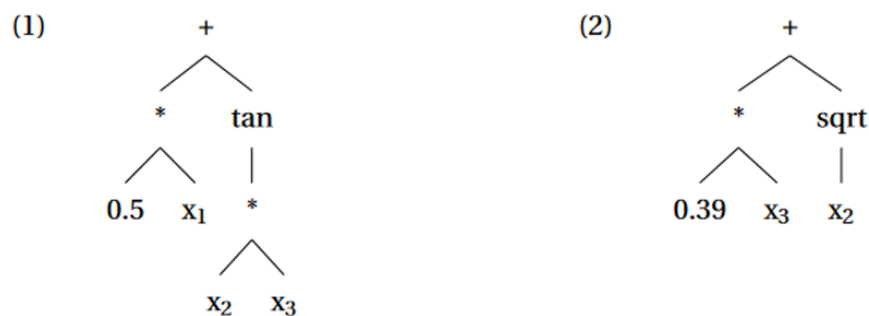


Figure 3.2 Example of a MSRGP model [119]

### 3.1.4.2 Fuzzy Logic

Fuzzy Logic (FL) is an CI technique that mimics human reasoning by handling imprecise, uncertain, and vague information. Unlike traditional boolean logic, which classifies data as strictly true (1) or false (0), fuzzy logic allows for degrees of truth





between 0 and 1, enabling more flexible decision-making. The study will consider this for inference of the quality parameters output based on created rules that set the output values into discrete grading levels. Fuzzy logic operates using membership functions and if-then rules that is then defuzzified to model human-like reasoning from the quality parameter estimations. The study will consider the Mamdani inference method for the fuzzification, degree of activation, implication, aggregation and defuzzification that is summarized in Table 3.4. There are several methods for defuzzification but the study will focus on utilizing the center of gravity method shown in Equation 3.7.

$C$  denotes the crisp value of the certain output variable which is the center of mass. The  $\mu(y)$  is the function of the aggregated output member functions of the output variable 86 from the results of the interference while  $y$  is the  $y$ -coordinate along the horizontal axis. The limits of integration depend on the range of the output variable. This formula can be interpreted as a weighted mean since the  $\mu(y)$  is weighed with  $y$  and it will be further normalized by the integral in the denominator.

Table 3.4 Summary of fuzzy logic terms and description

| Fuzzy Features       | Description                                                                                                   |
|----------------------|---------------------------------------------------------------------------------------------------------------|
| Rule Base            | Uses IF-THEN rules with fuzzy antecedents and fuzzy consequents                                               |
| Degree of Activation | Uses logical operations (AND = min, OR = max) to compute rule strength                                        |
| Implication          | Applying Rule Strength Uses min (clipping) or product (scaling) methods to modify output membership functions |
| Aggregation          | Uses max (union) operation to merge all rule outputs into a final fuzzy set                                   |
| Defuzzification      | Converts the fuzzy output into a single crisp value, typically using centroid (COG) method                    |

$$C = \frac{\int_{y_1}^{y_2} \mu(y)y \, dy}{\int_{y_1}^{y_2} \mu(y) \, dy} \quad 3.7$$

### 3.1.5 Performance Metrics

#### 3.1.5.1 Regression Metrics

Regression metrics help with evaluating how well the predicted values from the quality parameters by a given model equation match with actual ground truth values when



conducting validation for provided continuous values. The study will consider coefficient of determination ( $R^2$  score) as our main model performance metric. The mean absolute error (MAE) shown in Equation 3.9 measures the average absolute difference between the predicted quality score  $\hat{y}_i$  and the actual ground truth score  $y_i$ . Lower values indicate better accuracy. Root mean square error (RMSE) shown in equation 3.10 penalizes large errors attributed to the trained model and validated scores. Lower RMSE means the model has better precision in predicting quality scores from subtracting  $y_i$  from  $\hat{y}_i$ . Lastly, the  $R^2$  score measures how well the model predicts the values for the actual quality scores. The closer the value is to 1 corresponds to a perfect fit with a good causal relationship as shown in Equation 3.8. The variables are as follows:  $y_i$  is the actual observed values of the dependent variable where the true values are from the given dataset;  $\hat{y}_i$  is the predicted values from the regression model; Lastly,  $\bar{y}$  is the mean of the  $y_i$  observed values.

$$R^2 = \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2} \quad 3.8$$

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad 3.9$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad 3.10$$

### 3.1.5.2 Throughput Time

Time study is a systematic method used to measure the time required for the system to complete a specific task under standardized conditions. It involves analyzing the cycle time, normal time, throughput time, and theoretical time to optimize operational efficiency.

The average cycle time is calculated by taking the sum of observed times for performing a task and dividing it by the number of cycles recorded as define in Equation 3.11. This value is then adjusted using a rating factor to determine the normal time, which accounts for variations in system performance mentioned Equation 3.12. To account for



unavoidable delays such as malfunctions and interruptions, the throughput time is computed by adjusting the normal time with allowances defined in Equation 3.13. Additionally, in motion-based processes, the theoretical time is estimated using the relationship between distance and average speed, allowing for better planning and resource allocation given in Equation 3.14. This is needed to define the system efficiency for the current study of sorting iceberg lettuce.

$$\text{Average cycle time} = \sum \frac{\text{Times to perform each element}}{\text{Number of cycles observed}} \quad 3.11$$

$$\text{Normal time} = (\text{Average cycle time}) * \text{Rating Factor} \quad 3.12$$

$$\text{Throughput time} = \frac{\text{Normal Time}}{1 - \text{allowances}} \quad 3.13$$

$$\text{Theoretical Time} = \frac{\text{Distance(m)}}{\text{Average Speed}} \times 60 \quad 3.14$$

### 3.1.5.3 Repeatability

Repeatability measures the ability of the system to consistently assign the same quality grade to the same lettuce sample throughout multiple trials. High repeatability guarantees consistent and reliable classification under identical conditions. The evaluation process for repeatability involved the following steps: (1) selecting a fixed test sample set ( $N = 5$ ) containing lettuce samples of varying quality with known classifications; (2) performing multiple sorting trials where each lettuce sample was processed five times; and (3) comparing the assigned grades across trials, to verify whether each sample was correctly classified. The aggregate of these results determined the repeatability score, as defined in Equation 3.15. Additionally, an error rate—which is given in Equation 3.16—was considered to assess the performance of the system model in repeatability trials during validation.

$$\text{Repeatability (\%)} = \frac{\text{Correct Repeated Sorting}}{\text{Total Trials}} \times 100 \quad 3.15$$



$$\text{Error Rate} = \frac{|\text{Measure Value} - \text{Given Value}|}{\text{Given Value}} \times 100 \quad 3.16$$

The repeatability of the conveyor belt sorting system was assessed at two operating speeds: 70% and 100%. Five iceberg lettuce heads were loaded into the system, where each sample was assigned a quality grade and subsequently sorted. This procedure was conducted separately for both speed settings, with five trials per speed.

### 3.2 Conceptual Framework

The input-process-output (IPO) model of the study is shown in the Figure 3.3 below. For the input, basic knowledge regarding the specifics of the targeted parameters for machine vision, particularly the chlorophyll-*a*, moisture content, and the fresh head weight of an iceberg lettuce. This is essential to ensure a proper working classification model, and grading system that are being proposed by the study. For the hardware requirements, technical drawings of the proposed mini-production line wherein specific kinematic movements will be able to take images of one of an iceberg lettuce's sides for proper grading. Another essential component in hardware would be the multiple 1080p cameras that will aid the machine vision setup by taking images of the lettuces to be graded by the system. Fusion360 and SolidWorks are used for modeling and simulation of the design of the mini-production line for the grading system of iceberg lettuce, the lever arms used for the servo motors, and for CADs of the different components of the system. Aside from that, the use of MATLAB with different toolboxes such as the Arduino, GPTIPS, and the Fuzzy Logic toolboxes will aid in constructing the machine vision setup and models to be integrated into the mini-production line.

For the process part, after material requisition, the integration of both hardware and software of the mini-production line will take place. Based on the mini-production line's technical drawings and design aspect, the frames for the imaging chamber will be made using angled bars, with thin plywood acting as the roof and sides of the chamber. Aside from the cameras being placed in their respective places, other sensors and electronics are connected to an Arduino Mega 2560, which will act as the main source of



controls within the system, able to control the movement of the conveyors and the servo motors. This setup allows the system to generate a real-time grading of iceberg lettuces and sort them accordingly. Finally, after corresponding to the given inputs and processes, the achieved product of the study is a fully autonomous vision-based mini-production line for classifying and grading iceberg lettuce.

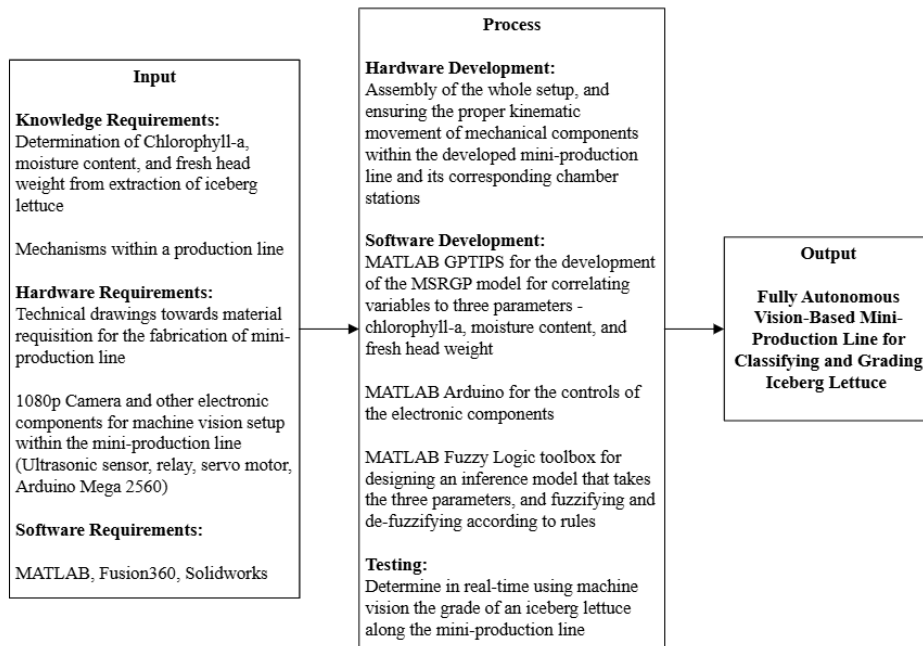


Figure 3.3 The Conceptual Framework



## Chapter 4

### Methodology

The study aims to evaluate the performance of the system's conveyor belt sorting system to classify and grade iceberg lettuce samples based on quality. Three key factors were conducted to assess the systems' reliability, throughput efficiency, and accuracy.

1. Accuracy Evaluation:

The classification model's accuracy were tested by comparing predicted data and actual data grades (High, Medium, Low) for iceberg lettuce using MSRGP.

2. Reliability Assessment:

The sorting trials were conducted at speeds of 70% and 100% with 5 trials and repeatability percentages are compared to evaluating the systems' consistency at different speeds.

3. Throughput Efficiency:

The system's efficiency was measured by evaluating 5 iceberg lettuce samples per batch, evaluating, sorting speed, actuation speeds, and overall performance of the system.

#### 4.1 Engineering Standards and Design Constraints

The engineering standards considered are shown in Table 4.1. Safety standards given by ANSI-CEMA 402-2003 and PAES 302-2000 cover the expected configuration and safety standards for conveyor belt machinery. The National Electrical Manufacturing Association (NEMA) provided the different frames and enclosures needed for motor selection and enclosure for external conditions. Lastly, for compliance with food-grade safety for conveyor belt systems, 3-A Sanitary Standards were considered and overall compliance with electricity were covered upon review of the Philippine National Electrical Code. These were all considered for the sorting system of the current study.



Table 4.1 Engineering standards relevant to the current study

| Standard/Organization                               | Description                                                                                                                           | Use for current study                                                                                                                                                                                    |
|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ANSI-CEMA 402-2003 [14],<br>PNS-PAES 302-2000 [15], | International and national safety standards and configuration for conveyor belt machinery.                                            | Guidelines to design safe electrical and mechanical systems for conveyor belt. For study, it will be considered given expected postharvest environment are in damp conditions.                           |
| NEMA Standards [16]                                 | Sets performance, testing, and safety standards for motors and actuators. NEMA also covers enclosures suited for external conditions. | The choosing and validating of electrical linear actuators with suitable performance and safety characteristics for study's automated sorting system. Study will consider NEMA 17 and 23 stepper motors. |
| 3-A Sanitary Standards [17, p. 30]                  | Food safety management system standard.                                                                                               | Compliance with 3-A Sanitary standards is considered for the conveyor belt that will conduct postharvest handling and processing of iceberg lettuce.                                                     |
| Philippine National Electrical Code [18]            | Local or national standards regulating electrical installations and machinery safety.                                                 | Known electrical wiring code and basic safety requirements for control panels will be considered for the study's system operations considering system may be in damp condition.                          |

## 4.1 System Overview Flowchart

The overall process of the system can be seen in Figure 4.1 below, which shows the flow of how inference of the captured images from the conveyor is then graded and sorted. This serves as an overview of the needed procedures needed for grade level determination of the given iceberg lettuce samples and how the lettuce was sorted accordingly through fuzzy logic.

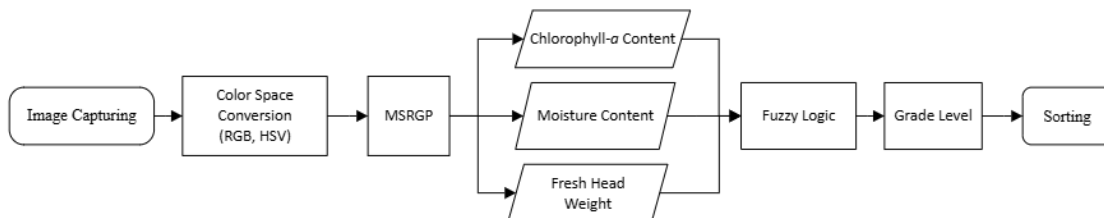


Figure 4.1. Overall system process flowchart





To achieve the wanted process, corresponding sub-processes are done and performed by the study. Shown in Figure 4.2 is the MSRGP block diagram that maps out how the regression models fit in the current system. To obtain the necessary data for modelling, extraction procedure for image capturing and laboratory-based extractions for chlorophyll-*a*, moisture content and fresh head weight are done after procurement of a total amount 80 lettuce samples that are divided into 60-20 train-test datasets for both image extraction and laboratory extractions.

For image capturing, each lettuce was placed in a chamber with an average light intensity of 400 lux, which was used for the conversion of color spaces by having found the mean RGB and HSV values of each image. Due to a scarcity of data, augmentation was done to create a wider variety of images, which helped increase the training dataset for the model. After augmentation was done, and color space values from each image were taken, extraction of chlorophyll-*a*, moisture content, and fresh head weight on each lettuce was also done, with each parameter having served as the outputs for the training data. The use of MSRGP output multiple different models for each parameter, wherein the models with the highest  $R^2$  value, and low model complexity, were chosen.

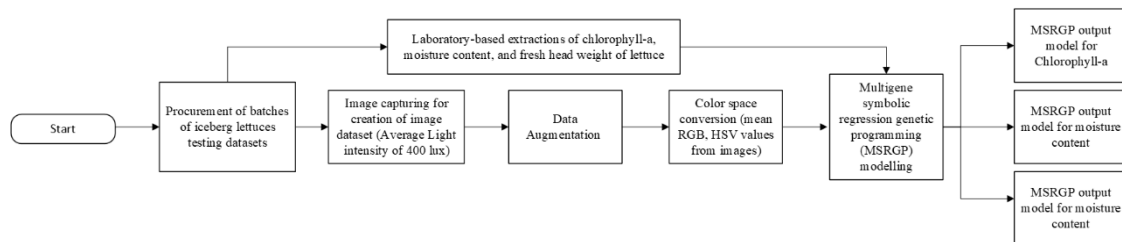


Figure 4.2. MSRGP block diagram for the current study

After retrieving the MSRGP models which provide estimated values of quality parameters from the images captured by the camera system setup in conveyor belt, it will then be processed into a fuzzy logic system, as illustrated in the block diagram shown in Figure 4.3. The study employs a Mamdani-Type 1 fuzzy logic system that will perform fuzzification on the incoming values and subsequently aggregate them to output three grading levels categorized as high, medium and low qualities. These are represented by linguistic values of ‘L1’, ‘L2’, and ‘L3’ respectively.

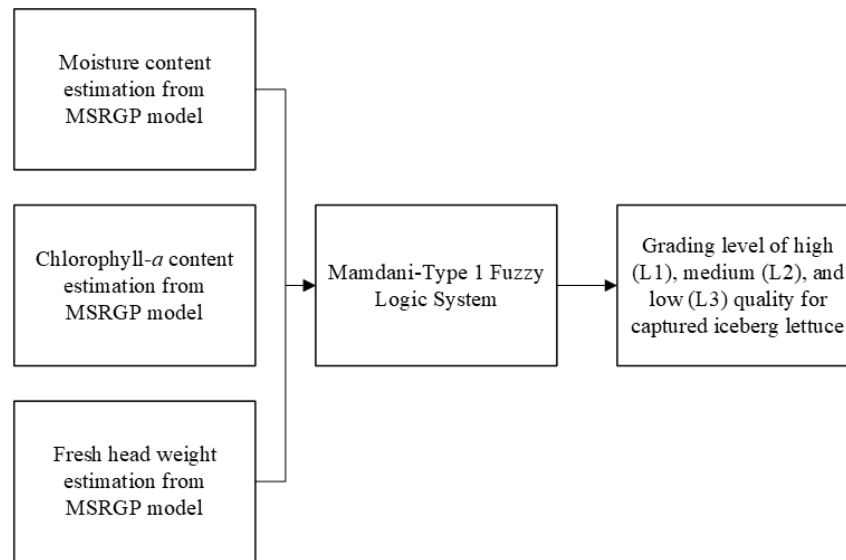


Figure 4.3 Fuzzy logic block diagram for the current study

Lastly, there is a need to design and construct the mini-production line needed for the study that considers a conveyor belt system, camera setup system, controls system and sorting system. This mini-production line is then constructed to then test the MSRGP and fuzzy logic systems for classifying and grading iceberg lettuce in terms of its test throughput time and the repeatability of the sorters capable of sorting upon fuzzy output. The creation of the mini-production line started with the considerations of possible designs, which changed as the prototype kept being built. These considerations consisted of the mechanical side, the electronic design, and the software and controls that were used for the system. The overall flow of the mini-production line can be seen in Figure 4.4.

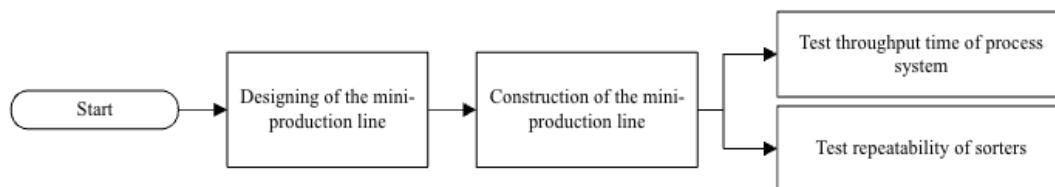


Figure 4.4 Mini-production line block diagram

The above block diagrams serve as the guide that the current study employed to achieve their specific objectives and wanted flow of their developed grading and sorting system. The succeeding subsections focus on the specifics of these enumerated processes.



## 4.2 MSRGP Implementation Procedures

### 4.2.1 Extraction Processes

#### 4.2.1.1 Image Capturing Process

This study required images of iceberg lettuce to be captured, which was utilized in training the genetic algorithm. The setup required iceberg lettuce samples, a 1080p webcam, a laptop, and a refrigerator. The lighting conditions of the refrigerator were replicated in the imaging chamber of the mini-production line that kept constant to have a 400-lux light intensity. Figure 4.5 illustrates the flow of the image-capturing process of lettuce.

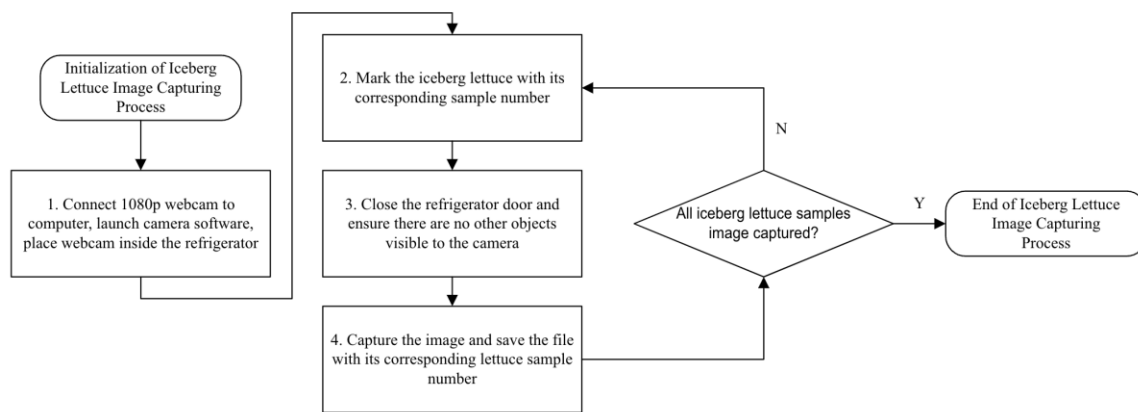


Figure 4.5. Image Capturing Process for Iceberg Lettuce Samples

During the initialization of the image-capturing process, a computer was configured with a 1080p webcam. The webcam was placed inside the refrigerator and positioned near the rear wall. Each iceberg lettuce sample was labeled with its corresponding sample number. After labeling, a sample was placed inside the refrigerator with its stem facing downwards. After that, the image was captured and saved with the sample's corresponding number in the file name for reference. Once the image was captured and saved, the lettuce was removed, and a new sample was placed until there are no remaining samples left for image capture.



## 4.2.1.2 Chlorophyll-*a* Extraction

Chlorophyll-*a* required parameter for one of the genetic programming models to be used in the fuzzy logic model. Consequently, chlorophyll-*a* needs to be extracted from iceberg lettuce, and this process is shown in Figure 4.6.

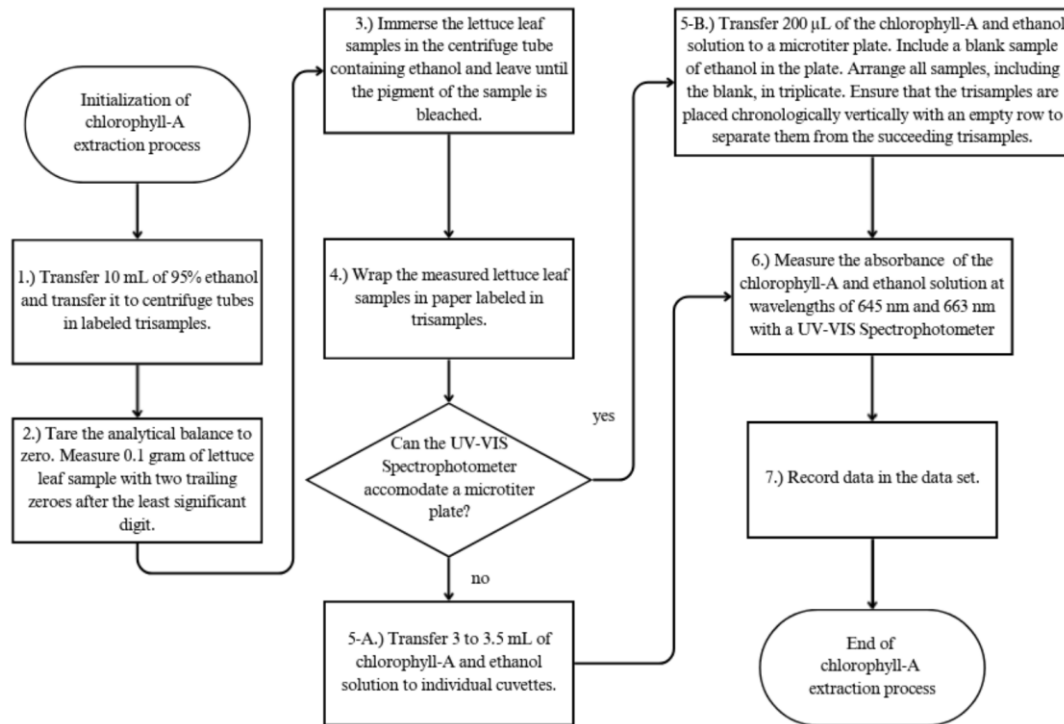


Figure 4.6 Chlorophyll extraction process. Adapted from [114]

The items needed for the extraction process are presented in Table 4.2 below. Chlorophyll-*a* extraction was conducted to determine the pigment concentration in lettuce leaves based on the protocol presented in [114]. The extraction process began with carefully cutting 0.1 g of lettuce leaf and placing it on a paper wrap to preserve its moisture. A total of 10 ml of ethanol was prepared in centrifuge tubes, which were labeled for trampling. The 0.1 g lettuce leaf sample was then immersed in the ethanol solution for 24 hours, or until the lettuce leaf sample was bleached of its color, allowing the chlorophyll to dissolve into the solution and produce a green, clear extract. Following the bleaching process, 200 uL was extracted from the 10 mL solution and transferred to 1 mL titer plates. These plates were then placed in a UV-VIS spectrophotometer to measure



absorbance at wavelengths of 645 nm and 663 nm. The obtained corrected absorbance values were subsequently processed, generating a dataset containing the chlorophyll-*a* concentrations of the lettuce samples.

Table 4.2 List of items for chlorophyll-*a* extraction process

| Item                             | Amount/Unit  | Purpose for the study                                                                                                                            |
|----------------------------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| 95% Analytical Grade Ethanol     | 2 liters (L) | The study utilized this as the solvent of choice for chlorophyll extraction.                                                                     |
| 15 ml Centrifuge Tubes           | 180 pcs      | To store extracted lettuce samples and ethanol amount for chlorophyll extraction.                                                                |
| Forceps-Dissecting Scissors Pair | 3 units      | For precise handling and cutting of lettuce tissue samples.                                                                                      |
| 10 ml Pipettes                   | 2 units      | Use to deliver precise amounts of ethanol into centrifuge tubes for chlorophyll-extraction                                                       |
| 200 $\mu$ L Pipettes             | 2 units      | Used to deliver precise amounts of obtained solution into titer plates or cuvettes into the spectrophotometer.                                   |
| UV-Vis spectrophotometer         | 1 unit       | Measures absorbance of chlorophyll- <i>a</i> at 645 nm and 663 nm, allowing quantification of pigment concentration.                             |
| Analytical Balance               | 1 unit       | Ensures accurate weighing of lettuce samples before extraction and moisture content analysis.                                                    |
| 1 ml Titer Plates/Cuvettes       | 60 pcs       | Depending on the spectrophotometer, a corresponding titer plate or cuvettes is needed to measure the absorbance light from dissolved chlorophyll |
| 60-hole tube rack                | 2 units      | Organizes and secures centrifuge tubes during sample preparation and storage.                                                                    |

#### 4.2.1.3 Moisture Content Extraction

The moisture content is another parameter that is required to be utilized by one of the genetic programming models in the fuzzy logic model. As a result, the moisture content needs to be extracted from iceberg lettuce, and this process is shown in Figure 4.7.

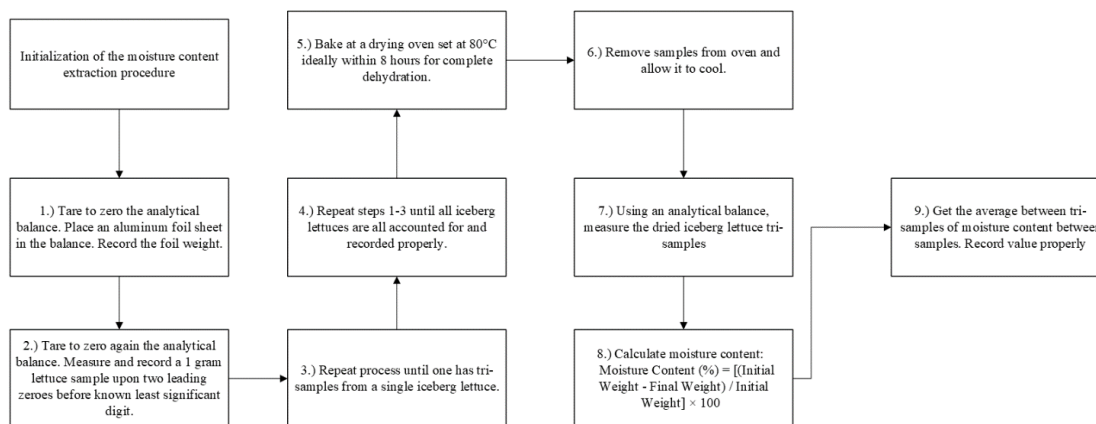


Figure 4.7. Moisture Content Extraction Process

A 1-gram sample of lettuce leaf was cut and placed in an aluminum foil sheet to preserve its moisture. The weight was measured using an analytical balance; ensuring two leading zeroes were present before the known least significant digit. This process was repeated for each lettuce sample three times and then repeated for all lettuce samples. Once completed, the samples were placed in an oven and baked at 80° C for 8 h to ensure complete dehydration. After the oven-drying process, the dried samples were weighed to determine the total water loss. The difference in weight before and after baking represented the moisture content of the lettuce leaves. The extracted values were recorded in a datasheet for further analysis. The materials required for this extraction can be seen below in Table 4.3.

Table 4.3 List of items for moisture content extraction process

| Item                             | Amount/Unit | Purpose for the study                                                                         |
|----------------------------------|-------------|-----------------------------------------------------------------------------------------------|
| Analytical Balance               | 1 unit      | Ensures accurate weighing of lettuce samples before extraction and moisture content analysis. |
| Aluminum Foil Sheets             | 1 roll      | Holds lettuce samples, preventing moisture loss before drying.                                |
| Drying Oven (80°C Capacity)      | 1 unit      | Dehydrate lettuce samples within 8 hours to remove all moisture content.                      |
| Forceps-Dissecting Scissors Pair | 3 units     | For precise handling and cutting of lettuce tissue samples.                                   |



## 4.2.2 MSRGP Modelling and Accuracy Testing

MSRGP modelling is done following the concepts given in Section 3.1.4.1 For the current study, the MSRGP modeling process in MATLAB is given in Figure 4.8. The hyperparameters of the MSRGP are initialized based on wanted the population size, number of generations, crossover probability, mutation probability, tournament size and tournament selection. Next, an initial population of mathematical expressions that are represented as tree structures is randomly generated as the first generation and converted into symbolic expressions. These candidate solutions are evaluated using a fitness function, often based on error metrics. Through genetic operators such as selection, crossover, and mutation, the population evolves over multiple generations, refining the models to improve their predictive accuracy until the end criterion is met. Once the iterations are done, the pareto model report undergoes checking by the user to determine model fitness to extract the best-performing model. The accuracy is then tested between the relationships between RGB HSV input values and the target quality parameters of the current study.

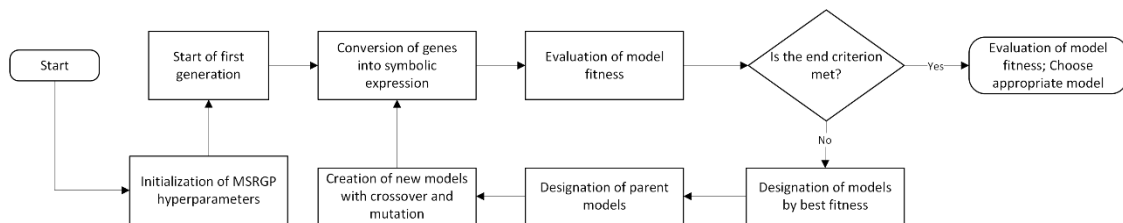


Figure 4.8 MSRGP Modelling Flowchart

To test the accuracy of the MSRGP models, the study utilized their test dataset with known RGB and HSV color space values and known three quality parameters values to be compared with the estimated values outputted by the created models.

## 4.2.3 MATLAB GPTIPS Toolbox

GPTIPS, an open-source genetic programming toolbox in MATLAB, was used as the main tool for programming as it offered the ability to use genetic programming, a machine learning model that allowed the generation of multiple equations correlating the input variables to the output. In the system, GPTIPS was utilized by creating a workspace





that included training and testing datasets, which was then ran through genetic programming with specific parameters of 50 generations through 50 runs with the other known run parameters in Table 4.4. Afterwards, a pareto report was generated that contained multiple equations of varying  $R^2$  and model complexity values, from which a specific model was chosen. This process was repeated once for every input parameter the study required, which were chlorophyll-*a*, moisture content, and fresh head weight. Multiple runs of at most four attempts were done to find the best fitting models considering  $R^2$  and model complexity, where run parameters remained constant for every model, with only generations changing while training was done for a better model.

Table 4.4 MSRGP run parameters for the current study

| Run Parameters         | Values                                                                 |
|------------------------|------------------------------------------------------------------------|
| Population Size        | 70                                                                     |
| Generations            | 50                                                                     |
| Input Variables        | 6 (RGB-HSV); 3 (HSV only)                                              |
| Training Instance      | 468                                                                    |
| Tournament Size        | 50                                                                     |
| Elite Fraction         | 0.1                                                                    |
| Tournament Probability | 0.5                                                                    |
| Max Genes              | 20                                                                     |
| Crossover Probability  | 0.84                                                                   |
| Mutation Probabilities | 0.14                                                                   |
| Function set           | TIMES MINUS PLUS SQRT SQUARE<br>SIN COS ADD3 MULT3 LOG CUBE<br>NEG ABS |

#### 4.2.4 MSRGP Model Considerations and Results

The function set for the current study needed to model complex nonlinear data for quality estimation from input variables of mean RGB, HSV values from an image therefore mathematical operations that the study's genetic programming needed to consider were LOG, EXP, POWER, SQRT, SIN, COS. These operations lessen the model complexity needed to achieve high  $R^2$  value. The target model complexity is within the value of 300 with gene expressions to be less than 30 to achieve an  $R^2$  value of more than 0.5. The study then considered two input possibilities which are six input values of RGB-HSV of the image and only three inputs of HSV values. These were considered as HSV is more stable under light changes and may improve the study's  $R^2$  value.



## 4.2.4.1 MSRGP models using RGB-HSV color space

The following outputs are the results of the best run and fitting models for the study's three quality parameters that considered the mean RGB-HSV values extracted from train dataset. The top 3 models from the pareto report were extracted and considered for the study, and afterwards, the bias and gene weight were then checked on how well the chosen regression equation fit towards the given train data provided predictions.

### 4.2.4.1.1 MSRGP output for chlorophyll-*a* using RGB-HSV input

Shown in Table 4.5 are the Top 3 models considered for the first MSRGP model for the estimation of chlorophyll-*a* using RGB-HSV input. The pareto report given showcase how all models have mainly used SIN and COS as mathematical operations to bring output regression line close to the given dataset. With this, the study chose model ID 241 as our main model for the chlorophyll-*a* given inputs as it had the least model complexity and is relatively close to the highest  $R^2$  value.

Table 4.5 Top 3 considered models for MSRGP chlorophyll-*a* from pareto report using RGB-HSV input

| Model ID | R <sup>2</sup> value | Model Complexity | Model                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------|----------------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2,186    | 0.748                | 412              | 11.2 x5 - 0.0692 x3 - 52.2 x4 - 0.00472 x1 - 8.29 x6 + 0.0569 cos(cos(x3) + log(x3)) - 0.00472 cos(x1 x6) - 0.0118 log(abs(x1 - 1.0 x3)) - 0.0417 cos(x3) + 12.7 cos(x4) - 13.3 cos(x6) + 4.09 log(x3) + 0.303 log(x5) - 12.4 sin(x5) + 0.0505 x62 (x3 - 1.0 x5 + log(x5)) + 0.347 x66 cos(x1 x6)2 + 0.0428 abs(x1) abs(x6)2 + 6.82e-5 x1 cos(x3 x64 cos(x1 x6)2 (cos(x4) + log(x5))2) - 0.0973 x1 x64 - 2.98 x4 log(abs(x1) abs(x6)2) + 13.7 x4 log(x3) + 0.0329 x42 cos(81.0 x6) (cos(x3) + log(x5)) - 8.82 |
| 223      | 0.722                | 291              | 0.0139 x2 - 0.0269 x1 + 0.0207 x3 - 4.03 x4 - 0.0335 x6 - 3.37 cos(x21/2) + 2.9 sin(cos(x21/2)) + 1.48 sin(cos(x21/2))3 - 0.0269 log(x2)1/2 + 0.0475 sin(x2)3 - 0.0396 sin(x2)9 + 0.0323 sin(x2)81 - 0.188 x2 x5 + 0.268 abs(x1) abs(x5)2 + 3.44 abs(x1)1/2 abs(x5) - 40.6 sin(x53)1/2 - 0.00457 sin(x2(9/2))9 + 0.0137 sin(cos(x3))3 - 14.7 sin(cos(x4))3 - 7.69 x43 - 0.227 x53 - 1.19 x63 - 0.13 x6 sin(sin(x2)3)3 + 8.58                                                                                  |
| 241      | 0.72                 | 265              | 0.0144 x2 - 0.0274 x1 + 0.0209 x3 - 4.0 x4 - 0.0289 x6 - 3.13 cos(x21/2) + 2.7 sin(cos(x21/2)) + 1.4 sin(cos(x21/2))3 - 0.0274 log(x2)1/2 + 0.0458 sin(x2)3 - 0.0376 sin(x2)9 + 3.17 sin(x6)9 + 0.032 sin(x2)81 - 0.193 x2 x5 + 0.277 abs(x1) abs(x5)2 + 3.53 abs(x1)1/2 abs(x5) - 42.1 sin(x53)1/2 + 0.0133 sin(cos(x3))3 - 14.4 sin(cos(x4))3 - 7.51 x43 - 0.235 x53 - 1.54 x63 - 0.129 x6 sin(sin(x2)3)3 + 8.41                                                                                            |



Shown in Table 4.6 is the summative GP tree structural properties for the chosen model ID 241. It has a total of 20 terms or gene expressions that was aggregated from the initial 265 expressions that it considered in the population with 96 mathematical operators, nodes, needed to define the regression equation. Furthermore, the depth is the maximum branch or chain that the gene tree made to make the model which is 7. Bias was low with a value of 8.41 indicating a good fit. Overall, it is considered a complex model but given the other models, this suits best the task needed.

Table 4.6 Model ID 241 GP tree structural properties

| Description             | Value                   |
|-------------------------|-------------------------|
| Genes                   | 20                      |
| Nodes                   | 96                      |
| Expressional complexity | 265                     |
| Depth                   | 7                       |
| Inputs                  | 6                       |
| Inputs used             | X1, X2, X3, X4, X5, X6, |

During the GP training, the initial augmented dataset was split into two parts for a (468) train and (166) test split to validate the model in a small portion of the data it was trained on. Shown in Figure 4.9 is the scatterplot that maps out the performance of the given MSRGP model in terms of its predictions from actual datapoints using the train split. Given the spread of the data, it performed within the expected  $R^2$  value aforementioned. Further testing on new datapoints was done in section 5.2

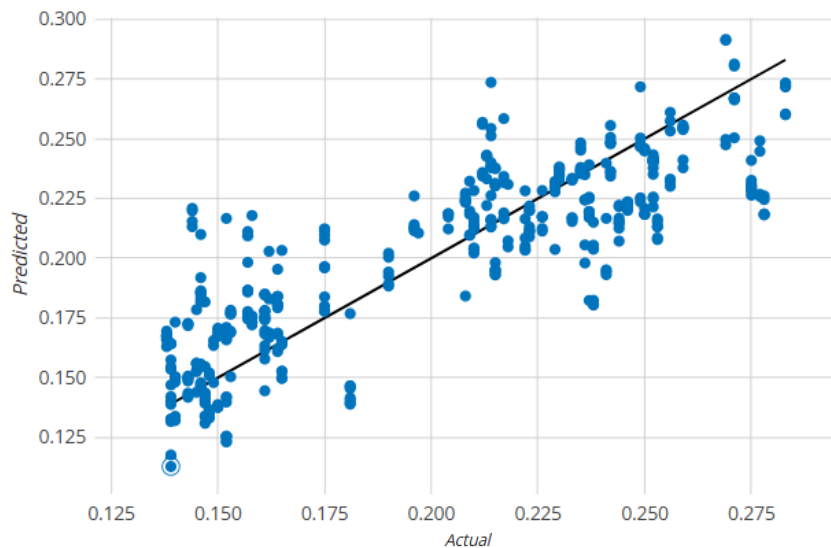


Figure 4.9 Predicted vs. actual scatterplot for train dataset on Model ID 241

#### 4.2.4.1.2 MSRGP output for moisture content using RGB-HSV input

Shown in Table 4.7 are the Top 3 models considered for the second MSRGP model for the estimation of moisture content using RGB-HSV input. The pareto report showcase how all models have mainly used LOG and ABS as mathematical operations to bring output regression line close to the given dataset. With this, the study chose model ID 2248 as our main model for the moisture content given inputs as it had the least model complexity and is relatively close to the highest  $R^2$  value.

Table 4.7 Top 3 considered models for MSRGP moisture content from pareto report using RGB-HSV input

| Model ID | $R^2$ value | Model Complexity | Model                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------|-------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2,248    | 0.431       | 496              | 1310.0 x6 - 2.38 x2 - 1.34 x3 - 1.34 x1 + 3.23 log(abs(sin(x3 log(abs(x4)))))) - 2.92 log(abs(sin(x3 log(abs(x6)))))) - 2.38 sin(x1 + cos(abs(x3)) + sin(x2)) - 2.92 sin(x6 + cos(abs(x3)) + sin(x3)) + 3.48 cos(x3 log(abs(x6))) - 5.63 abs(sin(x3 abs(x6))) + 6.7 abs(sin(x3 cos(x5))) - 1.34 sin(x3 log(abs(x6))) - 3.99 abs(x3 x5 log(x1)) + 778.0 abs(x5 x6 log(x3)) - 38.1 abs(x4) - 1.64 cos(x1) + 0.426 log(x6) - 1.69 sin(x32 log(abs(x6))) + 4.1 abs(sin(x3 log(abs(x1)))) + 3.23 sin(x1 + x3 + sin(x3 log(x6))) + 14.4 cos(sin(x3 log(abs(x4)))) + 0.426 x3 cos(x5) + 44.0 |
| 2,252    | 0.432       | 510              | 1270.0 x6 - 2.37 x2 - 1.17 x3 - 1.17 x1 + 3.3 log(abs(sin(x3 log(abs(x4)))))) - 2.85 log(abs(sin(x3 log(abs(x6)))))) - 2.37 sin(x1 + cos(abs(x3)) + sin(x2)) - 2.85 sin(x6 + cos(abs(x3)) + sin(x3)) + 3.66 cos(x3 log(abs(x6))) - 5.69 abs(sin(x3                                                                                                                                                                                                                                                                                                                                    |



|     |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----|-------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |       |     | $\begin{aligned} & \text{abs}(x6))) + 6.76 \text{ abs}(\sin(x3 \cos(x5))) + 0.256 \sin(x3 \\ & \log(\text{abs}(x1))) - 1.17 \sin(x3 \log(\text{abs}(x6))) - 3.81 \text{ abs}(x3 x5 \\ & \log(x1)) + 736.0 \text{ abs}(x5 x6 \log(x3)) - 30.7 \text{ abs}(x4) - 1.66 \\ & \cos(x1) - 1.57 \sin(x32 \log(\text{abs}(x6))) + 3.87 \text{ abs}(\sin(x3 \\ & \log(\text{abs}(x1)))) + 3.3 \sin(x1 + x3 + \sin(x3 \log(x6))) + 14.5 \\ & \cos(\sin(x3 \log(\text{abs}(x4)))) + 0.256 x3 \text{ abs}(x6) + 55.0 \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                    |
| 842 | 0.349 | 465 | $\begin{aligned} & 9.81 \cos(\cos(\text{abs}(x2 + x3))) - 4.3 \cos(\text{abs}(2.0 x2 + \cos(x3))) \\ & + 1.76 \cos(\text{abs}(\cos(x3) \log(x6)) \text{ abs}(x3)^{1/2}) + 4.55 \\ & \cos(\text{abs}(2.0 x2 + x4)) + 2.47 \cos(\text{abs}(x2 + \cos(x1)^{1/2})) - \\ & 366.0 \cos(x3) - 0.797 \text{ abs}(\cos(x1) \log(x6)) \text{ abs}(x3)^{1/2} + 1.03 \\ & \text{abs}(x3)^{1/2} \text{ abs}(\cos(\cos(x3)) \log(x6)) + 8.89 x1^{1/2} - 3.21 \\ & x3^{1/2} - 0.572 x1^{1/2} \cos(\text{abs}(x1 + x2)) \cos(x3) - 0.193 x1^{1/2} \\ & \cos(x1^{1/2}) \log(x2 + x4) + 11.5 \text{ abs}(2.0 x2 + \cos(x1))^{1/2} \\ & \cos(x3) \log(x3) + 0.296 \cos(\text{abs}(x2 + \cos(x1)^{1/2})) \cos(x3) \\ & (x2 + x4)^{1/2} - 1.39 x1^{1/2} x6 \cos(\cos(x1)) + 2.41 x1^{1/2} \\ & \cos(x3) \log(x4) - 12.5 x3^{1/2} \cos(x3) \log(x3) - 24.5 x3^{1/2} \\ & \cos(x3) \log(x6) + 26.5 \end{aligned}$ |

Shown in Table 4.8 is the summative GP tree structural properties for the chosen model ID 2248. It has a total of 15 terms or gene expressions that were aggregated from the initial 496 expressions that it considered in the population with 134 mathematical operators, nodes, needed to define the regression equation. Bias was high of 44 indicating a bad fit and was needed to shift the equation to match datapoints. Furthermore, the depth is the maximum branch or chain that the gene tree made to make the model which is 9. Overall, it is considered a very complex model but given the other models, this suits best the needed task for the current study.

Table 4.8 Model ID 2248 GP tree structural properties

| Description             | Value                   |
|-------------------------|-------------------------|
| Genes                   | 15                      |
| Nodes                   | 134                     |
| Expressional complexity | 496                     |
| Depth                   | 9                       |
| Inputs                  | 6                       |
| Inputs used             | X1, X2, X3, X4, X5, X6, |

Shown in Figure 4.10Figure 4.9 is the scatterplot that maps out the performance of the given MSRGP model in terms of its predictions from actual datapoints using the train split. Datapoints are very skewed to the right resulting to low  $R^2$  value. Despite the spread of the data, model is chosen given the inputs. Further testing on new datapoints was done in section 5.2

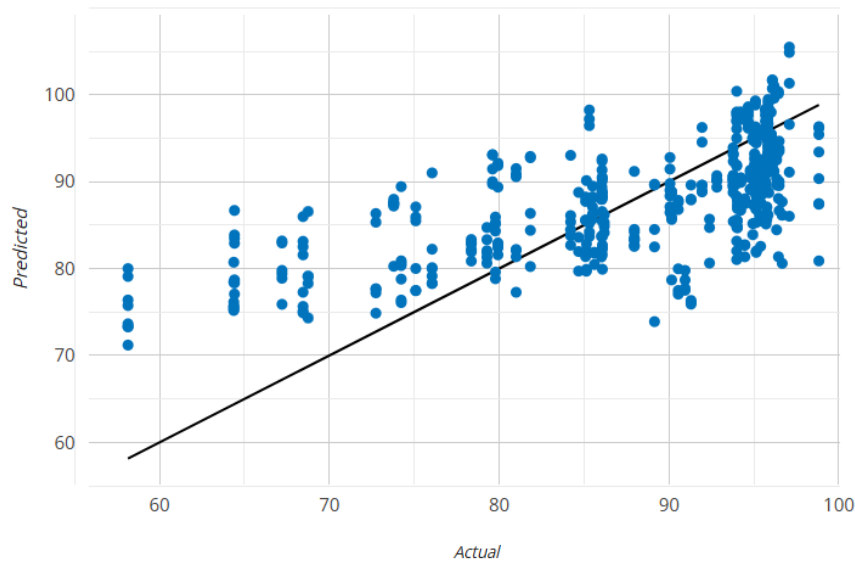


Figure 4.10 Predicted vs. actual scatterplot for train dataset on Model ID 2248

#### 4.2.4.1.3 MSRGP models for fresh head weight using RGB-HSV input

Shown in Table 4.9 are the Top 3 models considered for the third MSRGP model for the estimation of fresh head weight using RGB-HSV input. The pareto report showcase how all models have mainly used a variety of COS, SIN, LOG and ABS as mathematical operations to bring output regression line close to the given dataset. With this, the study chose model ID 967 as our main model for the fresh head weight given inputs as it had the least model complexity and is relatively close to the highest  $R^2$  value.

Table 4.9 Top 3 considered models for MSRGP fresh head weight from pareto report using RGB-HSV input

| Model ID | $R^2$ value | Model Complexity | Model                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------|-------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 925      | 0.567       | 497              | 69.0 x2 - 68.9 x1 - 88.0 x3 + 2130.0 x4 + 1.6e+4 x5 + 174.0 cos(cos(log(x3)3)) - 455.0 cos(log(x1 - 1.0 x5)2) - 37.4 cos(log(-1.0 x51/2 (x5 - 1.0 x1 + x53))3) + 46.6 cos(log(x2 x5)3) + 34.1 sin(x21/2 x3 x6) + 86.9 cos(cos(x1)2) + 18.9 abs(sin(sin(x3))) + 18.9 cos(x3) + 96.6 cos(log(x3 x52)3) + 227.0 cos(cos(log(x2 x5)3)) + 143.0 log(x3) + 86.9 abs(sin(abs(x53 - 1.0 x2 x5))) - 2370.0 log(x6 (2.0 x5 + cos(x52)) - 1.0 x53)2 - 0.34 x1 x2 + 143.0 x1 x6 + 525.0 log(cos(cos(log(x2 x5)3)))2 - 1.95e+4 x51/2 - 1410.0 x53 + 106.0 x63 + 1.1e+4 |
| 967      | 0.565       | 495              | 74.4 x2 - 69.3 x1 - 86.9 x3 + 2140.0 x4 + 1.61e+4 x5 - 2140.0 x6 + 174.0 cos(cos(log(x3)3)) - 454.0 cos(log(x1 - 1.0 x5)2) + 17.7 abs(sin(sin(abs(x1)))) - 37.7 cos(log(x51/2 (x1 -                                                                                                                                                                                                                                                                                                                                                                       |



|       |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------|-------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |       |     | $2.0 x_5)) + 45.6 \cos(\log(x_2 x_5)^3) + 34.1 \sin(x_2^{1/2} x_3 x_6) + 79.6 \cos(\cos(x_1)^2) + 17.7 \cos(x_3) + 95.0 \cos(\log(x_3 x_5^2)^3) + 251.0 \cos(\cos(\log(x_2 x_5)^3)) + 152.0 \log(x_3) + 79.6 \text{abs}(\sin(\text{abs}(x_5^3 - 1.0 x_2 x_5))) - 2410.0 \log(x_6 (2.0 x_5 + \cos(x_5^2)) - 1.0 x_5^3)^2 - 0.369 x_1 x_2 + 153.0 x_1 x_6 + 556.0 \log(\cos(\cos(\log(x_2 x_5)^3)))^2 - 1.95e+4 x_5^{1/2} - 3380.0 x_5^3 + 97.3 x_6^3 + 1.11e+4$                               |
| 1,562 | 0.501 | 391 | $157.0 x_2 - 57.8 x_1 - 140.0 x_3 + 2620.0 x_4 + 2450.0 x_5 + 2.94e+5 x_6 - 139.0 \sin(x_2 x_5) + 147.0 \sin(x_3 x_5) + 8.16 \cos(x_1) + 1.23e+5 \cos(x_6) + 19.9 \sin(x_3) - 2.55e+5 \sin(x_6) + 195.0 \sin(x_2 x_5)^3 - 513.0 x_2 x_5 + 665.0 x_3 x_5 + 25.8 x_2 x_4^4 + 25.8 x_2 x_4^6 + 86.9 \cos(\sin(x_2 + \text{abs}(x_5 + x_6)))^2 + 1290.0 x_4^{1/2} + 46.1 x_1 \log(x_5^3) \sin(x_5) + 1.49e+4 x_1 x_4^3 x_5 - 1.51e+4 x_2 x_4^3 x_5 - 8.16 x_1 \sin(x_3 x_5) \sin(x_5) - 1.28e+5$ |

Shown in Table 4.10 is the summative GP tree structural properties for the chosen model ID 968. It has a total of 19 terms or gene expressions that were aggregated from the initial 495 expressions that it considered in the population with 133 mathematical operators, nodes needed to define the regression equation. Bias was very low high of  $1.11e+4$ , indicating a good fit with little adjustments needed to shift the equation to match datapoints. Furthermore, the depth is the maximum branch or chain that the gene tree made to make the model which is 9. Overall, it is considered a very complex model but given the other models, this suits best the needed task for the current study.

Table 4.10 Model ID 967 GP tree structural properties

| Description             | Value                          |
|-------------------------|--------------------------------|
| Genes                   | 19                             |
| Nodes                   | 133                            |
| Expressional complexity | 495                            |
| Depth                   | 9                              |
| Inputs                  | 6                              |
| Inputs used             | $x_1, x_2, x_3, x_4, x_5, x_6$ |

Shown in Figure 4.11 is the scatterplot that maps out the performance of the given MSRGF model in terms of its predictions from actual datapoints using the train split. Datapoints are very varied resulting to low  $R^2$  value. Despite the spread of the data, model is chosen given the inputs. Further testing on new datapoints was done in section 5.2



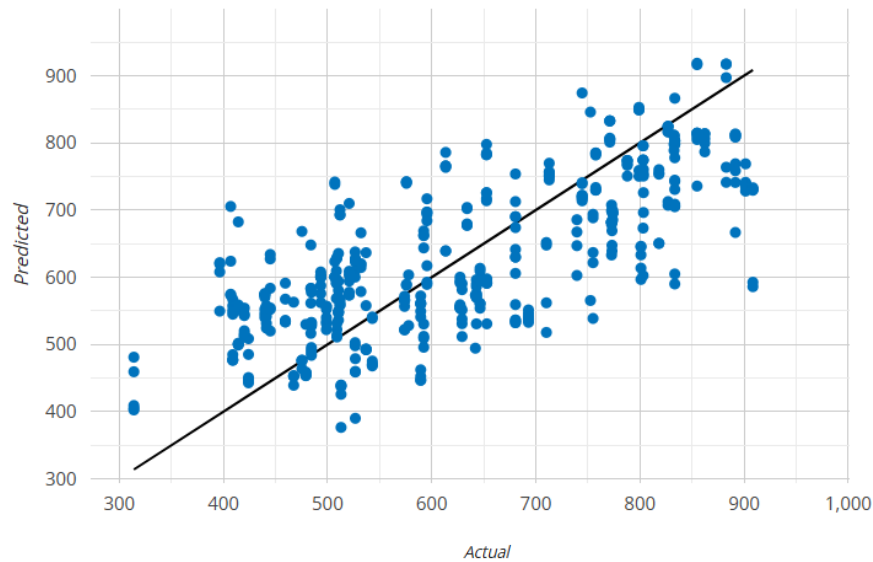


Figure 4.11 Predicted vs. actual scatterplot for train dataset on Model ID 967

#### 4.2.4.2 MSRGP models using HSV only color space

The following outputs are the results of the best run with the best-fitting model for the study's three quality parameters that considered only the mean HSV values extracted from train dataset. The top 3 models from the pareto report were extracted and considered for the study and afterwards the bias and gene weight is then checked on how well the chosen regression equation fit towards the given train data provided predictions.

##### 4.2.4.2.1 MSRGP output for chlorophyll-*a* using HSV input only

Shown in Table 4.11 are the Top 3 models considered for the second MSRGP model for the estimation of chlorophyll-*a* using HSV only input only. The pareto report shows how all models have low  $R^2$  value despite those high expressions needed to define a regression equation for the given dataset. Despite this, the study checked the bias and gene weights to choose model ID 1147 for given condition and input.

Table 4.11 Top 3 considered models for MSRGP chlorophyll-*a* from pareto report using HSV input only

| Model ID | $R^2$ value | Model Complexity | Model                                                                                                                                                                                                                                                                                                                                                   |
|----------|-------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1,167    | 0.573       | 441              | $65.5 x_1 + 374.0 x_2 + 0.262 \sin(\sin(\text{abs}(x_3) (x_3 + 7.0) (x_1 + x_3 + 10.0))) - 8530.0 \cos(\sin(x_3^2)^2) + 382.0 \cos(\cos(\cos(x_2))^3 + x_1 x_3^2) - 0.247 \sin(x_3 (x_3 + 7.0) (x_1 + x_3 + 10.0)) - 179.0 \cos(x_1 x_3^2 + x_1^2/2) + 351.0 \sin(x_3^2) + 260.0 \sin(\sin(x_3^2)) - 0.951 \text{abs}(x_1 + x_3 + x_2^2 - 1.0) - 362.0$ |



|       |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------|-------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |       |     | $\begin{aligned} & \text{abs}(x_2) + 190.0 \cos(x_1) + 204.0 \cos(x_2) - 74.7 \sin(x_3) + \\ & 0.0606 \cos(\sin(\text{abs}(x_3)) (x_3 + 7.0) (x_1 + x_3 + 10.0))) + 1370.0 \\ & \sin(x_3)^6 + 7.19 x_1 x_2 + 3390.0 \cos(\sin(x_3)^2)^3 + 116.0 \\ & \cos(x_1 x_3^2 + x_1/2)^3 - 97.9 \cos(\sin(x_1 x_3 (x_1 - 1.0 x_2))) - 1.0 \\ & x_1 x_3^2)^3 - 374.0 x_3^3 + 105.0 x_1 x_3 \cos(x_1) + 4530.0 \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                       |
| 1,147 | 0.568 | 430 | $\begin{aligned} & 86.7 x_1 + 87.1 x_2 + 0.25 \sin(\sin(\text{abs}(x_3)) (x_3 + 7.0) (x_1 + x_3 \\ & + 10.0))) - 6650.0 \cos(\sin(x_3)^2) - 0.232 \sin(x_3 (x_3 + 7.0) \\ & (x_1 + x_3 + 10.0)) + 1380.0 \cos(\cos(x_2)) + 366.0 \sin(x_2^3) - \\ & 80.2 \cos(\sin(x_3^2)) - 0.951 \text{abs}(x_1 + x_3 + x_2^2 - 1.0) - 74.4 \\ & \text{abs}(x_2) + 96.5 \cos(x_1) + 1460.0 \cos(x_2) + 12.7 \sin((x_2 - 1.0 \\ & x_1 \sin(x_3^2) \cos(x_1))^2) + 3.88 \sin(x_3) + 0.0564 \\ & \cos(\sin(\text{abs}(x_3)) (x_3 + 7.0) (x_1 + x_3 + 10.0))) + 560.0 \\ & \sin(x_3)^6 + 9.23 x_1 x_2 + 2600.0 \cos(\sin(x_3)^2)^3 + 71.6 \\ & \cos(x_1 x_3^2 + x_1/2)^3 - 84.3 \cos(\sin(x_1 x_3 (x_1 - 1.0 x_2))) - 1.0 \\ & x_1 x_3^2)^3 - 87.1 x_3^3 + 84.0 x_1 x_3 \cos(x_1) + 1850.0 \end{aligned}$ |
| 1,258 | 0.548 | 385 | $\begin{aligned} & 2410.0 \cos(\sin(x_1 - 0.452)) - 1070.0 x_3 - 546.0 \cos(x_1^2 x_3^2 \\ & (x_3 + x_3^3)) - 1070.0 x_1 - 3630.0 \cos(\sin(x_2 - 0.904)) + \\ & 1790.0 \sin(x_2 - 0.452) - 2.04 \log(x_2) + 43.6 \log(x_3) + 650.0 \\ & \log(x_3 + 3.0)^3 + 0.82 \log(x_1)^3 - 2040.0 x_2^2 \text{abs}(x_1)^2 - 123.0 \\ & x_1 x_3^2 + 13.9 x_1 \log(x_2) + 1140.0 x_1^2 + 45.8 x_3^3 + 109.0 x_3 \\ & (0.999 x_1 + 0.999 x_3) + 3980.0 x_2^2 \text{abs}(x_1)^2 \text{abs}(x_3) - 75.3 \\ & x_3^3 \sin(\sin(x_3)) (x_1 + x_2 + x_3) - 21.6 x_1 \text{abs}(x_1) (\log(x_2) + \\ & x_1 x_3^3 (x_1 + x_3 - 0.452)) - 1080.0 x_2^3 \sin(x_3) (x_1 + x_3 - \\ & 0.0923) + 983.0 \end{aligned}$                                                                                              |

Shown in Table 4.12 is the summative GP tree structural properties for the chosen model ID 1147. It has a total of 20 terms or gene expressions that were aggregated from the initial 441 expressions that it considered in the population with 132 mathematical operators, nodes, needed to define the regression equation. Bias was considerably high of 4530 indicating a big shift was needed for the equation to match datapoints. Furthermore, the depth is the maximum branch or chain that the gene tree made to make the terms which is 7. Overall, it is considered a very complex model but given the other models, this suits best the needed task for the current study.

Table 4.12 Model ID 1147 GP tree structural properties

| Description             | Value                                            |
|-------------------------|--------------------------------------------------|
| Genes                   | 20                                               |
| Nodes                   | 132                                              |
| Expressional complexity | 441                                              |
| Depth                   | 7                                                |
| Inputs                  | 3                                                |
| Inputs used             | X <sub>1</sub> , X <sub>2</sub> , X <sub>3</sub> |



Shown in Figure 4.9 is the scatterplot that maps out the performance of the given MSRGP model in terms of its predictions from actual datapoints using the train split. Datapoints are very skewed to the right resulting to low  $R^2$  value and predictions do not align that with a regression line with predictions off by 0.1 mg/g. Despite the spread of the data, model is chosen given the inputs. Further testing on new datapoints was done in section 5.2

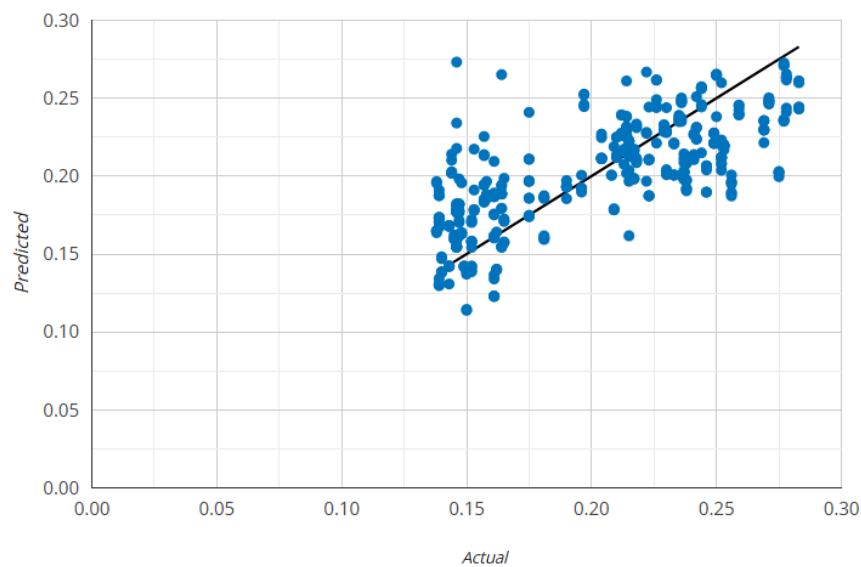


Figure 4.12 Predicted vs. actual scatterplot for train dataset on Model ID 1147

#### 4.2.4.2.2 MSRGP output for moisture content using HSV input only

Shown in Table 4.13 are the Top 3 models considered for the second MSRGP model for the estimation of moisture content using HSV only input. The pareto report shows how all models have low  $R^2$  value despite those high expressions needed to define a regression equation for the given dataset. Despite this, the study checked the bias and gene weights to choose model ID 2852 for given condition and input.

Table 4.13 Top 3 considered models for MSRGP moisture content from pareto report using HSV input only

| Model ID | $R^2$ value | Model Complexity | Model                                                                                                                                                                                                                                                                                             |
|----------|-------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2,836    | 0.254       | 566              | $8240.0 x_1 + 5.43e+5 x_2 - 199.0 x_3 - 4.12 \cos(\log(x_1 x_2)^3) - 12.2 \text{abs}(\cos(\log(x_2)^3)) + 186.0 \text{abs}(x_2 + x_1 x_3 \log(x_3)) - 5.75 \cos(\cos(\log(x_2 + x_1 x_2 x_3)^3)^3) - 12.2 \log(-1.0 \log(x_1 + x_2)) + 4.63e+4 \cos(x_1) - 350.0 \log(x_2) - 5.32e+5 \sin(x_2) -$ |



|       |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------|-------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |       |     | $5.56 \text{ abs}(x_1 + x_3 \cos(\log(x_{12})^3) \log(x_2)) + 2430.0 \cos(\sin(x_1 + x_2 + x_3)^3) - 5780.0 x_{12} \log(x_3) - 91.9 \log(x_1)^3 + 8.54 \text{ abs}(\cos(\log(x_2)^3))^2 + 5.56 \text{ abs}(\cos(\log(x_3)^3))^3 + 4.83e+4 x_{12} x_3 - 5.55e+4 x_{13} x_3 - 5.52e+5 \text{ abs}(\sin(x_1) - 1.0 x_1 (\sin(x_1) + x_1 x_2))^4 + 8130.0 x_{12} - 5.34e+4 x_{22} + 5.89e+4 x_1 x_{22} x_3 - 5.55e+4 x_{12} x_2 x_3 + 2.95e+4 x_{12} x_{22} x_{32} - 5.18e+4$                                                                                                                                                                                                                                                                                                         |
| 2,852 | 0.241 | 501 | $7860.0 x_1 + 4.76e+5 x_2 - 150.0 x_3 - 2.45 \cos(\log(x_{12})^3) - 13.7 \text{ abs}(\cos(\log(x_2)^3)) - 3160.0 \cos(\cos(x_1 x_3 \log(x_3))^3) + 155.0 \text{ abs}(x_2 + x_1 x_3 \log(x_3)) + 4.04e+4 \cos(x_1) - 337.0 \log(x_2) - 4.65e+5 \sin(x_2) - 4.54 \text{ abs}(x_2 + x_3 \cos(\log(x_{12})^3) \log(x_2)) + 2500.0 \cos(\sin(x_1 + x_2 + x_3)^3) - 5190.0 x_{12} \log(x_3) - 89.9 \log(x_1)^3 + 9.37 \text{ abs}(\cos(\log(x_2)^3))^2 + 4.54 \text{ abs}(\cos(\log(x_3)^3))^3 + 4.62e+4 x_{12} x_3 - 5.58e+4 x_{13} x_3 - 5.47e+5 \text{ abs}(\sin(x_1) - 1.0 x_1 (\sin(x_1) + x_1 x_2))^4 + 7810.0 x_{12} - 4.88e+4 x_{22} + 6.0e+4 x_1 x_{22} x_3 - 5.58e+4 x_{12} x_2 x_3 + 3.0e+4 x_{12} x_{22} x_{32} - 4.42e+4$                                                  |
| 1,074 | 0.217 | 496 | $3.62e+4 x_1 + 7230.0 x_2 + 4.62e+5 x_3 - 1.53 \sin(\log(x_{13})^3) - 2.41e+5 \cos(\text{abs}(x_1)) - 3.69e+5 \cos(\cos(\text{abs}(x_1))) - 1.34 \sin(\log(x_2)^3) - 6160.0 \text{ abs}(x_3) + 4.8e+4 \log(x_3) - 1570.0 x_{22} x_{32} + 32.0 (x_1 + \text{abs}(\log(\sin(x_2)))) + \text{abs}(x_3)^2)^2 - 5.17e+5 \text{ abs}(x_3)^{1/2} + 1.19e+5 \text{ abs}(x_3)^4 - 1570.0 \text{ abs}(x_3)^{3/2} - 2.86e+4 x_{32} \text{ abs}(x_3)^8 - 4.64e+9 x_{38} \text{ abs}(x_1)^{20} - 1.66e+4 \text{ abs}(x_2) \text{ abs}(x_3)^2 - 3.47e+4 (\text{abs}(x_3) + \cos(x_2) + \text{abs}(x_3)^2)^2 + 1570.0 x_2 \sin(x_1) - 2.11e+4 x_{11/2} - 7.03e+4 x_{22} + 2730.0 x_1 x_2 x_3 - 8890.0 x_{23} x_{32} \text{ abs}(x_2) + 1570.0 x_1 \text{ abs}(x_1) \text{ abs}(x_3)^2 + 7.14e+5$ |

Shown in Table 4.14 is the summative GP tree structural properties for the chosen model ID 2852. It has a total of 20 terms or gene expressions that were aggregated from the initial 501 expressions that it considered in the population with 136 mathematical operators, nodes, needed to define the regression equation. Bias was considerably high of  $-4.42e+4$  indicating big offset to the equation to match datapoints. Furthermore, the depth is the maximum branch or chain that the gene tree made to make the terms which is 9. Overall, it is considered a very complex model but given the other models, this suits best the needed task for the current study.

Table 4.14 Model ID 2852 GP tree structural properties

| Description             | Value           |
|-------------------------|-----------------|
| Genes                   | 20              |
| Nodes                   | 136             |
| Expressional complexity | 501             |
| Depth                   | 9               |
| Inputs                  | 3               |
| Inputs used             | $x_1, x_2, x_3$ |



Shown in Figure 4.13 the scatterplot that maps out the performance of the given MSRGP model in terms of its predictions from actual datapoints using the train split. Datapoints are very skewed to the right resulting to low  $R^2$  value and predictions do not align that with regression line. Despite the spread of the data, model is chosen given the inputs. Further testing on new datapoints was done in section 5.2

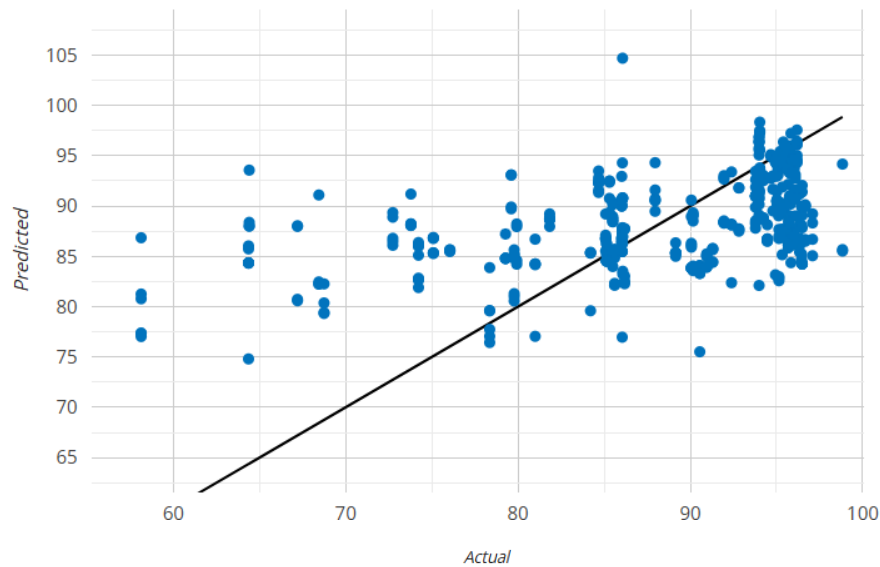


Figure 4.13 Predicted vs. actual scatterplot for train dataset on Model ID 2852

#### 4.2.4.2.3 MSRGP output for fresh head weight using HSV input only

Shown in Table 4.15 are the Top 3 models considered for the second MSRGP model for the estimation of moisture content using HSV only input. The pareto report shows how all models have low  $R^2$  value despite those moderate expressions needed to define a regression equation for the given dataset. Despite this, the study checked the bias and gene weights to choose model ID 3460 for given condition and input.

Table 4.15 Top 3 considered models for MSRGP fresh head weight from pareto report using HSV input only

| Model ID | $R^2$ value | Model Complexity | Model                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------|-------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2,859    | 0.545       | 495              | $451.0 x_1 + 19.5 x_2 - 420.0 x_3 + 2840.0 \cos(\cos(x_1 x_2)) - 93.1 \cos(\log(x_1)) - 1460.0 \log(\cos(x_3)) + 2840.0 \log(\sin(x_1)) - 0.882 \sin(6.0 x_3 + \cos(\log(x_3))) + \log(\sin(x_1)) + \cos(x_2)) - 2300.0 \cos(\log(\cos(x_1))) + 143.0 \cos(x_1) + 125.0 \cos(x_2) - 2790.0 \log(x_1) + 143.0 x_1 x_2 - 86.2 x_3^9 \cos(x_1)^3 + 42.8 \text{abs}(x_3 + \cos(x_1 + x_2 + x_3)) - 1.0 \cos(x_1)^{1/2} 1/2 + 11.0$ |



|       |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------|-------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |       |     | $\log(x_3^3)^3 + 21.0 x_2 x_3^3 - 295.0 \log(\sin(x_3))^3 - 504.0 x_3^3 + 237.0 \cos(x_2 \cos(\log(x_1)))^3 + 17.9 x_1 x_2 \log(\sin(x_3))^3 - 420.0 x_1 x_2 x_3 (x_3 + \cos(x_1 + x_2 + x_3)) - 1.0 \cos(x_1)^{1/2} 1/2 + 311.0$                                                                                                                                                                                                                                                                                                                                                 |
| 3,460 | 0.536 | 233 | $3130.0 x_1 + 13.5 x_2 - 506.0 x_3 + 6430.0 \cos(\cos(x_3)) - 1.0 x_2^3 - 1570.0 \log(\cos(x_3)) - 3.95 \log(\text{abs}(x_2)^2 + 8.0 x_1^3) + 0.978 \cos(\log(x_3)^3) + 3810.0 \cos(x_1) + 5840.0 \cos(x_3) - 709.0 \log(x_1) + 63.4 \log(x_3) - 2230.0 x_1^2 x_2^3 - 115.0 \log(x_1)^2 - 202.0 x_1 x_3^2 + 202.0 x_1^2 x_3 + 2230.0 x_1 x_2^6 + 53.1 \cos(x_1 - 1.0 x_3)^3 + 675.0 x_1^3 - 106.0 x_2^2 - 4170.0 x_2^3 + 67.3 x_3^3 - 742.0 x_2^9 + 1.56e+4 x_1^2 \text{abs}(x_2)^3 \text{abs}(x_3)^3 - 2530.0 x_1 \cos(x_3) \cos(\cos(\cos(\cos(x_3) - 1.0 x_2^3)))^3 - 1.43e+4$ |
| 3,453 | 0.532 | 226 | $2650.0 x_1 + 13.4 x_2 - 473.0 x_3 + 5400.0 \cos(\cos(x_3)) - 1.0 x_2^3 - 1320.0 \log(\cos(x_3)) - 3.54 \log(\text{abs}(x_2)^2 + 8.0 x_1^3) + 3230.0 \cos(x_1) + 4850.0 \cos(x_3) - 602.0 \log(x_1) + 66.9 \log(x_3) - 1890.0 x_1^2 x_2^3 - 97.5 \log(x_1)^2 - 170.0 x_1 x_3^2 + 170.0 x_1^2 x_3 + 1890.0 x_1 x_2^6 + 45.2 \cos(x_1 - 1.0 x_3)^3 + 574.0 x_1^3 - 104.0 x_2^2 - 3460.0 x_2^3 + 56.7 x_3^3 - 631.0 x_2^9 + 1.33e+4 x_1^2 \text{abs}(x_2)^3 \text{abs}(x_3)^3 - 2120.0 x_1 \cos(x_3) \cos(\cos(\cos(\cos(x_3) - 1.0 x_2^3)))^3 - 1.19e+4$                            |

Shown in Table 4.16 is the summative GP tree structural properties for the chosen model ID 3460. It has a total of 19 terms or gene expressions that were aggregated from the initial 233 expressions that it considered in the population with 83 mathematical operators, nodes, needed to define the regression equation. Bias was very high of - 1.43e+4, indicating that big offset to the equation to match datapoints. Furthermore, the depth is the maximum branch or chain that the gene tree made to make the terms which is 8. Overall, it is considered a very complex model but given the other models, this suits best the needed task for the current study.

Table 4.16 Model ID 3460 GP tree structural properties

| Description             | Value                                            |
|-------------------------|--------------------------------------------------|
| Genes                   | 19                                               |
| Nodes                   | 83                                               |
| Expressional complexity | 233                                              |
| Depth                   | 8                                                |
| Inputs                  | 3                                                |
| Inputs used             | X <sub>1</sub> , X <sub>2</sub> , X <sub>3</sub> |



Shown in Figure 4.14 the scatterplot that maps out the performance of the given MSRGP model in terms of its predictions from actual datapoints using the train split. Datapoints are very skewed to the right resulting to low  $R^2$  value and many of the predictions do not align that with regression line. Despite the spread of the data, model is chosen given the inputs. Further testing on new datapoints was done in section 5.2

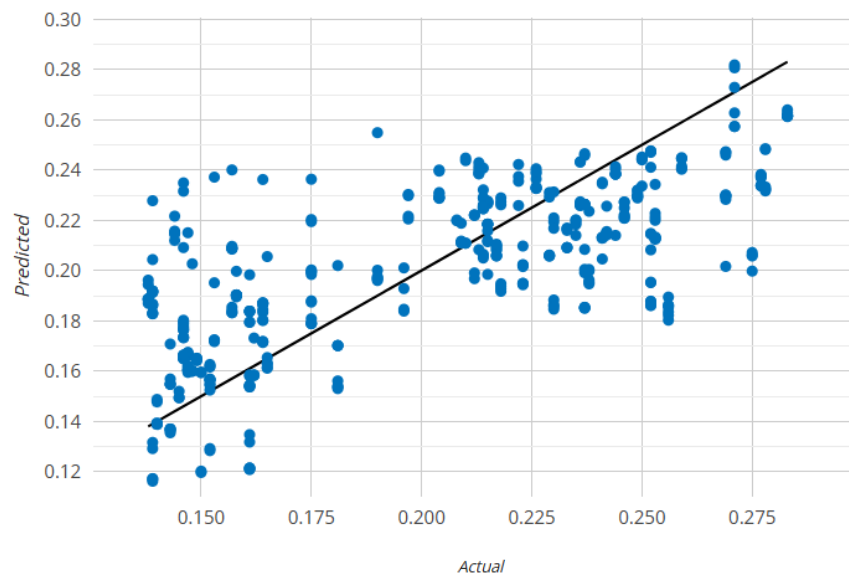


Figure 4.14 Predicted vs. actual scatterplot for train dataset on Model ID 3460

## 4.3 Fuzzy Logic Implementation Procedures

### 4.3.1 Fuzzy Logic Modelling and Accuracy Testing

The estimated values from the previous models were inputted into a custom fuzzy logic model based on the flowchart seen below in Figure 4.15, the selected fuzzy method was a Mamdani Type-1, a fuzzy inference system that was based on implication rules. The output parameters and membership functions (MFs) were first defined, this being the ‘Grading Level’ of the system with the parameters distributed equally for each MF having linguistic values of ‘L3’, ‘L2’, and ‘L1’ correlating to the quality grade of iceberg lettuce. Additionally, input parameters were defined as ‘chlorophyll- $a$ ’, ‘moisture content’, and ‘fresh head weight’. Each input had three membership functions each, having linguistic values ‘Low’, ‘Average’, and ‘High’, with their parameters set according to existing PNS



values for iceberg lettuce. Rules were then set for every possible input for inferencing, which totaled to 27 rules created for the model, which were de-fuzzified and graded as either 'L3', 'L2', or 'L1', depending on the inputs. The process for obtaining the fuzzy logic model can be seen in Figure 4.15.

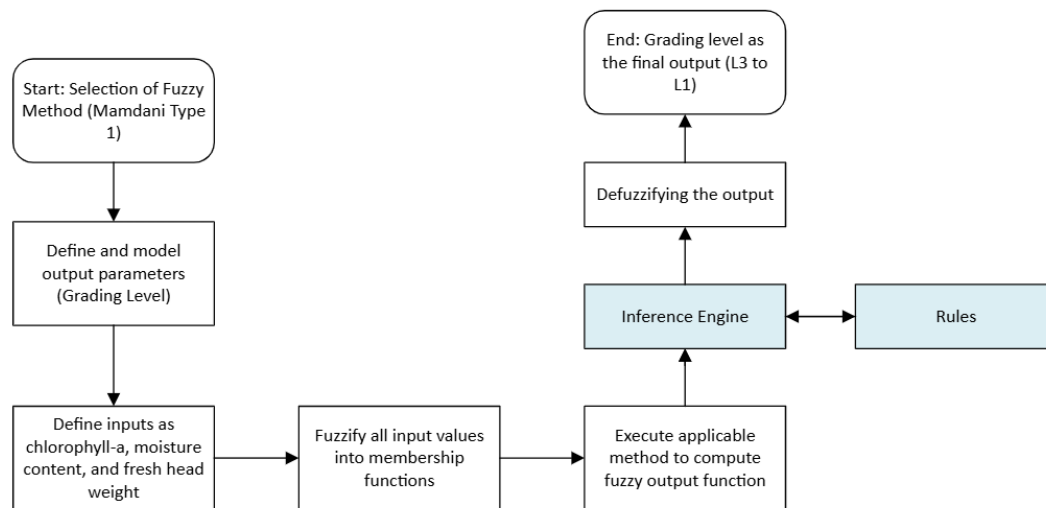


Figure 4.15 Fuzzy Logic Modelling Flowchart

The model was tested by taking the test dataset from extractions of chlorophyll-*a*, moisture content, and fresh head weight, wherein the grading level is based off standard, using these values as inputs to the fuzzy inference system. A MATLAB program was run to determine which grading level the data falls under after the model was made. In cases where the model accuracy is not high enough, the parameters of the inputs and outputs can be changed within the Fuzzy Logic Design toolbox, and membership function types can be changed, depending on the range of parameters for said input to achieve desired fuzzy logic accuracy.

#### 4.3.2 MATLAB Fuzzy Logic Designer

The fuzzy logic toolbox in MATLAB included a simple method of creating a simple fuzzy inference system through the fuzzy logic designer application, which allowed the users to create their own inputs that will get fuzzified, and de-fuzzified according to the parameters and rules set by the user. For the system used, the inputs





created were moisture content, chlorophyll-*a*, and fresh head weight, which all had three membership functions (MF) that consisted of a trapezoidal ‘Low’, a triangular ‘Average’, and a trapezoidal ‘High’. The output membership functions, meanwhile, consisted of three gaussian types ‘L3’, ‘L2’, and ‘L1’, which served as the final output values for the whole system that determined the movement of the sorting process. These membership functions are shown in Figure 4.16 where A.) features the membership function done for moisture content, B.) features the membership function done for chlorophyll-*a*, C.) features the membership function done for fresh head weight and D.) features the membership function done for the output grading level.

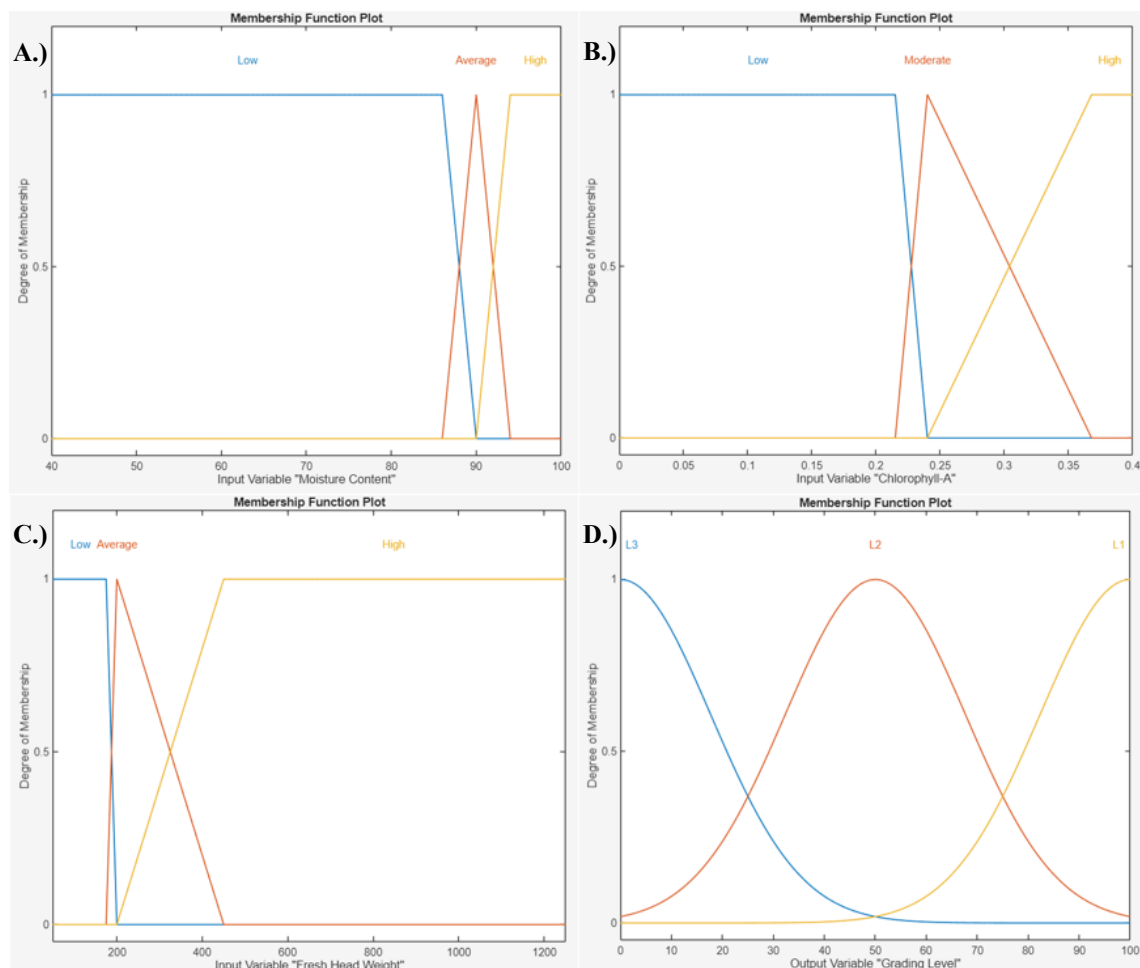


Figure 4.16 Membership functions of current study



### 4.3.3 Fuzzy Logic Model Considerations and Results

The study mapped out the different possible grading levels among the three quality parameters and is shown in Table 4.17. The common logic is that, if at least two parameters have a “Low” linguistic value, then the lettuce quality is deemed an overall low quality lettuce or an “L3” quality lettuce. If there is at least one parameter that has a “Low” linguistic value and one parameter that has an “Average” linguistic value then it is deemed an average quality lettuce or an “L2” quality lettuce. Lastly if there is no “Low” linguistic value among the quality parameters and at least two parameters have a “High” linguistic value then it is deemed as a high quality lettuce marked as “L1” quality. Certain exemptions to this reasoning is applied like when all parameters have “High” linguistic values is determined as an “L2” quality lettuce due to overabundance of chlorophyll-*a* concentration and water content that may suggest possible stress due to overhydration and rapid spoilage indication. Similarly, if all quality parameters have an “Average” linguistic value that it is deemed an “L1” quality lettuce as it there is no “Low” linguistic value and is an indication of harvest maturity that is suitable for selling into the market.

Table 4.17 Associative matrix for the current study

| Chlorophyll-a | Moisture Content | Head Weight | Lettuce Quality |
|---------------|------------------|-------------|-----------------|
| Low           | Low              | Low         | L3              |
| Low           | Low              | Average     | L3              |
| Low           | Low              | High        | L3              |
| Low           | Average          | Low         | L3              |
| Low           | Average          | Average     | L2              |
| Low           | Average          | High        | L2              |
| Low           | High             | Low         | L3              |
| Low           | High             | Average     | L2              |
| Low           | High             | High        | L2              |
| Average       | Low              | Low         | L3              |
| Average       | Low              | Average     | L2              |
| Average       | Low              | High        | L2              |
| Average       | Average          | Low         | L2              |
| Average       | Average          | Average     | L1              |
| Average       | Average          | High        | L1              |
| Average       | High             | Low         | L2              |



|         |         |         |           |
|---------|---------|---------|-----------|
| Average | High    | Average | <b>L1</b> |
| Average | High    | High    | <b>L1</b> |
| High    | Low     | Low     | <b>L3</b> |
| High    | Low     | Average | <b>L1</b> |
| High    | Low     | High    | <b>L1</b> |
| High    | Average | Low     | <b>L1</b> |
| High    | Average | Average | <b>L1</b> |
| High    | Average | High    | <b>L1</b> |
| High    | High    | Low     | <b>L1</b> |
| High    | High    | Average | <b>L1</b> |
| High    | High    | High    | <b>L2</b> |

Triangular functions were initially considered for the membership function (MF) parameters, with each one having been set to a low value, a peak, and a high value, thus forming the triangle that gets de-fuzzified. However, testing the model with the study's test dataset yielded a low accuracy, only reaching 73% accuracy. These were due to the study's grading criteria for the current study evinced in Table 3.2 that is in conformity with the PNS/BAFS 19:2005 standard. It was noticed that all iceberg lettuces procured were above average, especially with the given fresh head weight averaging 500 grams or higher and noticeable outliers in the dataset for the other quality parameters. For the improvement of the model, trapezoidal functions were used for each 'Low' and 'High' MF to distinguish that values within the said functions were guaranteed to fall under these classifications. While the 'Average' MF remained triangular to help with model classification of the inputs based on their proximity to the functions.

Shown in Figure 4.17 are the resulting surface graphs for the current study that correlate the corresponding quality parameters into the output grading level. A.) shows the relation of fresh head weight to chlorophyll-*a* to the grading level values where a cutoff of 400 grams or lower for fresh head weight was considered to cutoff a zero-grade level value. B.) shows the relation of moisture content and fresh head weight to the grading level values where a value of 80% is deemed the zero value for grading level. Lastly, C.) shows the relation of moisture content and chlorophyll-*a* where a value of 0.2



mg/g for chlorophyll-*a* is the cutoff for a zero value for grading level. Blue indicates a low grading level value associated with ‘L3’ lettuce grade, bluish-green to green indicate values where the fuzzy system associates an ‘L2’ lettuce grade and values that are indicated in yellow are associated as ‘L1’ lettuce grade.

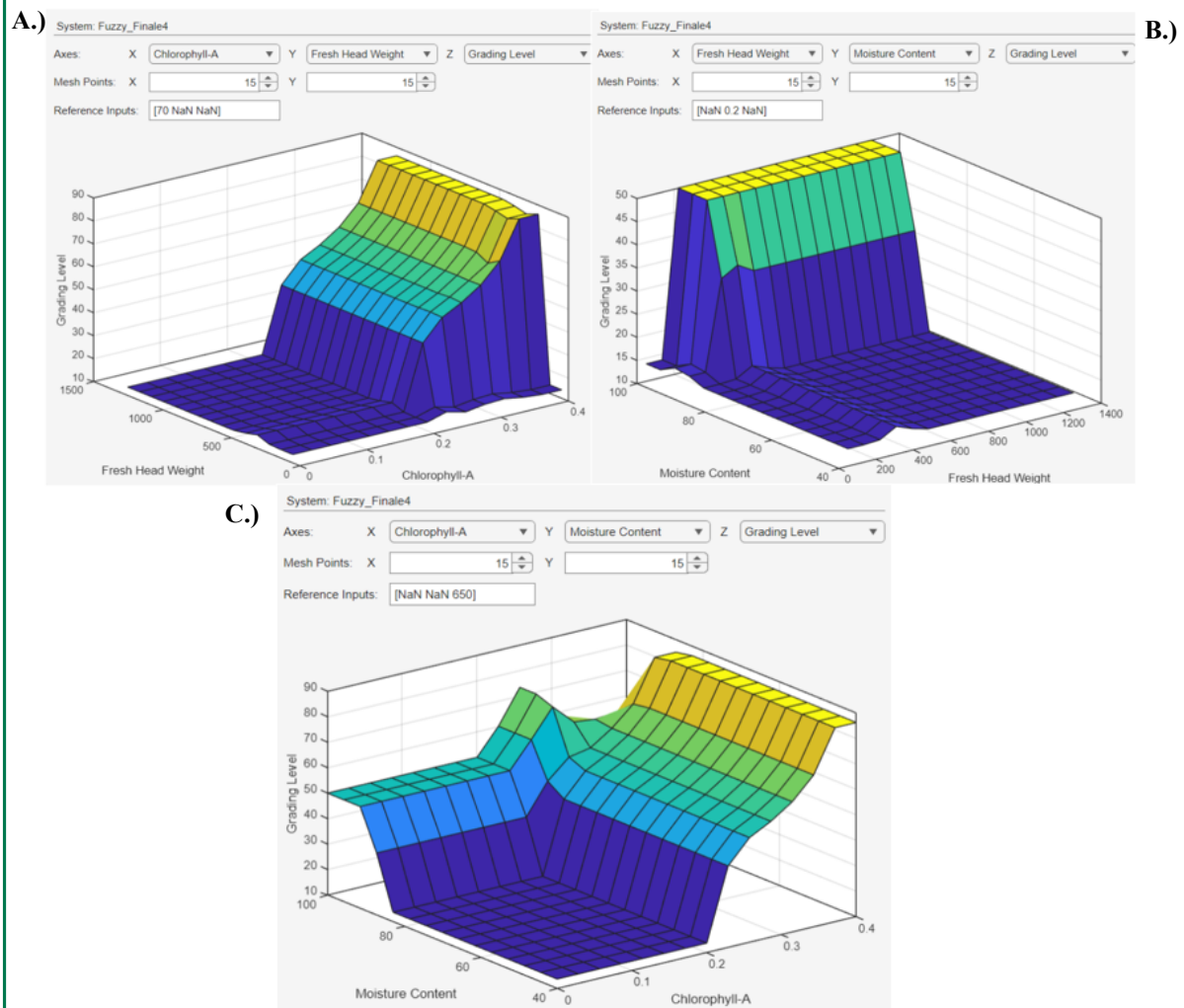


Figure 4.17 Fuzzy surface graphs for the current study



## 4.4 Mini-Production Line Implementation Procedures

### 4.4.1 Design Considerations

#### 4.4.1.1 Mechanical Design

The mechanical design of the mini-production line consists of the design considerations and calculations for the imaging chamber, sorting conveyor, conveyor belt selection, dimensional calculations, weight calculations, and motor selection for the sorting mechanism. CAD modelling software and calculations were used to get mechanical design.

##### 4.4.1.1.1 Imaging Chamber

The imaging chamber was designed in such a way that it provides a controlled environment for accurate image capturing of the iceberg lettuce batches of five, as these are transported throughout the image-capturing station. For high-quality image acquisition for further processing, the design considerations centered on background uniformity and seamless integration with the conveyor system. Figure 4.18 provides a visual representation of the CAD model of the imaging chamber where all units are in millimeters (mm).

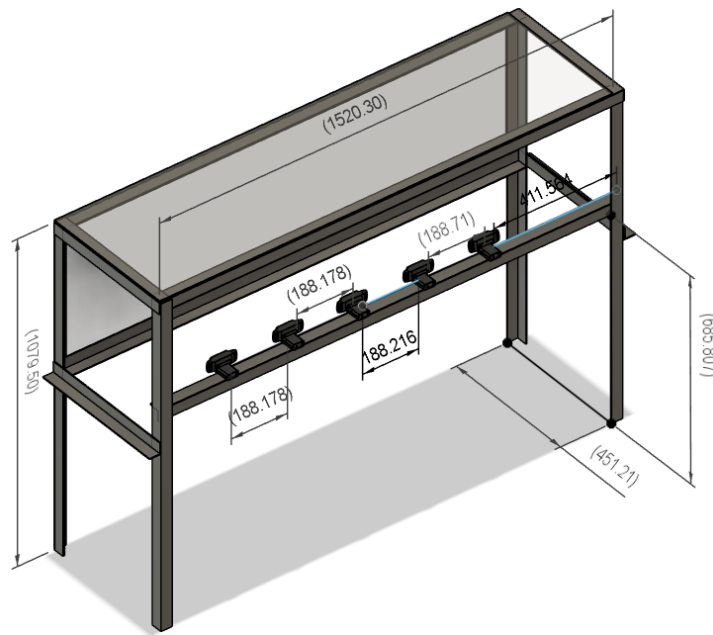


Figure 4.18. Imaging Chamber CAD model



The frame of the imaging chamber was constructed using slotted steel angled bars and 8 mm nuts and bolts for the purpose of modularity. The available slots on the angled bars allowed for the five 1080p web cameras to be mounted on the imaging chamber with 6.35 mm nuts and bolts without the use of destructive methods, which necessitated its selection. The mounted cameras were positioned at intervals of 0.22 m. Figure 4.19 illustrates the CAD model of the camera mounting section of the imaging chamber.

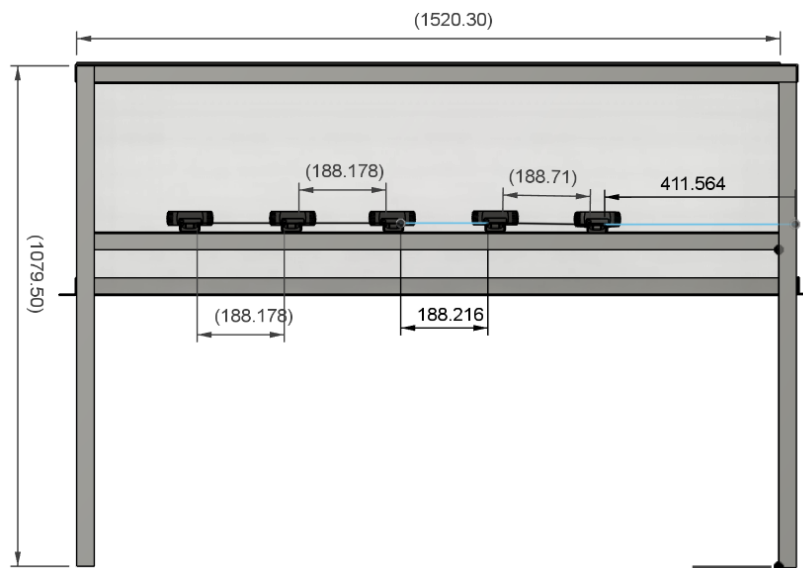


Figure 4.19. Camera Placement on Imaging Chamber CAD Model

The internal walls of the imaging chamber were colored white to alleviate background noise during image capturing to improve the image quality of the iceberg lettuce head photos. Figure 4.20 shows an image of a batch of lettuce inside the imaging chamber. Five cameras were chosen instead of one or two cameras. Using fewer cameras would decrease the image quality due to the increased distance from the camera to the lettuce. An increase in distance for the camera-to-lettuce would also require a larger imaging chamber chassis. Therefore, five cameras were used for the design featuring five lettuces.

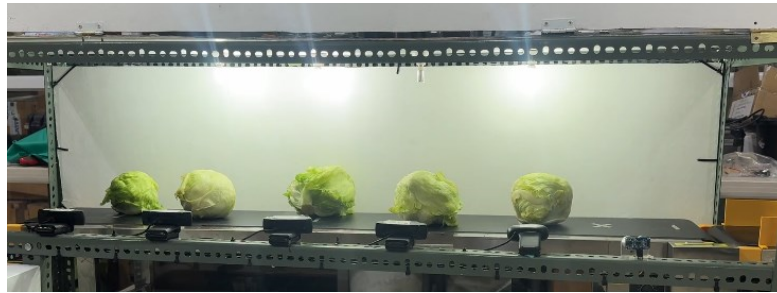


Figure 4.20. Image of the imaging chamber with a batch of lettuce

Additionally, each camera's light intensity was measured, which was reconstructed as close as possible to the extraction setup for the images to be easily read, and accurate predicted by the system. Seen in Figure 4.21 below are the light intensities of each camera, averaging out to 400 lux across all cameras.

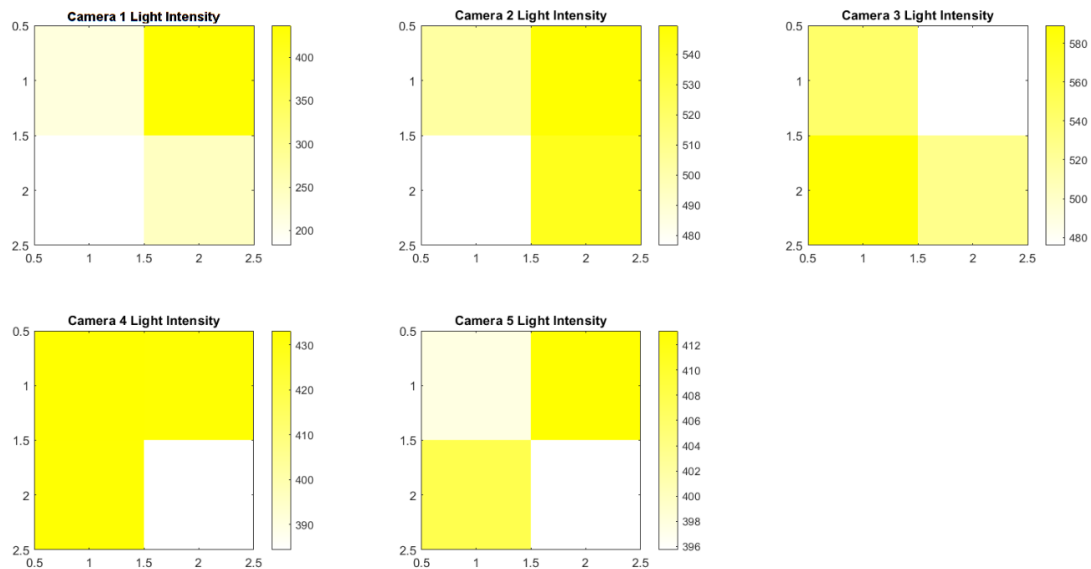


Figure 4.21. Light intensity chart for imaging chamber of the current study

#### 4.4.1.1.2 Sorting Conveyor

The sorting conveyor is used to sort the iceberg lettuce samples into their respective grading bins. The sorting conveyor consists of the conveyor itself, guide rails, servo motors, servo motor mounts, and servo lever arm. CAD modelling software was used to design the layout of the sorting conveyor's components seen in Figure 4.22 below.

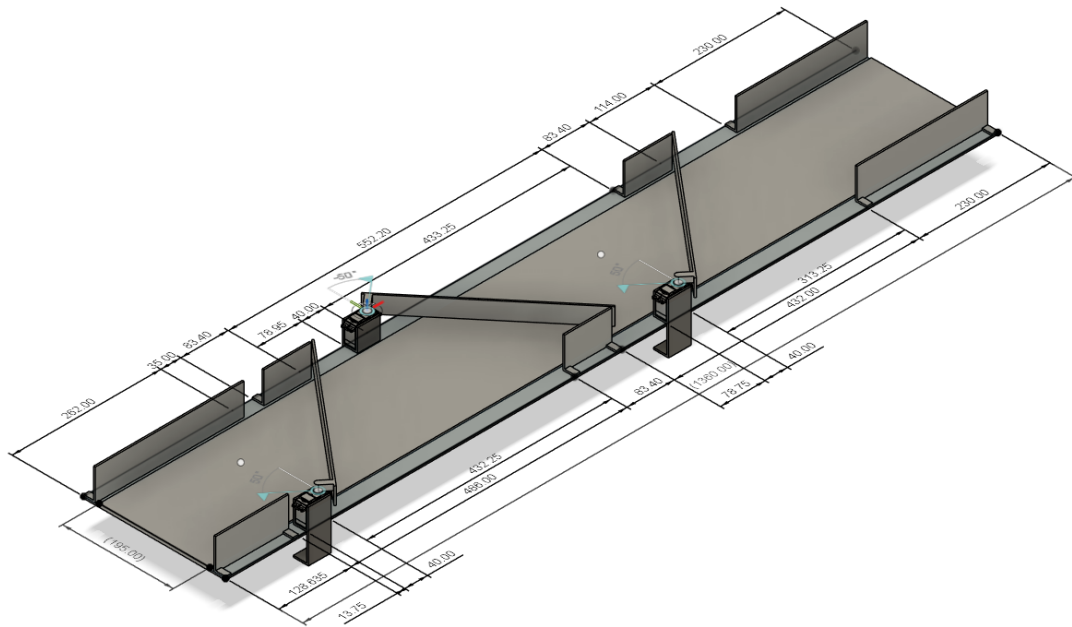


Figure 4.22. Isometric view of CAD drawing the sorting conveyor with known placement of sorting arms.

A mount for the servo was modeled which featured a c-clamp design for both the servo motor itself and where the mount would attach to the conveyor itself. The thickness of the conveyor chassis was taken and applied into the design of the mount. Dimensions of the MG996R servo motor were also taken and applied into the design of the mount. Figure 4.23 shows the CAD model of the servo mount.

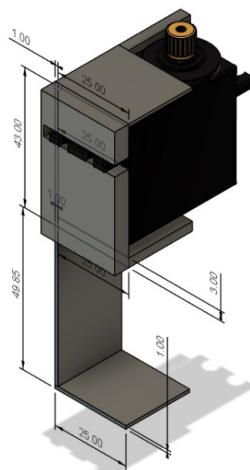


Figure 4.23. Isometric view of servo mount with servo motor in place





A lever arm was also modelled which would be actuated by the servo motor to guide and sort iceberg lettuce onto their designated sorting bins. The length of the lever arm was based on the width of the conveyor belt. The CAD model of the lever arm is shown in Figure 4.24 below with the overall 3D-printed sorter arm attached to the conveyor seen in Figure 4.25.

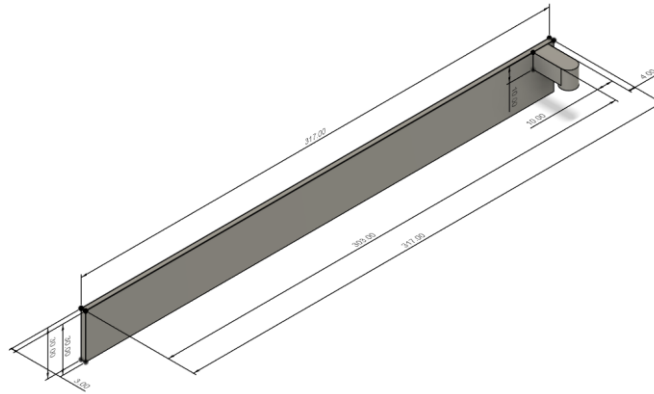


Figure 4.24. Isometric view of 3D-printed sorter arm to secure in place the MG966R servo motor on the conveyor.

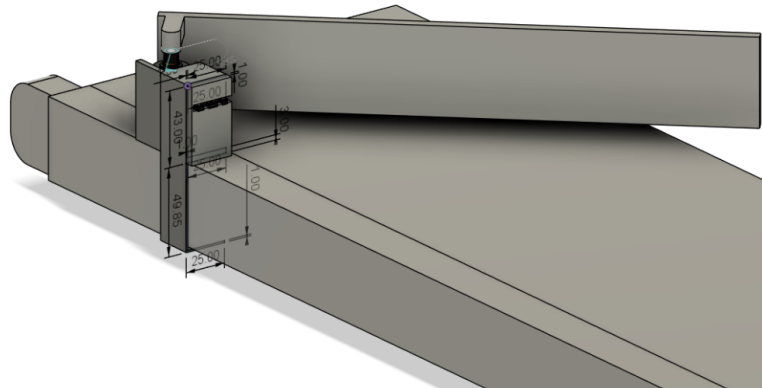


Figure 4.25. Isometric view of a 3D-printed sorter arm with the MG966R servo motor placed on conveyor.

Guide rails were also modelled which would be used to prevent lettuce from falling off the conveyor belt or prevent them from hitting the servo motors. Guide rails were positioned at the start and end of the conveyor, as well as in between designated sorting slots. The lengths of the guide rail varied depending on its placement on the



conveyor. A total of seven guide rails were manufactured: three 84.40 mm, two 230 mm, one 262 mm, and one 128.6mm. The CAD model of the guide rail is shown in Figure 4.26 below.

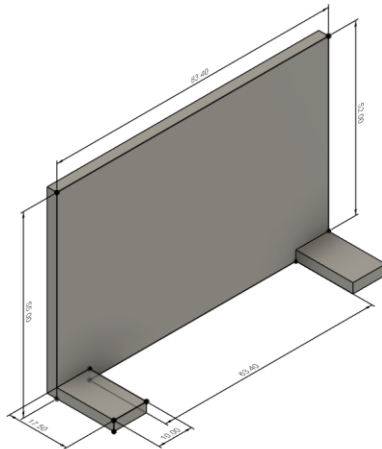


Figure 4.26. Isometric view of the 83.40 mm guide rail

#### 4.4.1.1.3 Conveyor Belt Selection

Two conveyor belts of identical specifications were utilized for the mini production line as these aligned with the objectives of the imaging and sorting requirements. The specification of the conveyor is shown below in Table 4.18.

Table 4.18. Conveyor Specifications

| Specification          | Value                                  |
|------------------------|----------------------------------------|
| Motor Power            | 60 watts                               |
| Motor Type             | Adjustable Speed Motor                 |
| Belt Length            | 118 inches (300 cm)                    |
| Belt Width             | 8 inches (20 cm)                       |
| Conveyor Length        | 59 inches (150 cm)                     |
| PVC Belt Width         | 7.8 inches (20 cm)                     |
| PVC Belt Thickness     | 0.07 inches (0.2 cm)                   |
| Conveyor Height        | 29.5 inches (75 cm)                    |
| Overall Size           | 63 × 18 × 12 inches (160 × 44 × 29 cm) |
| Weight                 | 76 lbs (22 kg)                         |
| Material               | Stainless Steel + PVC                  |
| Usage                  | Packing, adding mosaic, etc.           |
| Application Industries | Food, Beverage, Pharmaceutical.        |



Both conveyors were powered by a 60 watts adjustable speed motor, which allowed precise control over belt movement to accommodate different processing speeds. For the handling of the lettuce, the consideration for selecting the specifications for the conveyor belt was that the PVC belt had a width of 20 cm and thickness of 0.2 cm. It also had a length of 150 cm, and a 75 cm height, which ensured the handling of lettuce was non-invasive. The material of the conveyor belt's frame consisted of stainless steel, which was ideal due to being food grade. Using two similar conveyors, one for imaging and another for sorting, allowed efficient processing of five lettuces, which improved workflow, added automation, and ensured accurate, quality ratings.

#### 4.4.1.1.4 Dimensional Calculation

Two conveyor belts of identical specifications were utilized for the mini-production line: one for the image-capturing station and one for the sorting system. The space required by the conveyor was determined based on the diameter of the iceberg lettuce and the spacing between the lettuces in each batch of five. The no. of load  $N$  that can be applied to the conveyor belt is defined by:

$$N = \frac{L}{d + s} \quad 4.1$$

Where:

$L$  = length of conveyor

$s$  = space in between loads

$d$  = diameter of iceberg lettuce

It was assumed that  $d = 0.20$  m and  $s = 0.04$  m

The assumed  $d$  is the maximum diameter of an iceberg lettuce. The assumed  $s$  is the assumed space required by the camera to capture an image of the lettuce without disruption from the adjacent lettuces.



It was assumed that  $N = 5$ , therefore:

$$5 = \frac{L}{0.20[m] + 0.04[m]} \quad 4.2$$
$$L = 1.2 \text{ m}$$

The entry and exit points of the image-capturing station were also considered. This was implemented to ensure adequate spacing for the first and last lettuce within the imaging chamber, preventing external obstacles outside the imaging chamber from interfering with the captured images.

$$\text{Minimum length requirement of conveyor} = L + 2s$$

$$\text{Minimum length requirement of conveyor} = 1.2[m] + 2 \cdot 0.04[m]$$

$$\text{Minimum length requirement of conveyor} = 1.28 \text{ m}$$

Figure 4.27 depicts the visual representation for the dimensional requirements of the conveyor at the image-capturing station.

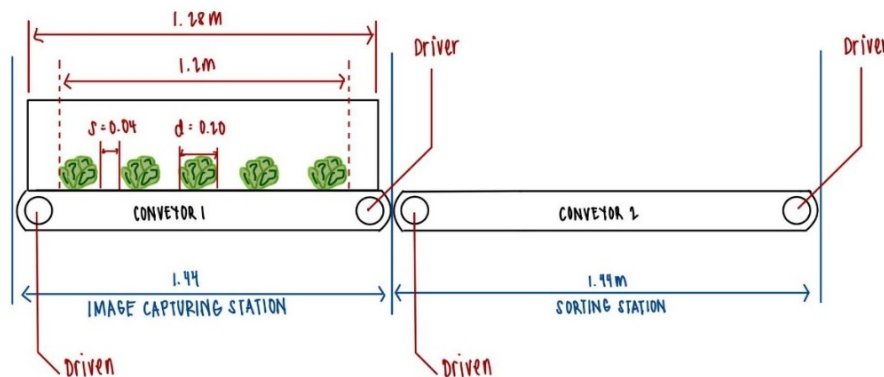


Figure 4.27. Dimensional Requirements for the conveyor at the image-capturing station

After the initial dimensional computations, it was determined that a conveyor with a length greater than 1.28 meters is suitable for the conveyor stationed for image-capturing. The length of the conveyor chosen was 1.44 meters.

However, during the construction of the whole system, the distance between the loads was reduced. Markings were placed on the conveyor belt to denote where the lettuce heads would be positioned when loading these onto the conveyor. During loading, the lettuce heads were placed in such a way that the center was aligned with the markings on



the conveyor belt. The midpoint-to-midpoint distance of the lettuce heads was used to determine the placement of these markings, which is computed by the following equation:

$$\begin{aligned} m &= d + s \\ m &= 0.20[m] + 0.04[m] \\ m &= 0.24 \text{ m} \end{aligned} \tag{4.3}$$

Where:

$m$  = midpoint-to-midpoint distance between the lettuce heads.

The value of  $m$  was reduced to 0.22 meters as a result of the camera angle adjustment that was done during construction. Figure 4.28 shows an illustration of the actual midpoint-to-midpoint distance that was used in image-capturing conveyor.

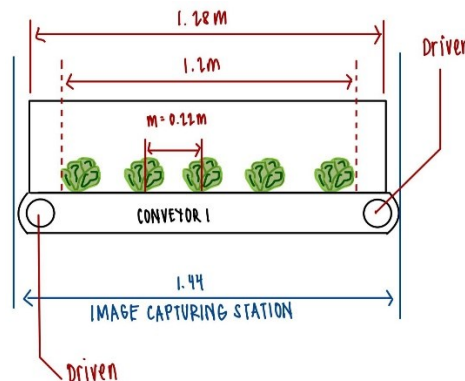


Figure 4.28. Midpoint-to-point distance used for the conveyor at the image-capturing station

#### 4.4.1.1.5 Motor Selection for Sorting Mechanism

For the sorting system of the mini-production line, three motors for three grading classifications with the capability to actuate lever arms had to be chosen. The specification required would be for it to be able to actuate a 100-gram 3D-printed lever arm made of PLA, be digitally controlled, and be digital. The selection of servo motors and their respective specifications can be seen in Table 4.19 below.

Table 4.19. Selection of Servo Motors

|        | Maximum Torque | Voltage   | Signal Type |
|--------|----------------|-----------|-------------|
| SG90   | 1.8-kgcm       | 4.6 – 6 V | Analog      |
| MG995  | 11-kgcm        | 4.6 – 6 V | Analog      |
| MG996R | 15-kgcm        | 4.6 – 6 V | Digital     |
| DS3218 | 20-kgcm        | 6 – 7.4 V | Digital     |



Mass of lever arm = 0.1 kg

The weight of the lever arm needs to be calculated using Newton to calculate the required torque.

5 = Weight of lever arm:

$$\begin{aligned} \text{Weight} &= mg \\ &= 0.100 \text{ kg} \times 9.81 \text{ m} \cdot \text{s}^{-2} \\ &= 0.981 \text{ N} \end{aligned} \quad 4.4$$

Where:

m = mass of lever arm

g = acceleration due to gravity

Using the calculated weight of the lever arm in Newtons, the required torque for the motor can be calculated using:

Calculate Required Torque:

$$\begin{aligned} \tau &= Fd \\ &= 0.981 \text{ N} \times 0.15 \text{ m} \\ &= 0.14715 \text{ N} \cdot \text{m} \end{aligned} \quad 4.5$$

Where:

$\tau$  = required torque

F = weight of the lever arm

d = distance from the pivot point of the motor

Next, the torque needs to be converted to  $\text{kg} \cdot \text{cm}$  using the equation:

Converting from  $\text{N} \cdot \text{m}$  to  $\text{kg} \cdot \text{cm}$ :

$$\begin{aligned} \tau_{\text{kg} \cdot \text{cm}} &= \tau_{\text{N} \cdot \text{m}} \times 10.1972 \\ &= 0.14715 \text{ N} \cdot \text{m} \times 10.1972 \\ &= 1.5 \text{ kg} \cdot \text{cm} \end{aligned} \quad 4.6$$

Where:

10.1972 = conversion factor

For dynamic loads and continuous operations, a safety factor (FoS) of 4 is applied:

Applying FoS of 4:

$$\begin{aligned} \tau_{\text{req}} &= \tau \times 4 \\ &= 1.5 \text{ kg} \cdot \text{cm} \times 4 \end{aligned} \quad 4.7$$



$$= 6 \text{ kg} \cdot \text{cm}$$

After computing the required torque, the motor specification needs 6-kgcm of torque and a digital signal. The MG996R servo motor fits this description and is most suitable for the required task.

#### **4.4.1.2 Electronic Design**

##### **4.4.1.2.1 Ultrasonic Sensor**

The ultrasonic sensor was chosen for its functionality to detect how far an object is from the sensor. The specific model HC-SR04 was used in the mini-production as it utilized sonar to determine the distance of the iceberg lettuce in relation to the cameras. Once the ultrasonic sonar detected an iceberg lettuce in alignment with the fifth camera, the sensor sent a signal to the microcontroller that informed it that it can halt the imaging conveyor and initiate the image processing sequence.

##### **4.4.1.2.2 Arduino Mega 2560**

Arduino Mega 2560 was the primary microcontroller for the setup. Its main purpose was to provide processing, input, and output capabilities to control the components in the system. The microcontroller has 16 analog inputs, 54 I/O pins, and has high memory capacity compared to other Arduino versions. The microcontroller was utilized for receiving signals and processing input data from various sensors, including ultrasonic sensors and servo motors. This functionality facilitates the control of DC-powered components within the overall system.

##### **4.4.1.2.3 Circuit Breaker**

The setup includes multiple electronic components which utilize both AC and DC power. A circuit breaker is utilized to serve as a protective barrier that can automatically interrupt power when irregularities are detected. Issues such as overheating, electronic failure, and fire hazards are addressed by interrupting the circuit when excessive current is detected. This protective measure not only protects sensitive electronics but also



ensures that the operator of the mini-production line is safe. Including a circuit breaker on the system improves system reliability, extends the lifespan of components, and ensures safe operation under a variety of conditions.

#### **4.4.1.2.4 AC-to-DC Power Supply**

The mini-production line utilizes an AC-to-DC power supply as multiple electronic components run at DC and at different voltages. AC voltage powers the conveyor and LED bulbs. Subsequently, DC power is used for the precise operation of electronics, such as sensors, microcontrollers, and control modules. The power supply also guarantees that a steady voltage is supplied instead of the varying voltage from AC voltage which could damage certain components of the system. Furthermore, the power supply provides protection against voltage spikes, thus increasing reliability.

#### **4.4.1.2.5 Control Box**

A control box was utilized to secure the electronic components in an enclosure. The enclosure protects the electronic components against accidental contact and debris. Additionally, the enclosure allows the cables and electronics to be easily accessible and organized, making it easy to troubleshoot issues that arise.

#### **4.4.1.2.6 Cameras**

The C920 Logitech webcam was chosen as the camera for the mini-production line for its capability to record 1080p 30fps, capture images in 1080p, and autofocus feature. The 1080p sensor allowed the camera to capture images of the iceberg lettuce in detail, which include color variations, texture, and overall condition. The autofocus feature of the webcam ensured that the iceberg lettuce will be in focus, which reduced the likelihood of blurry images; further minimizing possible errors in image processing.

#### **4.4.2 MATLAB Controls Flowchart**

With the help of multiple toolboxes, MATLAB remained the main component of the whole study, having acted as the brain of the system. Its ability to install different toolboxes for different occasions allowed it to be used for Arduino applications without





needing C++ coding, which made it possible to have the whole control process done solely on MATLAB, as can be seen in the Figure below. The components were first initialized, which allowed the Arduino pins to be called as a variable, during which both conveyors were turned on. With lettuces placed on the conveyor, it kept continuously moving until the ultrasonic sensor detected an object, after which the relay controlling the conveyor turned off, which stopped its movements. Each placed camera is then turned on for a moment for image taking before it is turned off to preserve memory and run the process faster. The images taken went through image processing before color space conversion was done to extract the mean RGB and HSV values for each. These were used as inputs for the MSRGP model, which output the three parameters – chlorophyll-*a*, moisture content, and fresh head weight. Lastly, these parameters went through the fuzzy logic model which gave an output of either ‘L3’, ‘L2’, or ‘L1’, which each turned on a different servo that helped the staggered release of the lettuces into the lever arms. The whole process then proceeded to loop from the imaging chamber process onwards until it stopped, either by the code being ended or the system being turned off manually. The overall process of the system controls can be seen in Figure 4.29 below.

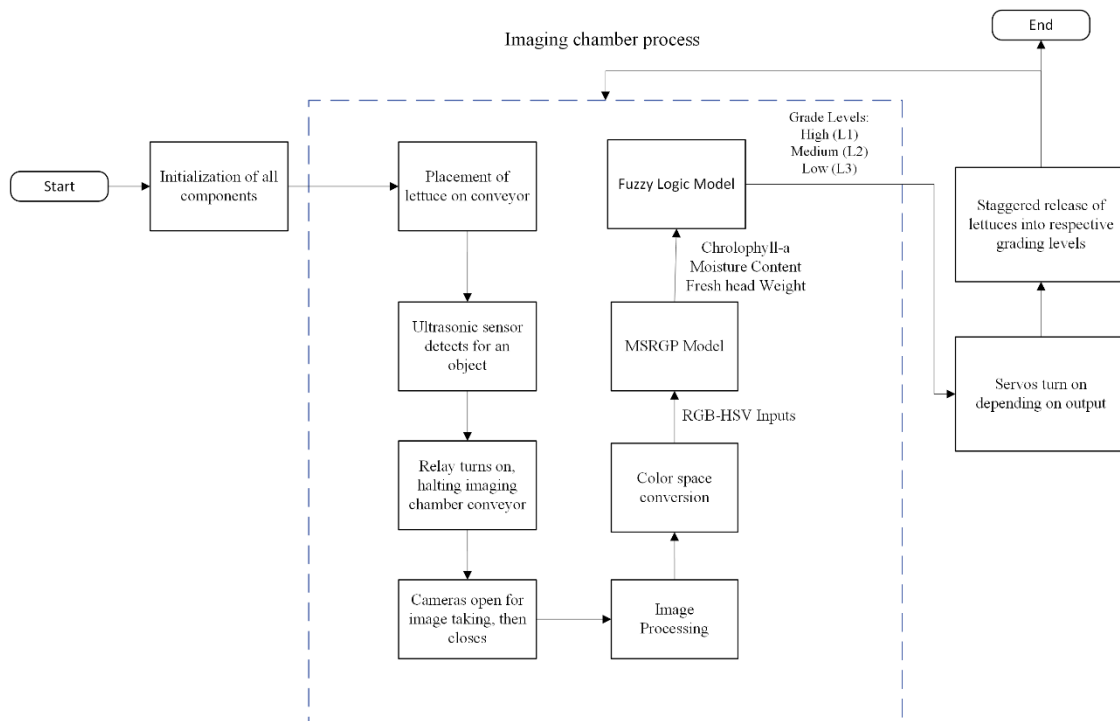


Figure 4.29. Overall system control logic flowchart

#### 4.4.3 Developed Prototype

An overview of the fully developed prototype can be seen in Figure 4.30 below. As mentioned, the system consisted of an imaging chamber with 5 cameras placed, a sorting conveyor wherein servo motors were responsible for sorting, and baskets that acted as boxes for the sorted lettuces. Located on the left side was the control box that was responsible for connecting every electronic and electrical parts together, with the Arduino, circuit breaker, power supply, and other electricals located inside the box.

A more detailed view is seen in Figure 4.31 that shows the driver and driven sides of the conveyor system. Shown in A.) is the starting position for the user to place the iceberg lettuce inside the imaging chamber. Shown in B.) is the driver side and end position of the iceberg lettuces where designated crates are placed to catch the iceberg lettuces upon sorting. All wirings are funneled into the given control box at A.) wherein a thorough continuity test check was performed to ensure that wires are functioning properly and voltage passed are adequate for the peripherals placed on other B.) side.



Figure 4.30. Developed prototype of the current study



Figure 4.31. Known driven and driver of the developed prototype. Shown is the a.) driven side of the prototype displaying the imaging chamber and start position for passing iceberg lettuce; b.) drive side detailing the placement of the crates that will catch the iceberg lettuce upon sorting.

The imaging conveyor was marked with tape that acted as guidelines for the manual input of the lettuces, which were placed such that each camera was able to see a lettuce without having other lettuces interfere with their field-of-view. The lights located at the top were able to output an average of 400 lux across each camera, which sufficiently provided the system an accurate reading of its RGB and HSV values. After these values were read by the system, they were inputted into the MSRGP models which provided



outputs of chlorophyll-*a*, moisture content, and fresh head weight, which acted as inputs for the fuzzy logic model. A more detailed view of the imaging chamber can be seen on the Figure 4.32 below.



Figure 4.32. Imaging chamber with lettuce sample

After the imaging chamber, the lettuces were sorted one-by-one in a staggered release format, onto the next conveyor that was applied with Teflon coating for smoother transition to the lever arms that were being controlled by the servo motors. The sorters were also labelled properly, with the first lever arm having led to the “L2” sorter, the second lever arm to the “L3” sorter, and the final lever arm led to the “L1” sorter. Should the system fail to detect properly, an error was shown, and the lever arms do not rotate, leading the lettuce to a fourth basket at the end for unidentified lettuce grades. Seen in the Figure 4.33 below is the sorting chamber, and conveyor for the system where the lefthand picture displays the handling of the iceberg lettuce from the imaging chamber into the sorting process of the system. In the righthand picture, it is displayed top view of the sorting conveyor with a sorting arm actuated to visualize how the sorting would look like for the current developed prototype.



A.)



B.)



Figure 4.33. Sorter system of the current study. Shown in a.) is the transition of the iceberg lettuce from going to the grading conveyor to sorting conveyor; b.) is the top view of sorting conveyor with sorting arm actuated into crate.



## Chapter 5

### RESULTS AND DISCUSSION

The results and discussion chapter presents an in-depth analysis of the dataset retrieved from the extraction processes and the result testing after implementation for the following: the MSRGF accuracy after modelling; the fuzzy logic accuracy after modelling and considerations; and the system's throughput time and repeatability trials.

#### 5.1 Resulting Datasets

##### 5.1.1 Training Dataset

The extracted training dataset is presented on Table 5.1. The study conducted the extraction processes mentioned in Section 4.2.1 to achieve these samples. The extraction procedure for this training dataset was done in two batches in the dates of November 14-15 and November 21-22, 2024. The iceberg lettuces originated in Atok, Benguet on standard coordinates of 16.58355, 120.69949 latitude and longitude respectively that was harvested on an elevation 2,294.5 meters above sea level. It was then outsourced from a vendor on the Benguet-Vizcaya FTI trading post, located in Western Bicutan Taguig. This training dataset consists of 56 samples with recorded values for chlorophyll-*a* content (mg/g), moisture content (%), and head weight (g). Additionally, it includes the corresponding linguistic levels for each feature and a final grade level of L1, L2, or L3 for fuzzy logic inferred from literature. It is important to base the grade levels from extraction process as the study does not consider visual cues for the creation of their ground truths but rather on laboratory results presented in current section. From the dataset, it is evident that most samples have a medium to low chlorophyll-*a* level, suggesting a generally consistent pigmentation among samples. The moisture content varies significantly, with some samples classified as low, average, or high. Most samples have a high moisture content, correlating with higher head weights, which is expected as moisture contributes to the total mass. Meanwhile, fresh head weight levels are predominantly high, indicating that most samples have well-developed structures.





From an analytical perspective, L1 samples (highest grade) correspond to higher chlorophyll-*a* levels (above 0.25 mg/g), high moisture content, and high head weights, indicating superior quality. On the other hand, L2 samples exhibit a mix of low to average chlorophyll-*a* levels but retain relatively good moisture and head weight properties. L3 samples, which have the lowest classification, tend to have low chlorophyll-*a* and moisture content, often associated with reduced head weight. These served as ground truths for the study for MSRGP and fuzzy logic modelling. Data augmentation increased this training dataset to 617 training samples along with RGB-HSV values for MSRGP modelling.

Table 5.1 Study's training dataset from extraction processes

| sample # | chlorophyll- <i>a</i> (mg/g) | moisture content (%) | head weight (g) | chlorophyll- <i>a</i> level | Moisture content level | head weight level | grade |
|----------|------------------------------|----------------------|-----------------|-----------------------------|------------------------|-------------------|-------|
| 1        | 0.146                        | 92.82                | 744.5           | Low                         | Average                | High              | L2    |
| 2        | 0.146                        | 64.41                | 901.5           | Low                         | Low                    | High              | L3    |
| 3        | 0.157                        | 68.45                | 752.5           | Low                         | Low                    | High              | L3    |
| 4        | 0.165                        | 85.31                | 833.5           | Low                         | Low                    | High              | L3    |
| 5        | 0.140                        | 94.23                | 512.0           | Low                         | High                   | High              | L2    |
| 6        | 0.145                        | 73.79                | 407.0           | Low                         | Low                    | Average           | L3    |
| 7        | 0.161                        | 91.95                | 680.5           | Low                         | Average                | High              | L2    |
| 8        | 0.139                        | 68.75                | 396.5           | Low                         | Low                    | Average           | L3    |
| 9        | 0.138                        | 89.15                | 755.0           | Low                         | Low                    | High              | L3    |
| 10       | 0.164                        | 75.09                | 507.0           | Low                         | Low                    | High              | L3    |
| 11       | 0.139                        | 85.5                 | 744.5           | Low                         | Low                    | High              | L3    |
| 12       | 0.153                        | 91.3                 | 773.9           | Low                         | Average                | High              | L2    |
| 13       | 0.164                        | 95.12                | 803.4           | Low                         | High                   | High              | L2    |
| 14       | 0.175                        | 93.99                | 832.8           | Low                         | Average                | High              | L2    |
| 15       | 0.181                        | 94.94                | 862.3           | Low                         | High                   | High              | L2    |
| 16       | 0.143                        | 95.21                | 891.7           | Low                         | High                   | High              | L2    |
| 17       | 0.147                        | 95.02                | 882.9           | Low                         | High                   | High              | L2    |
| 18       | 0.148                        | 95.04                | 854.9           | Low                         | High                   | High              | L2    |



|    |       |       |       |         |         |         |    |
|----|-------|-------|-------|---------|---------|---------|----|
| 19 | 0.147 | 94.79 | 827.0 | Low     | High    | High    | L2 |
| 20 | 0.152 | 90.55 | 799.1 | Low     | Average | High    | L2 |
| 21 | 0.139 | 95.76 | 771.1 | Low     | High    | High    | L2 |
| 22 | 0.161 | 93.79 | 757.6 | Low     | Average | High    | L2 |
| 23 | 0.149 | 95.72 | 772.8 | Low     | High    | High    | L2 |
| 24 | 0.150 | 85.61 | 787.9 | Low     | Low     | High    | L3 |
| 25 | 0.143 | 95.82 | 803.1 | Low     | High    | High    | L2 |
| 26 | 0.146 | 92.39 | 818.3 | Low     | Average | High    | L2 |
| 27 | 0.162 | 81    | 833.5 | Low     | Low     | High    | L3 |
| 28 | 0.158 | 94.47 | 773.2 | Low     | High    | High    | L2 |
| 29 | 0.161 | 95.09 | 712.9 | Low     | High    | High    | L2 |
| 30 | 0.152 | 96.09 | 652.7 | Low     | High    | High    | L2 |
| 31 | 0.196 | 95.49 | 592.4 | Low     | High    | High    | L2 |
| 32 | 0.238 | 96.67 | 532.1 | Low     | High    | High    | L2 |
| 33 | 0.256 | 95.84 | 498.9 | Average | High    | High    | L1 |
| 34 | 0.249 | 96.34 | 479.2 | Average | High    | High    | L1 |
| 35 | 0.275 | 84.68 | 459.5 | Average | Low     | High    | L2 |
| 36 | 0.175 | 84.21 | 439.8 | Low     | Low     | Average | L3 |
| 37 | 0.223 | 90.16 | 420.1 | Low     | Average | Average | L2 |
| 38 | 0.253 | 95.88 | 424.1 | Average | High    | Average | L1 |
| 39 | 0.271 | 95.4  | 475.4 | Average | High    | High    | L1 |
| 40 | 0.269 | 93.78 | 526.7 | Average | Average | High    | L1 |
| 41 | 0.229 | 79.28 | 577.9 | Low     | Low     | High    | L3 |
| 42 | 0.242 | 74.23 | 629.2 | Average | Low     | High    | L2 |
| 43 | 0.283 | 93.99 | 680.5 | Average | Average | High    | L1 |
| 44 | 0.217 | 96.45 | 627.3 | Low     | High    | High    | L2 |
| 45 | 0.222 | 95.14 | 574.0 | Low     | High    | High    | L2 |
| 46 | 0.208 | 87.96 | 520.8 | Low     | Low     | High    | L3 |
| 47 | 0.235 | 86.19 | 467.5 | Low     | Low     | High    | L3 |
| 48 | 0.230 | 90.95 | 414.3 | Low     | Average | Average | L2 |





|    |       |       |       |         |     |         |    |
|----|-------|-------|-------|---------|-----|---------|----|
| 49 | 0.237 | 86.05 | 441.3 | Low     | Low | Average | L3 |
| 50 | 0.218 | 85.14 | 508.5 | Low     | Low | High    | L3 |
| 51 | 0.190 | 79.78 | 575.8 | Low     | Low | High    | L3 |
| 52 | 0.157 | 86.07 | 643.0 | Low     | Low | High    | L3 |
| 53 | 0.233 | 67.21 | 710.2 | Low     | Low | High    | L3 |
| 54 | 0.252 | 64.37 | 739.5 | Average | Low | High    | L2 |
| 55 | 0.144 | 72.74 | 693.0 | Low     | Low | High    | L3 |
| 56 | 0.209 | 79.93 | 646.5 | Low     | Low | High    | L3 |

Presented in Figure 5.1 are the boxplots of the extracted training data, which depicts the trends observed in relation to the grading level and criteria range. For chlorophyll-*a*, ‘L1’ quality grade lettuces have a median of 0.2625 mg/g and a Q3 percentile of 0.271 mg/g. These values are within expected range for an ‘L1’ grade iceberg lettuce based on literature. ‘L2’ quality grade lettuces for chlorophyll-*a* concentrations have a median value of 0.161 mg/g but have a longer box whiskers covering up to 0.21825 mg/g in Q3 percentile and a max of 0.275 mg/g. ‘L3’ lettuce quality grade amounts to 0.164 mg/g which has a Q3 percentile of 0.21125 mg/g and maximum value of 0.237 mg/g. These results demonstrate how the chlorophyll-*a* levels influence lettuce grading, with higher chlorophyll-*a* concentrations associated with higher quality grades of ‘L1’. All values are within expected ranges for the grading criteria despite deviation highlighted with longer box whiskers for ‘L2’ and ‘L3’ quality grades.

For moisture content, it has shorter box whiskers corresponding to how compact and skewed the data is towards the higher values which is expected given the grading criteria of the current study expect for ‘L3’ which has a long box whisker indicating deviation. For ‘L1’ grade, it has a median of 95.62% and a Q3 percentile on 95.88% that is within expected grade level for high quality lettuce. For ‘L2’ grade, it has a median of 94.79% with a Q3 percentile of 95.28%. In this grade, three outliers are visible shown with a value of 84.68%, 74.23% and 64.37% respectively. These values should be categorized under ‘L3’ but due to association to other grading criteria it became an ‘L2’



correspondence. This may show how the grading criteria affects outliers turning into lettuce samples into different overall grade level than its intended range. Lastly, for 'L3' grade, it has a median of 81% with a Q3 percentile of 85.72% that is within expectations of low quality lettuce moisture content grading criteria. These reveal how the moisture content distribution for the test dataset supports the quality grading criteria but with significant outliers affecting the classification of certain grades, particularly for 'L2'.

Lastly for fresh head weight, all values are well above the expected grade level ranges and procured iceberg lettuces may not align properly with the PNS standard. 'L1' quality grade noticeably has a shorter box whiskers compared to other grades with a low median value of 489.05 g. It has a Q3 percentile of 526.7 g and outlier of 680.5 g. The weight of the lettuce samples for this grade is still consider high quality as it exceeds set grading criteria range. This is a common occurrence also for 'L2' where it has a median value of 727.6 g with a Q3 percentile of 807.125 g. A longer box whisker is shown where the min-max values are 414.3 g and 891.7 g respectively. Lastly, 'L3' grade has a median value of 643 g with a Q3 percentile of 752.125 g which is outside the grading criteria range for 'L3' in terms of fresh head weight. Based on these fresh head weight values, it can be stated that the procured iceberg lettuces for creation of training dataset are heavy and outside the standard grading mentioned in PNS:BAFS 19-2005 for weight which indicates a skewness to higher values within fresh head weight and how intended grading of the lettuces are outside of the grading criteria ranges which may impact the quality estimation using MSRGP and discrete grading levels outputted by fuzzy logic.

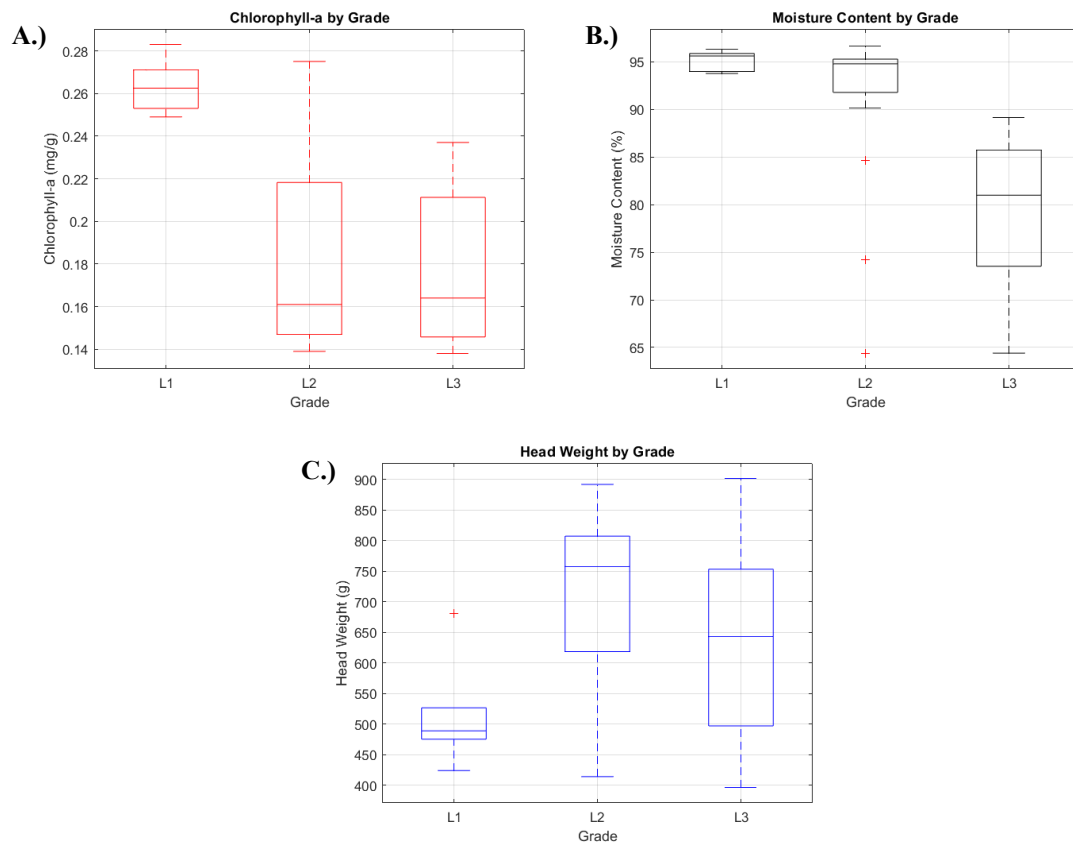


Figure 5.1 Boxplots of train dataset values from the current study based on quality grades for: a) chlorophyll-*a*, b) moisture content, and c) fresh head weight

## 5.1.2 Testing Dataset

The testing dataset from the extraction processes is presented in Table 5.2. The extraction process was conducted on March 11-12, 2025. The iceberg lettuces are outsourced from the same vendor on the Benguet-Vizcaya FTI trading post, located in Western Bicutan Taguig. Among the samples retrieved, a key observation is that samples classified as L1 (highest quality) generally have "Average" chlorophyll-*a* levels ( $\geq 0.244$  mg/g), high moisture content, and high head weights for the lettuce samples presented at that time. On the other hand, L2 samples exhibit low to average chlorophyll-*a* levels, yet they tend to have high moisture content and head weight, indicating that moisture content alone is not a sufficient factor for top grading based on criteria. Lastly L3 lettuce grade had at least “Low” ratings on at least two parameters in terms of chlorophyll-*a*, and moisture



content levels. Additionally, head weight levels for each lettuce were “Average” to “High”, without a single “Low” level, which can imply that head weight does not affect the grade of the lettuce as much. This dataset served as ground truth for testing of the made MSRGP models on unknown samples to check its performance outside training.

Table 5.2 Study’s testing dataset from extraction processes

| sample # | chlorophyll- <i>a</i> (mg/g) | moisture content (%) | head weight (g) | chlorophyll- <i>a</i> level | moisture content level | head weight level | grade |
|----------|------------------------------|----------------------|-----------------|-----------------------------|------------------------|-------------------|-------|
| 1        | 0.244                        | 95.33                | 801             | Average                     | High                   | High              | L1    |
| 2        | 0.213                        | 98.83                | 908.5           | Low                         | High                   | High              | L2    |
| 3        | 0.237                        | 85.09                | 510.5           | Low                         | Low                    | High              | L3    |
| 4        | 0.278                        | 90.06                | 589.5           | Average                     | Average                | High              | L1    |
| 5        | 0.210                        | 78.36                | 634             | Low                         | Low                    | High              | L3    |
| 6        | 0.197                        | 95.68                | 509             | Low                         | High                   | High              | L2    |
| 7        | 0.246                        | 93.99                | 595.5           | Average                     | Average                | High              | L1    |
| 8        | 0.252                        | 94.03                | 484             | Average                     | High                   | High              | L1    |
| 9        | 0.226                        | 96.02                | 537             | Low                         | High                   | High              | L2    |
| 10       | 0.204                        | 86.06                | 445             | Low                         | Low                    | Average           | L3    |
| 11       | 0.215                        | 97.09                | 493.5           | Low                         | High                   | High              | L2    |
| 12       | 0.241                        | 86.02                | 484.5           | Low                         | Low                    | High              | L3    |
| 13       | 0.250                        | 96.49                | 409             | Average                     | High                   | Average           | L1    |
| 14       | 0.259                        | 86.08                | 314             | Average                     | Low                    | Average           | L2    |
| 15       | 0.212                        | 94.67                | 513             | Low                         | High                   | High              | L2    |
| 16       | 0.215                        | 81.84                | 526.5           | Low                         | Low                    | High              | L3    |
| 17       | 0.214                        | 79.61                | 613.5           | Low                         | Low                    | High              | L3    |
| 18       | 0.236                        | 76.05                | 653             | Low                         | Low                    | High              | L3    |
| 19       | 0.277                        | 58.14                | 543             | Average                     | Low                    | High              | L2    |
| 20       | 0.214                        | 96.19                | 642             | Low                         | High                   | High              | L2    |

Seen in Figure 5.2 below are boxplots of the test data, which depicts the trends observed from the given test dataset was extracted for the study. For chlorophyll-*a*, ‘L1’



quality grade lettuces have a median of 0.25 mg/g, 'L2' quality grade lettuces have a median value of 0.2145 mg/g but have a longer box whisker covering up to 0.2425 in Q3 percentile and a max of 0.277 mg/g. 'L3' lettuce quality grade amounts to 0.215 mg/g which has a Q3 percentile of 0.23675 mg/g and maximum value of 0.241 mg/g. These results demonstrate how the chlorophyll-a levels influence lettuce grading, with higher chlorophyll-a concentrations associated with higher quality grades of 'L1'. The comparison between the grades shows the varying chlorophyll-a levels that correspond to the different quality categories. A similar trend is shown in the training dataset where chl-a concentrations had longer box whiskers for 'L2' and 'L3' grading level. This may indicate repetitiveness of quality brought by procuring the same iceberg lettuces from the same vendor.

For moisture content, it has shorter box whiskers corresponding to how compact and skewed the data is towards the higher values which is expected given the grading criteria of the current study. For 'L1' grade, its has a median of 94.03% and a Q3 percentile on 95.62% that is within expectations. For 'L2' grade, it has a median of 95.85% with a Q3 percentile of 96.64%. In this grade, it is spotted how an outlier is shown with a value of 58.14% indicating that this value influences the grading level to be associated under medium quality lettuce. Lastly, 'L3' grade has a median of 81.84% with a Q3 percentile of 85.7875% that is within expectations of low quality lettuce moisture content grading criteria. These reveal how the moisture content distribution for the test dataset supports the quality grading criteria but with significant outliers affecting the classification of certain grades, particularly for 'L2'. A similar trend is noticed for the train dataset where in values are similar with also a known outlier in 'L2' which suggests that values within 'L2' quality grade range are supposed to be in 'L3' given criteria but is not due to association with different grading levels suggesting that the values are heavily reliant to one another to define the overall grade of the lettuce samples.

Lastly for fresh head weight, it has a well-distributed box whiskers but outliers are seen in 'L2' quality grade. For 'L1', the median value is 589.5 g with a Q3 percentile



value of 646.875 g. It's maximum is 801 grams which may be considered too high of a value but is within the expected range of what an 'L1' grade iceberg lettuce. For 'L2', it has a median value of 525 g with a Q3 percentile of 592.5 g. It has two outliers of 314 g and 908.5 g which is outside the intended percentile ranges. This may indicate how these datapoints have attributed low quality lettuce on other parameters resulting to this deviation of being graded as 'L2' which is outside the expected grading criteria range. Lastly, 'L3' grade has a median value of 526.5 g with a Q3 percentile of 628.875 g is outside the expected grading range for 'L3' lettuce grade. Based on these fresh head weight values, it can be stated that the procured iceberg lettuces for creation of testing dataset are heavy and outside the standard grading mentioned in PNS:BAFS 19-2005 which indicates a skewness to higher values within fresh head weight.

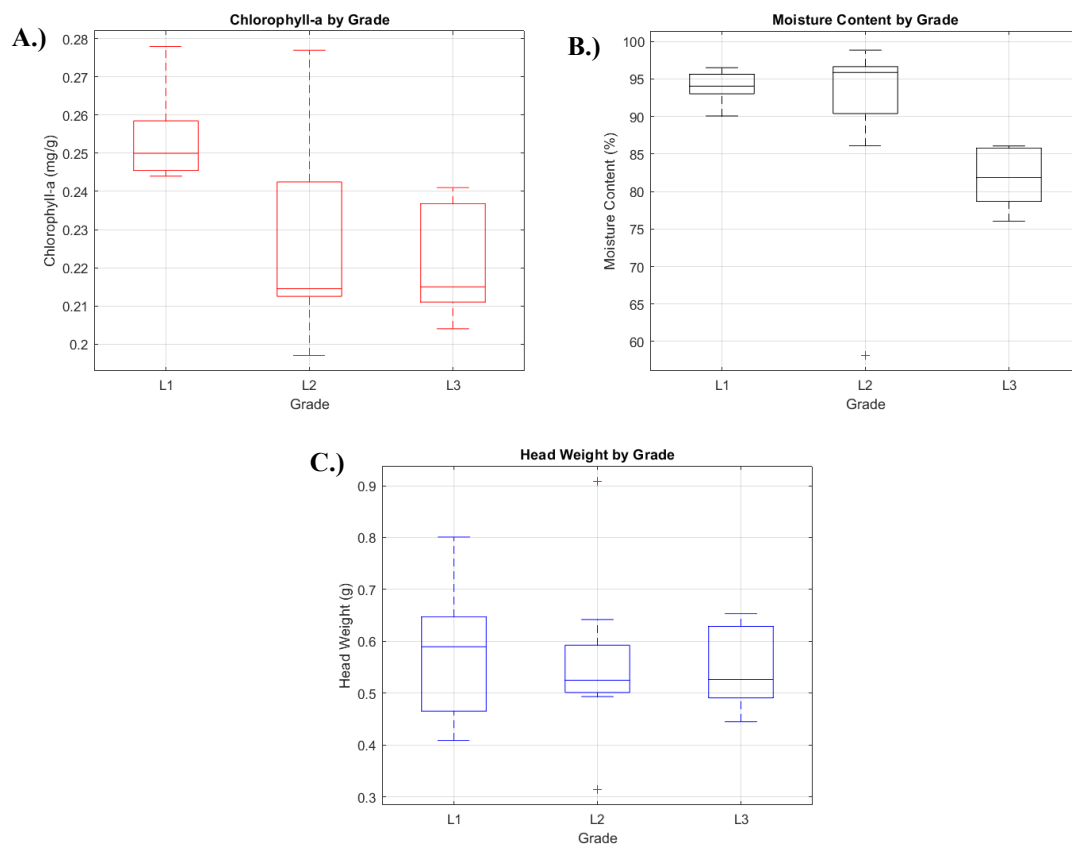


Figure 5.2 Boxplots of test dataset values from the current study based on quality grades for: a) chlorophyll-*a*, b) moisture content, and c) fresh head weight



## 5.2 MSRGP Output Models and Accuracy

Table 5.3 presents the model complexities and  $R^2$  values MSRGP models applied to predict chlorophyll-*a*, moisture content, and fresh head weight with mean RGB-HSV as input values. The  $R^2$  value indicates the model's goodness-of-fit, while model complexity represents the number of terms and operations in the symbolic regression equation considered in initial population until it is condensed into defined gene terms. These were considered among multiple attempts in finding the optimal model with the highest  $R^2$  with the least complexity and consideration on input highlighted in Section 4.2.4

Among the three models, the chlorophyll-*a* model exhibits the highest  $R^2$  value of 0.72, indicating a moderately strong correlation between the input parameters and chlorophyll-*a* levels. The model's complexity is relatively lower of 265 terms compared to the other models, suggesting that the relationship between the independent variables and chlorophyll-*a* levels has less spread and deviation.

The moisture content model has the lowest  $R^2$  value of 0.431, implying a weaker correlation and lower predictive accuracy. Based on the training dataset, the moisture content was concentrated at certain ranges that affected this value. Additionally, the high complexity of 496 suggests that the model has many nonlinear terms and transformations, needed to capture this variation in the data.

Lastly, the fresh head weight model has a moderate  $R^2$  value of 0.565, indicating a reasonable strong correlation. Similar to moisture content data, the range is concentrated at a certain range and thus very skewed to the right. The regression model aimed to fit as much as possible with a complexity of 495, making the model as highly intricate as the moisture content model. A summary of the chosen models for each parameter can be seen below in Table 5.3.



Table 5.3 Chosen MSRGP models of RGB-HSV R<sup>2</sup> Values and Complexities

| Input Parameters         | Model                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Model ID and Complexity     | R <sup>2</sup> Values |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-----------------------|
| Chlorophyll-<br><i>a</i> | 0.0144 x2 - 0.0274 x1 + 0.0209 x3 - 4.0 x4 - 0.0289 x6 - 3.13 cos(x21/2) + 2.7 sin(cos(x21/2)) + 1.4 sin(cos(x21/2))3 - 0.0274 log(x2)1/2 + 0.0458 sin(x2)3 - 0.0376 sin(x2)9 + 3.17 sin(x6)9 + 0.032 sin(x2)81 - 0.193 x2 x5 + 0.277 abs(x1) abs(x5)2 + 3.53 abs(x1)1/2 abs(x5) - 42.1 sin(x53)1/2 + 0.0133 sin(cos(x3))3 - 14.4 sin(cos(x4))3 - 7.51 x43 - 0.235 x53 - 1.54 x63 - 0.129 x6 sin(sin(x2)3)3 + 8.41                                                                                                                                                                                 | ID: 241<br>Complexity: 265  | 0.72                  |
| Moisture<br>Content      | 1270.0 x6 - 2.37 x2 - 1.17 x3 - 1.17 x1 + 3.3 log(abs(sin(x3 log(abs(x4)))))) - 2.85 log(abs(sin(x3 log(abs(x6)))))) - 2.37 sin(x1 + cos(abs(x3)) + sin(x2)) - 2.85 sin(x6 + cos(abs(x3)) + sin(x3)) + 3.66 cos(x3 log(abs(x6))) - 5.69 abs(sin(x3 abs(x6))) + 6.76 abs(sin(x3 cos(x5))) + 0.256 sin(x3 log(abs(x1))) - 1.17 sin(x3 log(abs(x6))) - 3.81 abs(x3 x5 log(x1)) + 736.0 abs(x5 x6 log(x3)) - 30.7 abs(x4) - 1.66 cos(x1) - 1.57 sin(x32 log(abs(x6))) + 3.87 abs(sin(x3 log(abs(x1)))) + 3.3 sin(x1 + x3 + sin(x3 log(x6))) + 14.5 cos(sin(x3 log(abs(x4)))) + 0.256 x3 abs(x6) + 55.0 | ID: 2248<br>Complexity: 496 | 0.432                 |
| Fresh Head<br>Weight     | 74.4 x2 - 69.3 x1 - 86.9 x3 + 2140.0 x4 + 1.61e+4 x5 - 2140.0 x6 + 174.0 cos(cos(log(x3)3)) - 454.0 cos(log(x1 - 1.0 x5)2) + 17.7 abs(sin(sin(abs(x1)))) - 37.7 cos(log(x51/2 (x1 - 2.0 x5))3) + 45.6 cos(log(x2 x5)3) + 34.1 sin(x21/2 x3 x6) + 79.6 cos(cos(x1)2) + 17.7 cos(x3) + 95.0 cos(log(x3 x52)3) + 251.0 cos(cos(log(x2 x5)3)) + 152.0 log(x3) + 79.6 abs(sin(abs(x53 - 1.0 x2 x5))) - 2410.0 log(x6 (2.0 x5 + cos(x52)) - 1.0 x53)2 - 0.369 x1 x2 + 153.0 x1 x6 + 556.0 log(cos(cos(log(x2 x5)3)))2 - 1.95e+4 x51/2 - 3380.0 x53 + 97.3 x63 + 1.11e+4                                  | ID: 967<br>Complexity: 495  | 0.565                 |

Shown in Figure 5.3 are the line graphs that provide the test data predictions from the actual data based on each data point from the train-test split to validate the output models. The trend shows that both chlorophyll-*a* and fresh head weight have decent estimation close to the actual datapoints given. On the other hand, due to the varied dataset from extracted moisture content, the model had a harder basis for estimation for the quality parameter with one point even going above the intended range of moisture content.



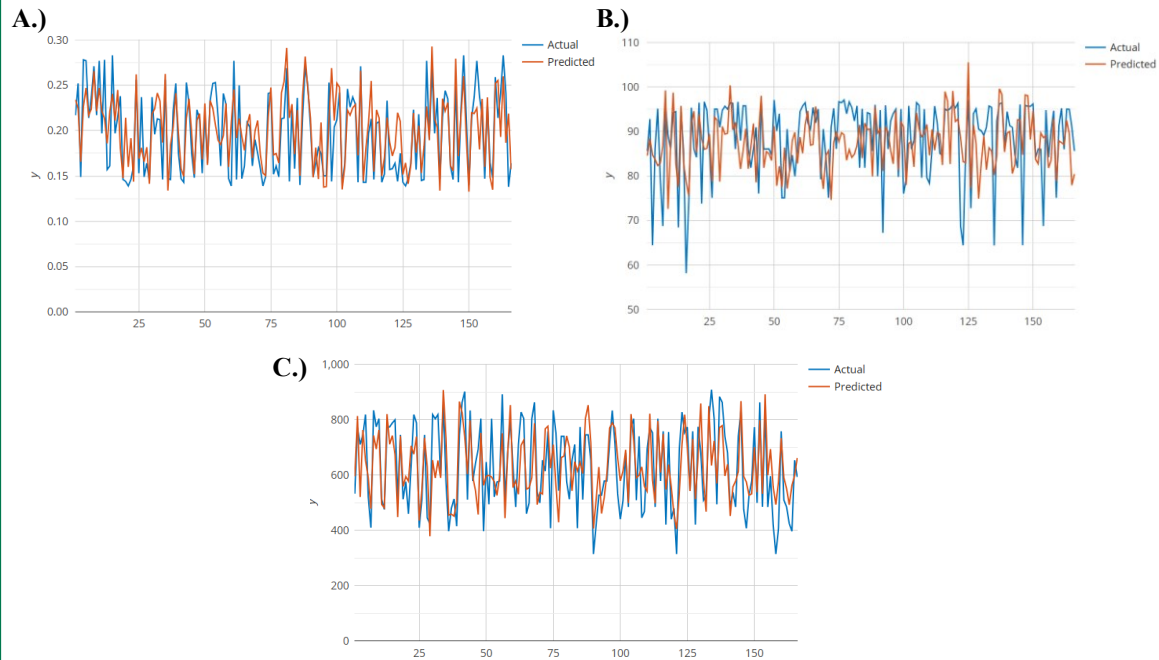


Figure 5.3 Test data predictions for MSRGP models for a.) chlorophyll-*a* b.) moisture content and c.) fresh head weight with RGB-HSV values as inputs.

Nevertheless, error metrics of RMSE and MAE given in these models indicate good coverage that are near to expected value ranges for study's quality parameter and thus values do not stray that far from study's grading criteria as shown in Table 5.4. This is known in Table 3.2 where chl-*a* ranges between 0.24 mg/g to 0.3691 mg/g thus error metrics indicate low error by 0.02 mg/g units. Moisture content ranges from 90%-96% thus error metrics indicate that predicted values are off to the actual values by 8.18% for RMSE and 6.57% for MAE. Lastly, fresh head weight ranges from 200-600 or higher grams thus error metrics suggest that predicted values may be off by 104.53 grams for RMSE and 87.64 grams for MAE from actual values. Given that all of these values are within the given ranges. These MSRGP models are thus acceptable for the given study to use as quality estimation for iceberg lettuce images.



Table 5.4 Known error metrics for MSRGP models of RGB-HSV input for the current study

| Quality Parameters and Model ID  | Error metric values            |
|----------------------------------|--------------------------------|
| Chlorophyll- <i>a</i><br>ID: 241 | RMSE: 0.0293<br>MAE: 0.0229    |
| Moisture content<br>ID: 2248     | RMSE: 8.1887<br>MAE: 6.5755    |
| Fresh head weight<br>ID: 967     | RMSE: 104.5386<br>MAE: 87.6449 |

In a similar fashion, the study also considered utilizing HSV values as the input only for their MSRGP modelling with known best models for each estimation provided in Table 5.5. Among the three models, the chlorophyll-*a* model still exhibits the highest  $R^2$  value of 0.573 for current iteration, indicating a moderately strong correlation between the input parameters and chlorophyll-*a* levels as indeed given for the study. The model's complexity is relatively higher of 441 than the 223 from the RGB-HSV input models.

The moisture content model still has the lowest  $R^2$  value of 0.241. A drastic change in the goodness-of-fit value implying a weaker correlation and lower predictive accuracy. Similar explanation that the moisture content was concentrated at certain ranges that affected this value for the chosen output model. Additionally, the high complexity of 501 suggests that the model has many nonlinear terms and transformations, needed to capture this variation in the data.

Lastly, the fresh head weight model has a moderate  $R^2$  value of 0.536, indicating a reasonable strong correlation but is still lower than previous RGB-HSV model made but has a lower complexity of 233.

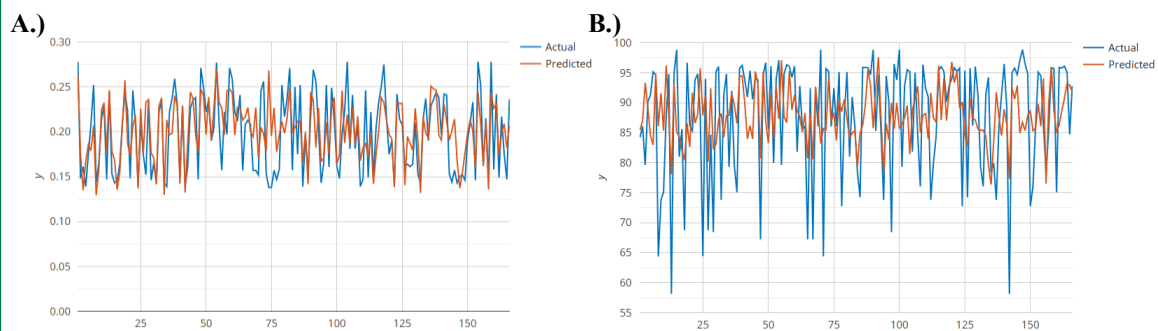
Table 5.5. Chosen MSRGP models of HSV inputs only  $R^2$  Values and Complexities

| Input Parameters      | Model                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Model ID and Complexity            | $R^2$ Values |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|--------------|
| Chlorophyll- <i>a</i> | $65.5 x_1 + 374.0 x_2 + 0.262 \sin(\sin(\text{abs}(x_3) (x_3 + 7.0) (x_1 + x_3 + 10.0))) - 8530.0 \cos(\sin(x_3)^2) + 382.0 \cos(\cos(\cos(x_2))^3 + x_1 x_3^2) - 0.247 \sin(x_3 (x_3 + 7.0) (x_1 + x_3 + 10.0)) - 179.0 \cos(x_1 x_3^2 + x_1^2/2) + 351.0 \sin(x_2^3) + 260.0 \sin(\sin(x_3)^2) - 0.951 \text{abs}(x_1 + x_3 + x_2^2 - 1.0) - 362.0 \text{abs}(x_2) + 190.0 \cos(x_1) + 204.0 \cos(x_2) - 74.7 \sin(x_3) + 0.0606 \cos(\sin(\text{abs}(x_3) (x_3 + 7.0) (x_1 + x_3 + 10.0))) + 1370.0 \sin(x_3)^6 + 7.19 x_1 x_2 + 3390.0 \cos(\sin(x_3)^2)^3 + 116.0 \cos(x_1 x_3^2 + x_1^2/2)^3 - 97.9 \cos(\sin(x_1$ | ID: 1167<br><br>Complexity:<br>441 | 0.573        |



|                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                             |       |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-------|
|                   | $x_3 (x_1 - 1.0 x_2)) - 1.0 x_1 x_3^2 - 374.0 x_3^3 + 105.0 x_1 x_3 \cos(x_1) + 4530.0$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                             |       |
| Moisture Content  | $7860.0 x_1 + 4.76e+5 x_2 - 150.0 x_3 - 2.45 \cos(\log(x_1^2)) - 13.7 \text{abs}(\cos(\log(x_2^3))) - 3160.0 \cos(\cos(x_1 x_3 \log(x_3))) + 155.0 \text{abs}(x_2 + x_1 x_3 \log(x_3)) + 4.04e+4 \cos(x_1) - 337.0 \log(x_2) - 4.65e+5 \sin(x_2) - 4.54 \text{abs}(x_2 + x_3 \cos(\log(x_1^2))) \log(x_2) + 2500.0 \cos(\sin(x_1 + x_2 + x_3^3)) - 5190.0 x_1^2 \log(x_3) - 89.9 \log(x_1^3) + 9.37 \text{abs}(\cos(\log(x_2^3)))^2 + 4.54 \text{abs}(\cos(\log(x_3^3)))^3 + 4.62e+4 x_1^2 x_3 - 5.58e+4 x_1^3 x_3 - 5.47e+5 \text{abs}(\sin(x_1) - 1.0 x_1 (\sin(x_1) + x_1 x_2))^4 + 7810.0 x_1^2 - 4.88e+4 x_2^2 + 6.0e+4 x_1 x_2^2 x_3 - 5.58e+4 x_1^2 x_2 x_3 + 3.0e+4 x_1^2 x_2^2 x_3^2 - 4.42e+4$ | ID: 2852<br>Complexity: 501 | 0.241 |
| Fresh Head Weight | $3130.0 x_1 + 13.5 x_2 - 506.0 x_3 + 6430.0 \cos(\cos(x_3) - 1.0 x_2^3) - 1570.0 \log(\cos(x_3)) - 3.95 \log(\text{abs}(x_2)^2 + 8.0 x_1^3) + 0.978 \cos(\log(x_3^3)) + 3810.0 \cos(x_1) + 5840.0 \cos(x_3) - 709.0 \log(x_1) + 63.4 \log(x_3) - 2230.0 x_1^2 x_2^3 - 115.0 \log(x_1)^2 - 202.0 x_1 x_3^2 + 202.0 x_1^2 x_3 + 2230.0 x_1 x_2^6 + 53.1 \cos(x_1 - 1.0 x_3^3) + 675.0 x_1^3 - 106.0 x_2^2 - 4170.0 x_2^3 + 67.3 x_3^3 - 742.0 x_2^9 + 1.56e+4 x_1^2 \text{abs}(x_2^3) \text{abs}(x_3^3) - 2530.0 x_1 \cos(x_3) \cos(\cos(\cos(\cos(x_3) - 1.0 x_2^3)))^3 - 1.43e+4$                                                                                                                        | ID: 3460<br>Complexity: 233 | 0.536 |

Shown in Figure 5.4 are the line graphs that provide the test data predictions from the actual data based on each data point from the train-test split to validate the output models. The trend shows that both chlorophyll-*a* and fresh head weight have decent estimation close to the actual datapoints given which is the same insight made from the previous MSRGP models that used RGB-HSV inputs. On the other hand, due to the varied dataset from extracted moisture content, the model had a harder basis for estimation for the quality parameter with points being varied where moisture content values 80% or below have drastic difference in predict to actual values.



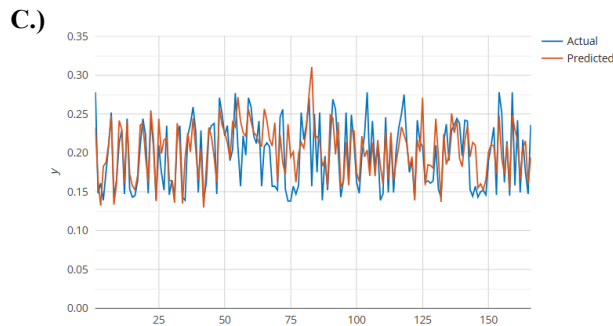


Figure 5.4 Test data predictions for MSRGP models for a.) chlorophyll-*a* b.) moisture content and c.) fresh head weight with HSV values as input only.

Nevertheless, error metrics of RMSE and MAE given in these models indicate good coverage that are near to expected value ranges for study's quality parameter and thus values do not stray that far from study's grading criteria as shown in Table 5.6. This is known in Table 3.2 where chl-*a* ranges between 0.24 mg/g to 0.3691 mg/g thus error metrics indicate low error by 0.03 mg/g units for RMSE and 0.02 for MAE. Moisture content ranges from 90%-96% thus error metrics indicate that predicted values are off to the actual values by 8.79% for RMSE and 7.06% for MAE. Lastly, fresh head weight ranges from 200-600 or higher grams thus error metrics suggest that predicted values may be off by 110.46 grams for RMSE and 90.20 grams for MAE from actual values. Given that all of these values are within the given ranges, these MSRGP models are thus acceptable for the given study to use as quality estimation for iceberg lettuce images in a similar way that the values from previous iteration with different input is acceptable as error metric is within close proximity from grading criteria range being considered by the current study.

Table 5.6 Known error metrics for MSRGP models of HSV input only for the current study

| Quality Parameters and Model ID   | Error metric values            |
|-----------------------------------|--------------------------------|
| Chlorophyll- <i>a</i><br>ID: 1167 | RMSE: 0.0313<br>MAE: 0.02436   |
| Moisture content<br>ID: 2852      | RMSE: 8.7959<br>MAE: 7.0661    |
| Fresh head weight<br>ID: 3460     | RMSE: 110.4681<br>MAE: 90.2091 |

Due to the similarities observed in the MSRGP output models generated from different input sets, this study conducted a comparative analysis to determine the most



appropriate model for quality estimation provided in Table 5.7. The different models shown that MSRGP models that used RGB-HSV values as inputs, denoted as MSRGP1, are better models for quality estimation than MSRGP models that used HSV value only denoted as MSRGP2, as these models have better metrics in terms of  $R^2$  value, RMSE and MAE.

Specifically, for chlorophyll-*a*, the MSRGP1 has a higher  $R^2$  value than MSRGP2 by 0.184, has a lower RMSE value by 0.002 and a lower MAE value by 0.00146. For moisture content, MSRGP1 has a better  $R^2$  value than MSRGP2 by 0.191, has a lower RMSE value by 0.6072 and a lower MAE value by 0.4906. Lastly, MSRGP1 for fresh head weight had a better  $R^2$  value than MSRGP2 by 0.008, a lower RMSE value by 5.9305 and a lower MAE by 2.5642. Thus, all metrics point out that MSRGP1 models are more suitable to be used for the current study as they receive better goodness-of-fit values and lower known errors of its predicted estimation from actual value.

Table 5.7 Summary comparison table of the output MSRGP models considered for the study

| Quality parameters    | MSRGP models using RGB-HSV values as inputs (MSRGP1)                                           | MSRGP models using HSV values as input only (MSRGP2)                                            |
|-----------------------|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Chlorophyll- <i>a</i> | Model ID: 241<br>$R^2$ value: 0.72<br>Model complexity: 496<br>RMSE: 0.0293<br>MAE: 0.0229     | Model ID: 1167<br>$R^2$ value: 0.536<br>Model complexity: 441<br>RMSE: 0.0313<br>MAE: 0.02436   |
| Moisture content      | Model ID: 2248<br>$R^2$ value: 0.432<br>Model complexity: 496<br>RMSE: 8.1887<br>MAE: 6.5755   | Model ID: 2852<br>$R^2$ value: 0.241<br>Model complexity: 501<br>RMSE: 8.7959<br>MAE: 7.0661    |
| Fresh head weight     | Model ID: 867<br>$R^2$ value: 0.565<br>Model complexity: 495<br>RMSE: 104.5386<br>MAE: 87.6449 | Model ID: 3460<br>$R^2$ value: 0.573<br>Model complexity: 233<br>RMSE: 110.4691<br>MAE: 90.2091 |

With the above stated, the current study will move forward and utilize their test dataset to check how well these values performed under unknown datapoints. Provided in Table 5.8 below shows the testing done for the chosen output MSRGP models made



to infer the parameters using RGB and HSV values as inputs to the equations. The model shows reasonably good predictive performance for chlorophyll-*a*, with the results of the prediction model show varying levels of accuracy across the three parameters based on the testing dataset with an average accuracy of 89.97% on chlorophyll-*a*, 92.8% on moisture content and 85.38% in fresh head weight. This can be due to the values from the testing data having similar range values on training data indicating that the lettuces procured for the current study can estimate accurately using the output MSRGP models.

Table 5.8 MSRGP Testing Accuracy

| Sample No. | Chlorophyll- <i>a</i> |        | Moisture Content |        | Fresh Weight |        |
|------------|-----------------------|--------|------------------|--------|--------------|--------|
|            | Predicted             | Actual | Predicted        | Actual | Predicted    | Actual |
| 1          | 0.184                 | 0.146  | 90.15            | 92.82  | 754.34       | 744.5  |
| 2          | 0.177                 | 0.157  | 78.70            | 64.41  | 747.43       | 901.5  |
| 3          | 0.153                 | 0.165  | 77.28            | 68.45  | 644.26       | 752.5  |
| 4          | 0.150                 | 0.14   | 98.24            | 85.31  | 736.16       | 833.5  |
| 5          | 0.156                 | 0.145  | 90.44            | 94.23  | 716.33       | 512    |
| 6          | 0.170                 | 0.161  | 87.60            | 73.79  | 756.64       | 407    |
| 7          | 0.164                 | 0.139  | 94.65            | 91.95  | 720.89       | 680.5  |
| 8          | 0.167                 | 0.138  | 79.10            | 68.75  | 633.09       | 396.5  |
| 9          | 0.162                 | 0.164  | 89.58            | 89.15  | 641.02       | 755    |
| 10         | 0.169                 | 0.153  | 77.50            | 75.09  | 663.40       | 507    |
| 11         | 0.180                 | 0.175  | 86.16            | 85.5   | 723.98       | 773.9  |
| 12         | 0.146                 | 0.181  | 87.86            | 91.3   | 659.16       | 803.4  |
| 13         | 0.150                 | 0.143  | 88.59            | 95.12  | 866.63       | 832.8  |
| 14         | 0.134                 | 0.147  | 81.03            | 93.99  | 834.98       | 862.3  |
| 15         | 0.137                 | 0.148  | 95.02            | 94.94  | 853.39       | 891.7  |
| 16         | 0.125                 | 0.152  | 87.52            | 95.21  | 837.10       | 882.9  |
| 17         | 0.166                 | 0.149  | 89.63            | 95.02  | 858.35       | 854.9  |
| 18         | 0.138                 | 0.15   | 92.64            | 95.04  | 757.64       | 827    |
| 19         | 0.183                 | 0.162  | 91.14            | 94.79  | 824.62       | 799.1  |
| 20         | 0.173                 | 0.158  | 77.08            | 90.55  | 852.61       | 771.1  |
| 21         | 0.214                 | 0.196  | 85.57            | 95.76  | 773.70       | 757.6  |
| 22         | 0.182                 | 0.238  | 89.46            | 93.79  | 722.71       | 772.8  |
| 23         | 0.261                 | 0.256  | 85.10            | 95.72  | 758.42       | 787.9  |
| 24         | 0.250                 | 0.249  | 82.39            | 85.61  | 751.46       | 803.1  |
| 25         | 0.233                 | 0.275  | 96.54            | 95.82  | 719.87       | 818.3  |
| 26         | 0.213                 | 0.223  | 85.68            | 92.39  | 614.70       | 773.2  |
| 27         | 0.216                 | 0.253  | 81.35            | 81     | 814.48       | 712.9  |



|                         |               |       |               |       |               |       |
|-------------------------|---------------|-------|---------------|-------|---------------|-------|
| 28                      | 0.266         | 0.271 | 82.74         | 94.47 | 650.51        | 652.7 |
| 29                      | 0.293         | 0.269 | 96.31         | 95.09 | 571.84        | 592.4 |
| 30                      | 0.230         | 0.229 | 101.71        | 96.09 | 658.63        | 532.1 |
| <b>Average Accuracy</b> | <b>89.97%</b> |       | <b>92.80%</b> |       | <b>85.38%</b> |       |

### 5.3 Fuzzy Logic Model and Accuracy

After choosing the appropriate MSRGP models, the current study then performed estimations from iceberg lettuce images that were then inputted into made fuzzy logic system using the test dataset and ten additional datapoints from the train dataset. The table below shows the data for testing the accuracy of the fuzzy logic model wherein ‘predicted’ column is the model’s value, while ‘actual’ is from extractions dataset, averaging to 90% accuracy or 27/30 correct. The table indicates that the model tended to be more skewed towards grading level ‘L2’ even if the actual level is ‘L3’, although not always, as seen in samples 5, 10, 16, 17, and 28, which were predicted correctly despite being ‘L3’. Furthermore, the system was able to correctly predict ‘L1’ indicating that the designed fuzzy logic made on Section 4.3 is working as intended where discrete grade levels values are being converted into intended linguistic values. These values are then put into the sorting system upon setting conditional statements that properly define distinction between grading levels.

Table 5.9 Fuzzy logic testing accuracy for the current study

| Sample No. | Predicted | Actual | Accuracy |
|------------|-----------|--------|----------|
| 1          | L1        | L1     | Correct  |
| 2          | L2        | L2     | Correct  |
| 3          | L2        | L3     | Wrong    |
| 4          | L1        | L1     | Correct  |
| 5          | L3        | L3     | Correct  |
| 6          | L2        | L2     | Correct  |
| 7          | L1        | L1     | Correct  |
| 8          | L1        | L1     | Correct  |
| 9          | L2        | L2     | Correct  |
| 10         | L3        | L3     | Correct  |
| 11         | L2        | L2     | Correct  |
| 12         | L2        | L3     | Wrong    |
| 13         | L1        | L1     | Correct  |
| 14         | L2        | L2     | Correct  |
| 15         | L2        | L2     | Correct  |



|          |    |    |         |
|----------|----|----|---------|
| 16       | L3 | L3 | Correct |
| 17       | L3 | L3 | Correct |
| 18       | L2 | L3 | Wrong   |
| 19       | L2 | L2 | Correct |
| 20       | L2 | L2 | Correct |
| 21       | L2 | L2 | Correct |
| 22       | L2 | L2 | Correct |
| 23       | L2 | L2 | Correct |
| 24       | L2 | L2 | Correct |
| 25       | L2 | L2 | Correct |
| 26       | L2 | L2 | Correct |
| 27       | L2 | L2 | Correct |
| 28       | L3 | L3 | Correct |
| 29       | L2 | L2 | Correct |
| 30       | L2 | L2 | Correct |
| Accuracy |    |    | 90%     |

## 5.4 Throughput Time Analysis

The findings show an inverse relationship between theoretical time and conveyor speed. The time needed to travel a set distance between Camera 5 and the different locations (Sorter 1, 2, 3) dramatically reduces as the conveyor speed rises. Across all tested speeds, from 5.33 m/min (10%) to 53.30 m/min (100%), this tendency is constant. For example, it takes about 16.21 seconds at 5.33 m/min to move from Camera 5 to Sorter 3 (1.44 m), whereas it only takes 2.95 seconds at 53.30 m/min. In a similar vein, throughout the same speed range, the time to reach Sorter 1 (0.825 m) drops from 9.29 to 1.69 seconds. This illustrates how shorter trip times and possibly higher throughput are facilitated by faster conveyor speeds.

Table 5.10. Overall throughput time per station for the current study

| Station               | Distance (cm) | Distance (m) | Speed % | Speed (m/min) | Theoretical Time (Sec) | 2nd Actuation Point (Sec) |
|-----------------------|---------------|--------------|---------|---------------|------------------------|---------------------------|
| Conveyor 1 (Baseline) | 40            | 0.4          | 10%     | 5.33          | 4.5                    | -                         |
| Camera 5 to Sorter 1  | 82.5          | 0.825        | 10%     | 5.33          | 9.29                   | -                         |
|                       |               |              | 20%     | 10.66         | 6.19                   | 6.62                      |
|                       |               |              | 30%     | 15.99         | 4.64                   | 4.97                      |
|                       |               |              | 40%     | 21.32         | 3.71                   | 3.97                      |
|                       |               |              | 50%     | 26.65         | 3.1                    | 3.31                      |
|                       |               |              | 100%    | 53.3          | 1.69                   | 1.81                      |
|                       | 115.5         | 1.155        | 10%     | 5.33          | 13                     | -                         |





|                         |     |      |      |       |       |       |
|-------------------------|-----|------|------|-------|-------|-------|
| Camera 5 to<br>Sorter 2 |     |      | 20%  | 10.66 | 8.67  | 9.27  |
|                         |     |      | 30%  | 15.99 | 6.5   | 6.96  |
|                         |     |      | 40%  | 21.32 | 5.2   | 5.56  |
|                         |     |      | 50%  | 26.65 | 4.33  | 4.64  |
|                         |     |      | 100% | 53.3  | 2.36  | 2.53  |
| Camera 5 to<br>Sorter 3 | 144 | 1.44 | 10%  | 5.33  | 16.21 | -     |
|                         |     |      | 20%  | 10.66 | 10.81 | 11.56 |
|                         |     |      | 30%  | 15.99 | 8.11  | 8.67  |
|                         |     |      | 40%  | 21.32 | 6.48  | 6.94  |
|                         |     |      | 50%  | 26.65 | 5.4   | 5.78  |
|                         |     |      | 100% | 53.3  | 2.95  | 3.15  |

To further add, adding a second actuation point shows that the speed increases even more, since greater speeds translate into faster actuation times. Although speed enhancement increases the efficient running of the system, it may also have an impact on safety, accuracy, and repeatability, all of which should be considered in ensuring optimization of the system. Another important part of examining conveyor efficiency is figuring out Average Sorts per Minute using the following formula is used to determine it:

Table 5.11. System repeatability trials for the current study

| Trial          | Pause to Grade<br>Camera (s) | Travel to<br>Camera<br>(s) | Sorting<br>Time (s) | Total<br>Cycle<br>Time (s) | Lettuces Per<br>Minute (Actual) | Lettuces Per<br>Minute<br>(Theoretical) |
|----------------|------------------------------|----------------------------|---------------------|----------------------------|---------------------------------|-----------------------------------------|
| 1              | 8                            | 6                          | 43                  | 57                         | 6                               | 8.3                                     |
| 2              | 8                            | 6                          | 44                  | 58                         | 8                               | 8.1                                     |
| 3              | 8                            | 6                          | 43                  | 57                         | 8                               | 7.7                                     |
| 4              | 8                            | 6                          | 44                  | 58                         | 9                               | 7.7                                     |
| 5              | 8                            | 6                          | 45                  | 59                         | 7                               | 8.1                                     |
| <b>Average</b> | <b>8</b>                     | <b>6</b>                   | <b>43.8</b>         | <b>57.8</b>                | <b>7.8</b>                      | <b>7.98</b>                             |

In an embedded system, there will be discrepancy between theoretical data and actual data due to various variables related to implementation, and hardware interaction. Theoretical data is rooted in systematic design specification and mathematical models, but this assumes that it will happen in an ideal environment, and hardware is working at ideal conditions and maximum capacity and taking note that the code will work with



errors. This estimation provides an upper limit of performance potential, such as processing 7.8 lettuces per minute in an embedded sorting system. However, actual performance is influenced by several factors, including hardware limitations and interruption handling. Inefficient coding practices or poorly optimized algorithms can significantly reduce performance, resulting in a lower throughput rate than anticipated. In this case, the actual performance measures around 5 lettuces per minute, considerably below the theoretical prediction.

The table presents the repeatability trial at 70% conveyor speed which yielded a 100% repeatability on all samples and trials. Each trial sorting results matches the assigned fuzzy grade accurately, showing a consistent performance in detection.

## 5.5 Repeatability Testing

The table below presents the data of the repeatability trial results for sorting conveyors operating at 100% speed. Five samples were tested across five runs. Each sample is assigned a fuzzy grade from the system (L1, L2, L3) to see if it was sorted correctly to its designated grade. Overall, repeatability has yielded 92%, calculated by dividing the number of correctly repeated sorting instances by several trials and multiplying by 100 to get the percent yield.

Table 5.12 Repeatability test at 70% conveyor speed

| Trial No.     | Trial #1    |           | Trial #2    |           | Trial #3    |           | Trial #4    |           | Trial #5    |           |
|---------------|-------------|-----------|-------------|-----------|-------------|-----------|-------------|-----------|-------------|-----------|
|               | Fuzzy Grade | Sorted in | Fuzzy Grade | Sorted in | Fuzzy Grade | Sorted in | Fuzzy Grade | Sorted in | Fuzzy Grade | Sorted in |
| 1             | L1          | L1        | L1          | L1        | L3          | L3        | L2          | L2        | L3          | L3        |
| 2             | L2          | L2        | L2          | L2        | L3          | L3        | L1          | L1        | L3          | L3        |
| 3             | L3          | L3        | L1          | L1        | L1          | L1        | L3          | L3        | L2          | L2        |
| 4             | L1          | L1        | L3          | L3        | L1          | L1        | L2          | L2        | L2          | L2        |
| 5             | L2          | L2        | L3          | L3        | L2          | L2        | L1          | L1        | L3          | L3        |
| Repeatability |             |           |             |           |             |           |             |           | 100%        |           |



The high repeatability rate shows that the system's sorting maintains a consistent performance at full speed. Although there are occasional discrepancies between the fuzzy and sorting results. (Sample 1 in Trial 3) However there are inconsistencies in detection or grading due to factors such as lighting, camera alignment, or speed related issues

The test shows the baseline for assessing sorting accuracy providing data on the reference before increasing the speed to 100%. The perfect performance at 70% speed shows that the system functions optimally when the system's conveyor operates at slower pace, which improved motion related errors.

Table 5.13 Repeatability test at 100% conveyor speed

| Sample No.    | Trial #1    |           | Trial #2    |           | Trial #3    |           | Trial #4    |           | Trial #5    |           |
|---------------|-------------|-----------|-------------|-----------|-------------|-----------|-------------|-----------|-------------|-----------|
|               | Fuzzy Grade | Sorted in | Fuzzy Grade | Sorted in | Fuzzy Grade | Sorted in | Fuzzy Grade | Sorted in | Fuzzy Grade | Sorted in |
| 1             | L2          | L2        | L1          | L1        | L2          | L2        | L3          | L3        | L1          | L1        |
| 2             | L3          | L2        | L2          | L1        | L3          | L3        | L1          | L1        | L2          | L2        |
| 3             | L1          | L1        | L1          | L1        | L2          | L2        | L2          | L2        | L3          | L3        |
| 4             | L2          | L2        | L2          | L2        | L3          | L3        | L1          | L1        | L3          | L3        |
| 5             | L1          | L1        | L3          | L3        | L2          | L2        | L1          | L1        | L2          | L2        |
| Repeatability |             |           |             |           |             |           |             |           | 92%         |           |

## 5.6 Results Synthesis

Overall, the study was able to meet their specific objectives through the results and deliverables that they performed and achieved summarized in Table 5.14. For SO1, the study was able to conduct laboratory extractions for their quality parameters of chlorophyll-*a*, moisture content and fresh head weight that was then act as input for MSRGP modelling. The study was then able to choose the appropriate of RGB-HSV values from their image dataset for the MSRGP models with an average testing accuracy of 89.97%, 92.8% and 85.38% chl-*a*, moisture content and weight respectively. For SO2, the study was able to make a Mamdani Type-1 fuzzy system that act as conversion of the predicted quality estimation values outputted by MSRGP models into discrete grading level values in line with the PNS/BAFS 19:2005 standard for iceberg lettuce that achieved a testing accuracy of 90% for correct grading into sorting bins. Lastly for SO3, the



conveyor belt system achieved 100% repeatability at 70% speed and 92% repeatability at 100% speed based on five trials.

Table 5.14 Summary table of results and deliverables for the current study

| Specific Objectives                                                                                                                                                                                                                                                                           | Results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SO1.) To develop and validate a non-invasive method for estimating chlorophyll- <i>a</i> concentrations, moisture content, and fresh head weight of freshly harvested iceberg lettuce samples that were harvested within two weeks, using multi-gene symbolic regression programming (MSRGP). | <p>After a comparative analysis where RGB-HSV input are deemed better inputs, the models developed for chlorophyll-<i>a</i>, moisture content, and fresh head weight estimation achieved <math>R^2</math> values of 0.72, 0.431, and 0.565, respectively.</p> <p>Also, RMSE values are 0.293, 8.1887 and 104.5386 respectively that are within acceptable error range for grading criteria. MAE values are 0.0229, 6.5755 and 87.6449 respectively also acceptable error range.</p> <p>The corresponding prediction accuracies were 89.97% for chlorophyll-<i>a</i>, 92.80% for Moisture Content, and 85.38% for fresh head weight.</p> |
| SO2.) To create an equivalent quality classification model that maps out three quality grading criteria (high, medium, and low) for iceberg lettuce.                                                                                                                                          | The fuzzy logic prediction model achieved an average accuracy of 90%, correctly classifying 27 out of 30 samples and misclassifying 3 samples.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| SO3.) To develop and evaluate a conveyor belt sorting system achieving five lettuces per batch sampling.                                                                                                                                                                                      | The sorting conveyor system achieved 100% repeatability at 70% speed and 92% repeatability at 100% speed, based on five trials for each speed setting. The decrease in repeatability at 100% speed indicates the presence of sorting errors in some trials.                                                                                                                                                                                                                                                                                                                                                                             |

In comparison to existing studies where overall accuracy, repeatability, and throughput time are known, provided in Table 5.15 are the metrics for it where the current study fall flat in terms of the throughput time it achieves per inference of quality but is substantial and on par in terms of overall accuracy and repeatability of the current system with other existing studies. The limitation of the study is in terms of grading, whereas the current study explored the use of interpretable models of MSRGP and fuzzy logic for their grading. Furthermore, the novelty of this study lies in its focus on the quality parameters of iceberg lettuce that are rarely utilized in conventional grading and sorting systems. While most existing studies primarily consider weight and morphological defects as key indicators of quality, this study expanded the evaluation to include: the biochemical property of moisture content, the physical property of fresh head weight, and



the phytochemical property of chlorophyll-a. Although these attributes are often discussed in the context of grading, they are rarely incorporated as active parameters within automated sorting systems. With this, the current study was able to provide a vision-based grading and sorting suited for iceberg lettuce with the only limitation in regard to the throughput time it can achieve to grade and sort iceberg lettuce per minute.

Table 5.15 Comparison table of existing studies to current study's metrics

| Known Metrics                           | Chakraborty et al., 2023 [42]                   | Wang et al., 2018 [99]           | Zhang et al., 2021 [100]                    | <b>Current Study's Mini-Production Line</b>                             |
|-----------------------------------------|-------------------------------------------------|----------------------------------|---------------------------------------------|-------------------------------------------------------------------------|
| Selected produce and quality parameters | Citrus Fruits; Weight and morphological defects | Mushrooms; morphological defects | Apples; weight and morphological defects    | Iceberg lettuce; chlorophyll-a, moisture content and fresh head weight. |
| Overall accuracy                        | 97.6%                                           | 97.42%                           | 99%                                         | 90% (fuzzy logic accuracy)                                              |
| Repeatability                           | 97.7%                                           | 99.9%                            | 81%                                         | 92% (100% speed)                                                        |
| Throughput Time                         | 102 citrus fruits per minute                    | 102.41 mushrooms per minute      | 450 apples per minute (from 7.5 per second) | 8 lettuce per minute                                                    |



## Chapter 6

### CONCLUSIONS AND RECOMMENDATIONS

#### 6.1 Conclusion

The main objective of this study was to develop a vision-based quality grading and sorting system for iceberg lettuce (*Lactuca sativa* var. *capitata*) within a mini-production line and to integrate computer vision and automation to enhance postharvest handling efficiency. Three quality parameters were analyzed with the use of multi-gene symbolic regression programming (MSRGP): chlorophyll-*a* content, moisture content, and weight estimation. Following this, an equivalent quality classification model for iceberg lettuce was implemented to map out three quality grading criteria: Grade 1 (high quality), Grade 2 (medium quality), and Grade 3 (low quality). Lastly, a conveyor belt sorting system that accommodates five iceberg lettuce heads per batch was developed and evaluated for repeatability.

The predictive models developed for chlorophyll-*a*, moisture content, and fresh head weight exhibited varying levels of accuracy, with  $R^2$  values of 0.72, 0.431, and 0.565, respectively. The prediction accuracies were determined to be 89.97% for chlorophyll-*a*, 85.38% for fresh head weight, and 92.80% for moisture content. Notably, the model exhibited the highest accuracy in predicting moisture content. Variation with other suppliers of iceberg lettuce or iceberg lettuce with changing quality parameters and increasing further the training dataset are required to further consider other possibilities where MSRGP model can further increase goodness-of-fit.

The three quality parameters were designated as the input variables for fuzzification in the Mamdani-Type 1 fuzzy logic system. The established grading criterion for iceberg lettuce based on the association matrix in Table 4.17 indicated that grade 1 includes lettuce with moisture content of at least 94%, chlorophyll-*a* concentration of at least 0.3691 mg/g, and a fresh head weight of at least 451 grams; Grade 2 consists of moisture content between 90%–94%, chlorophyll-*a* concentration



between 0.2884–0.3681 mg/g, and fresh head weight between 201–450 grams; and Grade 3 includes samples with moisture content of 90% or lower, chlorophyll-*a* concentration of 0.241 mg/g or lower, and fresh head weight of 200 grams or less. The fuzzy logic prediction model achieved an average classification accuracy of 90%, correctly identifying 27 out of 30 samples.

The sorting conveyor system demonstrated a repeatability rate of 100% at 70% operational speed and 92% at 100% speed based on five trials conducted for each speed setting. The reduction in repeatability at higher speed suggests increased operational variability, leading to errors in some trials.

## 6.2 Recommendations

Future research could consider other compatible model algorithms that can infer the iceberg grading levels extracted from quality parameters such as chlorophyll-*a*, moisture content, and fresh head weight. Additionally, the incorporation of other bio- and phytochemical properties, including vitamin content and morphological defects, should be considered, particularly those aligned with the PNS/BAFPS 19:2005 standard. Variation of iceberg lettuce should also be considered for the study.

Further investigation into different machine learning techniques and fuzzy-like methodologies is recommended to enhance the discrete grading process from quality estimation. For example, defuzzification methods could be examined to determine whether reliance on the highest membership function provides optimal grading accuracy.

The sole use of the HSV color space could be further investigated with the use of other color spaces as HSV is derived from RGB values. Additionally, HSV and other color spaces like CIELab is less sensitive to light intensity, color separation is clearer, and thresholding is easier which could imply that it is better for classification.

To improve the inference of quality grades, the use of multiple cameras, varied camera angles, and advanced image pre-processing techniques should be considered. The use of only one camera could be investigated, wherein, after a lettuce is graded by a



camera, it is moved to the sorting process; subsequently, while the lettuce is being sorted, another lettuce is being graded by the camera. Additionally, the integration of embedded systems for grade processing and system control should be considered to eliminate the reliance on an external computer for data processing.

The physical design of the system can be improved with the addition of a dedicated loading area at the start of the imaging conveyor, along with elevated spacers attached to the belt, could help ensure that only a single iceberg lettuce is positioned for grading, preventing obstructions within the camera's field of view. A loading area can make the system be more autonomous, as there will be no need for manual placement of lettuces onto the markings, which also ensures more reliable placement on the cameras for image taking.

Lastly, future research should consider the use of a slide, rollers, or an inclined plane instead of another conveyor for the sorter to minimize the costs.





## REFERENCES

- [1] M. Palumbo, M. Cefola, B. Pace, G. Attolico, and G. Colelli, "Computer vision system based on conventional imaging for non-destructively evaluating quality attributes in fresh and packaged fruit and vegetables," *Postharvest Biology and Technology*, vol. 200, p. 112332, Jun. 2023, doi: 10.1016/j.postharvbio.2023.112332.
- [2] Z. Zhou *et al.*, "Advancement in artificial intelligence for on-farm fruit sorting and transportation," *Front Plant Sci*, vol. 14, p. 1082860, Apr. 2023, doi: 10.3389/fpls.2023.1082860.
- [3] N. Ismail and O. A. Malik, "Real-time visual inspection system for grading fruits using computer vision and deep learning techniques," *Information Processing in Agriculture*, vol. 9, no. 1, pp. 24–37, Mar. 2022, doi: 10.1016/j.inpa.2021.01.005.
- [4] A. Bhargava and A. Bansal, "Fruits and vegetables quality evaluation using computer vision: A review," *Journal of King Saud University - Computer and Information Sciences*, vol. 33, no. 3, pp. 243–257, Mar. 2021, doi: 10.1016/j.jksuci.2018.06.002.
- [5] R. Mahendran, G. Jayashree, and K. Alagusundaram, "Application of Computer Vision Technique on Sorting and Grading of Fruits and Vegetables," *J Food Process Technol*, vol. s1, 2015, doi: 10.4172/2157-7110.S1-001.
- [6] B. Zhang *et al.*, "Principles, developments and applications of computer vision for external quality inspection of fruits and vegetables: A review," *Food Research International*, vol. 62, pp. 326–343, Aug. 2014, doi: 10.1016/j.foodres.2014.03.012.
- [7] A. Hayat, F. Morgado-Dias, T. Choudhury, T. P. Singh, and K. Kotecha, "FruitVision: A deep learning based automatic fruit grading system," *Open Agriculture*, vol. 9, no. 1, Jan. 2024, doi: 10.1515/opag-2022-0276.
- [8] K. K. Patel, A. Kar, S. N. Jha, and M. A. Khan, "Machine vision system: a tool for quality inspection of food and agricultural products," *J Food Sci Technol*, vol. 49, no. 2, pp. 123–141, Apr. 2012, doi: 10.1007/s13197-011-0321-4.
- [9] Bureau of Agriculture and Fisheries Standard, *PNS-BAFS19-2005 Head Lettuce*. Accessed: Mar. 05, 2025. [Online]. Available: <http://spsissuances.da.gov.ph/attachments/article/1084/PNS-BAFS19-2005HeadLettuce.pdf>
- [10] Food and Agriculture Organization, "Post-Harvest System and Food Losses." Accessed: Jul. 09, 2024. [Online]. Available: <https://www.fao.org/4/ac301e/AC301e03.htm>
- [11] F. F. Olayemi, J. A. Adegbola, E. I. Bamishaiye, and A. M. Daura, "Assessment of post-harvest challenges of small scale farm holders of tomatoes, bell and hot pepper in some local government areas of Kano State, Nigeria," *Bayero Journal of Pure and Applied Sciences*, vol. 3, no. 2, Art. no. 2, 2010, doi: 10.4314/bajopas.v3i2.63217.



- [12] S. Castro, “Post-harvest Technology in the Philippines,” Centre for Sustainable Agricultural Mechanization, 2004. Accessed: Mar. 10, 2025. [Online]. Available: <https://un-csam.org/node/1224>
- [13] E. Tisserant, L. Bessard, and M. de Sousa, “An Open Source IEC 61131-3 Integrated Development Environment,” in *2007 5th IEEE International Conference on Industrial Informatics*, Jun. 2007, pp. 183–187. doi: 10.1109/INDIN.2007.4384753.
- [14] American National Standards Institute, *ANSI-CEMA-402*. Accessed: Nov. 17, 2024. [Online]. Available: <https://cemanet.org/wp-content/uploads/2019/06/ANSI-CEMA-402-REVIEW-2019.pdf>
- [15] Agricultural Machinery Testing and Evaluation Center, *PHILIPPINE AGRICULTURAL ENGINEERING STANDARD PAES 302: 2000*. [Online]. Available: <https://amtec.uplb.edu.ph/wp-content/uploads/2020/06/PNS-PAES-302-2000-Engineering-Materials-Flat-Belts-and-Pulleys-Specifications-and-Applicati.pdf>
- [16] T. Dang and N. Sainato, “NEMA and IEC standards: A practical approach,” in *2013 IEEE-IAS/PCA Cement Industry Technical Conference*, Apr. 2013, pp. 1–12. doi: 10.1109/CITCON.2013.6525285.
- [17] F. Moerman, J. Kastelein, and T. Rugh, “Chapter 30 - Hygienic Design of Food Processing Equipment,” in *Food Safety Management (Second Edition)*, V. Andersen, H. Lelieveld, and Y. Motarjemi, Eds., San Diego: Academic Press, 2023, pp. 623–678. doi: 10.1016/B978-0-12-820013-1.00014-0.
- [18] Electrician Philippines, “Philippine Electrical Code Part 1 2017 Edition, New Provisions and Impacts | Electrician Philippines.” Accessed: Mar. 10, 2025. [Online]. Available: <https://electricianphilippines.com/philippine-electrical-code-part-1-2017-edition-new-provisions-and-impacts/>
- [19] T. U. Rehman, M. S. Mahmud, Y. K. Chang, J. Jin, and J. Shin, “Current and future applications of statistical machine learning algorithms for agricultural machine vision systems,” *Computers and Electronics in Agriculture*, vol. 156, pp. 585–605, 2019, doi: 10.1016/j.compag.2018.12.006.
- [20] H. A. M. Williams *et al.*, “Robotic kiwifruit harvesting using machine vision, convolutional neural networks, and robotic arms,” *Biosystems Engineering*, vol. 181, pp. 140–156, 2019, doi: 10.1016/j.biosystemseng.2019.03.007.
- [21] A. Signoroni, M. Savardi, A. Baronio, and S. Benini, “Deep learning meets hyperspectral image analysis: A multidisciplinary review,” *Journal of Imaging*, vol. 5, no. 5, 2019, doi: 10.3390/jimaging5050052.
- [22] Y. Lu, W. Saeys, M. Kim, Y. Peng, and R. Lu, “Hyperspectral imaging technology for quality and safety evaluation of horticultural products: A review and celebration of the past 20-year progress,” *Postharvest Biology and Technology*, vol. 170, 2020, doi: 10.1016/j.postharvbio.2020.111318.



- [23] A. Subeesh and C. R. Mehta, "Automation and digitization of agriculture using artificial intelligence and internet of things," *Artificial Intelligence in Agriculture*, vol. 5, pp. 278–291, 2021, doi: 10.1016/j.aiia.2021.11.004.
- [24] K. Hameed, D. Chai, and A. Rassau, "A comprehensive review of fruit and vegetable classification techniques," *Image and Vision Computing*, vol. 80, pp. 24–44, Dec. 2018, doi: 10.1016/j.imavis.2018.09.016.
- [25] Z. Wang, M. Hu, and G. Zhai, "Application of deep learning architectures for accurate and rapid detection of internal mechanical damage of blueberry using hyperspectral transmittance data," *Sensors (Switzerland)*, vol. 18, no. 4, 2018, doi: 10.3390/s18041126.
- [26] M. K. Tripathi and D. D. Maktedar, "A role of computer vision in fruits and vegetables among various horticulture products of agriculture fields: A survey," *Information Processing in Agriculture*, vol. 7, no. 2, pp. 183–203, Jun. 2020, doi: 10.1016/j.inpa.2019.07.003.
- [27] A. Bauer *et al.*, "Combining computer vision and deep learning to enable ultra-scale aerial phenotyping and precision agriculture: A case study of lettuce production," *Horticulture Research*, vol. 6, no. 1, 2019, doi: 10.1038/s41438-019-0151-5.
- [28] Y. Bai, B. Zhang, N. Xu, J. Zhou, J. Shi, and Z. Diao, "Vision-based navigation and guidance for agricultural autonomous vehicles and robots: A review," *Computers and Electronics in Agriculture*, vol. 205, p. 107584, Feb. 2023, doi: 10.1016/j.compag.2022.107584.
- [29] I. Nyalala *et al.*, "Tomato volume and mass estimation using computer vision and machine learning algorithms: Cherry tomato model," *Journal of Food Engineering*, vol. 263, pp. 288–298, Dec. 2019, doi: 10.1016/j.jfoodeng.2019.07.012.
- [30] J. Li, W. Luo, Z. Wang, and S. Fan, "Early detection of decay on apples using hyperspectral reflectance imaging combining both principal component analysis and improved watershed segmentation method," *Postharvest Biology and Technology*, vol. 149, pp. 235–246, Mar. 2019, doi: 10.1016/j.postharvbio.2018.12.007.
- [31] H. N. J. Schifferstein, T. Wehrle, and C.-C. Carbon, "Consumer expectations for vegetables with typical and atypical colors: The case of carrots," *Food Quality and Preference*, vol. 72, pp. 98–108, Mar. 2019, doi: 10.1016/j.foodqual.2018.10.002.
- [32] B. Zhang, B. Gu, G. Tian, J. Zhou, J. Huang, and Y. Xiong, "Challenges and solutions of optical-based nondestructive quality inspection for robotic fruit and vegetable grading systems: A technical review," *Trends in Food Science & Technology*, vol. 81, pp. 213–231, Nov. 2018, doi: 10.1016/j.tifs.2018.09.018.
- [33] P. Verboven, T. Defraeye, A. K. Datta, and B. Nicolai, "Digital twins of food process operations: the next step for food process models?," *Current Opinion in Food Science*, vol. 35, pp. 79–87, Oct. 2020, doi: 10.1016/j.cofs.2020.03.002.



- [34] J. F. I. Nturambirwe and U. L. Opara, "Machine learning applications to non-destructive defect detection in horticultural products," *Biosystems Engineering*, vol. 189, pp. 60–83, Jan. 2020, doi: 10.1016/j.biosystemseng.2019.11.011.
- [35] S. Abasi, S. Minaei, B. Jamshidi, and D. Fathi, "Dedicated non-destructive devices for food quality measurement: A review," *Trends in Food Science & Technology*, vol. 78, pp. 197–205, Aug. 2018, doi: 10.1016/j.tifs.2018.05.009.
- [36] D. I. Onwude, G. Chen, N. Eke-emezie, A. Kabutey, A. Y. Khaled, and B. Sturm, "Recent Advances in Reducing Food Losses in the Supply Chain of Fresh Agricultural Produce," *Processes*, vol. 8, no. 11, Art. no. 11, Nov. 2020, doi: 10.3390/pr8111431.
- [37] Z. Du, X. Zeng, X. Li, X. Ding, J. Cao, and W. Jiang, "Recent advances in imaging techniques for bruise detection in fruits and vegetables," *Trends in Food Science & Technology*, vol. 99, pp. 133–141, May 2020, doi: 10.1016/j.tifs.2020.02.024.
- [38] T. Gill, S. K. Gill, D. K. Saini, Y. Chopra, J. P. de Koff, and K. S. Sandhu, "A Comprehensive Review of High Throughput Phenotyping and Machine Learning for Plant Stress Phenotyping," *Phenomics*, vol. 2, no. 3, pp. 156–183, Jun. 2022, doi: 10.1007/s43657-022-00048-z.
- [39] M. Meenu, C. Kurade, B. C. Neelapu, S. Kalra, H. S. Ramaswamy, and Y. Yu, "A concise review on food quality assessment using digital image processing," *Trends in Food Science & Technology*, vol. 118, pp. 106–124, Dec. 2021, doi: 10.1016/j.tifs.2021.09.014.
- [40] A. Lazaro, M. Boada, R. Villarino, and D. Girbau, "Color Measurement and Analysis of Fruit with a Battery-Less NFC Sensor," *Sensors*, vol. 19, no. 7, Art. no. 7, Jan. 2019, doi: 10.3390/s19071741.
- [41] T. Y. Melesse, M. Bollo, V. D. Pasquale, F. Centro, and S. Riemma, "Machine Learning-Based Digital Twin for Monitoring Fruit Quality Evolution," *Procedia Computer Science*, vol. 200, pp. 13–20, Jan. 2022, doi: 10.1016/j.procs.2022.01.200.
- [42] S. K. Chakraborty *et al.*, "Development of an optimally designed real-time automatic citrus fruit grading–sorting machine leveraging computer vision-based adaptive deep learning model," *Engineering Applications of Artificial Intelligence*, vol. 120, p. 105826, Apr. 2023, doi: 10.1016/j.engappai.2023.105826.
- [43] K. Utai, M. Nagle, S. Hämmerle, W. Spreer, B. Mahayothee, and J. Müller, "Mass estimation of mango fruits (*Mangifera indica* L., cv. 'Nam Dokmai') by linking image processing and artificial neural network," *Engineering in Agriculture, Environment and Food*, vol. 12, no. 1, pp. 103–110, 2019, doi: 10.1016/j.eaef.2018.10.003.
- [44] S. Sabzi, H. Javadikia, and J. I. Arribas, "A three-variety automatic and non-intrusive computer vision system for the estimation of orange fruit pH value," *Measurement*, vol. 152, p. 107298, Feb. 2020, doi: 10.1016/j.measurement.2019.107298.



- [45] S. Haque *et al.*, “Computer vision approach to characterize size and shape phenotypes of horticultural crops using high-throughput imagery,” *Computers and Electronics in Agriculture*, vol. 182, p. 106011, Mar. 2021, doi: 10.1016/j.compag.2021.106011.
- [46] R. R. Sola-Guirado, S. Bayano-Tejero, F. Aragón-Rodríguez, B. Bernardi, S. Benalia, and S. Castro-García, “A smart system for the automatic evaluation of green olives visual quality in the field,” *Computers and Electronics in Agriculture*, vol. 179, p. 105858, Dec. 2020, doi: 10.1016/j.compag.2020.105858.
- [47] J. Xu and Y. Lu, “Prototyping and evaluation of a novel machine vision system for real-time, automated quality grading of sweetpotatoes,” *Computers and Electronics in Agriculture*, vol. 219, p. 108826, Apr. 2024, doi: 10.1016/j.compag.2024.108826.
- [48] R. Santos-Naressi, L. Santos-Pimentel-Oliveira, E. H. Santos, J. P. Francisco, and A. D. Lopes, “Iceberg lettuce cultivated in different systems of planting and sources of fertilizer,” *Braz. J. Biol.*, vol. 84, p. e255431, Mar. 2022, doi: <https://doi.org/10.1590/1519-6984.255431>.
- [49] E. Křístková, D. Ivana, A. Lebeda, V. Vinter, and A. Novotna, “Description of morphological characters of lettuce (*Lactuca sativa* L.) genetic resources,” *Horticultural Science*, vol. 35, pp. 113–129, Sep. 2008, doi: 10.17221/4/2008-HORTSCI.
- [50] S. P. Salgado, A. J. Pearlstein, Y. Luo, and H. Feng, “Quality of Iceberg (*Lactuca sativa* L.) and Romaine (*L. sativa* L. var. *longifolia*) lettuce treated by combinations of sanitizer, surfactant, and ultrasound,” *LWT - Food Science and Technology*, vol. 56, no. 2, pp. 261–268, May 2014, doi: 10.1016/j.lwt.2013.11.038.
- [51] I. Simko, R. J. Hayes, B. Mou, and J. D. McCreight, “Lettuce and spinach,” in *Yield Gains in Major U.S. Field Crops*, 2015, pp. 53–85. doi: 10.2135/cssaspecpub33.c4.
- [52] J. A. Tudela, N. Hernández, A. Pérez-Vicente, and M. I. Gil, “Growing season climates affect quality of fresh-cut lettuce,” *Postharvest Biology and Technology*, vol. 123, pp. 60–68, 2017, doi: 10.1016/j.postharvbio.2016.08.013.
- [53] M. Stojanović *et al.*, “Influence of the cultivar, fertiliser, and irrigation on iceberg lettuce morphology - single factor vs interaction effects,” *Book of proceedings XV International Scientific Agricultural Symposium “Agrosym 2024” Jahorina, October 10-13, 2024, Bosnia and Herzegovina*, pp. 339–345, 2024.
- [54] L. F. Johnson *et al.*, “Evapotranspiration-based Irrigation Scheduling of Head Lettuce and Broccoli,” *HortScience*, vol. 51, no. 7, pp. 935–940, Jul. 2016, doi: 10.21273/HORTSCI.51.7.935.
- [55] M. Aliste *et al.*, “Reclamation of agro-wastewater polluted with thirteen pesticides by solar photocatalysis to reuse in irrigation of greenhouse lettuce grown,” *Journal of Environmental Management*, vol. 266, 2020, doi: 10.1016/j.jenvman.2020.110565.





- [56] Y. Rouphael *et al.*, “GENOTYPIC VARIATION IN NUTRITIONAL AND ANTIOXIDANT PROFILE AMONG ICEBERG LETTUCE CULTIVARS,” *Acta Scientiarum Polonorum: Hortorum Cultus*, vol. 16, no. 3, Jun. 2017, Accessed: Mar. 01, 2025. [Online]. Available: <https://czasopisma.up.lublin.pl/index.php/asphc/article/view/2321>
- [57] T. T. Alemu and V. T. K. Oanh, “Effect of Maturity Stages on the Quality of Cold Storage Iceberg Lettuce (*Lactuca sativa* var. capitata) for Export,” *vjas*, vol. 7, no. 1, pp. 2052–2063, Mar. 2024, doi: 10.31817/vjas.2024.7.1.04.
- [58] S.-O. Tromp, R. Haijema, H. Rijgersberg, and J. G. A. J. van der Vorst, “A systematic approach to preventing chilled-food waste at the retail outlet,” *International Journal of Production Economics*, vol. 182, pp. 508–518, 2016, doi: 10.1016/j.ijpe.2016.10.003.
- [59] M. Cefola, C.-N. R. C. of I. Institute of Sciences of Food Production, B. Pace, A. Cardinali, I. D’Antuono, and F. Serio, “Relationship between Quality Parameters and the Overall Appearance in Lettuce during Storage,” *International Journal of Food Processing Technology*, vol. 1, no. 1, p. 18, 2014.
- [60] R. A. Schofield, J. R. DeEll, D. P. Murr, and S. Jenni, “Determining the storage potential of iceberg lettuce with chlorophyll fluorescence,” *Postharvest Biology and Technology*, vol. 38, no. 1, pp. 43–56, Oct. 2005, doi: 10.1016/j.postharvbio.2005.06.002.
- [61] E. J. Woltering and I. M. Witkowska, “Effects of pre-and postharvest lighting on quality and shelf life of fresh-cut lettuce,” *Acta Horticulturae*, vol. 1134, pp. 357–365, 2016, doi: 10.17660/ActaHortic.2016.1134.47.
- [62] R. Falcioni *et al.*, “Classification and Prediction by Pigment Content in Lettuce (*Lactuca sativa* L.) Varieties Using Machine Learning and ATR-FTIR Spectroscopy,” *Plants*, vol. 11, no. 24, 2022, doi: 10.3390/plants11243413.
- [63] R. S. Concepcion, E. P. Dadios, and J. Cuello, “Non-destructive in situ measurement of aquaponic lettuce leaf photosynthetic pigments and nutrient concentration using hybrid genetic programming,” *Agrivita*, 2021, doi: 10.17503/agrivita.v43i3.2961.
- [64] E. D. Cömert, B. A. Mogol, and V. Gökmen, “Relationship between color and antioxidant capacity of fruits and vegetables,” *Current Research in Food Science*, vol. 2, pp. 1–10, 2020, doi: 10.1016/j.crfs.2019.11.001.
- [65] J. M. Fonseca, “Postharvest Quality and Microbial Population of Head Lettuce as Affected by Moisture at Harvest,” *Journal of Food Science*, vol. 71, no. 2, pp. M45–M49, 2006, doi: 10.1111/j.1365-2621.2006.tb08906.x.
- [66] L. H. Vickers, I. G. Grove, and J. M. Monaghan, “IRRIGATION AFFECTS POSTHARVEST DISCOLOURATION AND YIELD IN ICEBERG LETTUCE,” *Acta Hortic.*, no. 1091, pp. 253–258, Aug. 2015, doi: 10.17660/ActaHortic.2015.1091.31.
- [67] J. M. Monaghan, L. H. Vickers, I. G. Grove, and A. M. Beacham, “Deficit irrigation reduces postharvest rib pinking in wholehead Iceberg lettuce, but at the



- expense of head fresh weight,” *Journal of the Science of Food and Agriculture*, vol. 97, no. 5, pp. 1524–1528, 2017, doi: 10.1002/jsfa.7895.
- [68] J. Poore and T. Nemecek, “Reducing food’s environmental impacts through producers and consumers,” *Science*, vol. 360, no. 6392, pp. 987–992, Jun. 2018, doi: 10.1126/science.aag0216.
- [69] Food and Agriculture Organization, “Seeking end to loss and waste of food along production chain,” Food and Agriculture Organization of the United Nations. Accessed: Jul. 09, 2024. [Online]. Available: <http://www.fao.org/in-action/seeking-end-to-loss-and-waste-of-food-along-production-chain/en/>
- [70] B. Pokhrel, “Review on post-harvest handling to reduce loss of fruits and vegetables,” vol. 2, pp. 48–52, May 2021, doi: 10.33545/26631067.2020.v2.i2a.52.
- [71] E. Esguerra, R. Rapusas, and R. S. Rolle, “Post-harvest management of lettuce to ensure quality and safety.” Accessed: Jul. 09, 2024. [Online]. Available: <https://openknowledge.fao.org/items/9a351185-d3a8-49c0-befc-01947e159b66>
- [72] L. Kitinoja, V. Y. Tokala, and A. Brondy, “Challenges and opportunities for improved postharvest loss measurements in plant-based food crops,” 2018.
- [73] S. G. Castro, “Post-Harvest Technology in the Philippines,” 2002.
- [74] D. Jose, “Lettuce Production Guide.” Accessed: Mar. 03, 2025. [Online]. Available: <https://library.buplant.da.gov.ph/images/1640931738Lettuce%20Production%20Guide.pdf>
- [75] M. L. D. Fang-asan, M. K. T. Dagupen, and L. N. Digal, Eds., “Lettuce Supply Chains and Marketing Margins in Benguet, Philippines,” *BANWA: A Multidisciplinary Journal*, 2009, doi: 10.22004/ag.econ.145552.
- [76] V. G. Monterde, J. Ekman, and E. R. V. Bayogan, “Food Handling Practices for Fresh-Cut Vegetables at Wet Markets and Supermarkets in Davao City, Philippines,” 2021.
- [77] Department of Agriculture, “Philippine Vegetable Industry Roadmap.” Accessed: Mar. 03, 2025. [Online]. Available: <https://www.da.gov.ph/wp-content/uploads/2023/05/Philippine-Vegetable-Industry-Roadmap.pdf>
- [78] Department of Science and Technology, “DOST’s Harmonized National Research and Development Agenda 2022-2028 - OVCRE.” Accessed: Jul. 09, 2024. [Online]. Available: <https://ovcre.uplb.edu.ph/press/memos/item/647-dost-s-harmonized-national-research-and-development-agenda-2022-2028>
- [79] Association of Southeast Asian Nations, “ASEAN Standard for Head Lettuce.” Accessed: Jul. 09, 2024. [Online]. Available: <https://asean.org/wp-content/uploads/2012/05/50-ASEAN-Standard-for-Head-lettuce.pdf>
- [80] EU Council Regulation, “marketing-standard-lettuces-curved-leaved-broad-leaved-endives.pdf.” Accessed: Jul. 09, 2024. [Online]. Available: <https://assets.publishing.service.gov.uk/media/5e57f543e90e07111640409/marketing-standard-lettuces-curved-leaved-broad-leaved-endives.pdf>



- [81] Association of Southeast Asian Nations, *Recommended-ASEAN-Standard-for-Head-lettuce.pdf*, 2016. Accessed: Mar. 06, 2025. [Online]. Available: <https://asean.org/wp-content/uploads/2016/10/Recommended-ASEAN-Standard-for-Head-lettuce.pdf>
- [82] United Nations Economic Commission for Europe, *UNECE STANDARD FFV-22 LETTUCES, CURLED-LEAVED*, 2017. Accessed: Mar. 06, 2025. [Online]. Available: [https://unece.org/fileadmin/DAM/trade/agr/standard/standard/fresh/FFV-Std/English/22\\_Lettuces.pdf](https://unece.org/fileadmin/DAM/trade/agr/standard/standard/fresh/FFV-Std/English/22_Lettuces.pdf)
- [83] J. Yang, J. Song, and J. Liu, "Prolonged Post-Harvest Preservation in Lettuce (*Lactuca sativa* L.) by Reducing Water Loss Rate and Chlorophyll Degradation Regulated through Lighting Direction-Induced Morphophysiological Improvements", Accessed: Mar. 07, 2025. [Online]. Available: <https://www.mdpi.com/2223-7747/13/18/2564>
- [84] M. V. Agüero, A. G. Ponce, M. R. Moreira, and S. I. Roura, "Lettuce quality loss under conditions that favor the wilting phenomenon," *Postharvest Biology and Technology*, vol. 59, no. 2, pp. 124–131, Feb. 2011, doi: 10.1016/j.postharvbio.2010.08.018.
- [85] P. B. Pathare, U. L. Opara, and F. A.-J. Al-Said, "Colour Measurement and Analysis in Fresh and Processed Foods: A Review," *Food Bioprocess Technol*, vol. 6, no. 1, pp. 36–60, Jan. 2013, doi: 10.1007/s11947-012-0867-9.
- [86] S. Sapkota, S. Sapkota, and Z. Liu, "Effects of Nutrient Composition and Lettuce Cultivar on Crop Production in Hydroponic Culture," *Horticulturae*, vol. 5, no. 4, Art. no. 4, Dec. 2019, doi: 10.3390/horticulturae5040072.
- [87] J.-S. Lee, D. Chandra, and J. Son, "Growth, Physicochemical, Nutritional, and Postharvest Qualities of Leaf Lettuce (*Lactuca sativa* L.) as Affected by Cultivar and Amount of Applied Nutrient Solution." Accessed: Mar. 06, 2025. [Online]. Available: <https://www.mdpi.com/2311-7524/8/5/436>
- [88] M. v. Agüero, M. v. Barg, A. Yommi, A. Camelo, and S. i. Roura, "Postharvest Changes in Water Status and Chlorophyll Content of Lettuce (*Lactuca Sativa* L.) and their Relationship with Overall Visual Quality," *Journal of Food Science*, vol. 73, no. 1, pp. S47–S55, 2008, doi: 10.1111/j.1750-3841.2007.00604.x.
- [89] J. Kold and C. Silverman, "Chapter 27 - Conveyors Used in the Food Industry," in *Handbook of Hygiene Control in the Food Industry (Second Edition)*, H. Lelieveld, J. Holah, and D. Gabrić, Eds., in Woodhead Publishing Series in Food Science, Technology and Nutrition. , San Diego: Woodhead Publishing, 2016, pp. 367–382. doi: 10.1016/B978-0-08-100155-4.00027-3.
- [90] A. C. Joaquin and R. P. Avila, "Automation of Mango Sorting using Computer Vision System," *Asian Journal of Postharvest and Mechanization*, vol. 6, no. 2, pp. 27–35.
- [91] J. M. Novelero, "Automated Pili Fruit Sorting Device using L-SVM Image Classification Model," *International Journal of Mechanical Engineering*, 2022.





- [92] L. Huimin, "Design and Implementation of the Automatic Sorting System Based on PLC," in *2023 IEEE 3rd International Conference on Power, Electronics and Computer Applications (ICPECA)*, Jan. 2023, pp. 467–471. doi: 10.1109/ICPECA56706.2023.10075963.
- [93] M. M. Nehete, A. P. Tadmalle, and D. S. Patil, "Design and Fabrication of PLC Based Conveyor System with Programmable Stations," in *International Journal of Analytical, Experimental and Finite Element Analysis (IJAEFEA)*, Oct. 2017. doi: 10.26706/IJAEFEA.3.4.20170906.
- [94] Y. Majeed and M. Waseem, "Postharvest Handling Systems," in *Encyclopedia of Digital Agricultural Technologies*, Q. Zhang, Ed., Cham: Springer International Publishing, 2023, pp. 993–1002. doi: 10.1007/978-3-031-24861-0\_125.
- [95] J. Miranda, P. Ponce, A. Molina, and P. Wright, "Sensing, smart and sustainable technologies for Agri-Food 4.0," *Computers in Industry*, vol. 108, pp. 21–36, Jun. 2019, doi: 10.1016/j.compind.2019.02.002.
- [96] C. S. Nandi, B. Tudu, and C. Koley, "A Machine Vision-Based Maturity Prediction System for Sorting of Harvested Mangoes," *IEEE Transactions on Instrumentation and Measurement*, vol. 63, no. 7, pp. 1722–1730, Jul. 2014, doi: 10.1109/TIM.2014.2299527.
- [97] X. Guo, L. Song, J. Huang, and F. Xiong, "Automatic Walnut Sorting System Based on Adaptive Fuzzy Control," in *2019 6th International Conference on Information Science and Control Engineering (ICISCE)*, Dec. 2019, pp. 685–689. doi: 10.1109/ICISCE48695.2019.00141.
- [98] Qin Zhang, "Encyclopedia of Digital Agricultural Technologies." Springer Nature, 2023. Accessed: Mar. 03, 2025. [Online]. Available: <https://link.springer.com/referencework/10.1007/978-3-031-24861-0>
- [99] F. Wang, J. Zheng, X. Tian, J. Wang, L. Niu, and W. Feng, "An automatic sorting system for fresh white button mushrooms based on image processing," *Computers and Electronics in Agriculture*, vol. 151, pp. 416–425, Aug. 2018, doi: 10.1016/j.compag.2018.06.022.
- [100] Z. Zhang, Y. Lu, and R. Lu, "Development and evaluation of an apple infield grading and sorting system," *Postharvest Biology and Technology*, vol. 180, p. 111588, Oct. 2021, doi: 10.1016/j.postharvbio.2021.111588.
- [101] N. Ismail and O. A. Malik, "Real-time visual inspection system for grading fruits using computer vision and deep learning techniques," *Information Processing in Agriculture*, vol. 9, no. 1, pp. 24–37, Mar. 2022, doi: 10.1016/j.inpa.2021.01.005.
- [102] H. Azarmdel, A. Jahanbakhshi, S. S. Mohtasebi, and A. R. Muñoz, "Evaluation of image processing technique as an expert system in mulberry fruit grading based on ripeness level using artificial neural networks (ANNs) and support vector machine (SVM)," *Postharvest Biology and Technology*, vol. 166, p. 111201, Aug. 2020, doi: 10.1016/j.postharvbio.2020.111201.



- [103] D. P. Cavallo, M. Cefola, B. Pace, A. F. Logrieco, and G. Attolico, “Non-destructive automatic quality evaluation of fresh-cut iceberg lettuce through packaging material,” *Journal of Food Engineering*, vol. 223, pp. 46–52, 2018, doi: 10.1016/j.jfoodeng.2017.11.042.
- [104] Y. Xu, Y. Zhai, Q. Chen, S. Kong, and Y. Zhou, “Improved Residual Network for Automatic Classification Grading of Lettuce Freshness,” *IEEE Access*, vol. 10, pp. 44315–44325, 2022, doi: 10.1109/ACCESS.2022.3169159.
- [105] R. Concepcion II *et al.*, “Adaptive Fertigation System Using Hybrid Vision-Based Lettuce Phenotyping and Fuzzy Logic Valve Controller Towards Sustainable Aquaponics,” *Journal of Advanced Computational Intelligence and Intelligent Informatics*, vol. 25, pp. 610–617, Sep. 2021, doi: 10.20965/jaciii.2021.p0610.
- [106] R. Concepcion II *et al.*, “Towards the Integration of Computer Vision and Applied Artificial Intelligence in Postharvest Storage Systems: Non-invasive Harvested Crop Monitoring,” in *2021 IEEE 13th International Conference on Humanoid, Nanotechnology, Information Technology, Communication and Control, Environment, and Management (HNICEM)*, Nov. 2021, pp. 1–6. doi: 10.1109/HNICEM54116.2021.9731973.
- [107] Z. Wang, T. Li, R. Du, N. Yang, and J. Ping, “A high-efficiency lettuce quality detection system based on FPGA,” *Computers and Electronics in Agriculture*, vol. 231, p. 109978, Apr. 2025, doi: 10.1016/j.compag.2025.109978.
- [108] D. Jayas, “Machine Vision System in postharvest technology,” *Stew. Posthar. Rev.*, vol. 1, no. 2, Aug. 2005, doi: 10.2212/spr.2005.2.2.
- [109] D. Ghadiri, “Food Imaging and Processing Systems and Methods,” US2024208098A1, Jun. 27, 2024
- [110] Y. Tao, D. Wang, R. Vinson, X. Cheng, M. Holmes, and G. E. Seibel, “Automatic vision guided intelligent fruits and vegetables processing system and method,” US11531317B2, Dec. 20, 2022
- [111] B. F. Taylor and R. S. Lapeyre, “Vision-based grading with automatic weight calibration,” 9886752, Feb. 06, 2018 Accessed: Mar. 07, 2025. [Online]. Available: <https://www.freepatentsonline.com/9886752.html>
- [112] D. He and T. Niu, “Small fruit and vegetable pneumatic automatic sorting machine,” CN110681603A, Jan. 14, 2020
- [113] “CN111069094 Multi-index comprehensive spherical fruit and vegetable quality grading-sorting device.” Accessed: Dec. 19, 2024. [Online]. Available: <https://patentscope.wipo.int/search/en/detail.jsf?docId=CN294744153>
- [114] U.S. Environmental Protection Agency, *Chlorophyll Determination*, Standard Operating Procedure 2030, Nov. 17, 1994.
- [115] “Methods for Determination of Moisture-Oven Drying | Analytical Chemistry.” Accessed: Mar. 14, 2025. [Online]. Available: <https://pubs.acs.org/doi/pdf/10.1021/ac60056a003>



- [116] F. Mendoza, P. Dejmek, and J. M. Aguilera, "Calibrated color measurements of agricultural foods using image analysis," *Postharvest Biology and Technology*, vol. 41, no. 3, pp. 285–295, Sep. 2006, doi: 10.1016/j.postharvbio.2006.04.004.
- [117] H. Tuoremaa, *A multi-gene symbolic regression approach for predicting LGD : A benchmark comparative study*. 2023. Accessed: Mar. 07, 2025. [Online]. Available: <https://urn.kb.se/resolve?urn=urn:nbn:se:umu:diva-210413>
- [118] I. Valenzuela, R. Baldovino, A. Bandala, and E. Dadios, "Pre-Harvest Factors Optimization Using Genetic Algorithm for Lettuce," *Journal of Telecommunication, Electronic and Computer Engineering*, Jan. 2018, Accessed: Jul. 25, 2024. [Online]. Available: <https://www.semanticscholar.org/paper/Pre-Harvest-Factors-Optimization-Using-Genetic-for-Valenzuela-Baldovino/96f809e7b5e0af33de109c578352615fc67c9b2f>
- [119] A. Sheta, A. Abdel-Raouf, and K. Fraihat, "Evolutionary Design of a PSO-Tuned Multigene Symbolic Regression Genetic Programming Model for River Flow Forecasting," *ResearchGate*, Oct. 2024, doi: 10.14569/IJACSA.2023.0140489.
- [120] A. L. J. T. Velasquez *et al.*, "Vision-Based Chlorophyll-a Measurement for Iceberg Lettuce Using Levenberg-Marquardt-Optimized Shallow Neural Network," in *TENCON 2023 - 2023 IEEE Region 10 Conference (TENCON)*, Chiang Mai, Thailand: IEEE, Oct. 2023, pp. 1181–1186. doi: 10.1109/TENCON58879.2023.10322332.
- [121] S. Gertphol, P. Chulaka, and T. Changmai, "Predictive models for Lettuce quality from Internet of Things-based hydroponic farm," in *2018 22nd International Computer Science and Engineering Conference (ICSEC)*, Nov. 2018, pp. 1–5. doi: 10.1109/ICSEC.2018.8712676.



## APPENDIX A. Bill of Materials

| Qty | Description                 | Cost (₱)    |             |
|-----|-----------------------------|-------------|-------------|
|     |                             | Amount      | Total       |
| 2   | 1.5 meter Conveyor          | ₱ 22,000.00 | ₱ 44,000.00 |
| 5   | 1080p Webcam                | ₱ 990.00    | ₱ 4950.00   |
| 3   | MG996R Servo                | ₱ 170.00    | ₱ 510.00    |
| 1   | PLA 3D Printing Filament    | ₱ 750.00    | ₱ 750.00    |
| 5   | E12 Refrigerator LED        | ₱ 54.00     | ₱ 270.00    |
| 1   | Arduino Mega 2560           | ₱ 3,249.00  | ₱ 3249.00   |
| 1   | 24V 14.6A 350W Power Supply | ₱ 1,860.00  | ₱ 1860.00   |
| 1   | 4 Port USB Hub 4            | ₱ 507.00    | ₱ 507.00    |
| 9   | THHN Wires 12/7             | ₱ 36.00     | ₱ 324.00    |
| 12  | THHN Wires 14/7             | ₱ 23.00     | ₱ 276.00    |
| 1   | Insulated Terminal Lugs     | ₱ 175.00    | ₱ 175.00    |
| 1   | PVC Control Switch Box      | ₱ 95.00     | ₱ 95.00     |
| 1   | Emergency Push Button       | ₱ 245.00    | ₱ 245.00    |
| 4   | 25A Terminal Block          | ₱ 34.00     | ₱ 136.00    |
| 5   | E12 Bulb Socket             | ₱ 13.72     | ₱ 68.60     |
| 1   | Iceberg Lettuce Samples     | ₱ 2977.54   | ₱ 2977.54   |
| 2   | Ethanol                     | ₱ 1,200.00  | ₱ 2,400.00  |
| 1   | Falcon Tubes                | ₱ 1,890.00  | ₱ 1,890.00  |
| 1   | Pipette Tips 100pcs         | ₱ 350.00    | ₱ 350.00    |
| 1   | Slotted Angle Bars          | ₱ 1,500.00  | ₱ 1,500.00  |



# De La Salle University

|       |                |          |             |
|-------|----------------|----------|-------------|
| 1     | Nuts and Bolts | ₱ 500.00 | ₱ 500.00    |
| Total |                |          | ₱ 67,033.14 |



## APPENDIX B. Raw Datasets

### Raw Chlorophyll-*a* UV-VIS Spectrophotometry Train

#### Dataset

| Sample No. | Wavelength |        |       |
|------------|------------|--------|-------|
|            | 645        | 663    | mg/g  |
| 1          | 0.0052     | 0.0126 | 0.146 |
| 2          | 0.0059     | 0.0127 | 0.146 |
| 3          | 0.0056     | 0.0135 | 0.157 |
| 4          | 0.0059     | 0.0142 | 0.165 |
| 5          | 0.0047     | 0.012  | 0.14  |
| 6          | 0.0052     | 0.0125 | 0.145 |
| 7          | 0.0058     | 0.0139 | 0.161 |
| 8          | 0.0028     | 0.0115 | 0.139 |
| 9          | 0.0014     | 0.0112 | 0.138 |
| 10         | 0.006      | 0.0141 | 0.164 |
| 11         | 0.0028     | 0.0115 | 0.139 |
| 12         | 0.0055     | 0.0132 | 0.153 |
| 13         | 0.0059     | 0.0141 | 0.164 |
| 14         | 0.0059     | 0.015  | 0.175 |
| 15         | 0.0062     | 0.0155 | 0.181 |
| 16         | 0.0044     | 0.0121 | 0.143 |
| 17         | 0.0045     | 0.0125 | 0.147 |
| 18         | 0.0052     | 0.0127 | 0.148 |
| 19         | 0.0047     | 0.0125 | 0.147 |
| 20         | 0.0051     | 0.013  | 0.152 |
| 21         | 0.0033     | 0.0116 | 0.139 |
| 22         | 0.0056     | 0.0138 | 0.161 |
| 23         | 0.0052     | 0.0128 | 0.149 |
| 24         | 0.0052     | 0.0129 | 0.15  |
| 25         | 0.0042     | 0.0121 | 0.143 |
| 26         | 0.0051     | 0.0125 | 0.146 |
| 27         | 0.0058     | 0.0139 | 0.162 |
| 28         | 0.0055     | 0.0136 | 0.158 |
| 29         | 0.0056     | 0.0138 | 0.161 |
| 30         | 0.0051     | 0.013  | 0.152 |
| 31         | 0.0072     | 0.0169 | 0.196 |



|    |        |        |       |
|----|--------|--------|-------|
| 32 | 0.0089 | 0.0206 | 0.238 |
| 33 | 0.01   | 0.0222 | 0.256 |
| 34 | 0.0096 | 0.0216 | 0.249 |
| 35 | 0.0107 | 0.0238 | 0.275 |
| 36 | 0.0068 | 0.0152 | 0.175 |
| 37 | 0.0085 | 0.0193 | 0.223 |
| 38 | 0.0097 | 0.0219 | 0.253 |
| 39 | 0.0101 | 0.0234 | 0.271 |
| 40 | 0.01   | 0.0232 | 0.269 |
| 41 | 0.0088 | 0.0198 | 0.229 |
| 42 | 0.0096 | 0.021  | 0.242 |
| 43 | 0.0109 | 0.0245 | 0.283 |
| 44 | 0.0082 | 0.0188 | 0.217 |
| 45 | 0.0085 | 0.0192 | 0.222 |
| 46 | 0.0077 | 0.0179 | 0.208 |
| 47 | 0.0088 | 0.0203 | 0.235 |
| 48 | 0.0085 | 0.0198 | 0.23  |
| 49 | 0.009  | 0.0205 | 0.237 |
| 50 | 0.0083 | 0.0189 | 0.218 |
| 51 | 0.007  | 0.0164 | 0.19  |
| 52 | 0.0059 | 0.0136 | 0.157 |
| 53 | 0.0089 | 0.0202 | 0.233 |
| 54 | 0.0093 | 0.0217 | 0.252 |
| 55 | 0.0054 | 0.0124 | 0.144 |
| 56 | 0.0077 | 0.018  | 0.209 |

## Raw Chlorophyll-*a* UV-VIS Spectrophotometry Test Dataset

| Sample Number | Wavelength |       |       |
|---------------|------------|-------|-------|
|               | 645        | 663   | mg/g  |
| 1             | 0.009      | 0.021 | 0.244 |
| 2             | 0.008      | 0.018 | 0.213 |
| 3             | 0.008      | 0.02  | 0.237 |
| 4             | 0.01       | 0.024 | 0.278 |
| 5             | 0.009      | 0.018 | 0.21  |





|    |       |       |       |
|----|-------|-------|-------|
| 6  | 0.007 | 0.017 | 0.197 |
| 7  | 0.01  | 0.021 | 0.246 |
| 8  | 0.009 | 0.022 | 0.252 |
| 9  | 0.011 | 0.02  | 0.226 |
| 10 | 0.011 | 0.018 | 0.204 |
| 11 | 0.008 | 0.019 | 0.215 |
| 12 | 0.009 | 0.021 | 0.241 |
| 13 | 0.004 | 0.009 | 0.107 |
| 14 | 0.009 | 0.022 | 0.259 |
| 15 | 0.007 | 0.018 | 0.212 |
| 16 | 0.008 | 0.019 | 0.215 |
| 17 | 0.008 | 0.018 | 0.214 |
| 18 | 0.01  | 0.021 | 0.236 |
| 19 | 0.01  | 0.024 | 0.277 |
| 20 | 0.008 | 0.018 | 0.214 |

## Raw Moisture Content Oven Train Dataset Batch 1

| Before Oven   |        |        |        | After Oven    |        |        |        |
|---------------|--------|--------|--------|---------------|--------|--------|--------|
| Sample Number | 1      | 2      | 3      | Sample Number | 1      | 2      | 3      |
| 1             | 1.4642 | 1.3671 | 1.3019 | 1             | 0.598  | 0.4142 | 0.3353 |
| 2             | 1.3373 | 1.3509 | 1.3221 | 2             | 0.7036 | 0.8387 | 0.5363 |
| 3             | 1.3709 | 1.4395 | 1.347  | 3             | 0.9293 | 0.5049 | 0.6669 |
| 4             | 1.3486 | 1.3464 | 1.3159 | 4             | 0.3917 | 0.3909 | 0.6698 |
| 5             | 1.3971 | 1.3017 | 1.4383 | 5             | 0.4216 | 0.3986 | 0.4896 |
| 6             | 1.3405 | 1.3073 | 1.3029 | 6             | 0.8003 | 0.3523 | 0.5861 |
| 7             | 1.3393 | 1.3293 | 1.348  | 7             | 0.4    | 0.4412 | 0.4143 |
| 8             | 1.3249 | 1.3369 | 1.3535 | 8             | 0.4019 | 0.6926 | 0.8585 |
| 9             | 1.3435 | 1.3369 | 1.3258 | 9             | 0.4022 | 0.3833 | 0.5403 |
| 10            | 1.3156 | 1.3376 | 1.3689 | 10            | 0.9126 | 0.4292 | 0.4245 |
| 11            | 1.3472 | 1.3514 | 1.3344 | 11            | 0.626  | 0.4046 | 0.4387 |
| 12            | 1.3476 | 1.4041 | 1.3791 | 12            | 0.4858 | 0.4589 | 0.4451 |
| 13            | 1.3874 | 1.4749 | 1.4273 | 13            | 0.4399 | 0.5279 | 0.4651 |
| 14            | 1.4108 | 1.4086 | 1.3927 | 14            | 0.4585 | 0.4798 | 0.4533 |



|    |        |        |        |    |        |        |        |
|----|--------|--------|--------|----|--------|--------|--------|
| 15 | 1.3848 | 1.3959 | 1.4127 | 15 | 0.4417 | 0.4436 | 0.4567 |
| 16 | 1.4332 | 1.3807 | 1.3826 | 16 | 0.4788 | 0.4343 | 0.4258 |
| 17 | 1.4016 | 1.3523 | 1.3106 | 17 | 0.4534 | 0.4001 | 0.3601 |
| 18 | 1.3677 | 1.3791 | 1.3544 | 18 | 0.4329 | 0.4233 | 0.3925 |
| 19 | 1.367  | 1.3813 | 1.371  | 19 | 0.4216 | 0.4268 | 0.4268 |
| 20 | 1.3881 | 1.3754 | 1.3563 | 20 | 0.5888 | 0.4223 | 0.3921 |
| 21 | 1.3605 | 1.3491 | 1.369  | 21 | 0.4045 | 0.3853 | 0.4121 |
| 22 | 1.315  | 1.3951 | 1.3961 | 22 | 0.4283 | 0.4299 | 0.4307 |
| 23 | 1.3894 | 1.3935 | 1.3962 | 23 | 0.4275 | 0.4414 | 0.4326 |
| 24 | 1.3821 | 1.3918 | 1.395  | 24 | 0.4331 | 0.7312 | 0.4324 |
| 25 | 1.3883 | 1.4166 | 1.4015 | 25 | 0.4314 | 0.4564 | 0.4381 |
| 26 | 1.3882 | 1.7113 | 1.7111 | 26 | 0.448  | 0.4186 | 1.1712 |
| 27 | 1.3742 | 1.392  | 1.4198 | 27 | 0.4236 | 0.8668 | 0.4637 |
| 28 | 1.3807 | 1.3957 | 1.4149 | 28 | 0.4202 | 0.4652 | 0.4654 |
| 29 | 1.3718 | 1.4085 | 1.3936 | 29 | 0.4201 | 0.4611 | 0.4395 |
| 30 | 1.409  | 1.4082 | 1.4053 | 30 | 0.4442 | 0.4472 | 0.4439 |

## Raw Moisture Content Oven Train Dataset Batch 2

| Before Oven   |        |        |        | After Oven    |        |        |        |
|---------------|--------|--------|--------|---------------|--------|--------|--------|
| Sample Number | 1      | 2      | 3      | Sample Number | 1      | 2      | 3      |
| 1             | 1.3557 | 1.3494 | 1.3435 | 1             | 0.3805 | 0.4092 | 0.3891 |
| 2             | 1.3393 | 1.3513 | 1.3505 | 2             | 0.374  | 0.388  | 0.3741 |
| 3             | 1.3766 | 1.3325 | 1.3568 | 3             | 0.4264 | 0.3684 | 0.3903 |
| 4             | 1.3443 | 1.3351 | 1.3259 | 4             | 0.3833 | 0.3583 | 0.3698 |
| 5             | 1.3455 | 1.3483 | 1.3491 | 5             | 0.3987 | 0.3896 | 0.7144 |
| 6             | 1.3561 | 1.3341 | 1.3478 | 6             | 0.7558 | 0.3704 | 0.3861 |
| 7             | 1.333  | 1.3613 | 1.347  | 7             | 0.3732 | 0.3888 | 0.567  |
| 8             | 1.3863 | 1.3831 | 1.3501 | 8             | 0.4307 | 0.4182 | 0.3862 |
| 9             | 1.3317 | 1.316  | 1.3251 | 9             | 0.3816 | 0.3603 | 0.3629 |
| 10            | 1.3138 | 1.3438 | 1.2986 | 10            | 0.3608 | 0.3712 | 0.4033 |
| 11            | 1.3282 | 1.3426 | 1.3418 | 11            | 0.7052 | 0.4132 | 0.5116 |
| 12            | 1.3683 | 1.4004 | 1.3928 | 12            | 0.4334 | 0.445  | 1.0547 |
| 13            | 1.3449 | 1.3405 | 1.3781 | 13            | 0.3815 | 0.4645 | 0.3963 |
| 14            | 1.3706 | 1.3729 | 1.3405 | 14            | 0.4115 | 0.4046 | 0.3717 |
| 15            | 1.3656 | 1.3693 | 1.3739 | 15            | 0.3983 | 0.3863 | 0.4645 |
| 16            | 1.3987 | 1.3819 | 1.4222 | 16            | 0.4867 | 0.414  | 0.659  |



|    |        |        |        |    |        |        |        |
|----|--------|--------|--------|----|--------|--------|--------|
| 17 | 1.3747 | 1.3819 | 1.3158 | 17 | 0.4065 | 0.7326 | 0.3435 |
| 18 | 1.3497 | 1.3289 | 1.3183 | 18 | 0.4588 | 0.3665 | 0.4351 |
| 19 | 1.3693 | 1.3148 | 1.325  | 19 | 0.4009 | 0.6593 | 0.3644 |
| 20 | 1.3317 | 1.3239 | 1.3708 | 20 | 0.375  | 0.4069 | 0.6886 |
| 21 | 1.3505 | 1.3786 | 1.3623 | 21 | 0.6594 | 0.4908 | 0.5453 |
| 22 | 1.3671 | 1.3628 | 1.3382 | 22 | 0.4677 | 0.5471 | 0.4679 |
| 23 | 1.3543 | 1.356  | 1.374  | 23 | 0.3942 | 1.102  | 0.5726 |
| 24 | 1.3517 | 1.3351 | 1.3606 | 24 | 0.9385 | 0.5159 | 0.6631 |
| 25 | 1.3896 | 1.3607 | 1.3693 | 25 | 0.4306 | 0.9939 | 0.5109 |
| 26 | 1.3478 | 1.3825 | 1.383  | 26 | 0.8627 | 0.4346 | 0.4128 |
| 27 | 1.3847 | 1.3783 | 1.3866 | 27 | 0.4349 | 0.4124 | 0.9042 |
| 28 | 1.374  | 1.3742 | 1.3864 | 28 | 0.387  | 0.6607 | 0.7029 |
| 29 | 1.3823 | 1.3871 | 1.3835 | 29 | 0.4312 | 0.4715 | 0.4719 |

## Raw Moisture Content Oven Test Dataset Batch 1

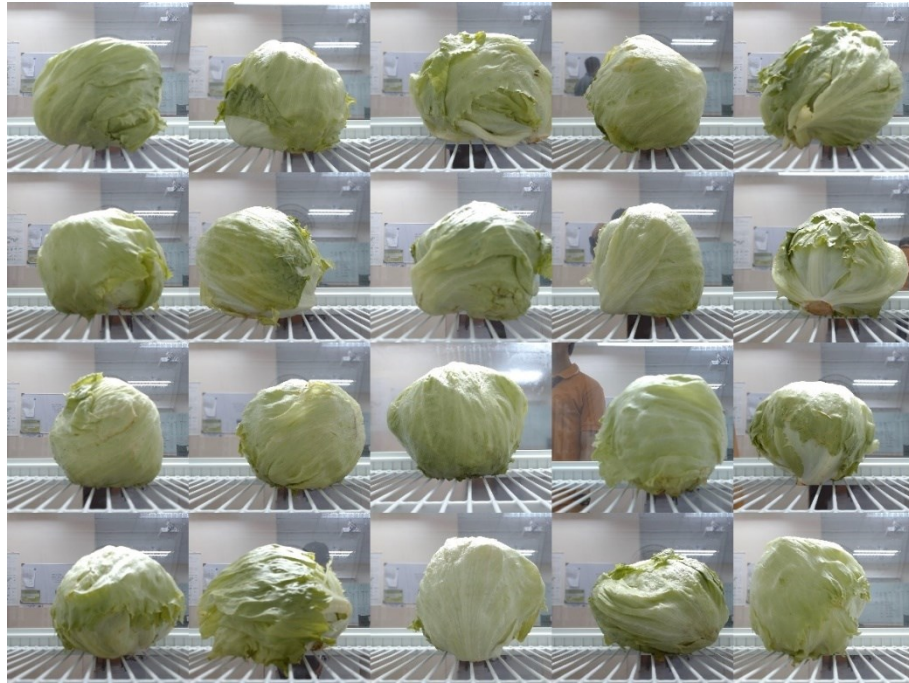
| Before Oven   |        |        |        | After Oven    |        |        |        |
|---------------|--------|--------|--------|---------------|--------|--------|--------|
| Sample Number | 1      | 2      | 3      | Sample Number | 1      | 2      | 3      |
| 1             | 1.2148 | 1.2331 | 1.2046 | 1             | 0.2626 | 0.2826 | 0.2417 |
| 2             | 1.2331 | 1.2388 | 1.2425 | 2             | 0.2373 | 0.2413 | 0.2721 |
| 3             | 1.2293 | 1.2448 | 1.2366 | 3             | 0.2602 | 0.2708 | 0.6239 |
| 4             | 1.2338 | 1.2107 | 1.2144 | 4             | 0.2727 | 0.4335 | 0.245  |
| 5             | 1.2348 | 1.2048 | 1.2185 | 5             | 0.2679 | 0.6498 | 0.3896 |
| 6             | 1.2249 | 1.2243 | 1.237  | 6             | 0.2637 | 0.2793 | 0.2731 |
| 7             | 1.2344 | 1.1978 | 1.228  | 7             | 0.2658 | 0.3127 | 0.2544 |
| 8             | 1.2074 | 1.1965 | 1.2033 | 8             | 0.267  | 0.2461 | 0.269  |
| 9             | 1.2338 | 1.2216 | 1.1971 | 9             | 0.2618 | 0.2767 | 0.227  |
| 10            | 1.172  | 1.1923 | 1.1931 | 10            | 0.2081 | 0.2289 | 0.536  |
| 11            | 1.2299 | 1.2434 | 1.2276 | 11            | 0.268  | 0.2731 | 0.2467 |
| 12            | 1.2359 | 1.2383 | 1.2188 | 12            | 0.2699 | 0.354  | 0.4855 |
| 13            | 1.2355 | 1.2188 | 1.2364 | 13            | 0.2675 | 0.2565 | 0.2718 |
| 14            | 1.2275 | 1.2188 | 1.1884 | 14            | 0.3852 | 0.2557 | 0.4128 |
| 15            | 1.2326 | 1.2069 | 1.202  | 15            | 0.2475 | 0.2361 | 0.3145 |
| 16            | 1.1729 | 1.2151 | 1.1972 | 16            | 0.2109 | 0.2563 | 0.6605 |
| 17            | 1.2167 | 1.2272 | 1.2086 | 17            | 0.2439 | 0.5955 | 0.4246 |
| 18            | 1.2316 | 1.2267 | 1.2312 | 18            | 0.6699 | 0.4648 | 0.2736 |
| 19            | 1.2383 | 1.2135 | 1.2309 | 19            | 0.6536 | 0.5627 | 0.7229 |
| 20            | 1.1989 | 1.2208 | 1.2166 | 20            | 0.2311 | 0.2496 | 0.261  |



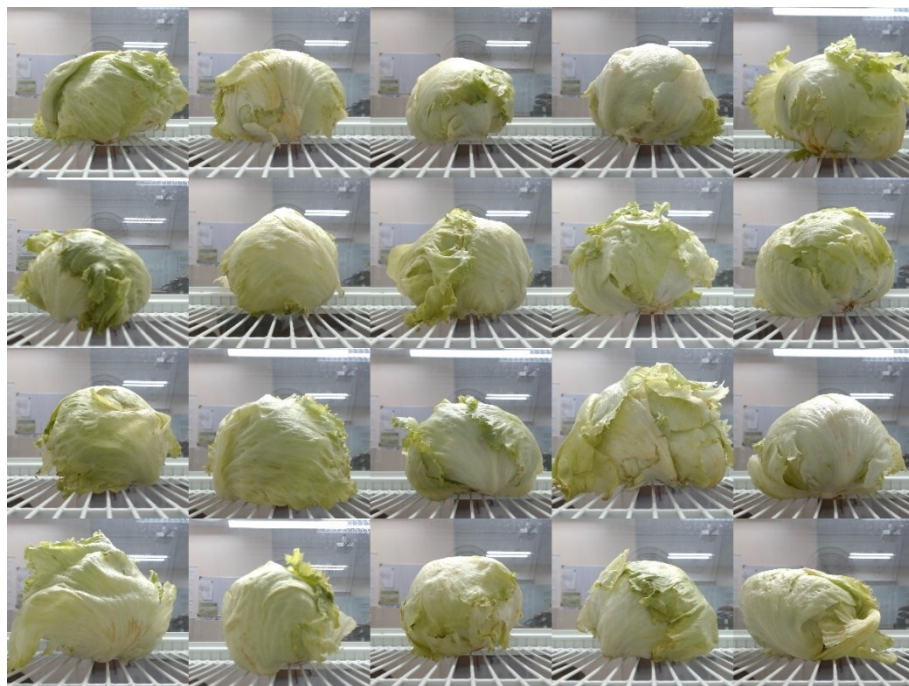
## Raw Image Training Dataset







**Raw Image Test Dataset**





## APPENDIX C. Source Code

### Systems Control MATLAB Code

```
%%  
function Comp_vis() % Opening and shutting camera  
% Selects the folder path  
folderPath = 'C:\Users\RAF  
Solutions\Documents\Romell\Thesis\MATLAB\Fuzzy\Lettuce_Image';  
  
% Creates if does not exist  
if ~isfolder(folderPath)  
    mkdir(folderPath);  
end  
  
% Initializes cameras  
cam1 = webcam(5);  
pause(1); % Gives time for the cameras to stabilize  
cam2 = webcam(1);  
pause(2.5);  
cam3 = webcam(3);  
cam4 = webcam(2);  
cam5 = webcam(4);  
  
% Takes image and closes  
img1 = snapshot(cam1);  
clear cam1;  
img2 = snapshot(cam2);  
clear cam2;  
img3 = snapshot(cam3);  
clear cam3;  
img4 = snapshot(cam4);  
clear cam4;  
img5 = snapshot(cam5);  
%psf= fspecial("gaussian",5,5);  
%img5 = deconvlucy(img5,psf);  
clear cam5;  
  
% Define file names for each image  
fileName1 = fullfile(folderPath, '1.jpg');  
fileName2 = fullfile(folderPath, '2.jpg');  
fileName3 = fullfile(folderPath, '3.jpg');
```



```
fileName4 = fullfile(folderPath, '4.jpg');
fileName5 = fullfile(folderPath, '5.jpg');

% Save images to the created folder
imwrite(img1, fileName1);
imwrite(img2, fileName2);
imwrite(img3, fileName3);
imwrite(img4, fileName4);
imwrite(img5, fileName5);

end

%%
function RGBHSV_extract() % Extraction of RGB and HSV values
%%
% Folder for screenshots
folder = 'C:\Users\RAF
Solutions\Documents\Romell\Thesis\MATLAB\Fuzzy\Lettuce_Image';

% Checks if the folder exists
if ~isfolder(folder)
    error('The specified folder does not exist: %s', folder);
end

%%
% Takes the file types from folder
jpgFiles = dir(fullfile(folder, '*.jpg'));
pngFiles = dir(fullfile(folder, '*.png'));
allFiles = [jpgFiles; pngFiles]; % Combine both lists

% Checks if images exist
if isempty(allFiles)
    error('No JPG or PNG files found in the specified folder.');
```

```
end

%%
% Arrays for RGB values and image names
numImages = numel(allFiles);
R = zeros(numImages, 1);
G = zeros(numImages, 1);
B = zeros(numImages, 1);
```



```
H = zeros(numImages, 1);
S = zeros(numImages, 1);
V = zeros(numImages, 1);

imageNames = cell(numImages, 1);

%% Extracting RGB HSV
% Loop through each image file and extract RGB values
for i = 1:numImages
    % Load the image
    filename = fullfile(folder, allFiles(i).name);
    image = imread(filename);

    % Converts to double
    dImage = double(image);

    % Extract the average RGB values
    R(i) = mean(dImage(:, :, 1), 'all'); % red channel
    G(i) = mean(dImage(:, :, 2), 'all'); % green channel
    B(i) = mean(dImage(:, :, 3), 'all'); % blue channel

    % Convert the image to HSV
    hsvImage = rgb2hsv(image);

    % Extract the HSV values
    H(i) = mean2(hsvImage(:, :, 1)); % hue channel
    S(i) = mean2(hsvImage(:, :, 2)); % saturation channel
    V(i) = mean2(hsvImage(:, :, 3)); % value channel

    % Save the image name
    imageNames{i} = allFiles(i).name;
end

%%
% Create a bar graph of the RGB values
%rgbData = [R, G, B];
%h = bar(rgbData);
%xlabel('Image Number')
%ylabel('RGB Value (0-255)')

% Set the colors of the bars and add a legend
%h(1).FaceColor = [1 0 0]; % Red
```





```
%h(2).FaceColor = [0 1 0]; % Green
%h(3).FaceColor = [0 0 1]; % Blue
%legend({'Red', 'Green', 'Blue'}, 'Location', 'northwest')

% Remove the image names from the x-axis
%set(gca, 'XTickLabel', {})

%%
% Save the RGB-HSV data to an Excel file
valueTable = table(imageNames, R, G, B, H, S, V, 'VariableNames', {'Image',
'Red', 'Green' ...
'Blue', 'Hue', 'Saturation', 'Value'});

writetable(valueTable, fullfile(folder, 'RGB_HSV_Values.xlsx'), 'FileType',
'spreadsheet');

disp('RGB and HSV values extracted and saved successfully!');

end

%%
function Fuzzy_logic(resultsMatrix) % Uses Fuzzy Logic on the Inputs
% Construct FIS
Fuzzy_Finale2 = mamfis(Name="Fuzzy_Finale2");

% Input 1
Fuzzy_Finale2 = addInput(Fuzzy_Finale2,[40 100],Name="Moisture Content");
Fuzzy_Finale2 = addMF(Fuzzy_Finale2,"Moisture Content","trapmf",[0 40 75 90], ...
    Name="Low",VariableType="input");
Fuzzy_Finale2 = addMF(Fuzzy_Finale2,"Moisture Content","trimf",[75 90 94], ...
    Name="Average",VariableType="input");
Fuzzy_Finale2 = addMF(Fuzzy_Finale2,"Moisture Content","trapmf",[90 94 100 125],
...
    Name="High",VariableType="input");

% Input 2
Fuzzy_Finale2 = addInput(Fuzzy_Finale2,[0.1 0.4],Name="Chlorophyll-A");
Fuzzy_Finale2 = addMF(Fuzzy_Finale2,"Chlorophyll-A","trapmf",[0 0.09 0.17 0.24],
...
    Name="Low",VariableType="input");
Fuzzy_Finale2 = addMF(Fuzzy_Finale2,"Chlorophyll-A","trimf",[0.17 0.24 0.368], ...
    Name="Moderate",VariableType="input");
```



```
Fuzzy_Finale2 = addMF(Fuzzy_Finale2,"Chlorophyll-A","trapmf",[0.24 0.368 0.4
0.45], ...
    Name="High",VariableType="input");

% Input 3
Fuzzy_Finale2 = addInput(Fuzzy_Finale2,[100 950],Name="Fresh Head Weight");
Fuzzy_Finale2 = addMF(Fuzzy_Finale2,"Fresh Head Weight","trapmf",[0 100 175
200], ...
    Name="Low",VariableType="input");
Fuzzy_Finale2 = addMF(Fuzzy_Finale2,"Fresh Head Weight","trimf",[150 200 450], ...
    Name="Average",VariableType="input");
Fuzzy_Finale2 = addMF(Fuzzy_Finale2,"Fresh Head Weight","trapmf",[200 450 950
1000], ...
    Name="High",VariableType="input");

% Output 1
Fuzzy_Finale2 = addOutput(Fuzzy_Finale2,[0 100],Name="Grading Level");
Fuzzy_Finale2 = addMF(Fuzzy_Finale2,"Grading Level","gaussmf",[17.69 4.441e-16],
...
    Name="L3",VariableType="output");
Fuzzy_Finale2 = addMF(Fuzzy_Finale2,"Grading Level","gaussmf",[17.69 50], ...
    Name="L2",VariableType="output");
Fuzzy_Finale2 = addMF(Fuzzy_Finale2,"Grading Level","gaussmf",[17.69 100], ...
    Name="L1",VariableType="output");

% Rules
Fuzzy_Finale2 = addRule(Fuzzy_Finale2,[1 1 1 1 1 1; ...
    1 1 2 1 1 1; ...
    1 1 3 1 1 1; ...
    2 1 1 1 1 1; ...
    2 1 2 2 1 1; ...
    2 1 3 2 1 1; ...
    3 1 1 1 1 1; ...
    3 1 2 2 1 1; ...
    3 1 3 2 1 1; ...
    1 2 1 1 1 1; ...
    1 2 2 2 1 1; ...
    1 2 3 2 1 1; ...
    2 2 1 2 1 1; ...
    2 2 2 3 1 1; ...
    2 2 3 3 1 1; ...
    3 2 1 2 1 1; ...
```



```
3 2 2 3 1 1; ...
3 2 3 3 1 1; ...
1 3 1 1 1 1; ...
1 3 2 3 1 1; ...
1 3 3 3 1 1; ...
2 3 1 3 1 1; ...
2 3 2 3 1 1; ...
2 3 3 3 1 1; ...
3 3 1 3 1 1; ...
3 3 2 3 1 1; ...
3 3 3 2 1 1]);

% Loop through each row of the resultsMatrix
for i = 1:size(resultsMatrix, 1)
    sample1 = resultsMatrix(i, :); % Get the required row of the matrix

    % Input values are rows from matrix
    inputValues = sample1; % [Moisture Chlorophyll Weight]
    output = evalfis(Fuzzy_Finale2, inputValues);

    varName = sprintf('output%d', i);
    assignin('base', varName, output); % Assigns var to the workspace

    % Display the output
    if output <= 33.33
        disp('Grade: L3');

    elseif (output > 33.33) && (output <= 66.67)
        disp('Grade: L2');

    elseif output > 66.6
        disp('Grade: L1');

    end
end
end

%%%
```



```
%% Opening conveyor
while true
    clear;
    a = arduino('COM3', 'Mega2560', 'Libraries', {'Servo', 'Ultrasonic'});
    % Pins for the ultrasonic sensor
    TrigPin = 'D2'; % Trigger pin connected to digital pin 2
    EchoPin = 'D3'; % Echo pin connected to digital pin 3
    RelayConveyorPin = 'D4'; % Relay pin connected to digital pin 4
    %RelayReleasePin = 'D5'; % Relay pin connected to digital pin 4

    % initializes all servos
    s1 = servo(a, 'D8', 'MinPulseDuration', 700e-6, 'MaxPulseDuration', 2300e-6);
    s2 = servo(a, 'D9', 'MinPulseDuration', 700e-6, 'MaxPulseDuration', 2300e-6);
    s3 = servo(a, 'D10', 'MinPulseDuration', 700e-6, 'MaxPulseDuration', 2300e-6);
    writePosition(s1, 0);
    writePosition(s2, 1);
    writePosition(s3, 0);

    clear sensor; % clears sensor in case it's already used

    % Creates pin for UTS
    sensor = ultrasonic(a, TrigPin, EchoPin);
    configurePin(a, RelayConveyorPin, 'DigitalOutput');

    % Continuously monitors the ultrasonic sensor for object detection
    while true

        distance = readDistance(sensor); % Measure distance in cm

        % Displays the measured distance (for troubleshoot)
        fprintf('Distance: %.2f cm\n', distance*100);

        % Checks for objects
        if distance*100 <= 29 % Object detected within 29 cm
            disp('Object Detected!');
            writeDigitalPin(a, RelayConveyorPin, 0); % Relay ON (activates the device)
            disp('Relay is on!');
            break;
        else
            disp('No Object Detected');
```



```
        writeDigitalPin(a, RelayConveyorPin, 1); % Relay OFF (deacts)
        disp('Relay is off');
    end

    % Pauses before the next measurement
    pause(0.05);
end
%%
% Create Arduino

%% Picture-taking
% Takes screenshot of the lettuces
Comp_vis();

%% Extraction
% Calls the extraction function
RGBHSV_extract();

% Folder Path
folderPath = 'C:\Users\RAF
Solutions\Documents\Romell\Thesis\MATLAB\Fuzzy\Lettuce_Image';

% Define file name and sheet name
filename = 'RGB_HSV_Values.xlsx';
sheetname = 'Sheet1';

% File Path
filePath = fullfile(folderPath, filename);

% Read data from the specified worksheet
data = readtable(filePath, 'Sheet', sheetname);

% Extract variables from the table
x1 = data.Red; % Red
x2 = data.Green; % Green
x3 = data.Blue; % Blue
x4 = data.Hue; % Hue
x5 = data.Saturation; % Saturation
x6 = data.Value; % Value

%% Moisture Content
% Compute - model 3237 50G50R
```



```
resultMoist = (...  
    0.408*x1 + 0.0193*x2 - 18.2*x4 - 18.2*x6 ...  
    - 1.78*sin((cos(x2) + 8.49).^2) + 3.26*cos(x3 - 1.0*log((x2 - 1.0*cos(x2)).^2).^2)  
...  
    + 1.83*cos(x2 - 4.6*x1) - 1.92*cos(x2 + 7.0*x1.*sin(x5)) ...  
    - 1.3*cos(49.0*x1.*x6.*sin(sin(x5))) + 7.05*cos(x3 - 1.0*log((cos(x2) +  
8.49).^2).^2) ...  
    + log(x2.^7.23e-6) + 11.0*cos(cos(x2 + 7.0*x1.*sin(x5))) + 42.6*cos(x2) ...  
    - 371.0*x1.*sin(sin(x5)) + 0.634*x2.*cos(x3 - 1.0*log((cos(x2) + 8.49).^2).^2) ...  
    - 0.703*x3.*cos(x3 - 1.0*log((cos(x2) + 8.49).^2).^2) ...  
    - 0.0248*x3.*cos(x3 - 1.0*sin(x5)) - 20.0*log((cos(x2) + 8.49).^2).^2 ...  
    + 368.0*x1.*sin(x5) - 0.0361*x2.*x6.*sin(x2) ...  
    - 8.76e-4*x2.*x3.*x5.*sin(x1 + cos(x2) - 7.0*x1.^2.*x6.*sin(x5) + 8.49) ...  
    - 5.06e-5*x1.*x3.*x4.*x6 - 1.02*x1.*x5.*sin(x1 + x3 + 2.32).*sin(x5) ...  
    + 434.0 ...  
);
```

%% Chlorophyll-A

```
resultChla = (...  
    0.0176 * x3 - 0.0252 * x1 + 0.0176 * x4 + 6.45 * x5 ...  
    + 0.0138 * sin(x2.^(1/2)) .* x4.^(1/2) .* (x1 + x4 + x5.^2).^(1/4)) ...  
    - 33.1 * cos(x1.^(1/2)) - 0.125 * cos(x3.^(1/2)) ...  
    - 0.0177 * cos(cos(x3)) ...  
    + 0.00851 * sin(x1.^(5/4)) .* cos(cos(x2)).^(1/2)) ...  
    + 0.0051 * sin(x1.^(1/2)) .* x2 .* x4 .* abs(x4)) ...  
    - 0.00666 * sin(3.0 * x3 .* x4 .* x5) ...  
    - 6.11 * cos(cos(x1.^(1/2))) ...  
    + 0.0215 * cos(x3 .* x4 .* x5 - 1.19) ...  
    - 66.6 * cos(cos(x1.^(1/2)).^(1/2)) ...  
    + 0.0441 * cos(cos(x2)).^(1/4)) ...  
    + 0.141 * x1.^(1/2) - 2.32 * x4.^(1/2) - 34.4 * x5.^2 + 344.0 * x5.^4 ...  
    + 0.0165 * x1.^(1/2) .* x4.^(1/2) .* (x1 + x4 + x5.^2).^(1/4)) ...  
    + 0.00114 * x1.^(1/2) .* x4.^(1/2) .* cos(x3) + 72.9 ...  
);
```

%% Weight

% Compute the expression - model id 917 50G50R

```
resultWeight = (...  
    2190.0 * x4 - 210.0 * x3 - 9480.0 * x6 ...  
    - 28.1 * sin(abs(x3 .* x6)) + 1570.0 * sin(sin(x3 .* x5)) ...  
    - 3940.0 * cos(sqrt(x5)) - 613.0 * sin(sin(x3)) ...  
);
```



```
+ 1970.0 .* cos(2.0 .* x5) - 35.0 .* sin(sin(31.3 .* abs(x1))) ...
- 219.0 .* sin(sin(31.3 .* abs(x6))) ...
- 1660.0 .* sin(sin(sin(x3 .* x5))) ...
- 8680.0 .* sin(sin(sqrt(x5))) ...
+ 19.7 .* sin(x1 + x4 + x5 + 2.0 .* sin(x3)) ...
- 3940.0 .* abs(x5) + 3000.0 .* cos(x5) ...
+ 674.0 .* sin(x3) + 5.91e4 .* sin(x6) ...
- 20.5 .* sin(cos(8.16 .* x2)) ...
- 8.53e4 .* (x1 + x3 + sin(x3)).^(1/4) ...
+ 3.72e4 .* (x1 + x4 + sin(x3)).^(1/4) ...
+ 1270.0 .* sqrt(x2) + 9480.0 .* sqrt(x3) ...
+ 9.76e4 ...
);

%% Switch to release
% Creates a matrix
resultsMatrix = [resultMoist, resultChla, resultWeight];
disp(resultsMatrix);
Fuzzy_logic(resultsMatrix);

%% Movement of actuators
% Restarting the conveyor staggered
% for debugging
% a = arduino('COM3', 'Mega2560', 'Libraries', {'Servo', 'Ultrasonic'});

%% Output 5
if (output5 > 33.33) && (output5 <= 66.67) && (output5 ~= 50)% For Level 2
    % Returns to Original Pos
    writePosition(s1, 0.25);
    current_Pos1 = readPosition(s1);
    writeDigitalPin(a, RelayConveyorPin, 1);
    pause(3.5);
    writeDigitalPin(a, RelayConveyorPin, 0);
    current_Pos1 = current_Pos1*180;
    fprintf('Current motor position is %d degrees\n', current_Pos1);

    % Stops the lettuce
    writePosition(s1, 0.25);
    current_Pos1 = readPosition(s1);
```





```
current_Pos1 = current_Pos1*180;
fprintf('Current motor position is %d degrees\n', current_Pos1);
pause(1);

% Returns to original pos
writePosition(s1, 0);
current_Pos1 = readPosition(s1)*180;
fprintf('Current motor position is %d degrees\n', current_Pos1);

elseif output5 <= 33% for level 3
% Returns to original pos
writePosition(s2, 0.75);
current_Pos2 = readPosition(s2);
current_Pos2 = current_Pos2*180;
fprintf('Current motor position is %d degrees\n', current_Pos2);
writeDigitalPin(a, RelayConveyorPin, 1);
pause(3.5);
writeDigitalPin(a, RelayConveyorPin, 0);

% Stops the lettuce
writePosition(s2, 0.75);
current_Pos2 = readPosition(s2);
current_Pos2 = current_Pos2*180;
fprintf('Current motor position is %d degrees\n', current_Pos2);
pause(2);

% Returns to original pos
writePosition(s2, 1);

elseif output5 > 66 % for level 1
% Returns to orig pos
writePosition(s3, 0.28);
current_Pos3 = readPosition(s3);
current_Pos3 = current_Pos3*180;
fprintf('Current motor position is %d degrees\n', current_Pos3);
writeDigitalPin(a, RelayConveyorPin, 1);
pause(3.5);
writeDigitalPin(a, RelayConveyorPin, 0);

% Stops the lettuce
writePosition(s3, 0.28);
current_Pos3 = readPosition(s3);
```



```
current_Pos3 = current_Pos3*180;
fprintf('Current motor position is %d degrees\n', current_Pos3);
pause(3.5);

% Returns to orig pos
writePosition(s3, 0);
current_Pos3 = readPosition(s3)*180;
fprintf('Current motor position is %d degrees\n', current_Pos3);

elseif output5 == 50 % for unidentified
    % Returns to original pos

    current_Pos2 = readPosition(s2);
    current_Pos2 = current_Pos2*180;
    fprintf('Current motor position is %d degrees\n', current_Pos2);
    writeDigitalPin(a, RelayConveyorPin, 1);
    pause(3.5);
    writeDigitalPin(a, RelayConveyorPin, 0);

    % Stops the lettuce

    current_Pos2 = readPosition(s2);
    current_Pos2 = current_Pos2*180;
    fprintf('Current motor position is %d degrees\n', current_Pos2);
    pause(5);

    % Returns to original pos
    writePosition(s2, 1);

end

pause(1);
%% Output 4
if (output4 > 33.33) && (output4 <= 66.67) && (output4 ~= 50)% For Level 2
    % Returns to Original Pos
    writePosition(s1, 0.25);
    current_Pos1 = readPosition(s1);
    writeDigitalPin(a, RelayConveyorPin, 1);
    pause(1.9);
    writeDigitalPin(a, RelayConveyorPin, 0);
    current_Pos1 = current_Pos1*180;
```



```
fprintf('Current motor position is %d degrees\n', current_Pos1);

% Stops the lettuce
writePosition(s1, 0.25);
current_Pos1 = readPosition(s1);
current_Pos1 = current_Pos1*180;
fprintf('Current motor position is %d degrees\n', current_Pos1);
pause(1);

% Returns to original pos
writePosition(s1, 0);
current_Pos1 = readPosition(s1)*180;
fprintf('Current motor position is %d degrees\n', current_Pos1);

elseif output4 <= 33 % for level 3
% Returns to original pos
writePosition(s2, 0.75);
current_Pos2 = readPosition(s2);
current_Pos2 = current_Pos2*180;
fprintf('Current motor position is %d degrees\n', current_Pos2);
writeDigitalPin(a, RelayConveyorPin, 1);
pause(1.9);
writeDigitalPin(a, RelayConveyorPin, 0);

% Stops the lettuce
writePosition(s2, 0.75);
current_Pos2 = readPosition(s2);
current_Pos2 = current_Pos2*180;
fprintf('Current motor position is %d degrees\n', current_Pos2);
pause(2);

% Returns to original pos
writePosition(s2, 1);

elseif output4 > 66 % for level 1
% Returns to orig pos
writePosition(s3, 0.28);
current_Pos3 = readPosition(s3);
current_Pos3 = current_Pos3*180;
fprintf('Current motor position is %d degrees\n', current_Pos3);
writeDigitalPin(a, RelayConveyorPin, 1);
pause(1.9);
```



```
writeDigitalPin(a, RelayConveyorPin, 0);

% Stops the lettuce
writePosition(s3, 0.28);
current_Pos3 = readPosition(s3);
current_Pos3 = current_Pos3*180;
fprintf('Current motor position is %d degrees\n', current_Pos3);
pause(3.5);

% Returns to orig pos
writePosition(s3, 0);
current_Pos3 = readPosition(s3)*180;
fprintf('Current motor position is %d degrees\n', current_Pos3);

elseif output4 == 50 % for unidentified
    % Returns to original pos

    current_Pos2 = readPosition(s2);
    current_Pos2 = current_Pos2*180;
    fprintf('Current motor position is %d degrees\n', current_Pos2);
    writeDigitalPin(a, RelayConveyorPin, 1);
    pause(1.9);
    writeDigitalPin(a, RelayConveyorPin, 0);

    % Stops the lettuce

    current_Pos2 = readPosition(s2);
    current_Pos2 = current_Pos2*180;
    fprintf('Current motor position is %d degrees\n', current_Pos2);
    pause(5);

    % Returns to original pos
    writePosition(s2, 1);

end

pause(1);
%% Output 3
if (output3 > 33.33) && (output3 <= 66.67) && (output3 ~= 50)% For Level 2
    % Returns to Original Pos
    writePosition(s1, 0.25);
```



```
current_Pos1 = readPosition(s1);
writeDigitalPin(a, RelayConveyorPin, 1);
pause(1.9);
writeDigitalPin(a, RelayConveyorPin, 0);
current_Pos1 = current_Pos1*180;
fprintf('Current motor position is %d degrees\n', current_Pos1);

% Stops the lettuce
writePosition(s1, 0.25);
current_Pos1 = readPosition(s1);
current_Pos1 = current_Pos1*180;
fprintf('Current motor position is %d degrees\n', current_Pos1);
pause(1);

% Returns to original pos
writePosition(s1, 0);
current_Pos1 = readPosition(s1)*180;
fprintf('Current motor position is %d degrees\n', current_Pos1);

elseif output3 <= 33 % for level 3
% Returns to original pos
writePosition(s2, 0.75);
current_Pos2 = readPosition(s2);
current_Pos2 = current_Pos2*180;
fprintf('Current motor position is %d degrees\n', current_Pos2);
writeDigitalPin(a, RelayConveyorPin, 1);
pause(1.9);
writeDigitalPin(a, RelayConveyorPin, 0);

% Stops the lettuce
writePosition(s2, 0.75);
current_Pos2 = readPosition(s2);
current_Pos2 = current_Pos2*180;
fprintf('Current motor position is %d degrees\n', current_Pos2);
pause(2);

% Returns to original pos
writePosition(s2, 1);

elseif output3 > 66 %output3 == 50 for level 1
% Returns to orig pos
writePosition(s3, 0.28);
```



```
current_Pos3 = readPosition(s3);
current_Pos3 = current_Pos3*180;
fprintf('Current motor position is %d degrees\n', current_Pos3);
writeDigitalPin(a, RelayConveyorPin, 1);
pause(1.9);
writeDigitalPin(a, RelayConveyorPin, 0);

% Stops the lettuce
writePosition(s3, 0.28);
current_Pos3 = readPosition(s3);
current_Pos3 = current_Pos3*180;
fprintf('Current motor position is %d degrees\n', current_Pos3);
pause(3.5);

% Returns to orig pos
writePosition(s3, 0);
current_Pos3 = readPosition(s3)*180;
fprintf('Current motor position is %d degrees\n', current_Pos3);

elseif output3 == 50 % for unidentified
    % Returns to original pos

    current_Pos2 = readPosition(s2);
    current_Pos2 = current_Pos2*180;
    fprintf('Current motor position is %d degrees\n', current_Pos2);
    writeDigitalPin(a, RelayConveyorPin, 1);
    pause(1.9);
    writeDigitalPin(a, RelayConveyorPin, 0);

    % Stops the lettuce

    current_Pos2 = readPosition(s2);
    current_Pos2 = current_Pos2*180;
    fprintf('Current motor position is %d degrees\n', current_Pos2);
    pause(4);

    % Returns to original pos
    writePosition(s2, 1);
end

pause(1);
%% Output 2
```



```
if (output2 > 33.33) && (output2 <= 66.67) && (output2 ~= 50)% For Level 2
    % Returns to Original Pos
    writePosition(s1, 0.25);
    current_Pos1 = readPosition(s1);
    writeDigitalPin(a, RelayConveyorPin, 1);
    pause(1.9);
    writeDigitalPin(a, RelayConveyorPin, 0);
    current_Pos1 = current_Pos1*180;
    fprintf('Current motor position is %d degrees\n', current_Pos1);

    % Stops the lettuce
    writePosition(s1, 0.25);
    current_Pos1 = readPosition(s1);
    current_Pos1 = current_Pos1*180;
    fprintf('Current motor position is %d degrees\n', current_Pos1);
    pause(1);

    % Returns to original pos
    writePosition(s1, 0);
    current_Pos1 = readPosition(s1)*180;
    fprintf('Current motor position is %d degrees\n', current_Pos1);

elseif output2 <= 33 % for level 3
    % Returns to original pos
    writePosition(s2, 0.75);
    current_Pos2 = readPosition(s2);
    current_Pos2 = current_Pos2*180;
    fprintf('Current motor position is %d degrees\n', current_Pos2);
    writeDigitalPin(a, RelayConveyorPin, 1);
    pause(1.9);
    writeDigitalPin(a, RelayConveyorPin, 0);

    % Stops the lettuce
    writePosition(s2, 0.75);
    current_Pos2 = readPosition(s2);
    current_Pos2 = current_Pos2*180;
    fprintf('Current motor position is %d degrees\n', current_Pos2);
    pause(2);

    % Returns to original pos
    writePosition(s2, 1);
```





```
elseif output2 > 66 % for level 1
    % Returns to orig pos
    writePosition(s3, 0.28);
    current_Pos3 = readPosition(s3);
    current_Pos3 = current_Pos3*180;
    fprintf('Current motor position is %d degrees\n', current_Pos3);
    writeDigitalPin(a, RelayConveyorPin, 1);
    pause(1.9);
    writeDigitalPin(a, RelayConveyorPin, 0);

    % Stops the lettuce
    writePosition(s3, 0.28);
    current_Pos3 = readPosition(s3);
    current_Pos3 = current_Pos3*180;
    fprintf('Current motor position is %d degrees\n', current_Pos3);
    pause(3.5);

    % Returns to orig pos
    writePosition(s3, 0);
    current_Pos3 = readPosition(s3)*180;
    fprintf('Current motor position is %d degrees\n', current_Pos3);

elseif output2 == 50 % for unidentified
    % Returns to original pos

    current_Pos2 = readPosition(s2);
    current_Pos2 = current_Pos2*180;
    fprintf('Current motor position is %d degrees\n', current_Pos2);
    writeDigitalPin(a, RelayConveyorPin, 1);
    pause(1.9);
    writeDigitalPin(a, RelayConveyorPin, 0);

    % Stops the lettuce

    current_Pos2 = readPosition(s2);
    current_Pos2 = current_Pos2*180;
    fprintf('Current motor position is %d degrees\n', current_Pos2);
    pause(5);

    % Returns to original pos
    writePosition(s2, 1);
```



```
end

pause(1);
%% Output 1
if (output1 > 33.33) && (output1 <= 66.67) && (output1 ~= 50) % For Level 2
    % Returns to Original Pos
    writePosition(s1, 0.25);
    current_Pos1 = readPosition(s1);
    writeDigitalPin(a, RelayConveyorPin, 1);
    pause(3);
    writeDigitalPin(a, RelayConveyorPin, 0);
    current_Pos1 = current_Pos1*180;
    fprintf('Current motor position is %d degrees\n', current_Pos1);

    % Stops the lettuce
    writePosition(s1, 0.25);
    current_Pos1 = readPosition(s1);
    current_Pos1 = current_Pos1*180;
    fprintf('Current motor position is %d degrees\n', current_Pos1);
    pause(1);

    % Returns to original pos
    writePosition(s1, 0);
    current_Pos1 = readPosition(s1)*180;
    fprintf('Current motor position is %d degrees\n', current_Pos1);

elseif output1 <= 33 % for level 3
    % Returns to original pos
    writePosition(s2, 0.75);
    current_Pos2 = readPosition(s2);
    current_Pos2 = current_Pos2*180;
    fprintf('Current motor position is %d degrees\n', current_Pos2);
    writeDigitalPin(a, RelayConveyorPin, 1);
    pause(3);
    writeDigitalPin(a, RelayConveyorPin, 0);

    % Stops the lettuce
    writePosition(s2, 0.75);
    current_Pos2 = readPosition(s2);
    current_Pos2 = current_Pos2*180;
```



```
fprintf('Current motor position is %d degrees\n', current_Pos2);
pause(2);

% Returns to original pos
writePosition(s2, 1);

elseif output1 > 66 % for level 1
    % Returns to orig pos
    writePosition(s3, 0.28);
    current_Pos3 = readPosition(s3);
    current_Pos3 = current_Pos3*180;
    fprintf('Current motor position is %d degrees\n', current_Pos3);
    writeDigitalPin(a, RelayConveyorPin, 1);
    pause(3);
    writeDigitalPin(a, RelayConveyorPin, 0);

    % Stops the lettuce
    writePosition(s3, 0.28);
    current_Pos3 = readPosition(s3);
    current_Pos3 = current_Pos3*180;
    fprintf('Current motor position is %d degrees\n', current_Pos3);
    pause(3.5);

    % Returns to orig pos
    writePosition(s3, 0);
    current_Pos3 = readPosition(s3)*180;
    fprintf('Current motor position is %d degrees\n', current_Pos3);

elseif output1 == 50 % for unidentified
    % Returns to original pos

    current_Pos2 = readPosition(s2);
    current_Pos2 = current_Pos2*180;
    fprintf('Current motor position is %d degrees\n', current_Pos2);
    writeDigitalPin(a, RelayConveyorPin, 1);
    pause(3);
    writeDigitalPin(a, RelayConveyorPin, 0);

    % Stops the lettuce

    current_Pos2 = readPosition(s2);
```



```
current_Pos2 = current_Pos2*180;
fprintf('Current motor position is %d degrees\n', current_Pos2);
pause(2);

% Returns to original pos
writePosition(s2, 1);
end
```

```
writeDigitalPin(a, RelayConveyorPin, 1); % Starts conveyor again

end
```

## GPTIPS MATLAB Code

```
% System initialization
clear; clc;
```

```
%-----
% LOAD DATA
%-----
```

```
% Load complete predictor and response data from the.xlsx file
dataset = readtable('Lettuce_Values.xlsx','Sheet','RGBHSV_moist');
morphdata = dataset(:,1:7);
```

```
% Load training data and split 80% from the total count
[m,n] = size(morphdata);
P = 0.56;
idx = randperm(m);
trainingdata = morphdata(idx(1:round(P*m)),:);
```

```
% Load testing data and split 20% from the total count
testdata = morphdata(idx(round(P*m)+1:end),:);
[k,l] = size(testdata);
Q = 0.45;
idz = randperm(k);
testingdata = testdata(idz(1:round(Q*k)),:);
```

```
% Load training data and split 80% from the total count
validationdata = trainingdata(idz(round(Q*k)+1:end),:);
```



```
% Separate the variables for test images and corresponding target value
% Splitting to predictors and response
% XTrain = table2array(trainingdata(:,1:6));
% YTrain = table2array(trainingdata(:,7));
%
% XVal = table2array(validationdata(:,1:6));
% YVal = table2array(validationdata(:,7));
%
% XTest = table2array(testingdata(:,1:6));
% YTest = table2array(testingdata(:,7));
XTrain = (trainingdata(:,1:6));
YTrain = (trainingdata(:,7));

XVal = (validationdata(:,1:6));
YVal = (validationdata(:,7));

XTest = (testingdata(:,1:6));
YTest = (testingdata(:,7));

function gp = config_file(gp)

%run control parameters
gp.runcontrol.pop_size = 70;
gp.runcontrol.num_gen = 50;
gp.runcontrol.timeout = 30;
gp.runcontrol.runs = 50;

%fitness
gp.fitness.fitfun = @regressmulti_fitfun;
gp.fitness.minimisation = true;

%selection
gp.selection.tournament.size = 50;
gp.selection.tournament.p_pareto = 0.05;
gp.selection.elite_fraction = 0.1;
gp.nodes.const.p_int= 0.5;

%load in the raw x and y data
load workspace.mat; %%file name
gp.userdata.xtrain = table2array(XTrain);
```



```
gp.userdata.ytrain = table2array(YTrain);
gp.userdata.xvalidation = table2array(XVal);
gp.userdata.yvalidation = table2array(YVal);
gp.userdata.xtest = table2array(XTest);
gp.userdata.ytest = table2array(YTest);
gp.nodes.inputs.num_inp = size(gp.userdata.xtrain,6);

gp.genes.multigene = true;

%genes
gp.genes.max_genes = 20;
gp.treedef.max_depth = 10;

%define function nodes
gp.nodes.functions.name =
{'times','minus','plus','sqrt','square','sin','cos','add3','mult3','log','cube','neg','abs'};

%%
% gp = rungp(@config_file)
% %summary(gp)
% %popbrowser(gp,'train')
% %gppretty(gp,'testbest')
% %paretoreport(gp,'testbest')
% %drawtrees(gp,27)
% %gpmodelreport(gp,1182)
% %runtree(gp,'testbest')
% %gpmodel2mfile(gp,364,'night')
```



## APPENDIX D. Operations Manual

| Specification          | Value                                                                                                     |
|------------------------|-----------------------------------------------------------------------------------------------------------|
| Speed                  | 53.25 m/min                                                                                               |
| Motor Power            | 60W                                                                                                       |
| Motor                  | Adjustable speed motor                                                                                    |
| Belt Length            | 118 inches (300 cm)                                                                                       |
| Belt Width             | 8 inches (20 cm)                                                                                          |
| Conveyor Length        | 59 inches (150 cm)                                                                                        |
| PVC Belt Width         | 7.8 inches (20 cm)                                                                                        |
| PVC Belt Thickness     | 0.07 inches (0.2 cm)                                                                                      |
| Conveyor Height        | 29.5 inches (75 cm)                                                                                       |
| Size                   | 63 x 18 x 12 inches (160 x 44 x 29 cm)                                                                    |
| Weight                 | 76 lb (22 kg)                                                                                             |
| Material               | Stainless steel + PVC                                                                                     |
| Usage                  | Packing, adding mosaic, etc.                                                                              |
| Application Industries | Food, beverage, pharmaceutical, electronic parts, tobacco, chemical industry, cosmetics, auto parts, etc. |

| Step | Action/Description                                                                                                              | Key Component(s)                                                                   | Step Documentation                                                                   |
|------|---------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| 1    | Manually place up to five iceberg lettuce heads evenly spaced on the conveyor belt based on markings seen in the conveyor belt. | Conveyor belt with position markings<br><br>Position markers for lettuce alignment |  |





|   |                                                                                                                                                                                                       |                                                                     |                                                                                       |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 2 | <p>As the conveyor runs, the ultrasonic sensor detects the presence of the first lettuce head.</p>                                                                                                    | <p>Ultrasonic sensor, NC relay, control box with logic circuit.</p> |     |
| 3 | <p>High-resolution cameras capture images of each lettuce head.</p> <p>Output: Each lettuce is assigned a grade or category.</p>                                                                      | <p>Cameras, MGRSGP model, fuzzy logic</p>                           |     |
| 4 | <p>Once grading is complete, the main conveyor resumes motion and forwards the lettuce to the sorting section.</p> <p>Guidance: Sorting positions are synchronized with the assigned grades.</p>      | <p>Conveyor system, sorting arms</p>                                |    |
| 5 | <p>At the appropriate moment, the system sends a signal to actuate a servo motor that controls sorting arms or diverters.</p>                                                                         | <p>Servo motor</p>                                                  |   |
| 6 | <p>Sorted lettuces are deposited into labeled sorting bins for users to get graded iceberg lettuces.</p> <p>Classification: Bins are organized by quality grade (e.g., Grade A, B, C, or Reject).</p> | <p>Sorting bins, crates for iceberg lettuce</p>                     |  |

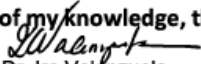


## APPENDIX E. Maintenance Manual

| Maintenance Task             | Frequency      | Description                                                               | Notes                                                                                          |
|------------------------------|----------------|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Oil-based Teflon Application | Weekly         | Apply oil-based Teflon to Conveyor 2 to reduce friction and wear.         | This helps to reduce friction and improve belt movement. Ensure even coverage on the PVC belt. |
| Motor Inspection             | Monthly        | Check the motor for wear and ensure it's functioning properly.            | Look for unusual noises or vibrations.                                                         |
| Belt Cleaning                | Weekly         | Clean the belt with a mild detergent and water.                           | Use a soft brush to prevent damaging the PVC.                                                  |
| Belt Tension Adjustment      | Every 3 months | Check and adjust the belt tension to ensure proper operation.             | Too much tension can cause motor strain.                                                       |
| Lubrication                  | Monthly        | Lubricate moving parts (e.g., pulleys, gears) with appropriate lubricant. | Use non-toxic lubricants for food industry usage.                                              |
| Check for Wear & Tear        | Monthly        | Inspect for any signs of wear or damage on the conveyor.                  | Replace any damaged parts promptly.                                                            |
| Conveyor Height Adjustment   | As needed      | Adjust the conveyor height to ensure ergonomic use.                       | Ensure the height is aligned with the workstation.                                             |



## APPENDIX F. Research Ethics Clearance From

| RESEARCH ETHICS CLEARANCE FORM <sup>1</sup><br>For Thesis Proposals                                                                                                                                                 |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>Names of Student Researcher(s):</b><br>Alberto, Alexi<br>Forones, Romell<br>Galang, Juan Gabriel<br>Magallanes, Fernando                                                                                         |  |
| <b>College:</b> GCOE                                                                                                                                                                                                |  |
| <b>Department:</b> MEM                                                                                                                                                                                              |  |
| <b>Course:</b> BS MEM-MRE                                                                                                                                                                                           |  |
| <b>Expected Duration of the Project:</b> from: March 2024 to:April 2025                                                                                                                                             |  |
| <b>Ethical considerations</b><br><br>(The <a href="#">Ethics Checklists</a> may be used as guides in determining areas for ethical concern/consideration)                                                           |  |
| <b>To the best of my knowledge, the ethical issues listed above have been addressed in the research.</b><br><br>Dr. Ira Valenzuela |  |
| <b>Name and Signature of Adviser/Mentor:</b><br><b>Date:</b>                                                                                                                                                        |  |
| <b>Noted by:</b><br><br><b>Name and Signature of the Department Chairperson:</b><br><b>Date:</b>                                                                                                                    |  |